

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

D8 MLB

LAST_MODIFIED=Mon Aug 27 13:24:34 2012

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
7	0001607307	ENGINEERING RELEASED	2012-08-28

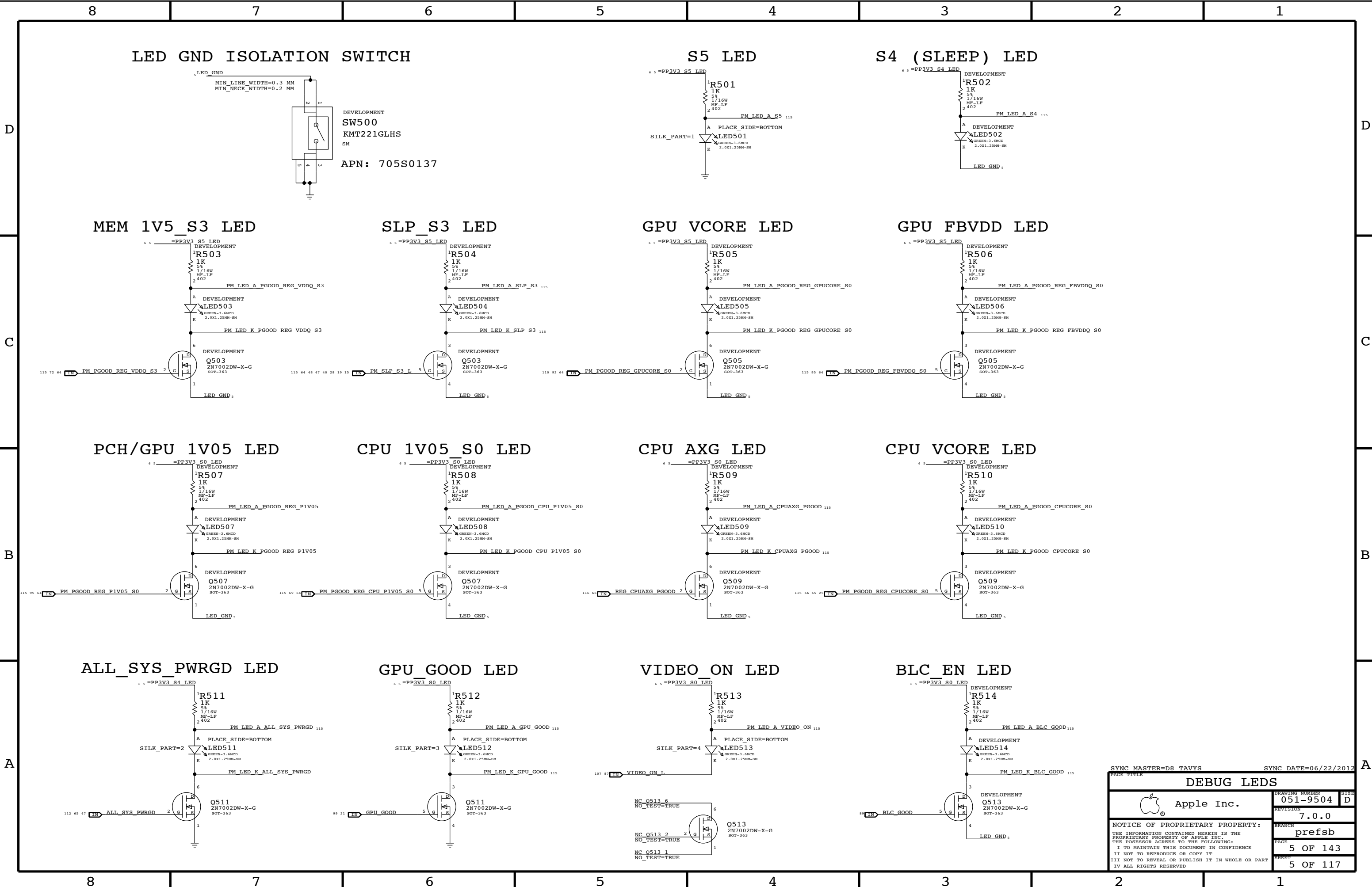
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3	Power Block Diagram	DB_HARK	04/23/2012
4	BOM Configuration	DB_MLB	12/19/2011
5	DEBUG LEDS	DB_TAVYS	06/22/2012
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8	Unused Signal Aliases	DB_MLB_ULTIMATE	04/02/2012
9	Signal Aliases	K70_MLB	08/23/2011
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11	CPU CLOCK/MISC/JTAG	D7_MLB	03/15/2012
12	CPU DDR3 INTERFACES	D7_MLB	03/15/2012
13	CPU POWER	D7_MLB	03/15/2012
14	CPU GROUNDS	D7_MLB	03/15/2012
15	STRAPS,PULL UPS,PULL DOWNS FOR PCH AND CPU	DB_MLB	03/29/2012
16	CPU NON-GFX DECOUPLING	D7_MLB	12/20/2011
17	GFX DECOUPLING & PCH PWR ALIAS	DB_KOSECOFF	01/26/2012
18	PCH SATA/PCIE/CLK/LPC/SPI	DB_MLB	N/A
19	PCH DMI/FDI/GRAPHICS	D7_MLB	03/15/2012
20	PCH PCI/USB	D7_MLB	03/15/2012
21	PCH MISC	DB_MLB	N/A
22	PCH POWER	DB_MLB	N/A
23	PCH GROUNDS	D7_MLB	03/15/2012
24	PCH DECOUPLING	DB_MLB	N/A
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26	CHIPSET SUPPORT	DB_MLB	N/A
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28	CPU Memory S3 Support	DB_HARK	04/23/2012
29	DDR3 SO-DIMM Connector A Slot0	DB_KOSECOFF	03/19/2012
30	DDR3 SO-DIMM Connector A Slot1	DB_KOSECOFF	03/19/2012
31	DDR3 SO-DIMM CONNECTOR B SLOT0	DB_KOSECOFF	03/19/2012
32	DDR3 SO-DIMM CONNECTOR B SLOT1	DB_KOSECOFF	03/19/2012
33	DDR3 ALIASES AND BITSWAPS	DB_KOSECOFF	03/19/2012
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35	AIRPORT/BT	DB_FYFIN	07/02/2012
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37	Thunderbolt Host (2 of 2)	D7_MLB	03/15/2012
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77	KEPLER FRAME BUFFER I/F	DB_YAN	05/15/2012
78	GPU SIGNAL & POWER ALIASES	DB_YAN	04/09/2012
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114	AUTO-CONSTRAINTS 3	DB_MARK	04/23/2012
115	AUTO-CONSTRAINTS 4	DB_MARK	04/23/2012
116	AUTO-CONSTRAINTS 5	DB_MARK	04/23/2012
117	AUTO-CONSTRAINTS 6	DB_MARK	04/23/2012

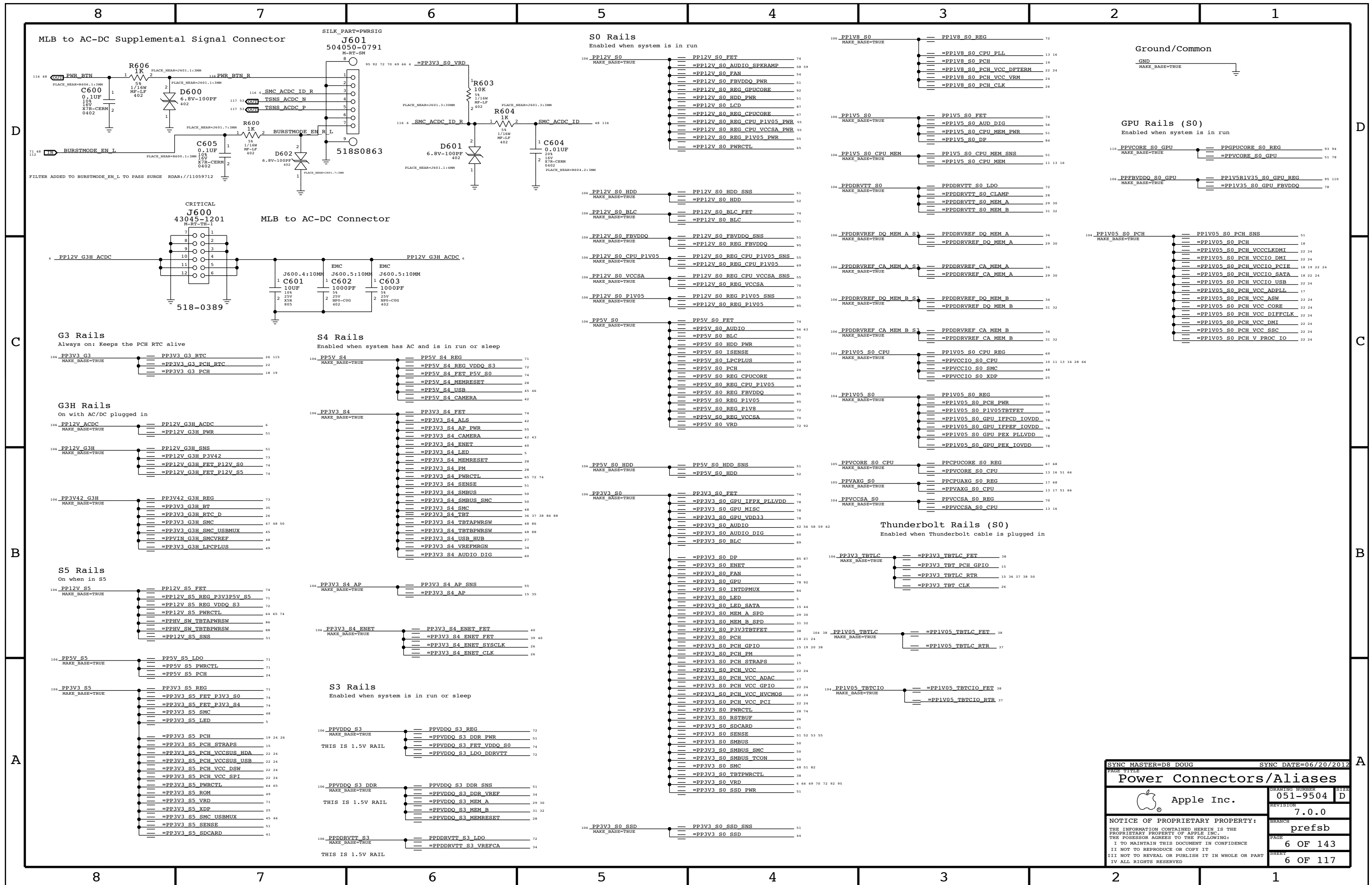
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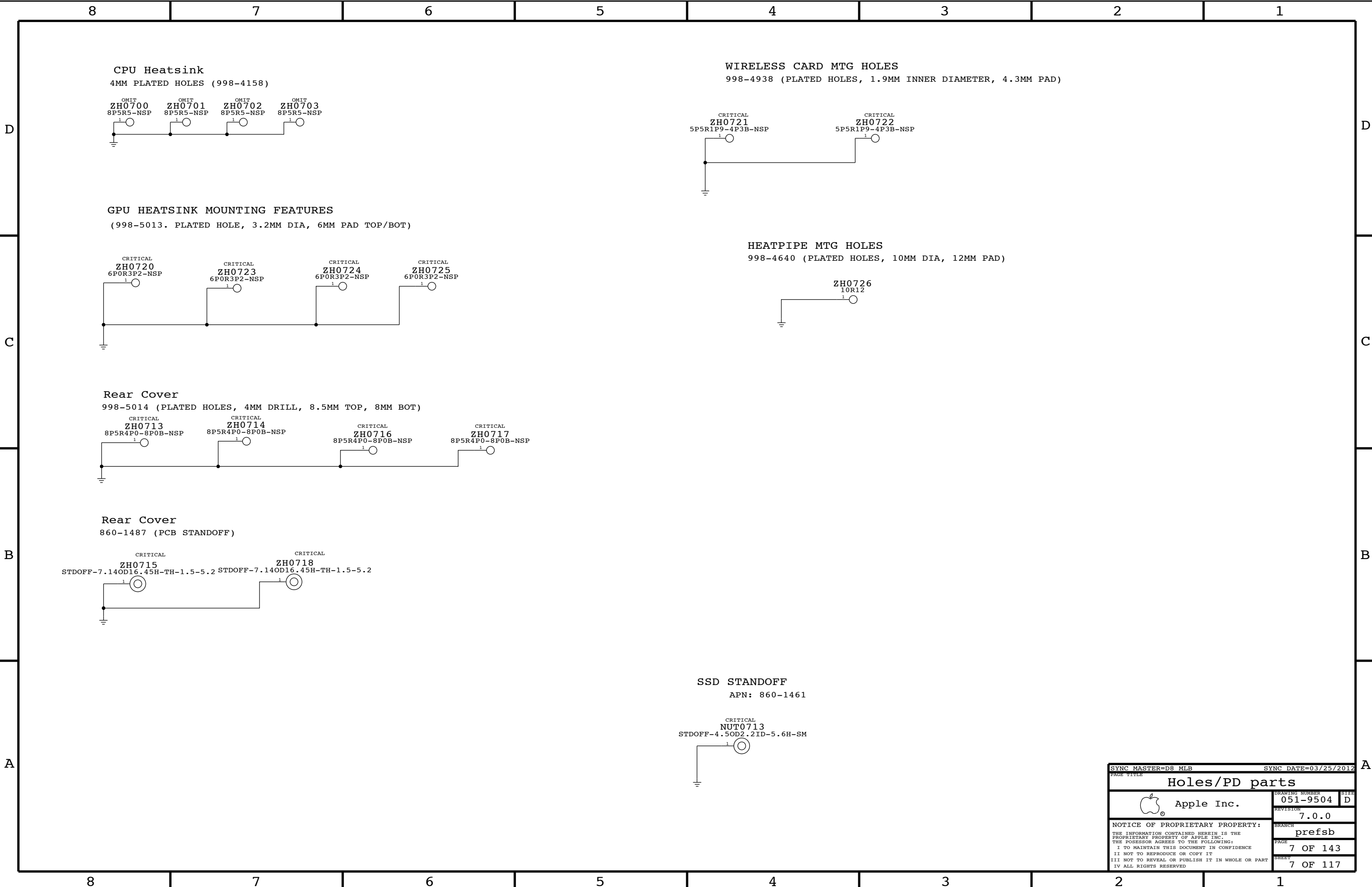
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D	BOM Variants																																																							
	<table><tr><th>BOM NUMBER</th><th colspan="2">BOM NAME</th><th colspan="5">BOM OPTIONS</th></tr><tr><td>639-3816</td><td colspan="2">PCBA,MLB,2.9G,4C,GK107,SAM,D8</td><td colspan="5">D8_COMMON,D8,CPU:4C_2P9GHZ,GPU:107EGE,FB:BOTH_SAMSUNG,EEEE:F2FR</td></tr><tr><td>639-3952</td><td colspan="2">PCBA,MLB,3.1G,4C,GK107,SAM,D8</td><td colspan="5">D8_COMMON,D8,CPU:4C_3P1GHZ,GPU:107EGE,FB:BOTH_SAMSUNG,EEEE:F49T</td></tr><tr><td>639-4092</td><td colspan="2">PCBA,MLB,2.9G,4C,GK107,HYN.D8</td><td colspan="5">D8_COMMON,D8,CPU:4C_2P9GHZ,GPU:107EGE,FB:BOTH_HYNIX,EEEE:F653</td></tr><tr><td>639-4093</td><td colspan="2">PCBA,MLB,3.1G,4C,GK107,HYN.D8</td><td colspan="5">D8_COMMON,D8,CPU:4C_3P1GHZ,GPU:107EGE,FB:BOTH_HYNIX,EEEE:F654</td></tr><tr><td>085-4433</td><td colspan="2">PCBA,MLB,DEV,D8</td><td colspan="5">DEVELOPMENT,D8_DEVEL</td></tr></table>								BOM NUMBER	BOM NAME		BOM OPTIONS					639-3816	PCBA,MLB,2.9G,4C,GK107,SAM,D8		D8_COMMON,D8,CPU:4C_2P9GHZ,GPU:107EGE,FB:BOTH_SAMSUNG,EEEE:F2FR					639-3952	PCBA,MLB,3.1G,4C,GK107,SAM,D8		D8_COMMON,D8,CPU:4C_3P1GHZ,GPU:107EGE,FB:BOTH_SAMSUNG,EEEE:F49T					639-4092	PCBA,MLB,2.9G,4C,GK107,HYN.D8		D8_COMMON,D8,CPU:4C_2P9GHZ,GPU:107EGE,FB:BOTH_HYNIX,EEEE:F653					639-4093	PCBA,MLB,3.1G,4C,GK107,HYN.D8		D8_COMMON,D8,CPU:4C_3P1GHZ,GPU:107EGE,FB:BOTH_HYNIX,EEEE:F654					085-4433	PCBA,MLB,DEV,D8		DEVELOPMENT,D8_DEVEL				
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C	Bar Code Labels / EEEE #'s																																																							
	<table><tr><th>PART#</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DESIGNATOR(S)</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>825-7896</td><td>1</td><td>LABEL,MLB,2D</td><td>EEEE_DHNM</td><td>CRITICAL</td><td>EEEE:DHNM</td></tr><tr><td>825-7896</td><td>1</td><td>LABEL,MLB,2D</td><td>EEEE_F2FR</td><td>CRITICAL</td><td>EEEE:F2FR</td></tr><tr><td>825-7896</td><td>1</td><td>LABEL,MLB,2D</td><td>EEEE_F49V</td><td>CRITICAL</td><td>EEEE:F49V</td></tr><tr><td>825-7896</td><td>1</td><td>LABEL,MLB,2D</td><td>EEEE_F49T</td><td>CRITICAL</td><td>EEEE:F49T</td></tr><tr><td>825-7896</td><td>1</td><td>LABEL,MLB,2D</td><td>EEEE_F653</td><td>CRITICAL</td><td>EEEE:F653</td></tr><tr><td>825-7896</td><td>1</td><td>LABEL,MLB,2D</td><td>EEEE_F654</td><td>CRITICAL</td><td>EEEE:F654</td></tr></table>								PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION	825-7896	1	LABEL,MLB,2D	EEEE_DHNM	CRITICAL	EEEE:DHNM	825-7896	1	LABEL,MLB,2D	EEEE_F2FR	CRITICAL	EEEE:F2FR	825-7896	1	LABEL,MLB,2D	EEEE_F49V	CRITICAL	EEEE:F49V	825-7896	1	LABEL,MLB,2D	EEEE_F49T	CRITICAL	EEEE:F49T	825-7896	1	LABEL,MLB,2D	EEEE_F653	CRITICAL	EEEE:F653	825-7896	1	LABEL,MLB,2D	EEEE_F654	CRITICAL	EEEE:F654						
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B	BOM Groups																																																							
	<table><tr><th>BOM GROUP</th><th colspan="7">BOM OPTIONS</th></tr><tr><td>D8_COMMON</td><td colspan="7">COMMON,ALTERNATE,D8_COMMON1,D8_PROGPARTS,D8_PRODUCTION</td></tr><tr><td>D8_COMMON1</td><td colspan="7">XDP,RSMRST:GATE,SPEAKERID,VREF:CPU,TBTHV:P12V,FBA,FBB</td></tr><tr><td>D8_PROGPARTS</td><td colspan="7">SMC:PROG,BOOTROM:PROG,TBTROM:PROG,CIVROM:PROG,CAMROM:PROG,BLCMCU:PROG</td></tr><tr><td>D8_DEVEL</td><td colspan="7">XDP_CONN,LPCPLUS,VREFMRGN:EXT,DEVEL_AUDIO,TEMPSNSDEV</td></tr><tr><td>D8_PRODUCTION</td><td colspan="7">VREFMRGN:N,PRODUCTION</td></tr></table>								BOM GROUP	BOM OPTIONS							D8_COMMON	COMMON,ALTERNATE,D8_COMMON1,D8_PROGPARTS,D8_PRODUCTION							D8_COMMON1	XDP,RSMRST:GATE,SPEAKERID,VREF:CPU,TBTHV:P12V,FBA,FBB							D8_PROGPARTS	SMC:PROG,BOOTROM:PROG,TBTROM:PROG,CIVROM:PROG,CAMROM:PROG,BLCMCU:PROG							D8_DEVEL	XDP_CONN,LPCPLUS,VREFMRGN:EXT,DEVEL_AUDIO,TEMPSNSDEV							D8_PRODUCTION	VREFMRGN:N,PRODUCTION						
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A	CPUs																																																							
	<table><tr><th>PART#</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DESIGNATOR(S)</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>337S4355</td><td>1</td><td>CVB,SR07A,PRO,N1,2.9,65W,4+1,1.1,6M,LGA</td><td>CPU</td><td>CRITICAL</td><td>CPU:4C_2P9GHZ</td></tr><tr><td>337S4372</td><td>1</td><td>CVB,SR079,PRO,N0,3.1,65W,4+1,1.1,6M,LGA</td><td>CPU</td><td>CRITICAL</td><td>CPU:4C_3P1GHZ</td></tr></table>								PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION	337S4355	1	CVB,SR07A,PRO,N1,2.9,65W,4+1,1.1,6M,LGA	CPU	CRITICAL	CPU:4C_2P9GHZ	337S4372	1	CVB,SR079,PRO,N0,3.1,65W,4+1,1.1,6M,LGA	CPU	CRITICAL	CPU:4C_3P1GHZ																														
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SYNC MASTER=D8 TAVVS		SYNC DATE=06/22/2012	
PAGE TITLE			
DEBUG LEDS			
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CPU Reserved		PCH PCIE		PCH Unused Display		PCH SATA		PCH Test Points	
10 TP CPU RSVD<16..1> == NC CPU RSVD<16..1> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCIE1 D2RN == NC PCIE1 D2RN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP CRT IG RED == NC CRT IG RED MAKE_BASE=TRUE NO_TEST=TRUE		10 TP SATA C R2D CN == NC SATA C R2D CNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP1 == NC PCH TP1 MAKE_BASE=TRUE NO_TEST=TRUE	
10 TP CPU RSVD<46..19> == NC CPU RSVD<46..19> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCIE1 D2RP == NC PCIE1 D2RPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP CRT IG GREEN == NC CRT IG GREEN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP SATA C R2D CP == NC SATA C R2D CPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP2 == NC PCH TP2 MAKE_BASE=TRUE NO_TEST=TRUE	
10 CPU CFG<15..12> == TP CPU CFG<15..12> MAKE_BASE=TRUE		10 TP PCIE1 R2D CN == NC PCIE1 R2D CNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP CRT IG BLUE == NC CRT IG BLUE MAKE_BASE=TRUE NO_TEST=TRUE		10 TP SATA C D2RN == NC SATA C D2RN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP3 == NC PCH TP3 MAKE_BASE=TRUE NO_TEST=TRUE	
CPU Memory		10 TP PCIE1 R2D CP == NC PCIE1 R2D CPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP CRT IG HSYNC == NC CRT IG HSYNC MAKE_BASE=TRUE NO_TEST=TRUE		10 TP SATA C D2RP == NC SATA C D2RPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP4 == NC PCH TP4 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP MEM A DO CB<7..0> == NC MEM A DO CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCIE2 D2RN == NC PCIE2 D2RN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP CRT IG VSYNC == NC CRT IG VSYNC MAKE_BASE=TRUE NO_TEST=TRUE		10 TP SATA D R2D CN == NC SATA D R2D CNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP5 == NC PCH TP5 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP MEM A DOS N<8> == NC MEM A DOSN<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCIE2 D2RP == NC PCIE2 D2RPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP CRT IG DDC CLK == NC CRT IG DDC CLK MAKE_BASE=TRUE NO_TEST=TRUE		10 TP SATA D R2D CP == NC SATA D R2D CPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP6 == NC PCH TP6 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP MEM A DOS P<8> == NC MEM A DOSPX<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCIE2 R2D CN == NC PCIE2 R2D CNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP CRT IG DDC DATA == NC CRT IG DDC DATA MAKE_BASE=TRUE NO_TEST=TRUE		10 TP SATA D D2RN == NC SATA D D2RN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP7 == NC PCH TP7 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP MEM B DO CB<7..0> == NC MEM B DO CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCIE2 R2D CP == NC PCIE2 R2D PNX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG B MLN<3..0> == NC DP IG B MLNX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP SATA D D2RP == NC SATA D D2RPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP8 == NC PCH TP8 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP MEM B DOS N<8> == NC MEM B DOSN<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 DMI MIDBUS CLK100M_N == NC DMI MIDBUS CLK100NX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG B MLP<3..0> == NC DP IG B MLPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP SATA E R2D CN == NC SATA E R2D CNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP9 == NC PCH TP9 MAKE_BASE=TRUE NO_TEST=TRUE	
12 TP MEM B DOS P<8> == NC MEM B DOSPX<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 DMI MIDBUS CLK100M_P == NC DMI MIDBUS CLK100PX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG B AUX N == NC DP IG B AUXNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP SATA E R2D CP == NC SATA E R2D CPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP10 == NC PCH TP10 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP PCIE CLK100M PE0N == NC PCIE CLK100M PE0NX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG B AUX P == NC DP IG B AUXPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP SATA E D2RN == NC SATA E D2RN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP11 == NC PCH TP11 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP PCIE CLK100M PE0P == NC PCIE CLK100M PE0PX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG B HPD == NC DP IG B HPDX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP SATA E D2RP == NC SATA E D2RPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP12 == NC PCH TP12 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP PCIE CLK100M PE4N == NC PCIE CLK100M PE4NX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG B DDC CLK == NC DP IG B DDC CLK MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP13 == NC PCH TP13 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP13 == NC PCH TP13 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP PCIE CLK100M PE4P == NC PCIE CLK100M PE4PX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG B DDC DATA == NC DP IG B DDC DATA MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP14 == NC PCH TP14 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP14 == NC PCH TP14 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP PCIE CLK100M PE5N == NC PCIE CLK100M PE5NX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG C MLN<3..0> == NC DP IG C MLNX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP15 == NC PCH TP15 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP15 == NC PCH TP15 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP PCIE CLK100M PE5P == NC PCIE CLK100M PE5PX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG C MLP<3..0> == NC DP IG C MLPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP16 == NC PCH TP16 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP16 == NC PCH TP16 MAKE_BASE=TRUE NO_TEST=TRUE	
		21 TP PCIE CLK100M PE6N == NC PCIE CLK100M PE6NX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG C AUX N == NC DP IG C AUXNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP17 == NC PCH TP17 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP17 == NC PCH TP17 MAKE_BASE=TRUE NO_TEST=TRUE	
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		21 TP PCIE CLK100M PE7N == NC PCIE CLK100M PE7NX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG C HPD == NC DP IG C HPDX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP19 == NC PCH TP19 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP19 == NC PCH TP19 MAKE_BASE=TRUE NO_TEST=TRUE	
		21 TP PCIE CLK100M PE7P == NC PCIE CLK100M PE7PX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG C CTRL CLK == NC DP IG C CTRL CLK MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP20 == NC PCH TP20 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH TP20 == NC PCH TP20 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP PE TX N<3..0> == NC PE TXNX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG D MLN<3..0> == NC DP IG D MLNX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		PCH Reserved		PCH PCI	
		10 TP PE TX P<3..0> == NC PE TPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG D MLP<3..0> == NC DP IG D MLPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH RESERVE 0 == NC PCH RESERVE 0 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCI AD<31..0> == NC PCI AD<31..0> MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP PE RX N<3..0> == NC PE RXNX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG D AUXN == NC DP IG D AUXNX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH RESERVE 1 == NC PCH RESERVE 1 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCI C BE L<3..0> == NC PCI C BE L<3..0> MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP PE RX P<3..0> == NC PE RPX<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG D AUXP == NC DP IG D AUXPX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH RESERVE 2 == NC PCH RESERVE 2 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCI PAR == NC PCI PAR MAKE_BASE=TRUE NO_TEST=TRUE	
		PCH USB		10 DP IG D HPD == NC DP IG D HPDX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH RESERVE 3 == NC PCH RESERVE 3 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCI RESET L == NC PCI RESET L MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_4_N == NC USB_PCH_4NX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG D CTRL CLK == NC DP IG D CTRL CLK MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH RESERVE 4 == NC PCH RESERVE 4 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH PCI GNT0 L == NC PCH PCI GNT0 L MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_4_P == NC USB_PCH_4PX MAKE_BASE=TRUE NO_TEST=TRUE		10 DP IG D CTRL DATA == NC DP IG D CTRL DATA MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH RESERVE 5 == NC PCH RESERVE 5 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH INIT3V3 L == NC PCH INIT3V3 L MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_5_N == NC USB_PCH_5NX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_TVCLKINN == NC_SDVO_TVCLKINN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH RESERVE 6 == NC PCH RESERVE 6 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP LPC DREQ0 L == NC LPC DREQ0 L MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_5_P == NC USB_PCH_5PX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_TVCLKINP == NC_SDVO_TVCLKINP MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH RESERVE 7 == NC PCH RESERVE 7 MAKE_BASE=TRUE NO_TEST=TRUE		PCH Miscellaneous	
		20 USB_PCH_6_N == NC USB_PCH_6NX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SDVO_STALLN == NC_SDVO_STALLN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH RESERVE 8 == NC PCH RESERVE 8 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_HDA_SDIN1 == NC_HDA_SDIN1 MAKE_BASE=TRUE NO_TEST=TRUE	
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		20 USB_PCH_12_N == NC USB_PCH_12NX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_L_BKLTCTL == NC_PCH_L_BKLTCTL MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH RESERVE 12 == NC PCH RESERVE 12 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_PWM1 == NC_PCH_PWM1 MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_12_P == NC USB_PCH_12PX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_L_BKLTEN == NC_PCH_L_BKLTEN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH RESERVE 13 == NC PCH RESERVE 13 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_PWM2 == NC_PCH_PWM2 MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_13_N == NC USB_PCH_13NX MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_L_VDD_EN == NC_PCH_L_VDD_EN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP PCH RESERVE 14 == NC PCH RESERVE 14 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_PWM3 == NC_PCH_PWM3 MAKE_BASE=TRUE NO_TEST=TRUE	
		20 USB_PCH_13_P == NC USB_PCH_13PX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP PCH RESERVE 15 == NC PCH RESERVE 15 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_SST == NC_PCH_SST MAKE_BASE=TRUE NO_TEST=TRUE	
		PCH Clocks				10 TP_PCH_RESERVE_16 == NC_PCH_RESERVE_16 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_CL_CLK1 == NC_PCH_CL_CLK1 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCH_CLKOUT_DPN == NC_PCH_CLKOUT_DPNX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_17 == NC_PCH_RESERVE_17 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_CL_DATA1 == NC_PCH_CL_DATA1 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCH_CLKOUT_DPP == NC_PCH_CLKOUT_DPPX MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_18 == NC_PCH_RESERVE_18 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCH_CL_RST1 == NC_PCH_CL_RST1 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 PCH_CLK25M_XTALOUT == NC_PCH_CLK25M_XTALOUT MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_19 == NC_PCH_RESERVE_19 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCI_CLK33M_OUT2 == NC_PCI_CLK33M_OUT2 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCH_GPIO64_CLKOUTFLEX0 == NC_PCH_GPIO64_CLKOUTFLEX0 MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_20 == NC_PCH_RESERVE_20 MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCI_CLK33M_OUT3 == NC_PCI_CLK33M_OUT3 MAKE_BASE=TRUE NO_TEST=TRUE	
		10 TP_PCH_GPIO65_CLKOUTFLEX1 == NC_PCH_GPIO65_CLKOUTFLEX1 MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_21 == NC_PCH_RESERVE_21 MAKE_BASE=TRUE NO_TEST=TRUE			
		10 TP_PCH_GPIO66_CLKOUTFLEX2 == NC_PCH_GPIO66_CLKOUTFLEX2 MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_22 == NC_PCH_RESERVE_22 MAKE_BASE=TRUE NO_TEST=TRUE			
		10 TP_PCH_GPIO67_CLKOUTFLEX3 == NC_PCH_GPIO67_CLKOUTFLEX3 MAKE_BASE=TRUE NO_TEST=TRUE				10 TP_PCH_RESERVE_23 == NC_PCH_RESERVE_23 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_24 == NC_PCH_RESERVE_24 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_25 == NC_PCH_RESERVE_25 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_26 == NC_PCH_RESERVE_26 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_27 == NC_PCH_RESERVE_27 MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_PCH_RESERVE_28 == NC_PCH_RESERVE_28 MAKE_BASE=TRUE NO_TEST=TRUE			
						PCH and CPU FDI			
						10 TP_CPU_FDI_TX_N<7..0> == NC_CPU_FDI_TXNX<7..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_CPU_FDI_TX_P<7..0> == NC_CPU_FDI_TPX<7..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 PCH_FDI_RX_N<7..0> == NC_PCH_FDI_RXNX<7..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 PCH_FDI_RX_P<7..0> == NC_PCH_FDI_RPX<7..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_CPU_FDI_FSYNC<1..0> == NC_CPU_FDI_FSYNC<1..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_CPU_FDI_LSYNC<1..0> == NC_CPU_FDI_LSYNC<1..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 TP_CPU_FDI_INT == NC_CPU_FDI_INT MAKE_BASE=TRUE NO_TEST=TRUE			
						10 PCH_FDI_FSYNC<1..0> == NC_PCH_FDI_FSYNC<1..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 PCH_FDI_LSYNC<1..0> == NC_PCH_FDI_LSYNC<1..0> MAKE_BASE=TRUE NO_TEST=TRUE			
						10 PCH_FDI_INT == NC_PCH_FDI_INT MAKE_BASE=TRUE NO_TEST=TRUE			
8	7	6	5	4	3	2	1		

SYNC MASTER=D8 MLB ULTIMATE

SYNC DATE=04/02/2012

Unused Signal Aliases

Apple Inc.

051-9504

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7.0.0

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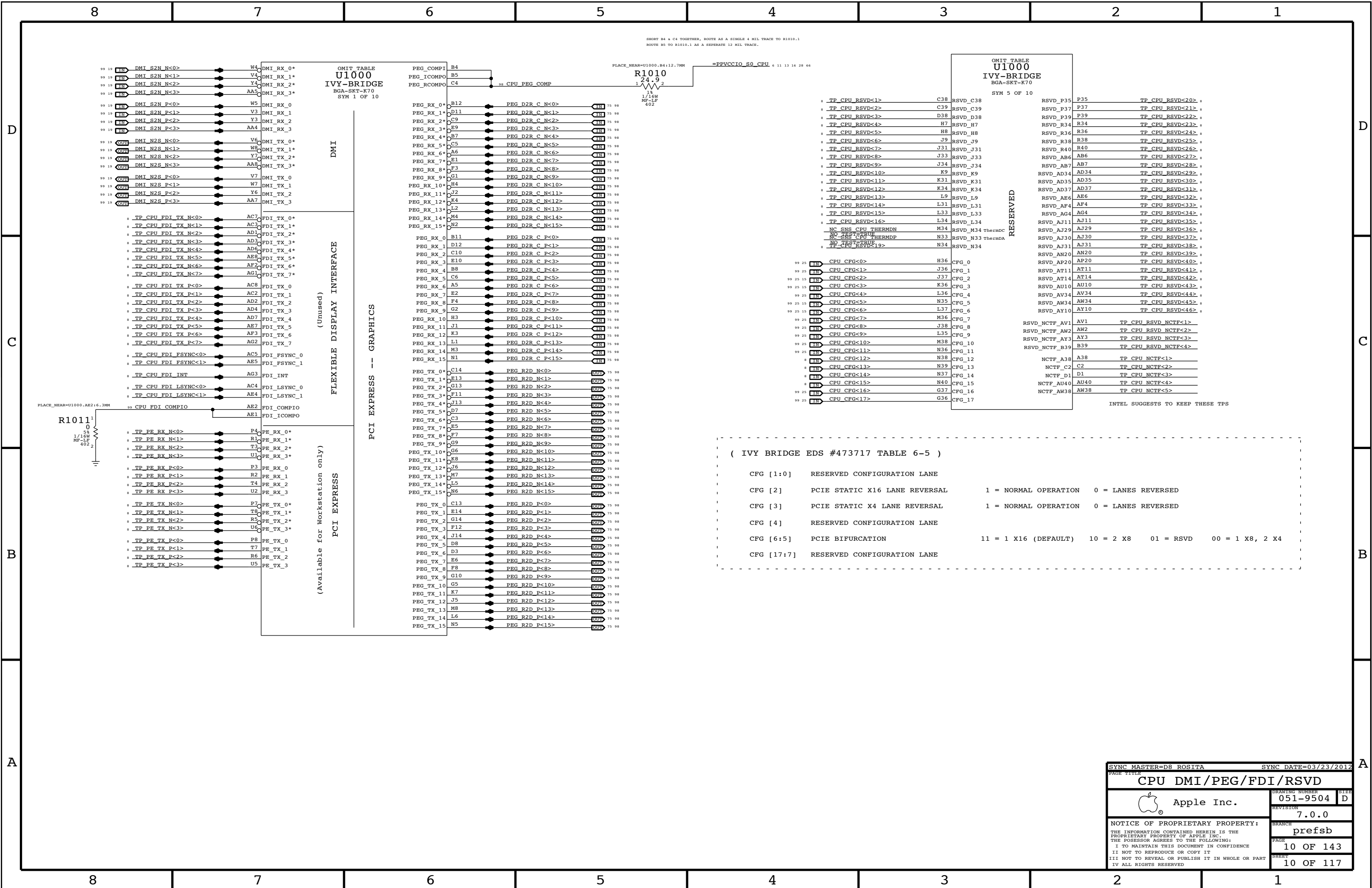
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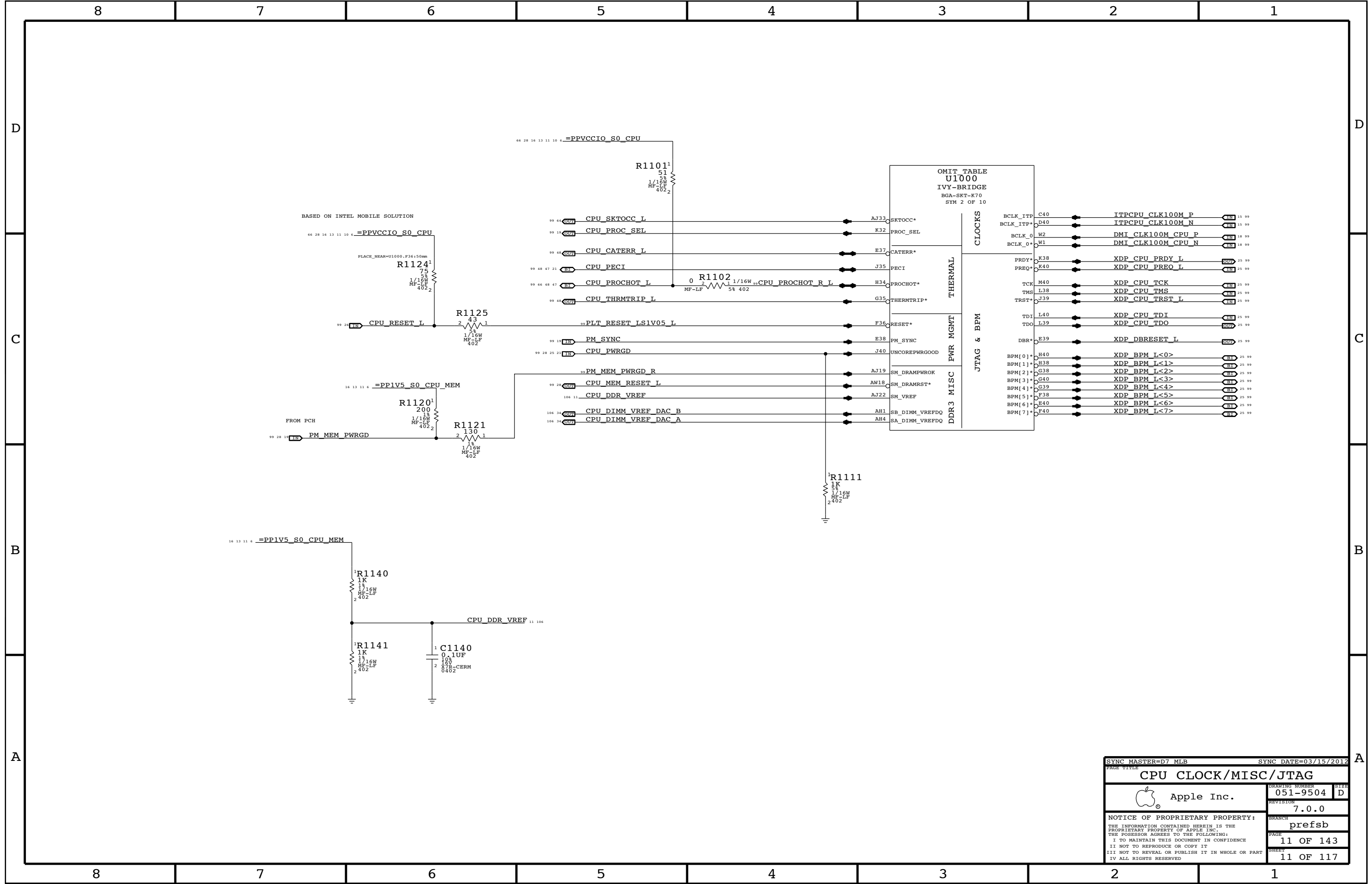
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D		ALIASES (BLANK)														D	
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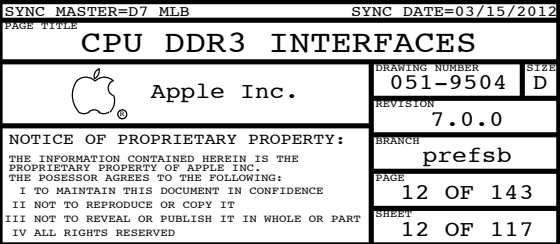
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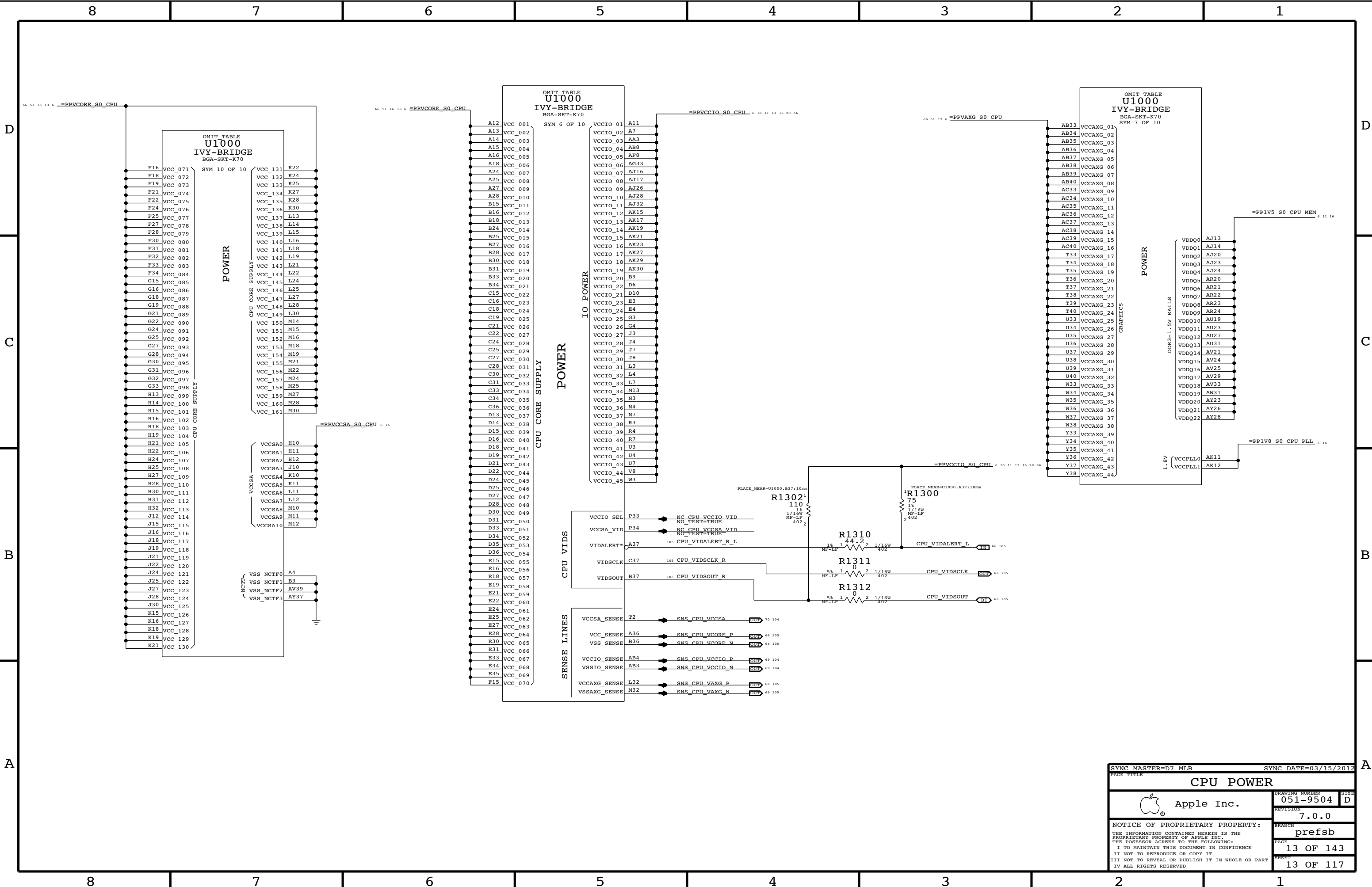
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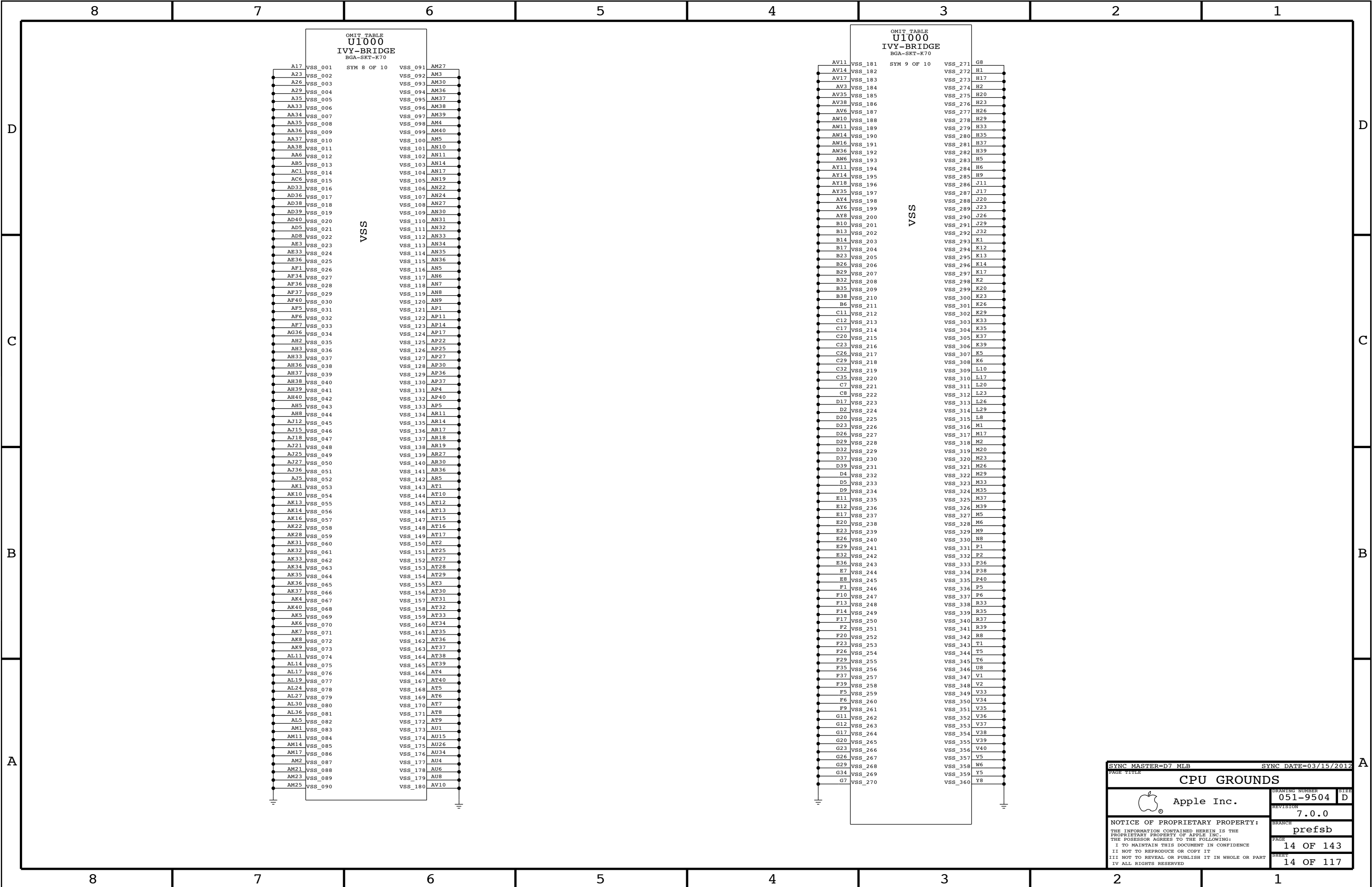
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SYNC MASTER=D7 MLB		SYNC DATE=03/15/2012	
PAGE TITLE			
CPU GROUNDS		DRAWING NUMBER	SIZE
 Apple Inc.		051-9504	D
		REVISION	
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		PAGE	14 OF 117
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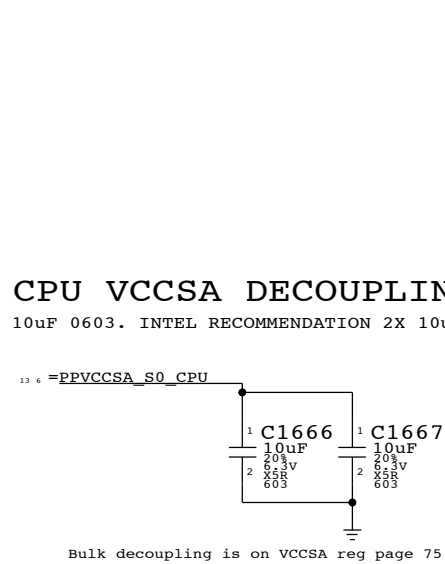
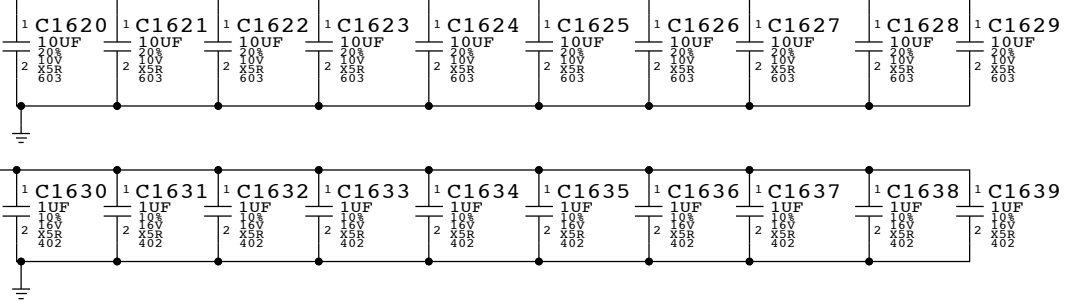
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10x 10UF and 10x 1UF CAPACITORS

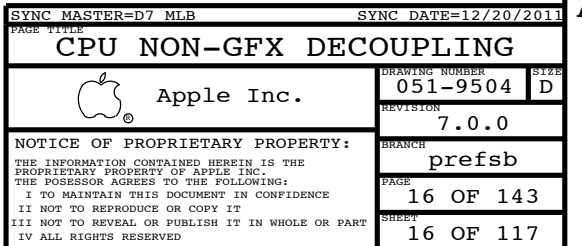
REPLACED WITH 603 PER RDAR://10700439

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66 51 16 13 6 =PPVCORE_S0_CPU
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Place inside socket cavity

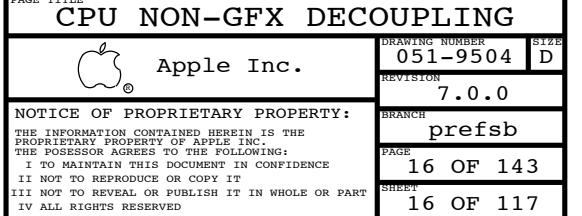


Memory (CPU VCCDDR) DECOUPLING



2x 47uF, 1x 22uF 0805, 1x 10uF 0603, 1x 4.7uF 0603, 1x 2.2uF 0402, 2x 1uF 0402. INTEL RECOMMENDATION 1x 10uF 0805

SYNC MASTER=D7 MLB	SYNC DATE=12/20/2011
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D

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C

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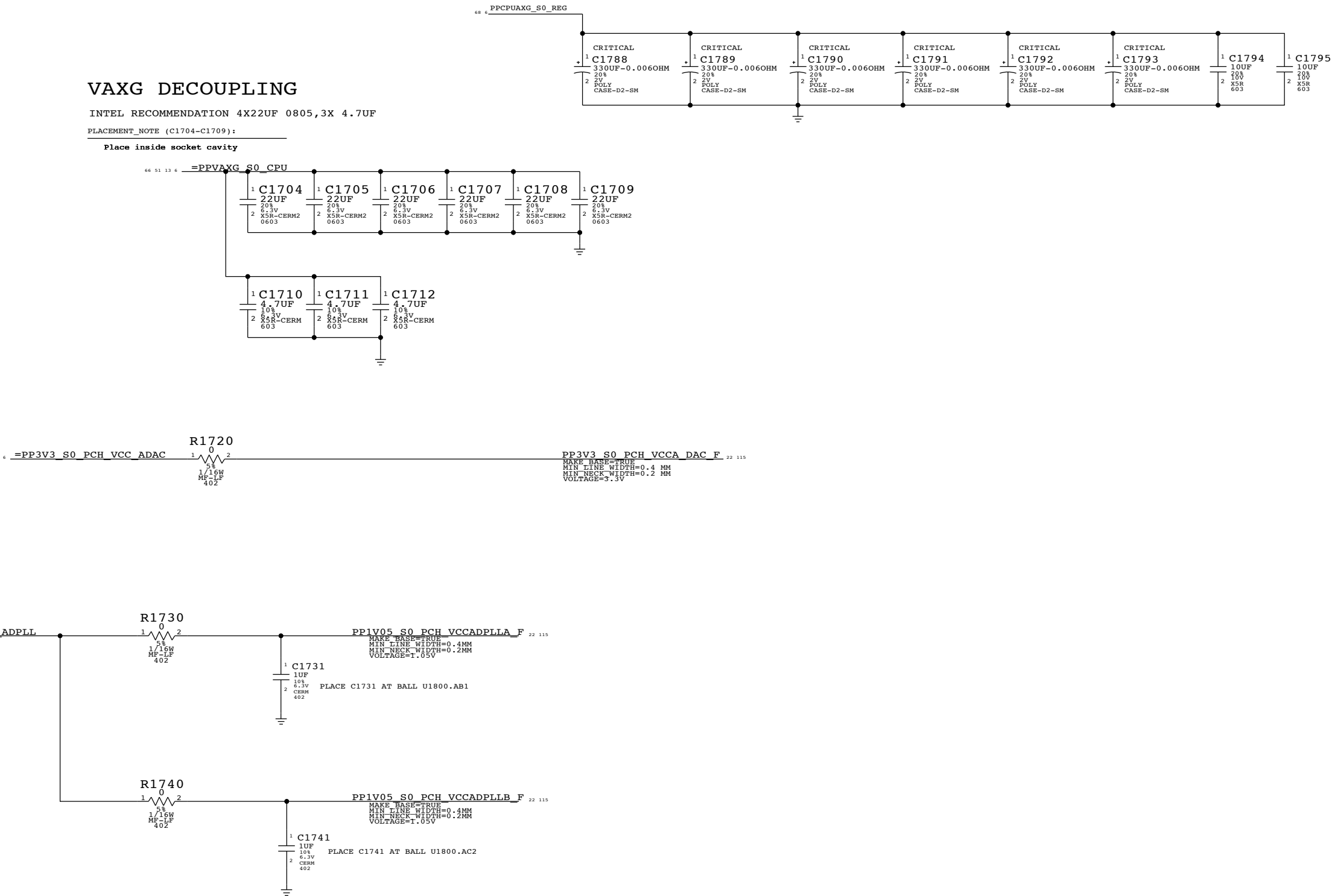
VAXG DECOUPLING

INTEL RECOMMENDATION 4X22UF 0805,3X 4.7UF

PLACEMENT_NOTE (C1704-C1709):

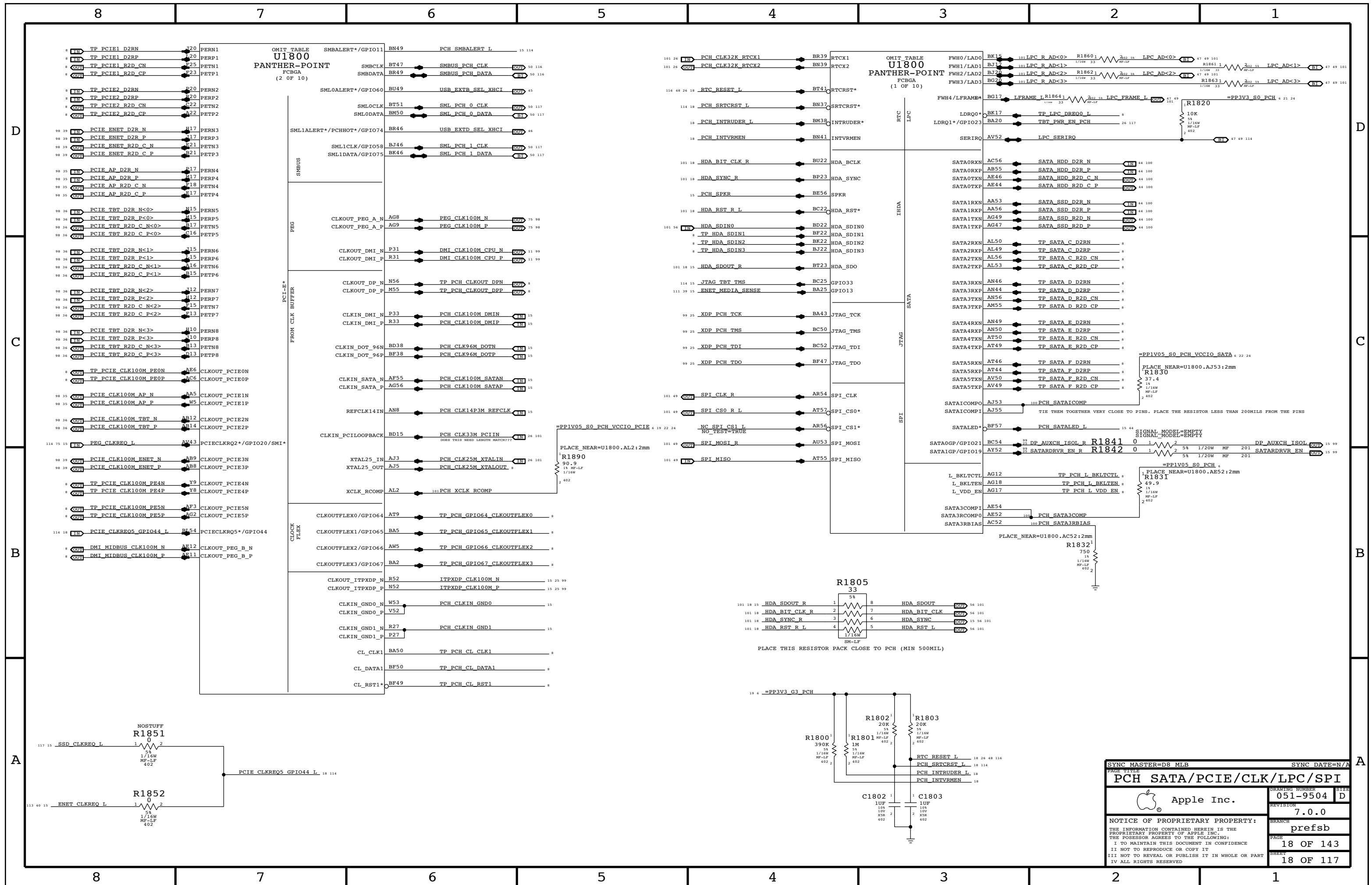
Place inside socket cavity

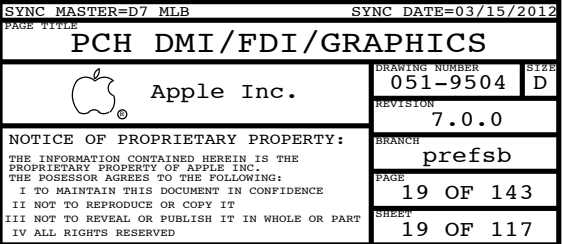
AXG BULK CAPS



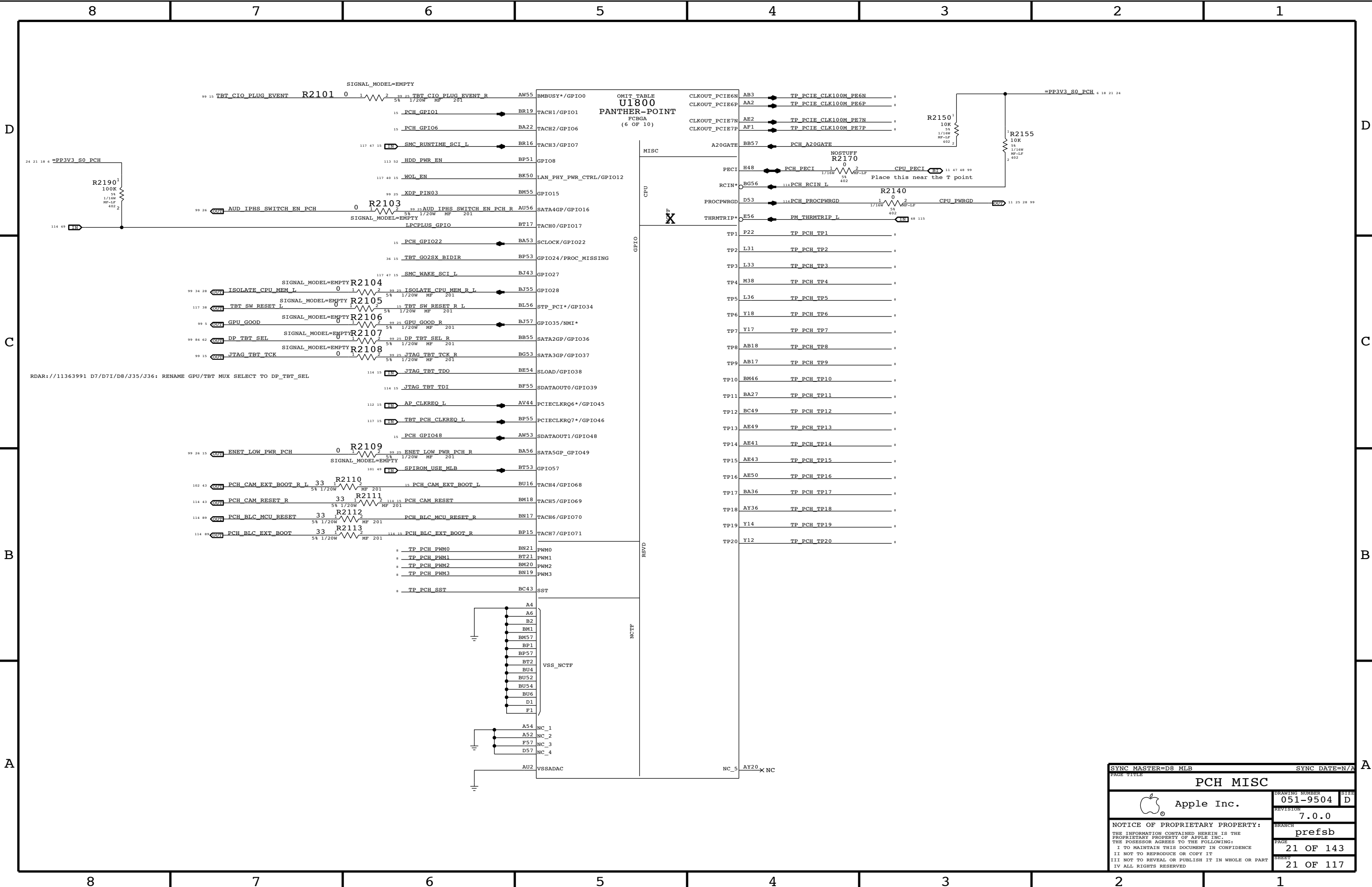
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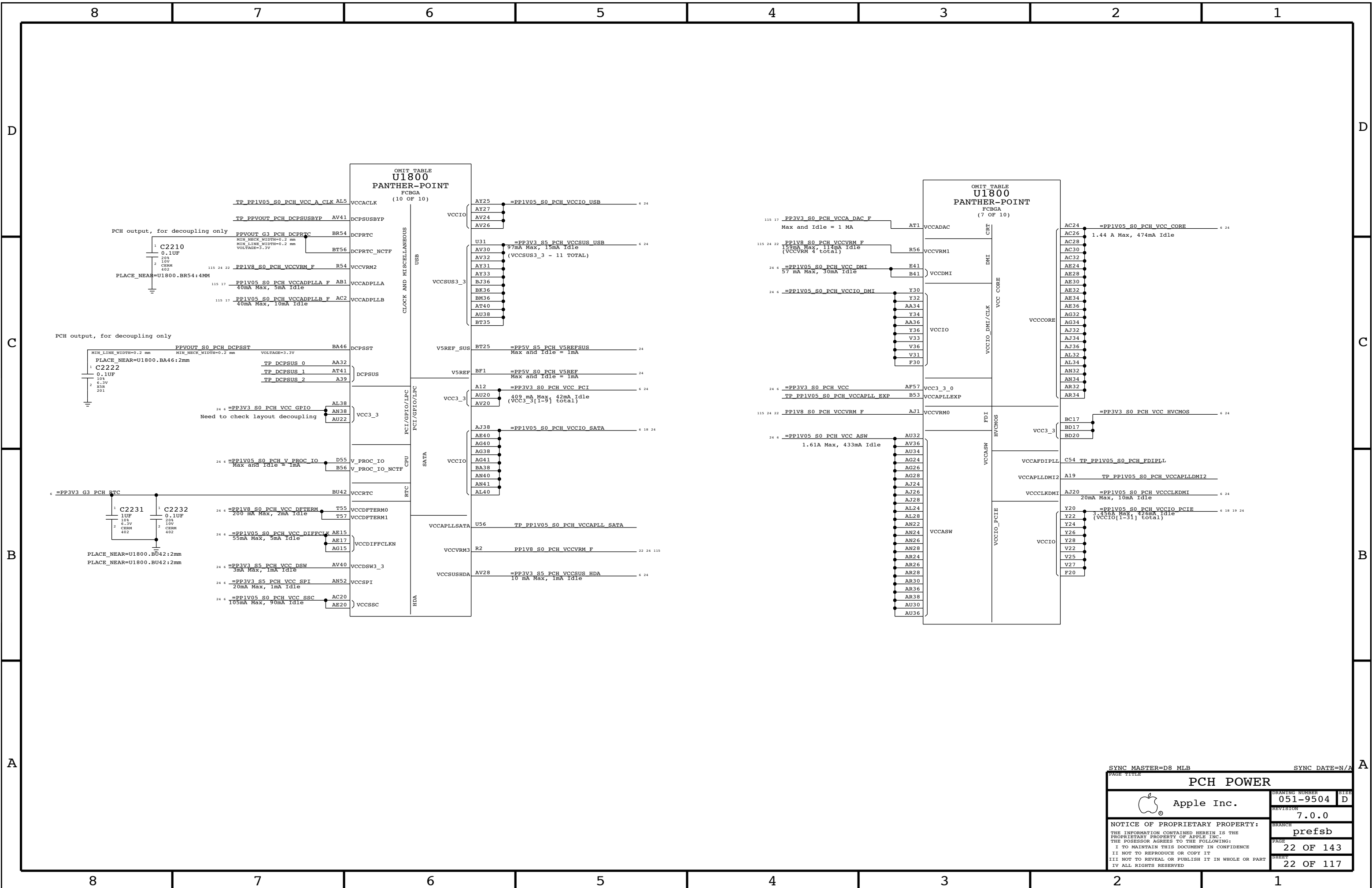
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GFX DECOUPLING & PCH PWR ALIAS		
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	REVISION	7.0.0
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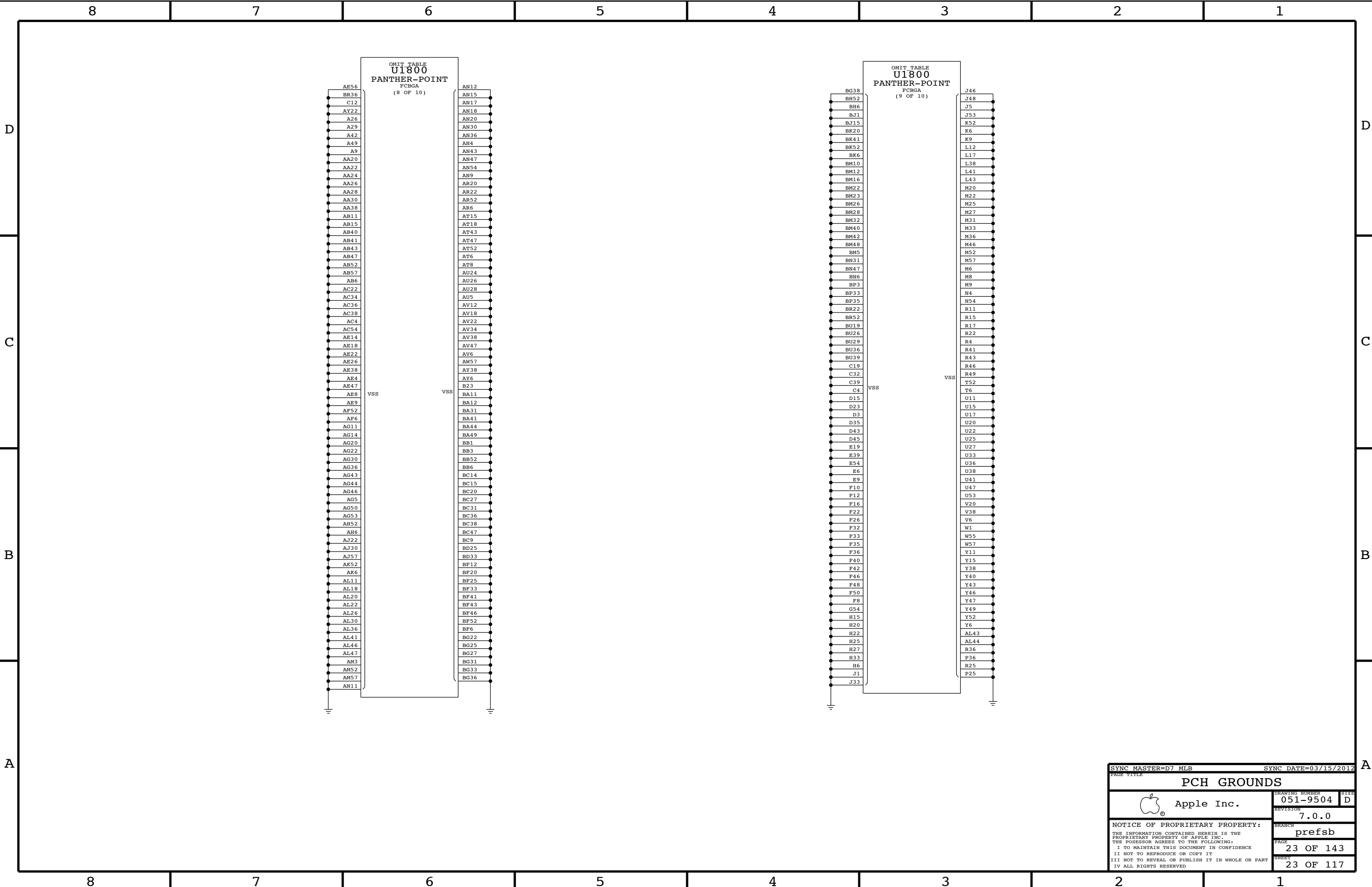












SYNC MASTER=D7 MLB

SYNC DATE=03/15/2012

PCH GROUNDS

 Apple Inc.

DRAWING NUMBER

051-9504

SIZE

D

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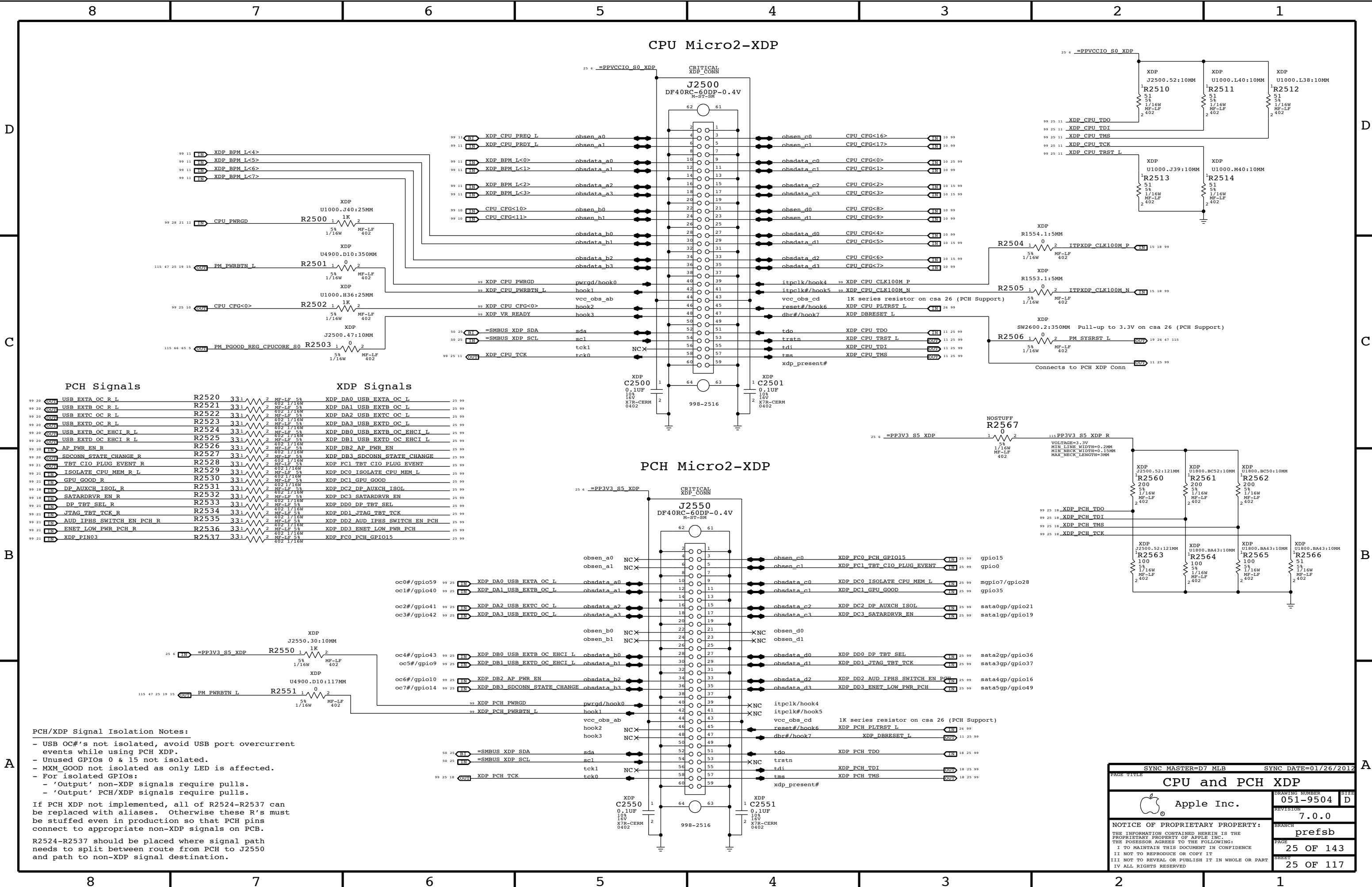
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PCH/XDP Signal Isolation Notes:

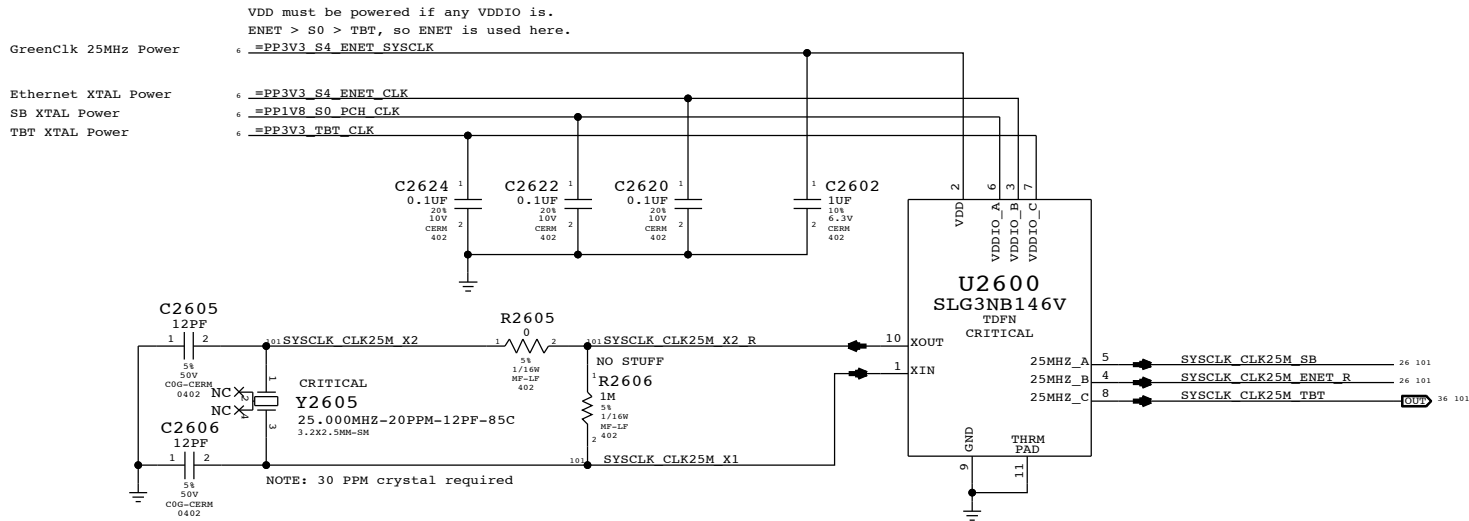
- USB OC#’s not isolated, avoid USB port overcurrent events while using PCH XDP.
- Unused GPIOs 0 & 15 not isolated.
- MXM GOOD not isolated as only LED is affected.
- For isolated GPIOs:
 - ‘Output’ non-XDP signals require pulls.
 - ‘Output’ PCH/XDP signals require pulls.

If PCH XDP not implemented, all of R2524-R2537 can be replaced with aliases. Otherwise these R’s must be stuffed even in production so that PCH pins connect to appropriate non-XDP signals on PCB.

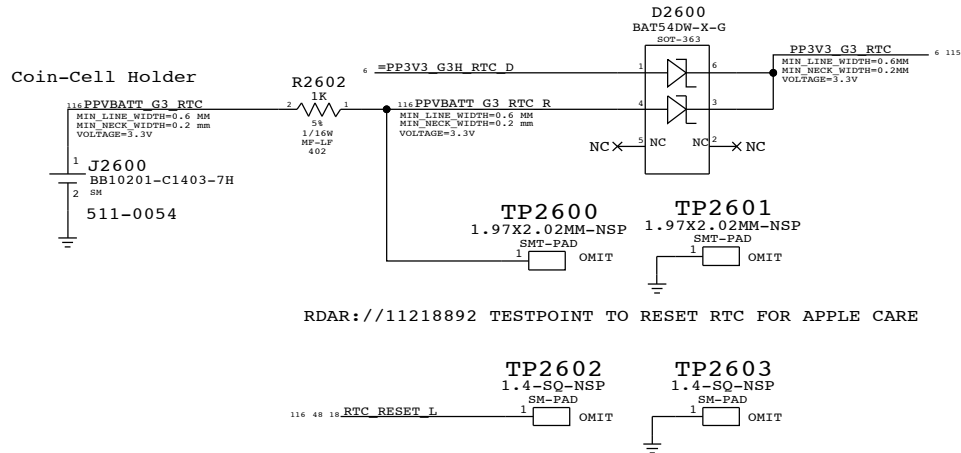
R2524-R2537 should be placed where signal path needs to split between route from PCH to J2550 and path to non-XDP signal destination.

SYNC MASTER=D7 MLB		SYNC DATE=01/26/2012	
CPU and PCH XDP			
DRAWING NUMBER		SIZE	
051-9504		D	
REVISION		7.0.0	
BRANCH		prefsb	
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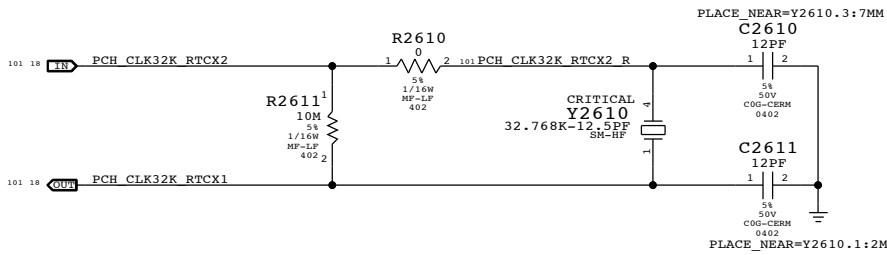
System 25MHz Clock Generator



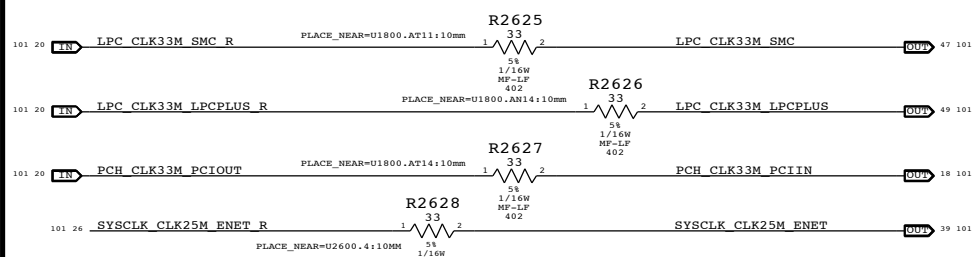
RTC Power Sources



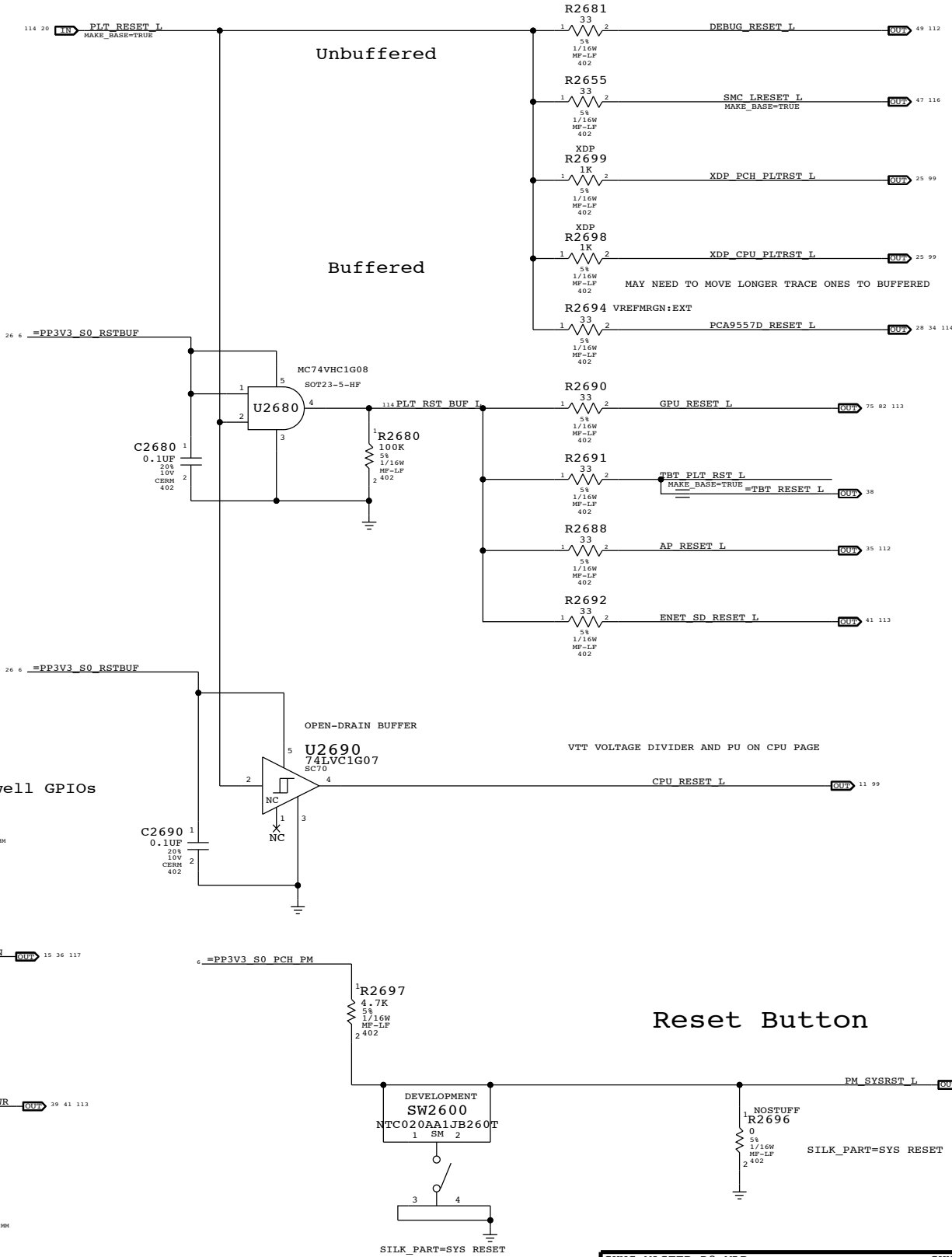
PCH RTC Crystal



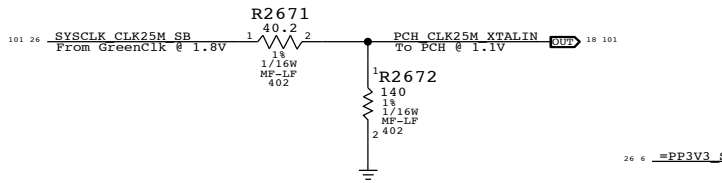
Clock series termination



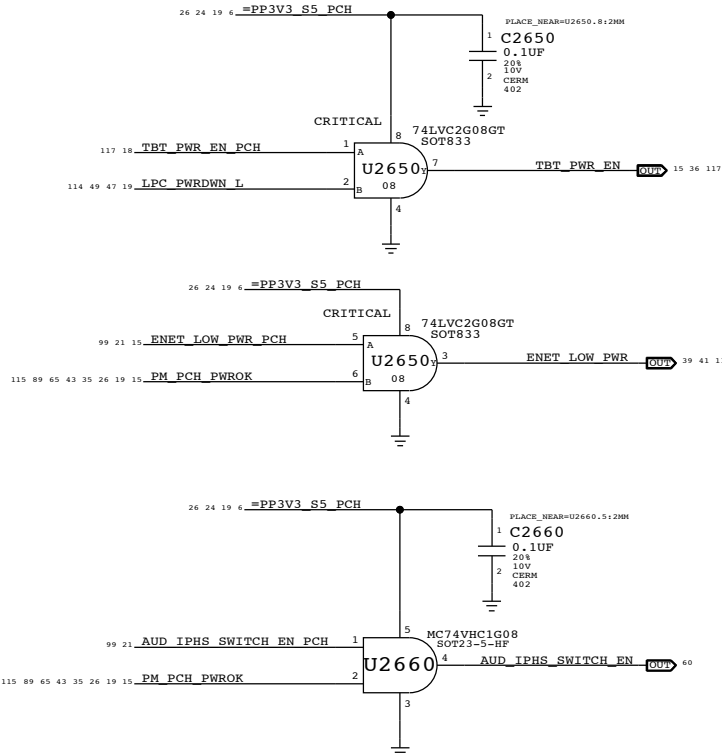
Platform Reset Connections



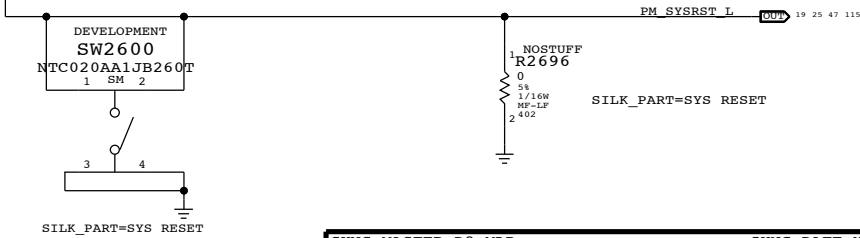
PCH 25MHZ CLOCK



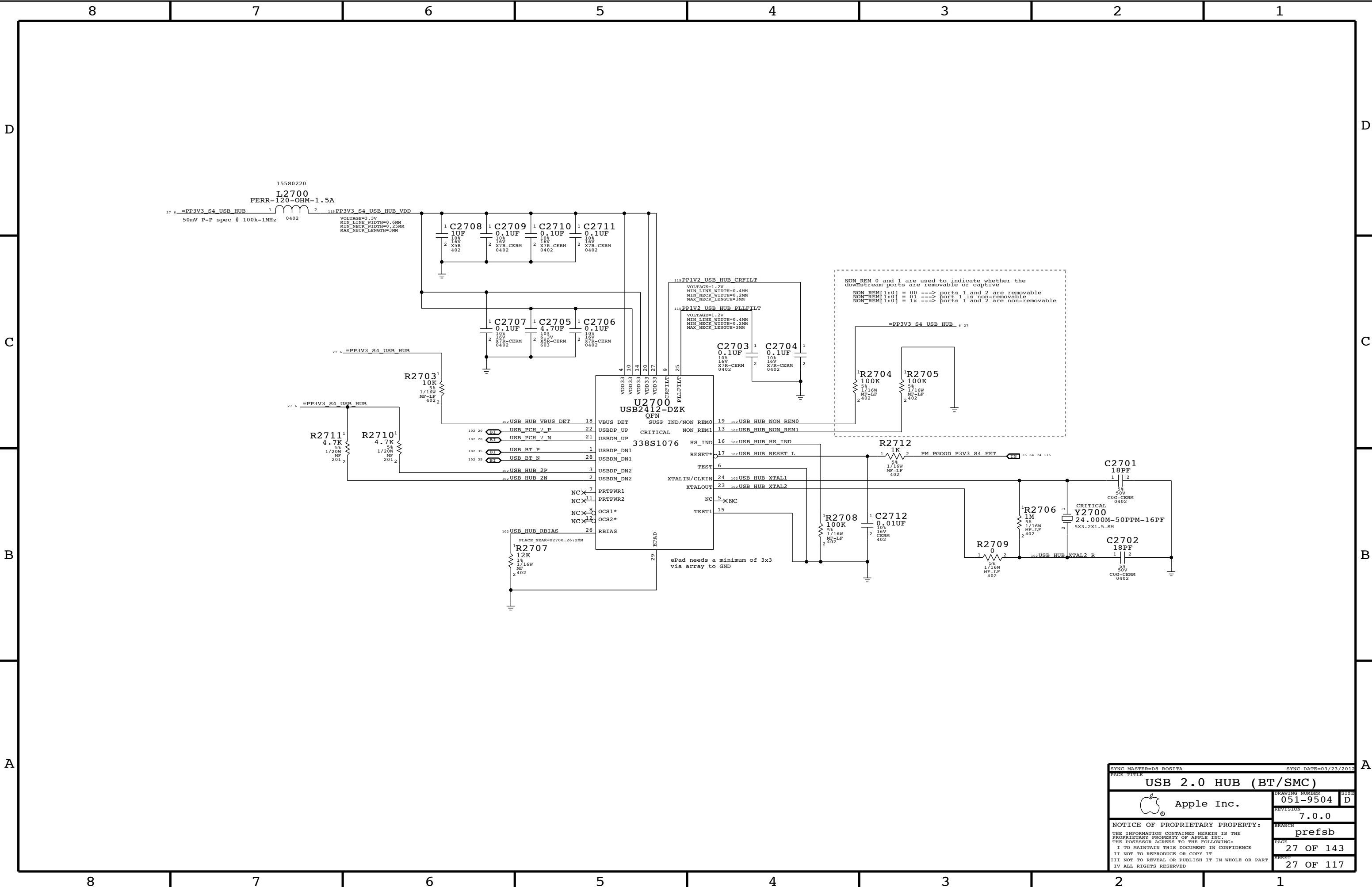
GPIO Isolation to prevent glitches on critical core well GPIOs



Reset Button



SYNC MASTER=D8 MLB		SYNC DATE=N/A	
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SYNC MASTER=D8 ROSITA		SYNC DATE=03/23/2012	
PAGE TITLE			
USB 2.0 HUB (BT/SMC)			
 Apple Inc.		DRAWING NUMBER	051-9504
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MEM_RESET_L Generator

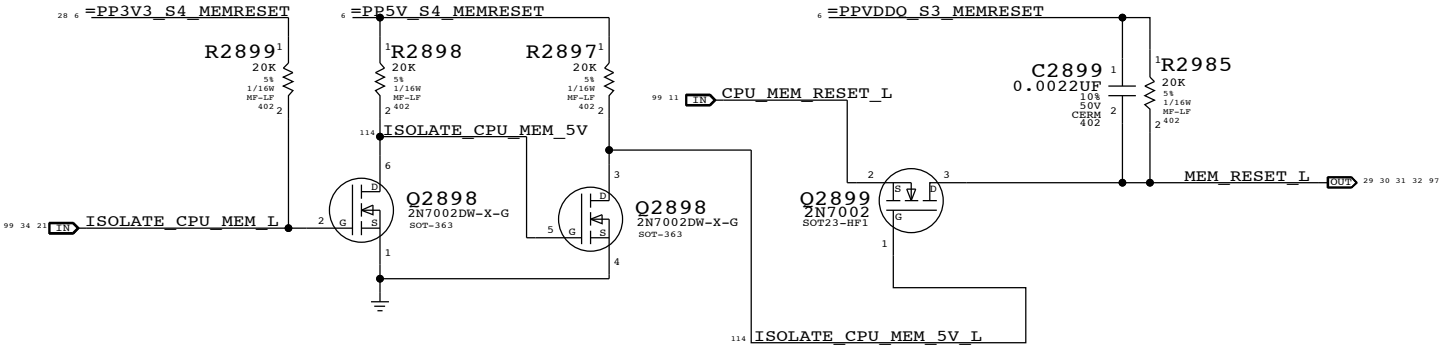
The circuits below handle MEMVTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the SO-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behaviour of signals.

WHEN HIGH: MEM_RESET_L NOT ISOLATED.

WHEN LOW: MEM_RESET_L IS ISOLATED.

MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L (Block CPU from driving MEM_RESET_L in S3)



MEMVTT_EN Generator

rdar://11117167

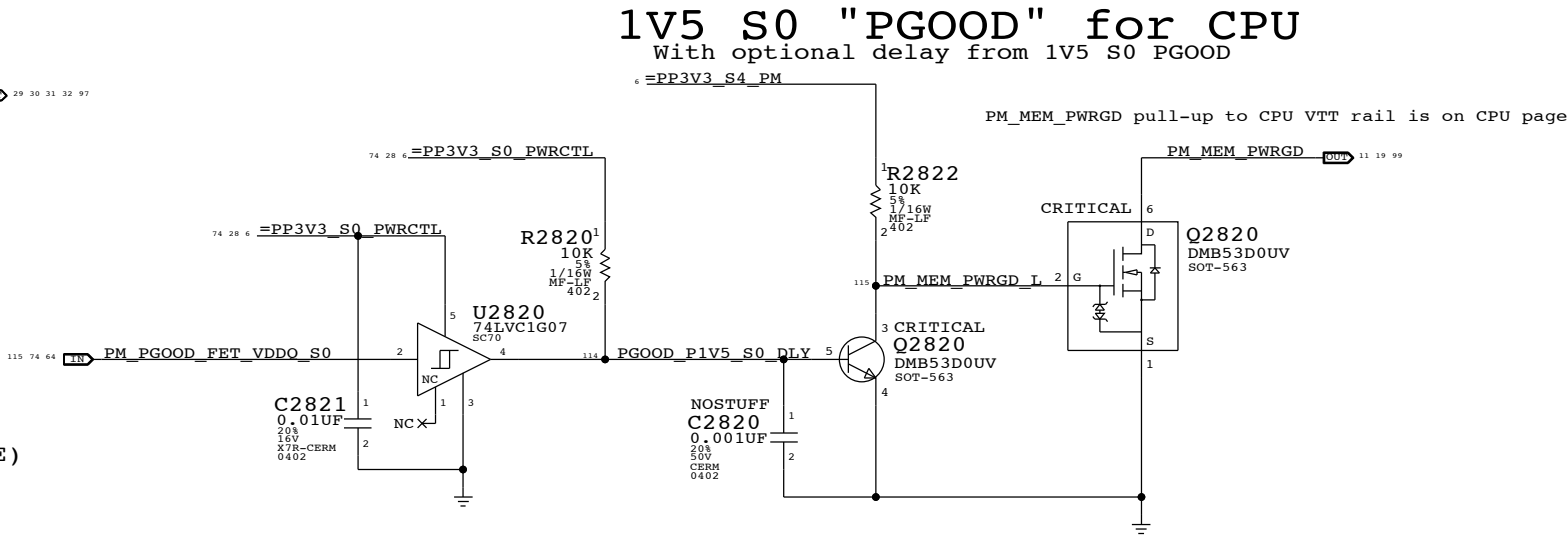
Enables MEMVTT when PCH drives CPU PWRGD.

CPU does not drive MEM_CKE until VCCORE activated but CPU 1V5 (VDDQ) leaks into it.

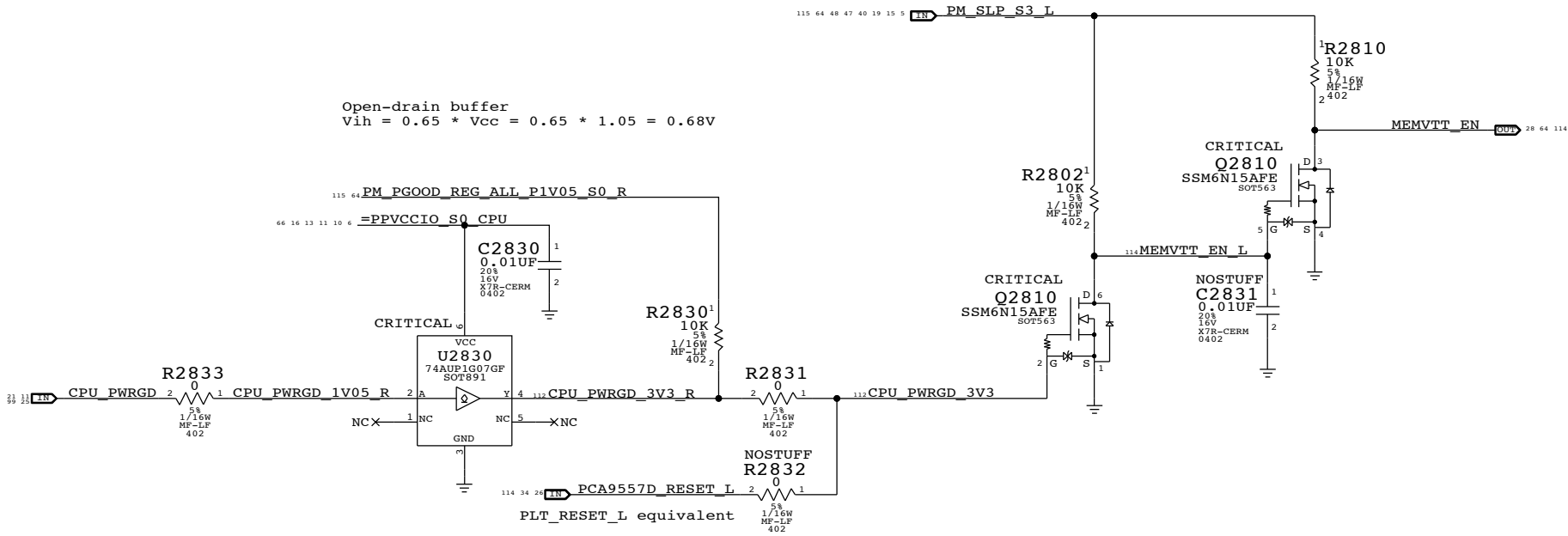
Clamping MEMVTT will keep the MEM_CKE low until CPU actively controls it.

MEMVTT Clamp actively holds MEMVTT rail low until MEMVTT is enabled.

MEMVTT_EN = CPU_PWRGD * PM_SLP_S3_L (VTT is enabled when PCH tells CPU to enable VCCORE)

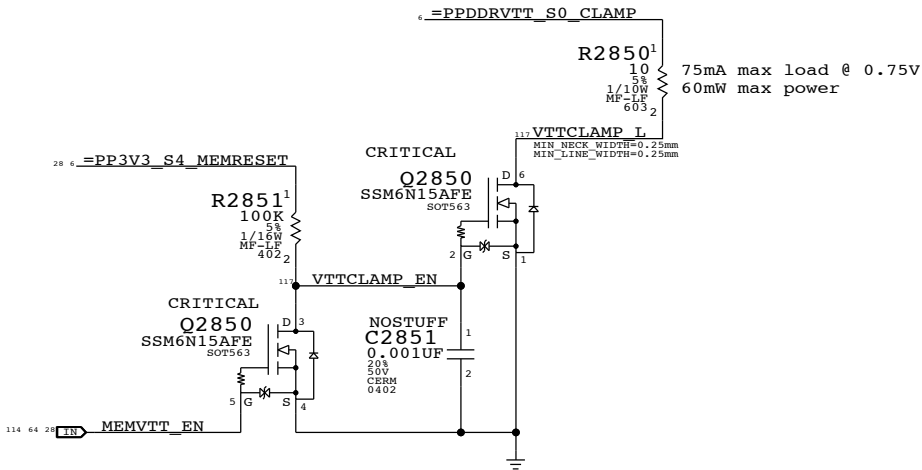


Open-drain buffer
 $V_{ih} = 0.65 * V_{cc} = 0.65 * 1.05 = 0.68V$



MEMVTT Clamp

Ensures CKE signals are held low in S3 and in S0 before CPU PWRGD



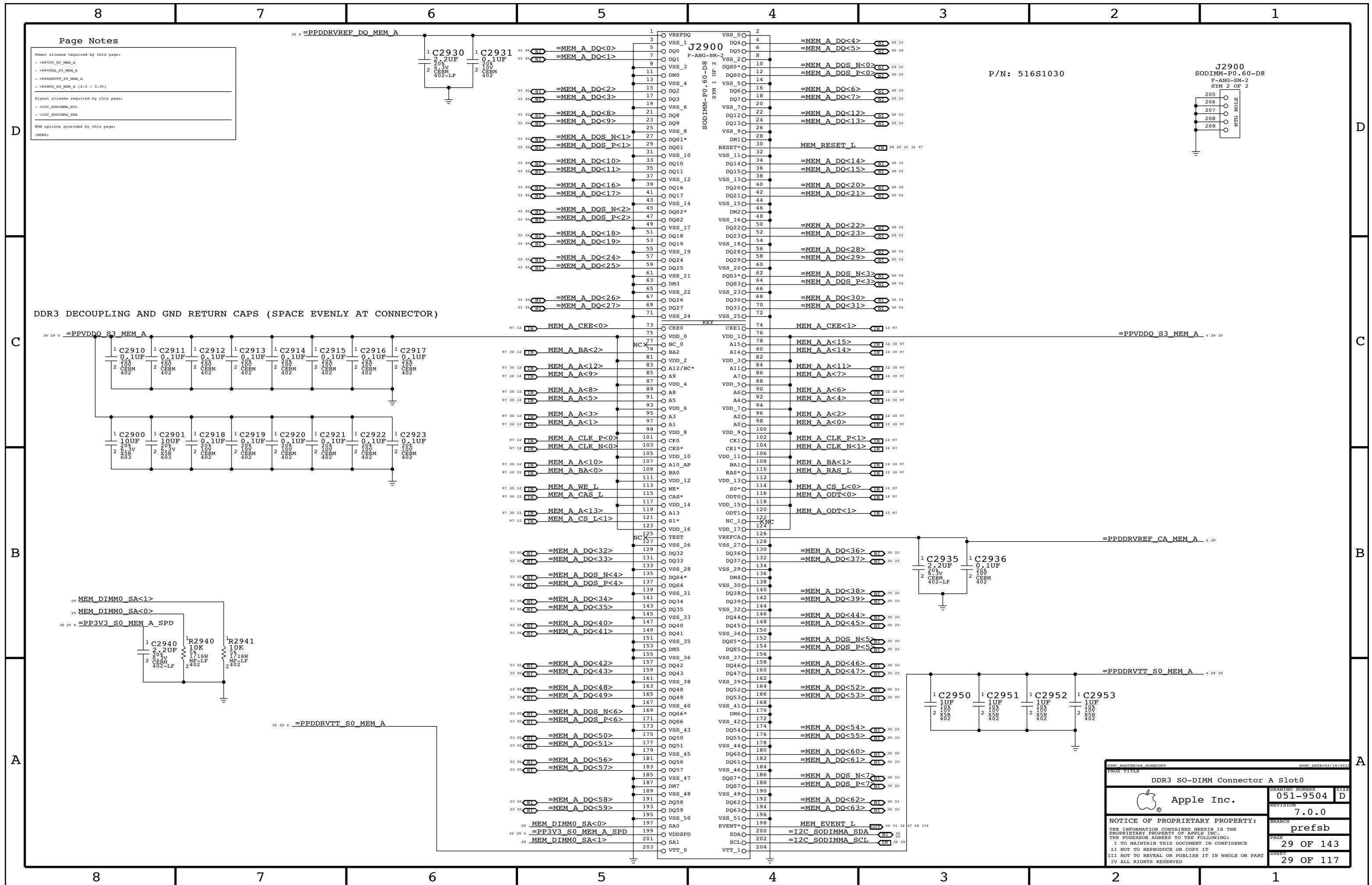
Step	ISOLATE_CPU_MEM_L	PM_SLP_S3_L	CPU_PWRGD	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
S0	0	1	1	CPU_MEM_RESET_L	1	1
to	1	1	1	1	1	1
2	0	1	1	1	1	1
3	0	0	0	X	1	0
4	0	0	0	X	1	0
5	0	1	1	0 (*)	1	1
6	0	1	1	1	1	1
S0	7	1	1	CPU_MEM_RESET_L	1	1

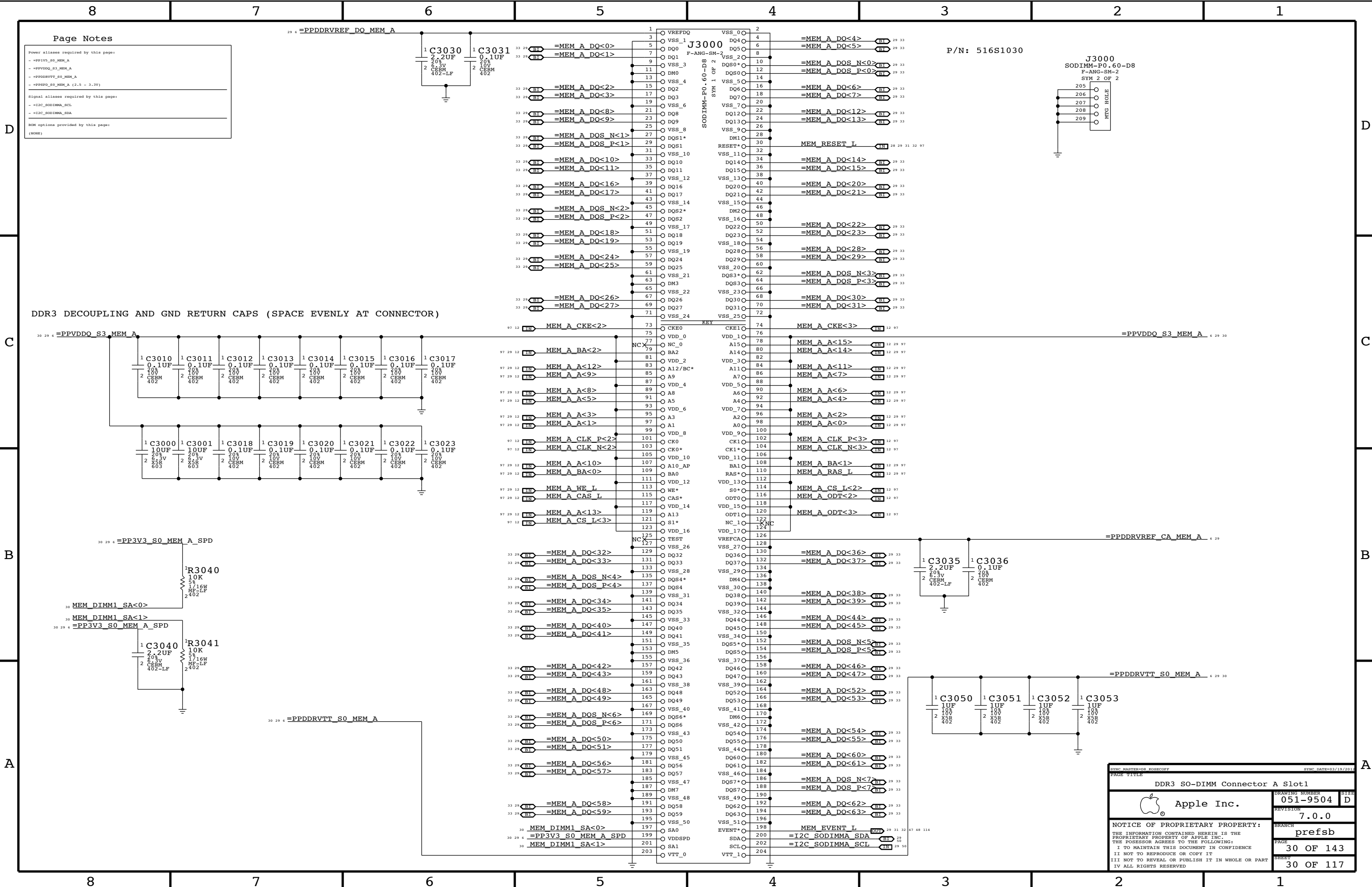
(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: On a S5->S0 transition, ISOLATE_CPU_MEM_L will default low.

Rails will power-up as if from S3, but MEM_RESET_L now needs to be asserted in S0. Software must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC MASTER=D8 MARK		SYNC DATE=04/23/2012	
PAGE TITLE			
CPU Memory S3 Support		DRAWING NUMBER	
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Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVTT_S0_MEM_A
- =PPSP0_S0_MEM_A (2.5 - 3.3V)

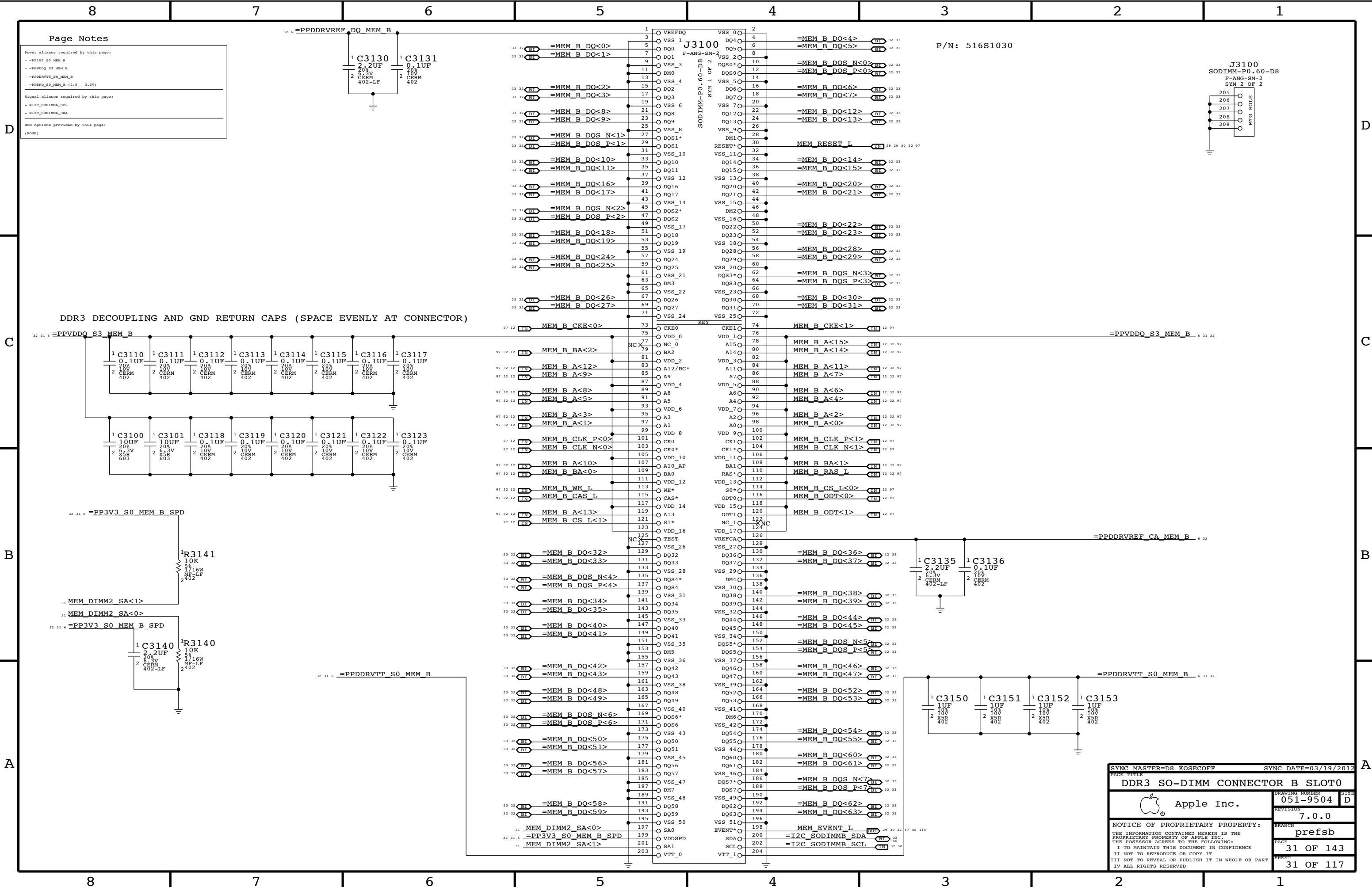
Signal aliases required by this page:

- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

ROM options provided by this page:

(NONE)

SYNCHARTING KORECOFF		SYNCHARTING KORECOFF	
PAGE TITLE		PAGE TITLE	
DDR3 S0-DIMM Connector A Slot1		DRAWING NUMBER	
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Page Notes

Power aliases required by this page:

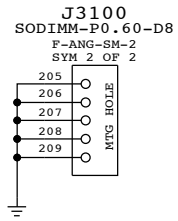
- =PP1V3_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVTT_S0_MEM_B
- =PPDRPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

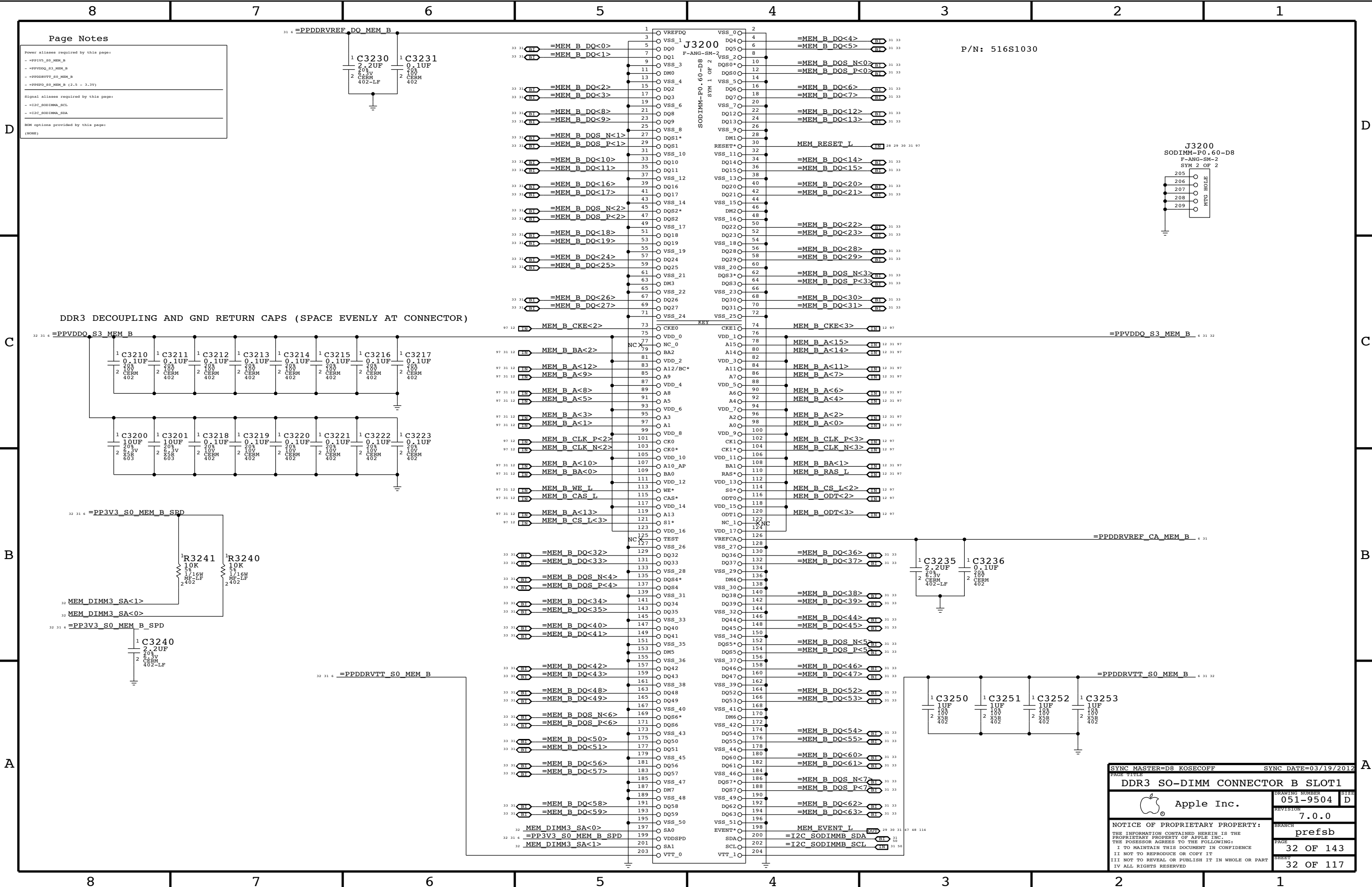
- =I2C_S0DIMM_A_SCL
- =I2C_S0DIMM_A_SDA

BOM options provided by this page:

(NONE)



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DDR3 SO-DIMM CONNECTOR B SLOT0		051-9504	
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Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVTT_S0_MEM_B
- =PPDRPD_S0_MEM_B (2.5 - 3.3V)

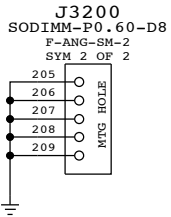
Signal aliases required by this page:

- =I2C_S0DIMM_A_SCL
- =I2C_S0DIMM_A_SDA

BOM options provided by this page:

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P/N: 516S1030



SYNC MASTER=D8 KOSECOFF		SYNC DATE=03/19/2012	
PAGE TITLE		DDR3 SO-DIMM CONNECTOR B SLOT1	
DRAWING NUMBER		051-9504	SIZE D
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8	7	6	5	4	3	2	1
D C B A	MEM A DQS N<0> MEM A DQS P<0> MEM A DQ<7> MEM A DQ<6> MEM A DQ<5> MEM A DQ<4> MEM A DQ<3> MEM A DQ<2> MEM A DQ<1> MEM A DQ<0> MEM A DQS N<1> MEM A DQS P<1> MEM A DQ<15> MEM A DQ<14> MEM A DQ<13> MEM A DQ<12> MEM A DQ<11> MEM A DQ<10> MEM A DQ<9> MEM A DQ<8> MEM A DQS N<2> MEM A DQS P<2> MEM A DQ<23> MEM A DQ<22> MEM A DQ<21> MEM A DQ<20> MEM A DQ<19> MEM A DQ<18> MEM A DQ<17> MEM A DQ<16> MEM A DQS N<3> MEM A DQS P<3> MEM A DQ<31> MEM A DQ<30> MEM A DQ<29> MEM A DQ<28> MEM A DQ<27> MEM A DQ<26> MEM A DQ<25> MEM A DQ<24> MEM A DQS N<4> MEM A DQS P<4> MEM A DQ<39> MEM A DQ<38> MEM A DQ<37> MEM A DQ<36> MEM A DQ<35> MEM A DQ<34> MEM A DQ<33> MEM A DQ<32> MEM A DQS N<5> MEM A DQS P<5> MEM A DQ<47> MEM A DQ<46> MEM A DQ<45> MEM A DQ<44> MEM A DQ<43> MEM A DQ<42> MEM A DQ<41> MEM A DQ<40> MEM A DQS N<6> MEM A DQS P<6> MEM A DQ<55> MEM A DQ<54> MEM A DQ<53> MEM A DQ<52> MEM A DQ<51> MEM A DQ<50> MEM A DQ<49> MEM A DQ<48> MEM A DQS N<7> MEM A DQS P<7> MEM A DQ<63> MEM A DQ<62> MEM A DQ<61> MEM A DQ<60> MEM A DQ<59> MEM A DQ<58> MEM A DQ<57> MEM A DQ<56>	MEM A DQS N<0> MEM A DQS P<0> MEM A DQ<7> MEM A DQ<6> MEM A DQ<5> MEM A DQ<4> MEM A DQ<3> MEM A DQ<2> MEM A DQ<1> MEM A DQ<0> MEM A DQS N<1> MEM A DQS P<1> MEM A DQ<15> MEM A DQ<14> MEM A DQ<13> MEM A DQ<12> MEM A DQ<11> MEM A DQ<10> MEM A DQ<9> MEM A DQ<8> MEM A DQS N<2> MEM A DQS P<2> MEM A DQ<23> MEM A DQ<22> MEM A DQ<21> MEM A DQ<20> MEM A DQ<19> MEM A DQ<18> MEM A DQ<17> MEM A DQ<16> MEM A DQS N<3> MEM A DQS P<3> MEM A DQ<31> MEM A DQ<30> MEM A DQ<29> MEM A DQ<28> MEM A DQ<27> MEM A DQ<26> MEM A DQ<25> MEM A DQ<24> MEM A DQS N<4> MEM A DQS P<4> MEM A DQ<39> MEM A DQ<38> MEM A DQ<37> MEM A DQ<36> MEM A DQ<35> MEM A DQ<34> MEM A DQ<33> MEM A DQ<32> MEM A DQS N<5> MEM A DQS P<5> MEM A DQ<47> MEM A DQ<46> MEM A DQ<45> MEM A DQ<44> MEM A DQ<43> MEM A DQ<42> MEM A DQ<41> MEM A DQ<40> MEM A DQS N<6> MEM A DQS P<6> MEM A DQ<55> MEM A DQ<54> MEM A DQ<53> MEM A DQ<52> MEM A DQ<51> MEM A DQ<50> MEM A DQ<49> MEM A DQ<48> MEM A DQS N<7> MEM A DQS P<7> MEM A DQ<63> MEM A DQ<62> MEM A DQ<61> MEM A DQ<60> MEM A DQ<59> MEM A DQ<58> MEM A DQ<57> MEM A DQ<56>	MEM B DQS N<0> MEM B DQS P<0> MEM B DQ<7> MEM B DQ<6> MEM B DQ<5> MEM B DQ<4> MEM B DQ<3> MEM B DQ<2> MEM B DQ<1> MEM B DQ<0> MEM B DQS N<1> MEM B DQS P<1> MEM B DQ<15> MEM B DQ<14> MEM B DQ<13> MEM B DQ<12> MEM B DQ<11> MEM B DQ<10> MEM B DQ<9> MEM B DQ<8> MEM B DQS N<2> MEM B DQS P<2> MEM B DQ<23> MEM B DQ<22> MEM B DQ<21> MEM B DQ<20> MEM B DQ<19> MEM B DQ<18> MEM B DQ<17> MEM B DQ<16> MEM B DQS N<3> MEM B DQS P<3> MEM B DQ<31> MEM B DQ<30> MEM B DQ<29> MEM B DQ<28> MEM B DQ<27> MEM B DQ<26> MEM B DQ<25> MEM B DQ<24> MEM B DQS N<4> MEM B DQS P<4> MEM B DQ<39> MEM B DQ<38> MEM B DQ<37> MEM B DQ<36> MEM B DQ<35> MEM B DQ<34> MEM B DQ<33> MEM B DQ<32> MEM B DQS N<5> MEM B DQS P<5> MEM B DQ<47> MEM B DQ<46> MEM B DQ<45> MEM B DQ<44> MEM B DQ<43> MEM B DQ<42> MEM B DQ<41> MEM B DQ<40> MEM B DQS N<6> MEM B DQS P<6> MEM B DQ<55> MEM B DQ<54> MEM B DQ<53> MEM B DQ<52> MEM B DQ<51> MEM B DQ<50> MEM B DQ<49> MEM B DQ<48> MEM B DQS N<7> MEM B DQS P<7> MEM B DQ<63> MEM B DQ<62> MEM B DQ<61> MEM B DQ<60> MEM B DQ<59> MEM B DQ<58> MEM B DQ<57> MEM B DQ<56>				

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SYNC DATE=03/19/2012

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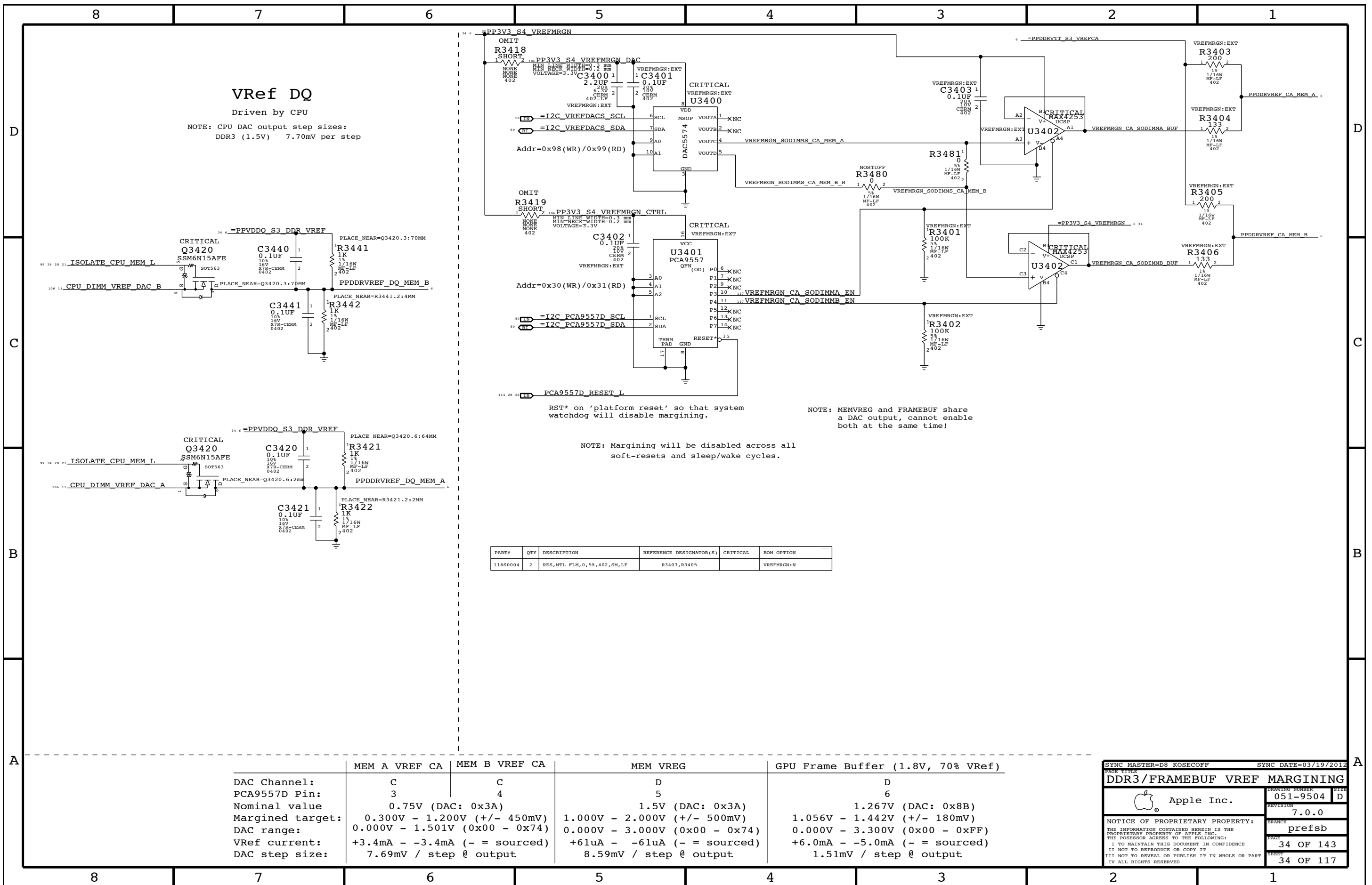
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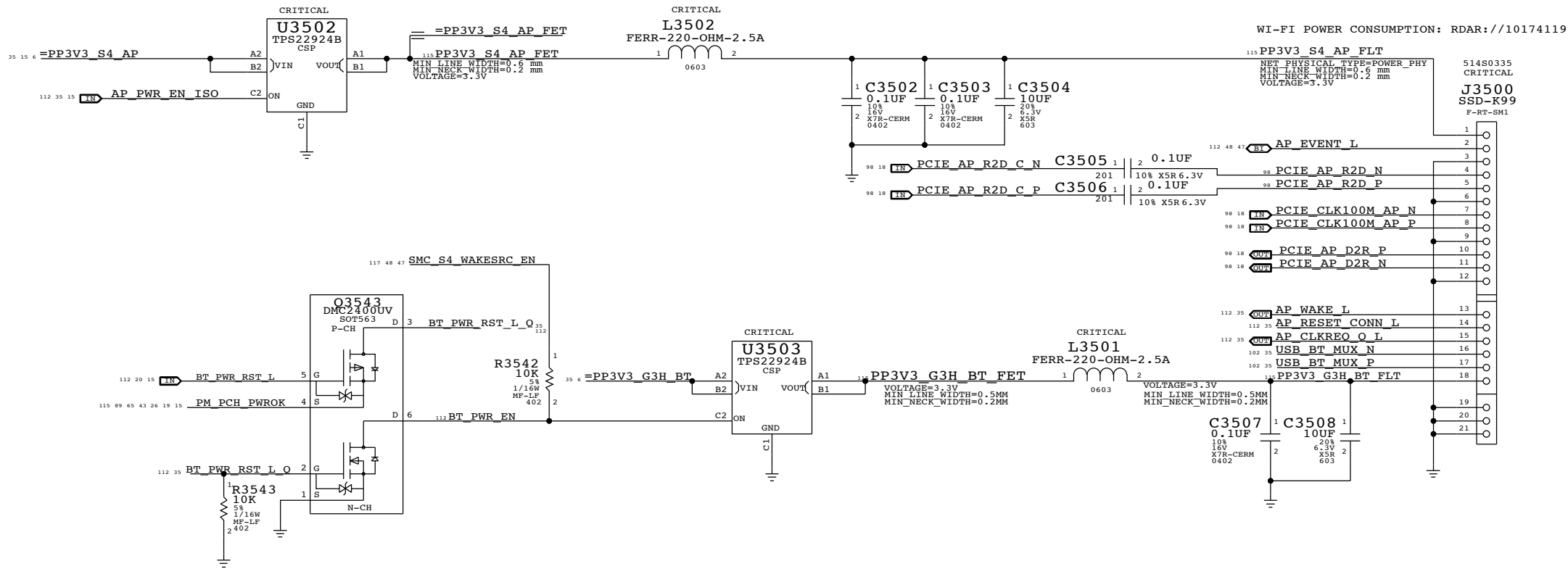
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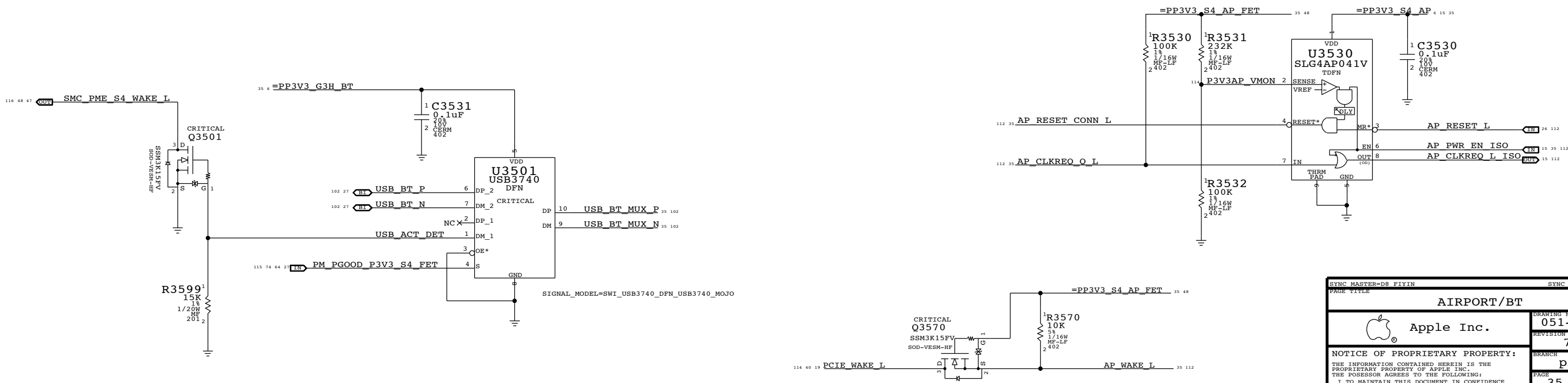
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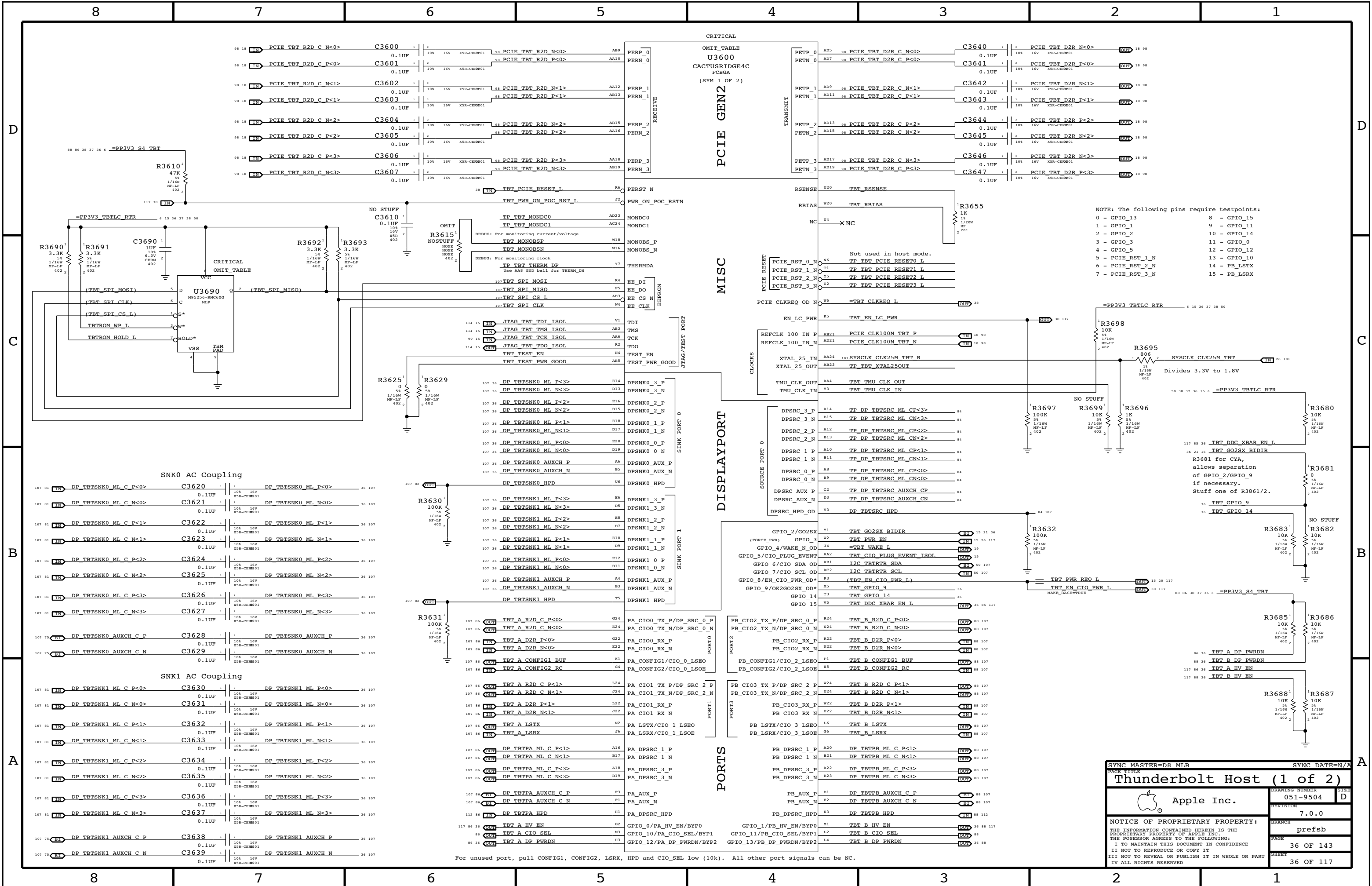
AIRPORT
BLUETOOTH



SUPERVISOR & CLKREG # ISOLATION
DELAY = 130 MS +/- 20%



SYNC MASTER=D8_FIVIN		SYNC DATE=07/02/2012	
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D

C

B

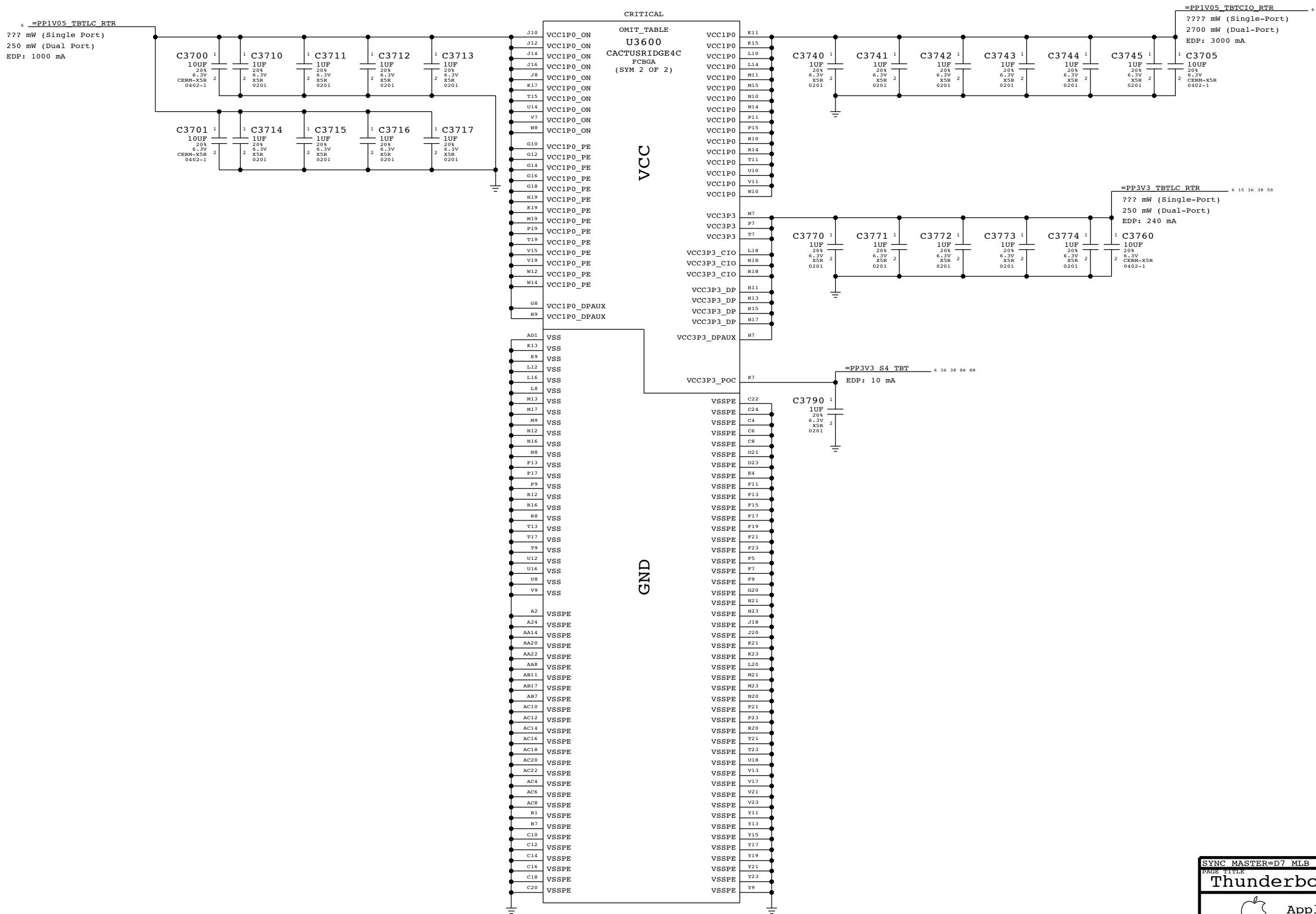
A

D

C

B

A



EDP current / power consumption figures from CR DG v0.57, IBL doc #472455.

SYNC MASTER=D7 MLB		SYNC DATE=03/15/2012	
PAGE TITLE		Thunderbolt Host (2 of 2)	
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Page Notes

Power aliases required by this page:

- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_TBT_P3V3TBTFFET (3.3V FET Input)
- =PP3V3_TBT_FET (3.3V FET Output)
- =PP3V3_S0_TBTFWRCTL
- =PP1V05_TBT_P1V05TBTFFET (1.05V FET Input)
- =PP1V05_TBT_FET (1.05V FET Output)

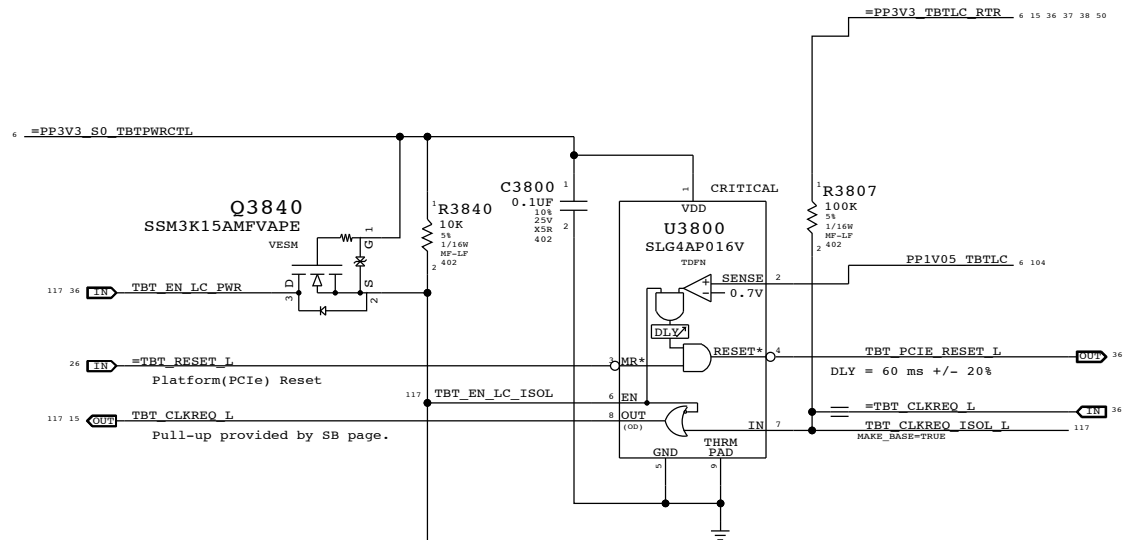
Signal aliases required by this page:

- =TBT_CLKREQ_L
- =TBT_RESET_L

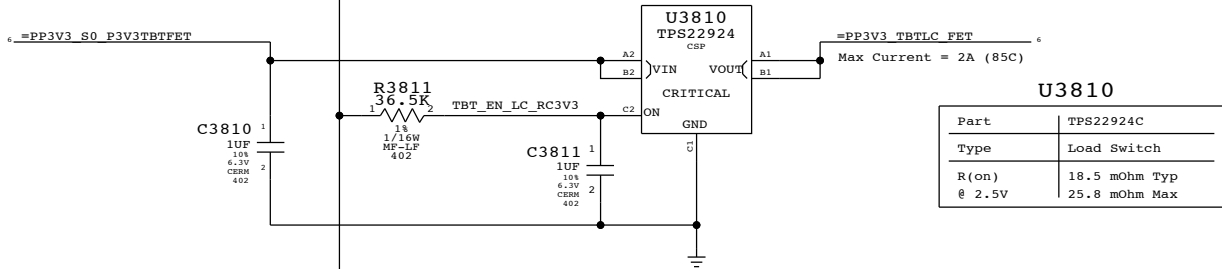
BOM options provided by this page:

TBTBST:Y - Stuffs 15V boost circuitry.

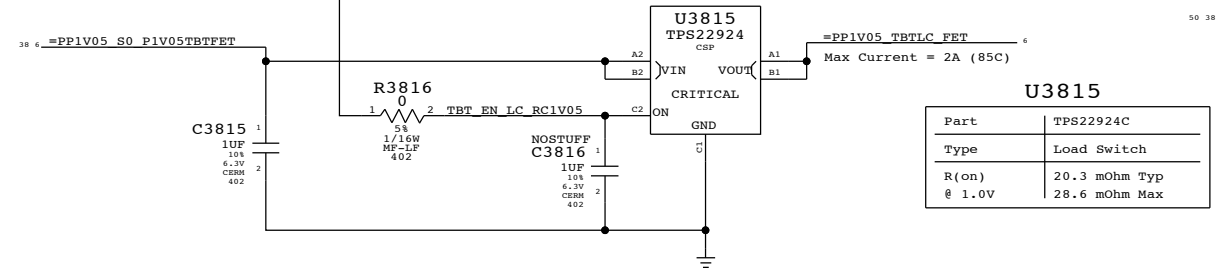
Supervisor & CLKREQ# Isolation



3.3V TBT "LC" Switch

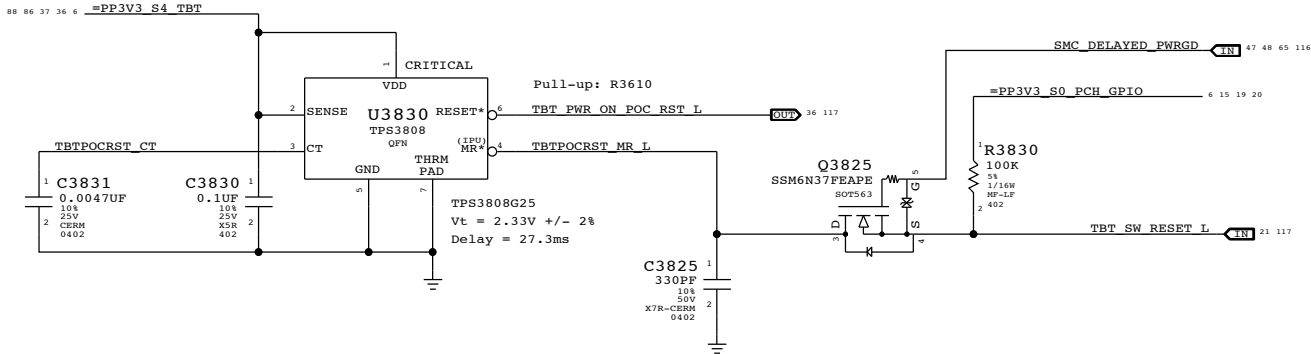


1.05V TBT "LC" Switch

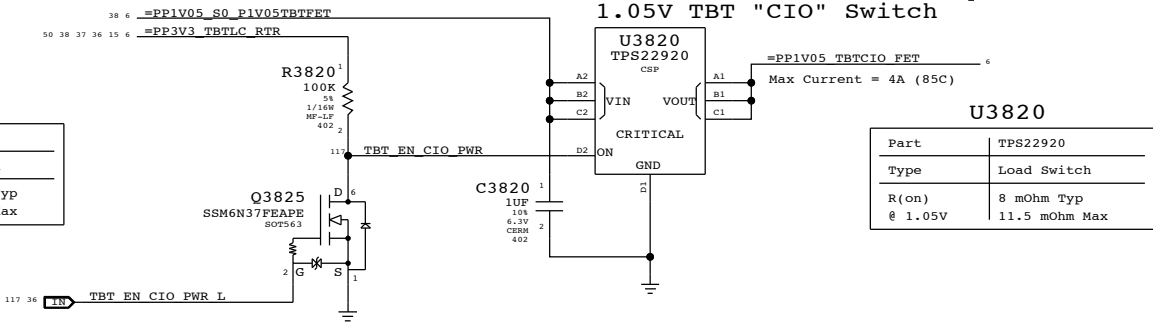


TBT "POC" Power-up Reset

Intel investigating whether RC is sufficient.



1.05V TBT "CIO" Switch



SYNC MASTER=D7 MLB SYNC DATE=03/15/2012

Thunderbolt Power Support

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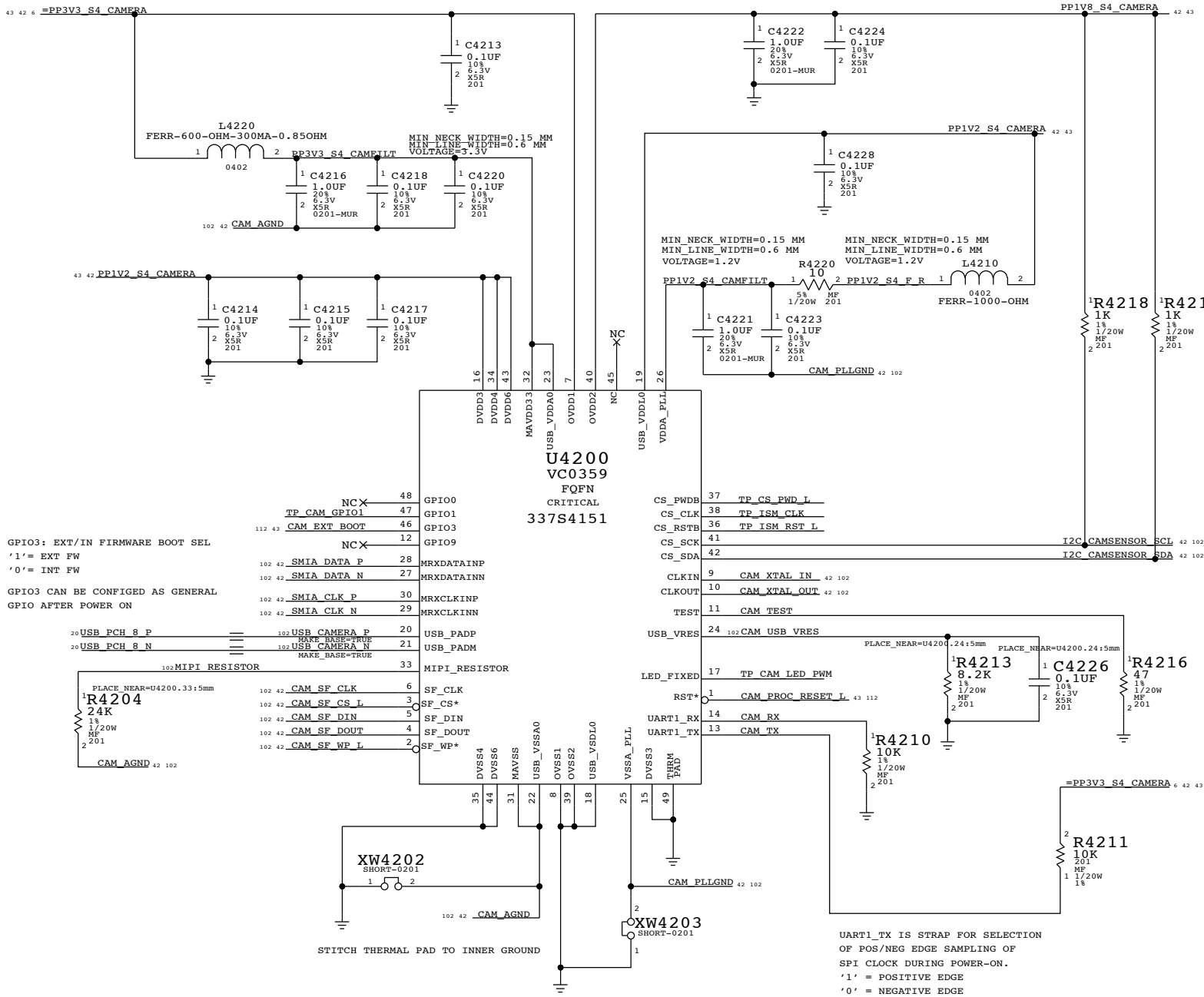
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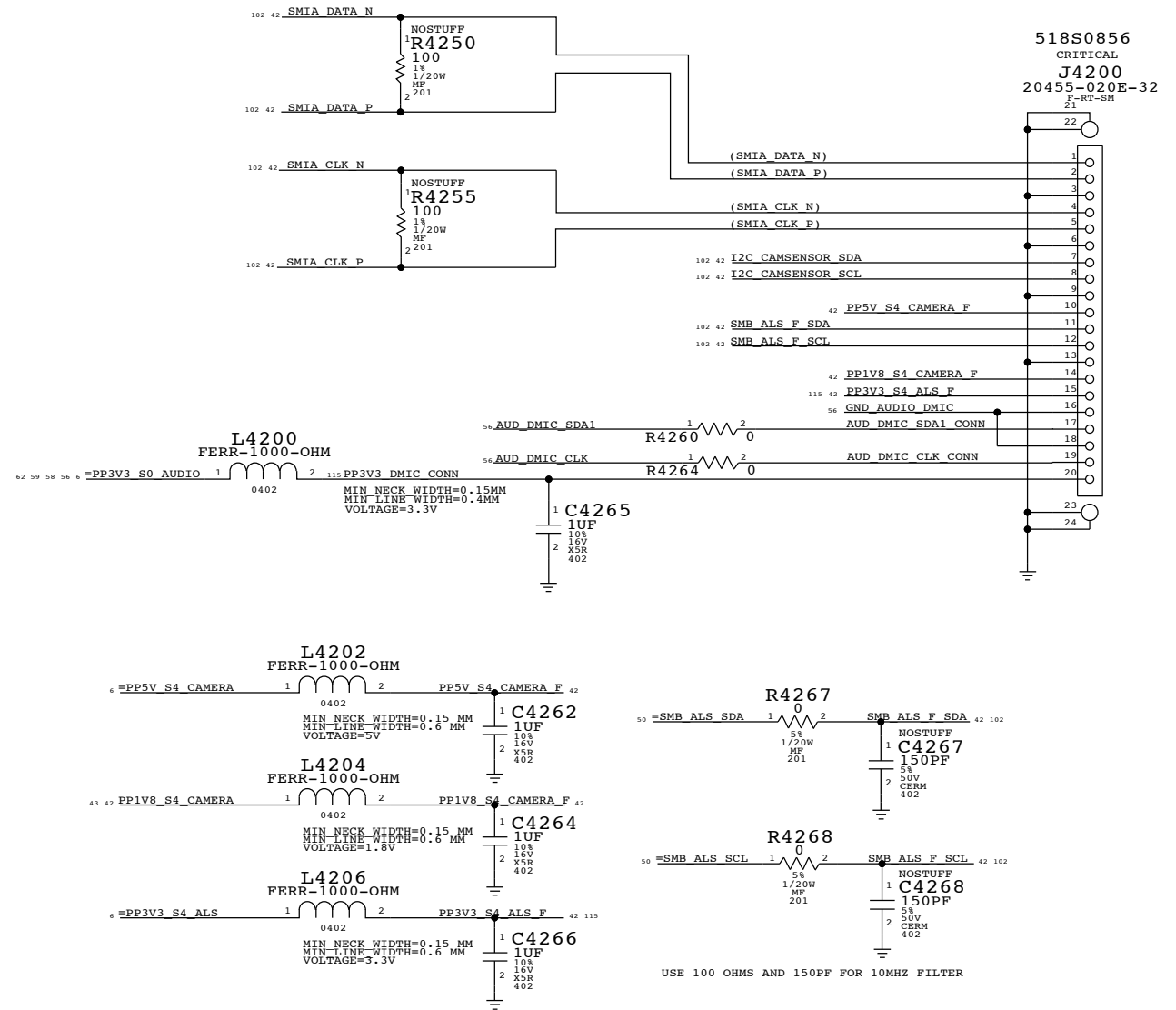
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A

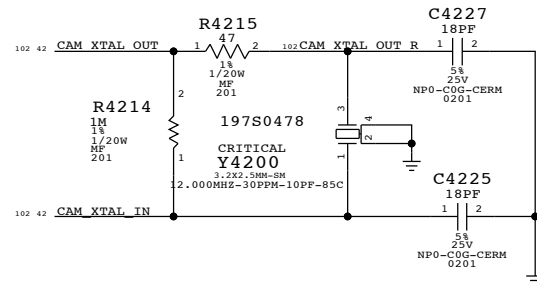
USB CAMERA CONTROLLER



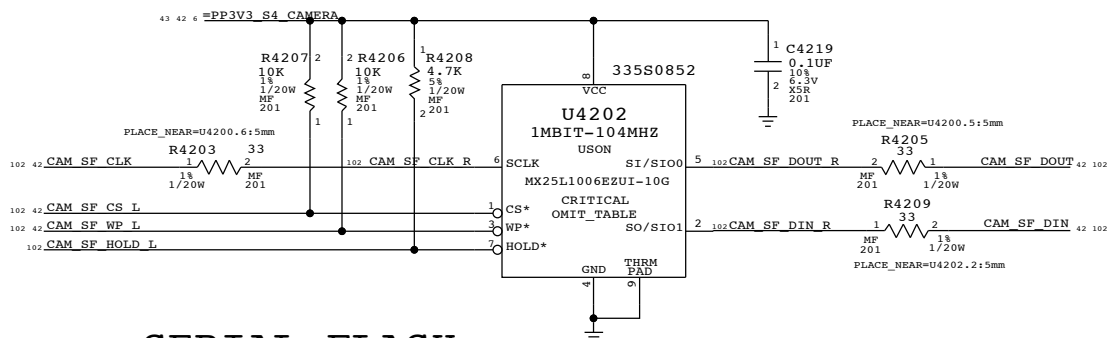
CAMERA/ALS/DMIC CONNECTOR



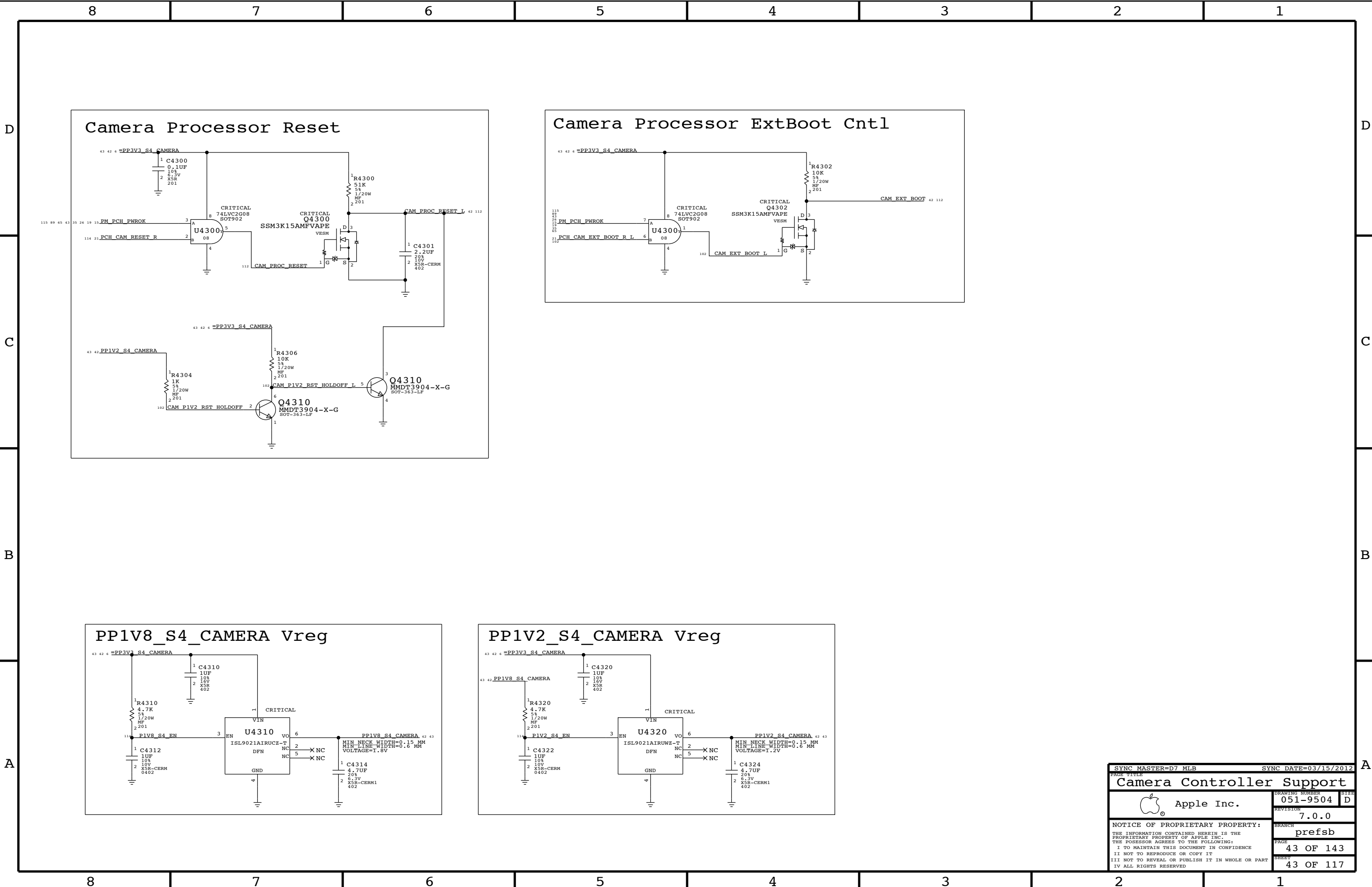
CRYSTAL



SERIAL FLASH



PAGE TITLE		PAGE NUMBER	
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D

C

B

A

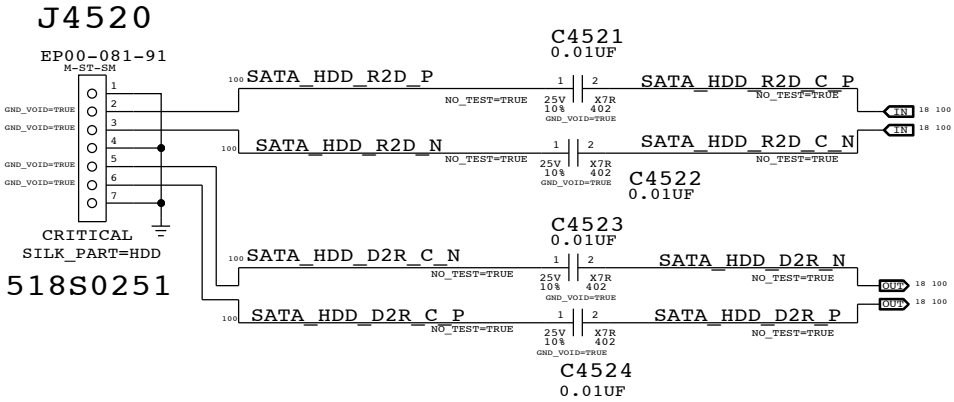
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C

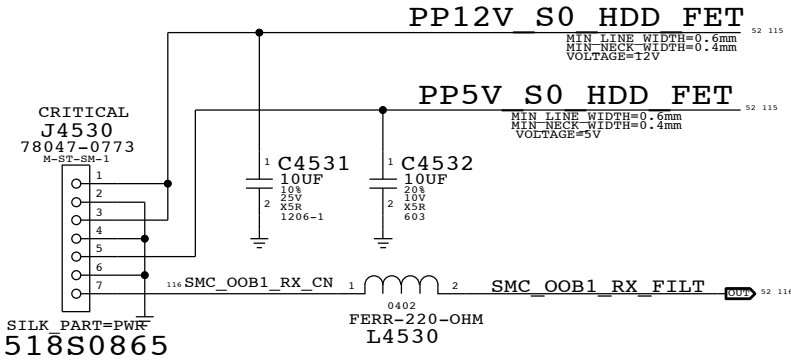
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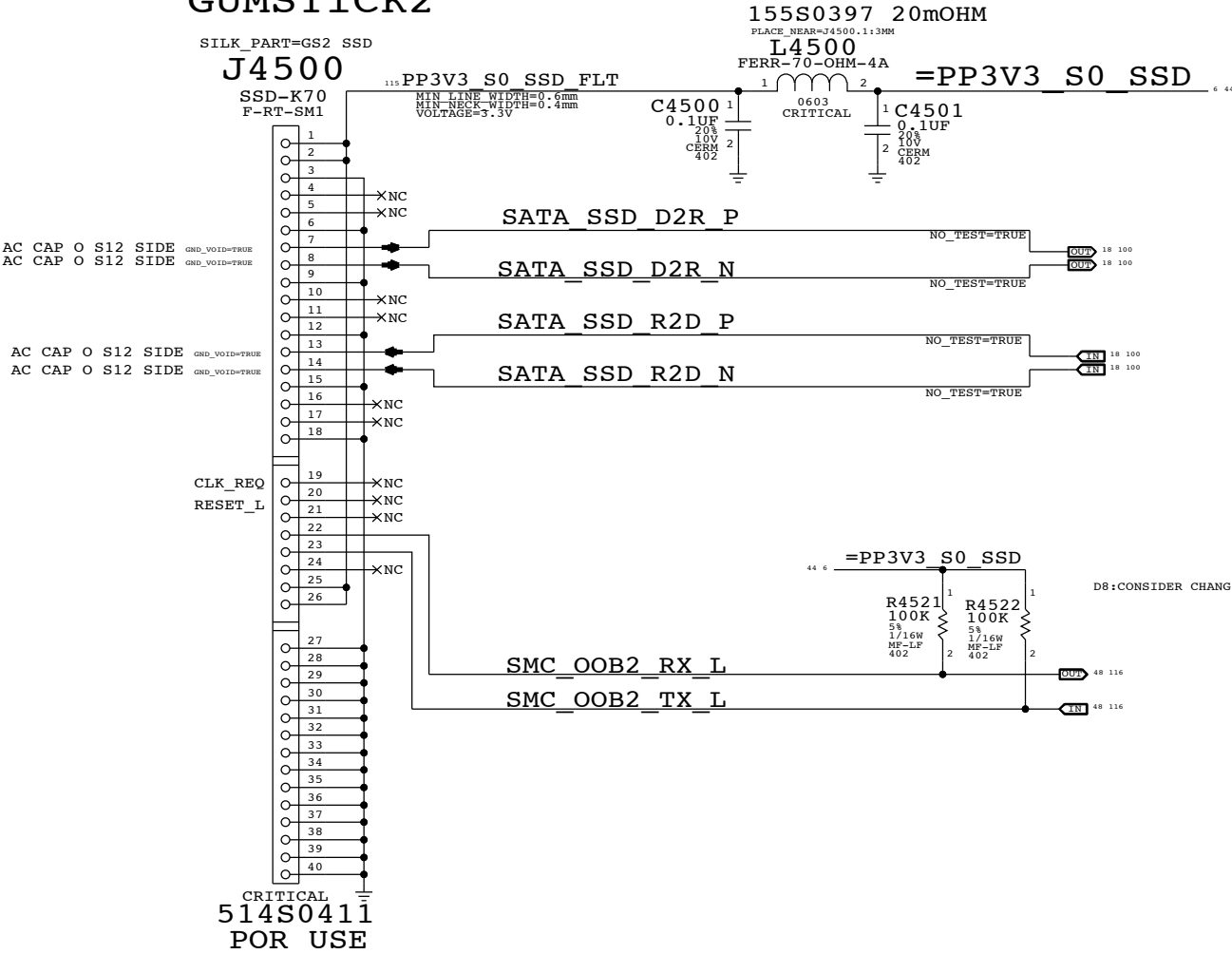
HDD DATA



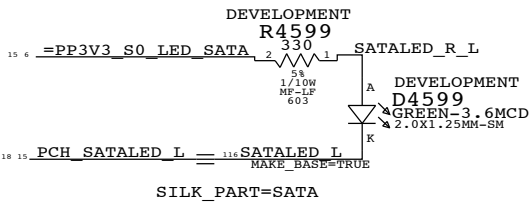
HDD POWER



GUMSTICK2

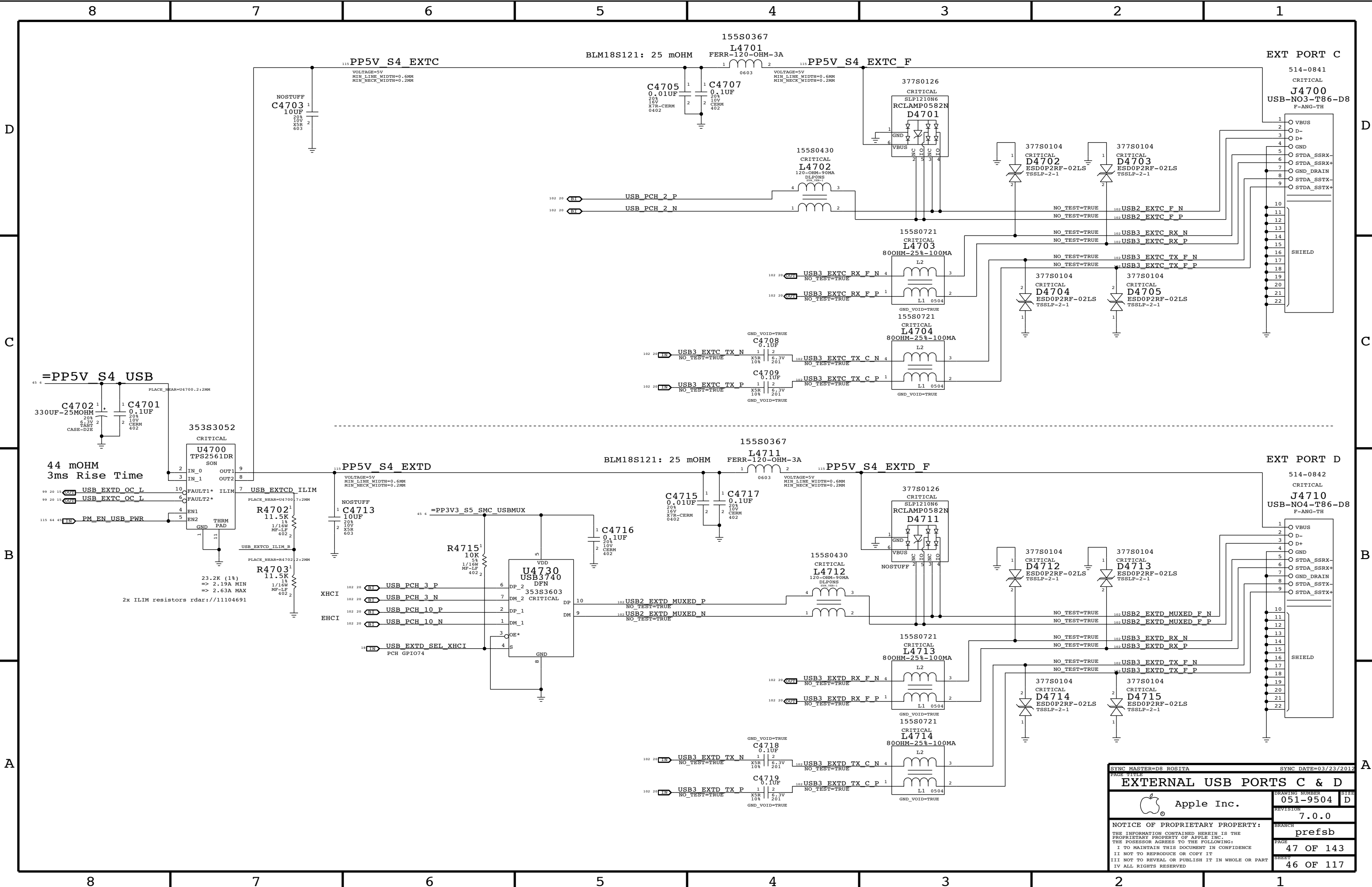


SATA Activity LED

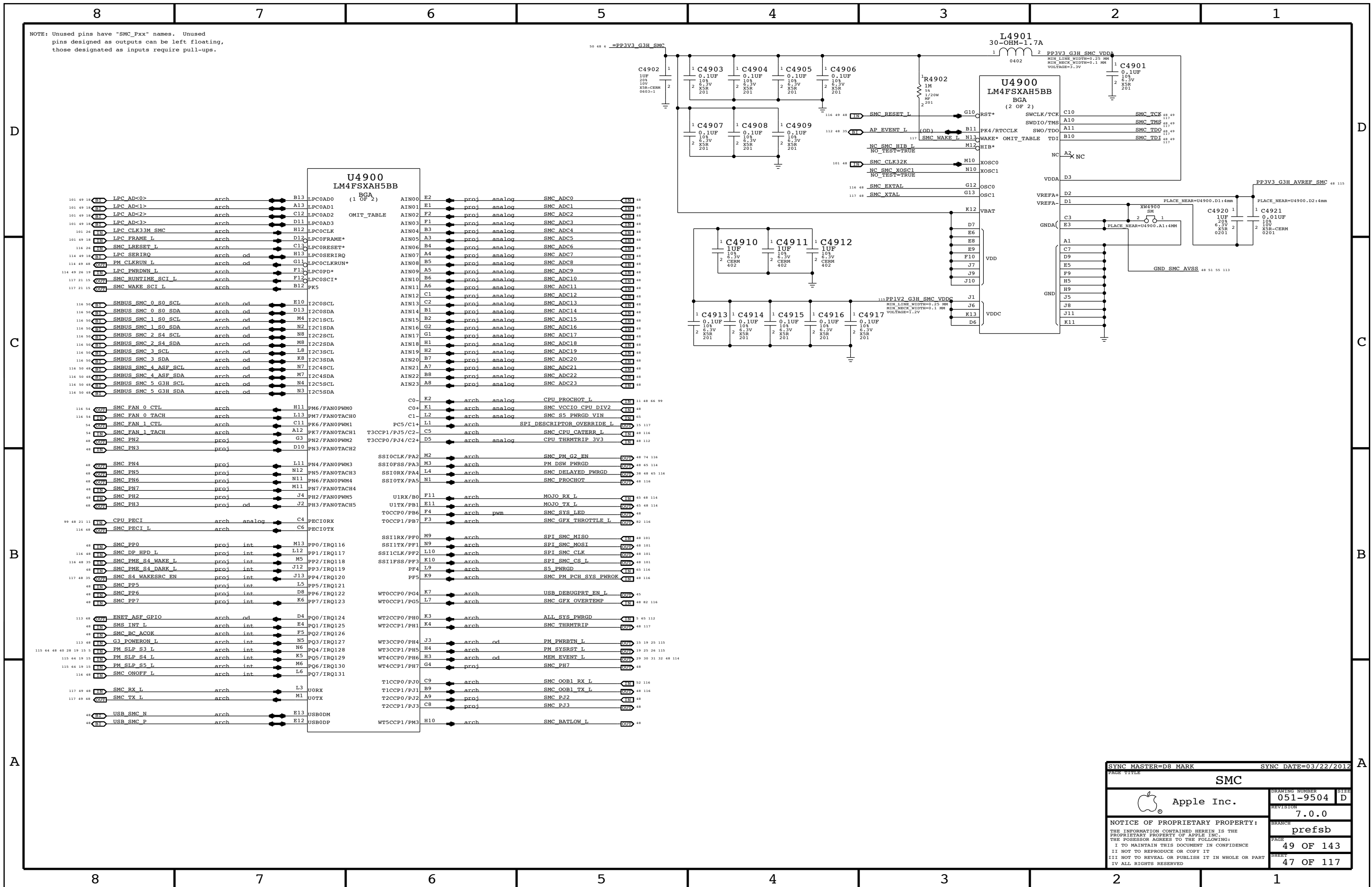


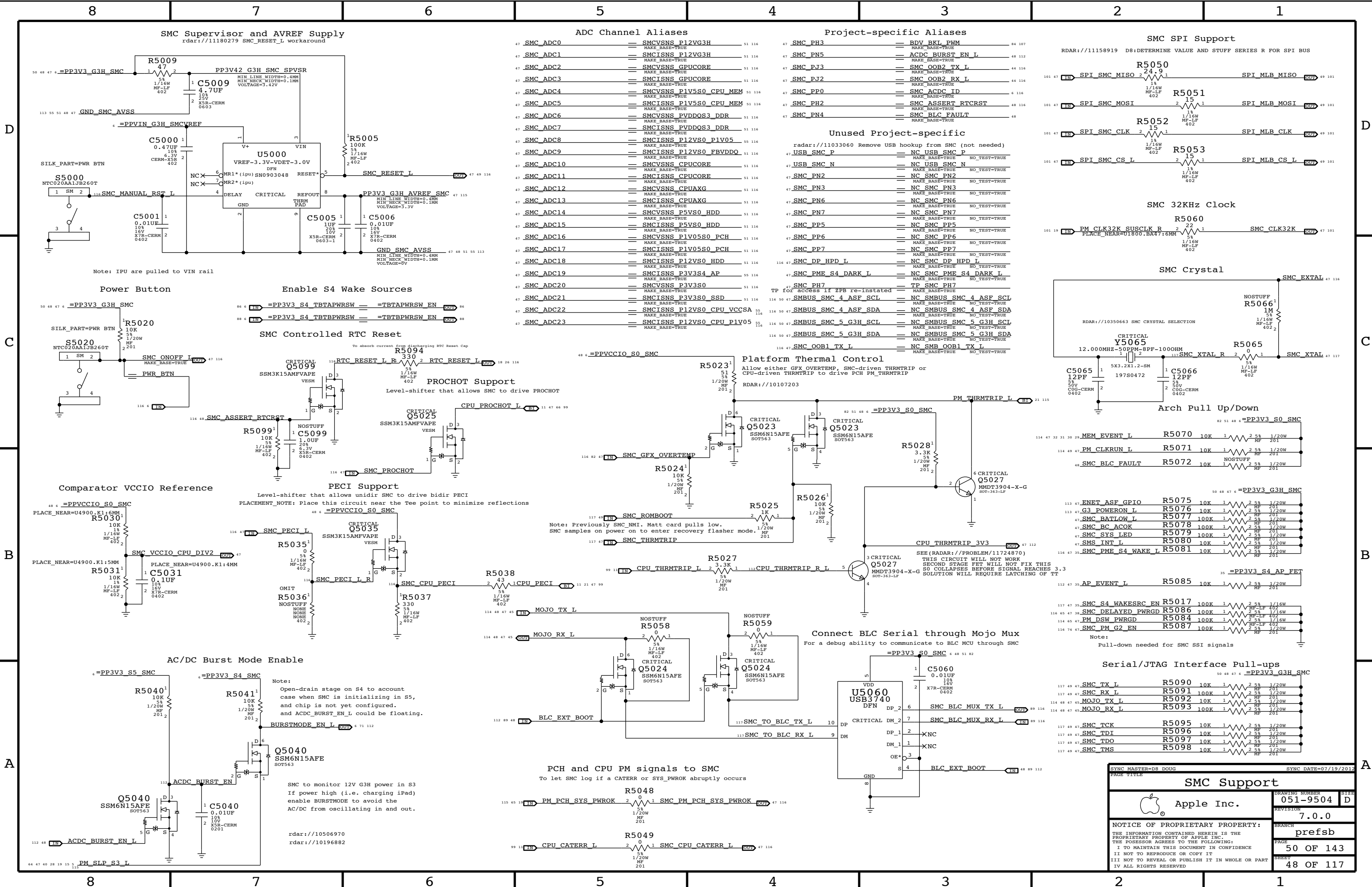
D8:CONSIDER CHANGING SMC_OOB2_RX/TX_L PULL-UP TO 100K (FROM 10K) RDAR://10817697

SYNC MASTER=D8 JERRY		SYNC DATE=01/31/2012	
PAGE TITLE			
SATA Connectors			
 Apple Inc.		DRAWING NUMBER	051-9504
		SIZE	D
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		BRANCH	prefsb
		PAGE	45 OF 143
		SHEET	44 OF 117



SYNC MASTER=D8 ROSITA		SYNC DATE=03/23/2012	
PAGE TITLE			
EXTERNAL USB PORTS C & D			
 Apple Inc.		DRAWING NUMBER	051-9504
		SIZE	D
		REVISION	7.0.0
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D

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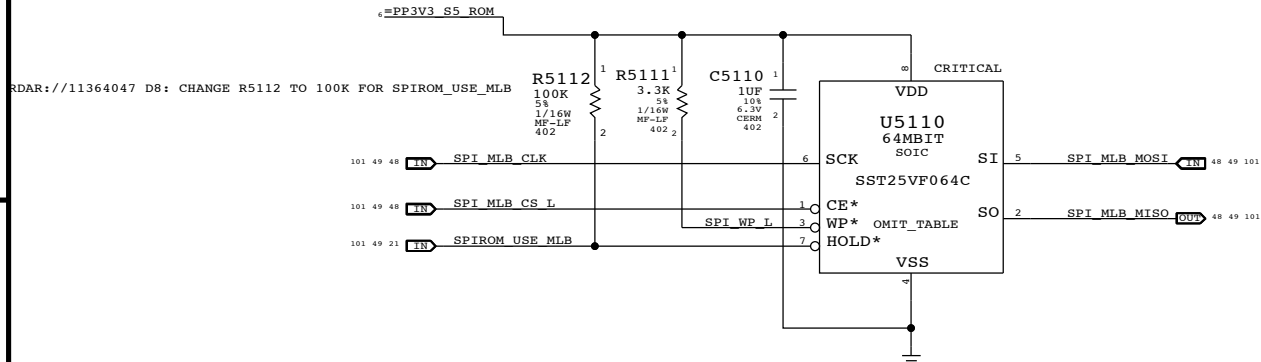
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C

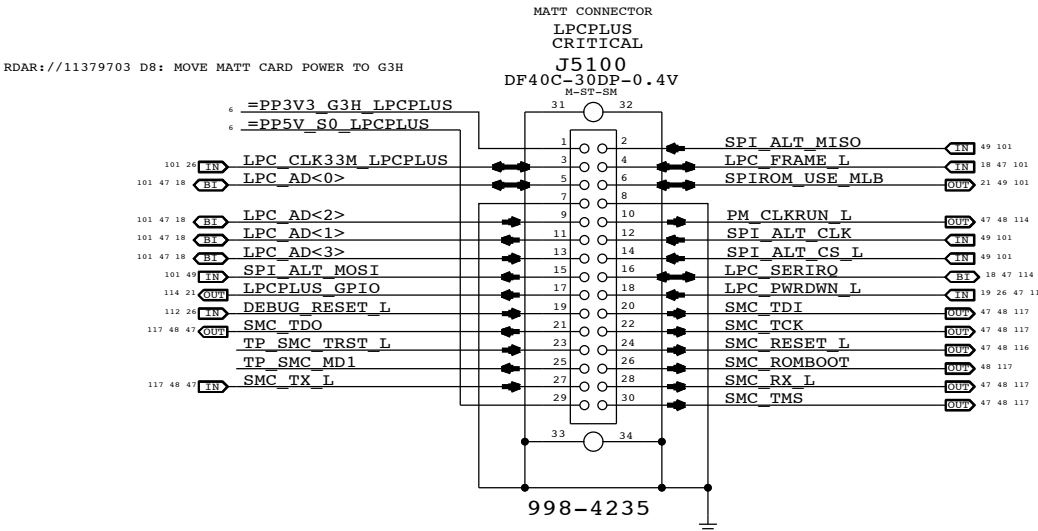
B

A

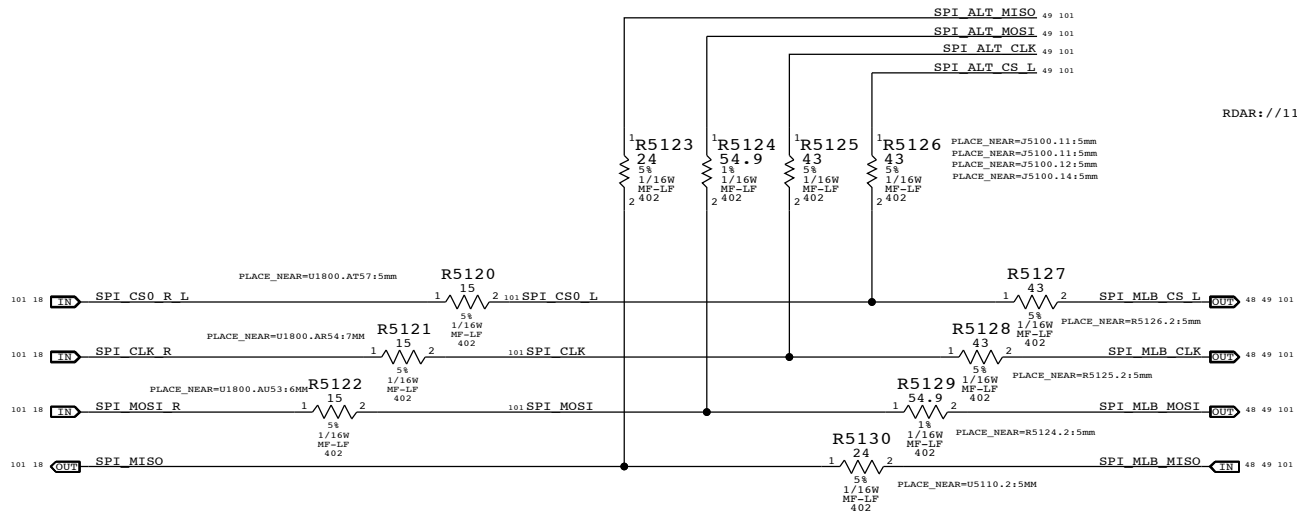
SPI BootROM



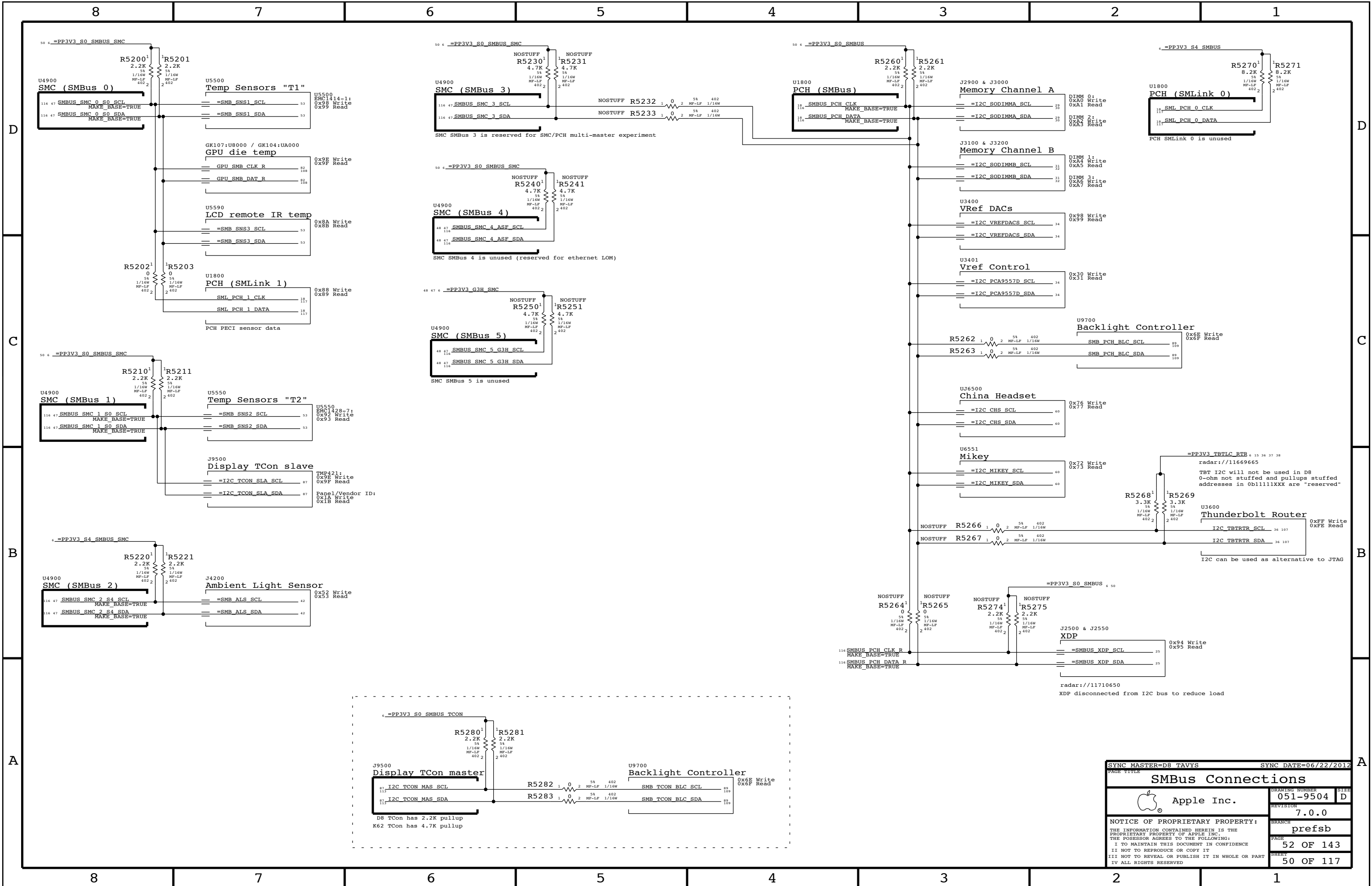
LPC+SPI Connector

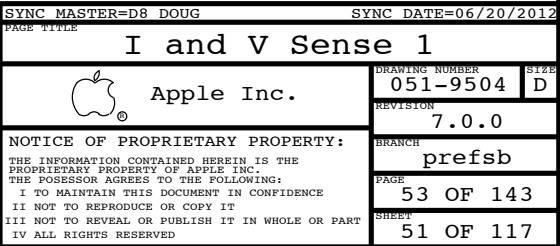


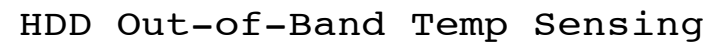
SPI Series Termination



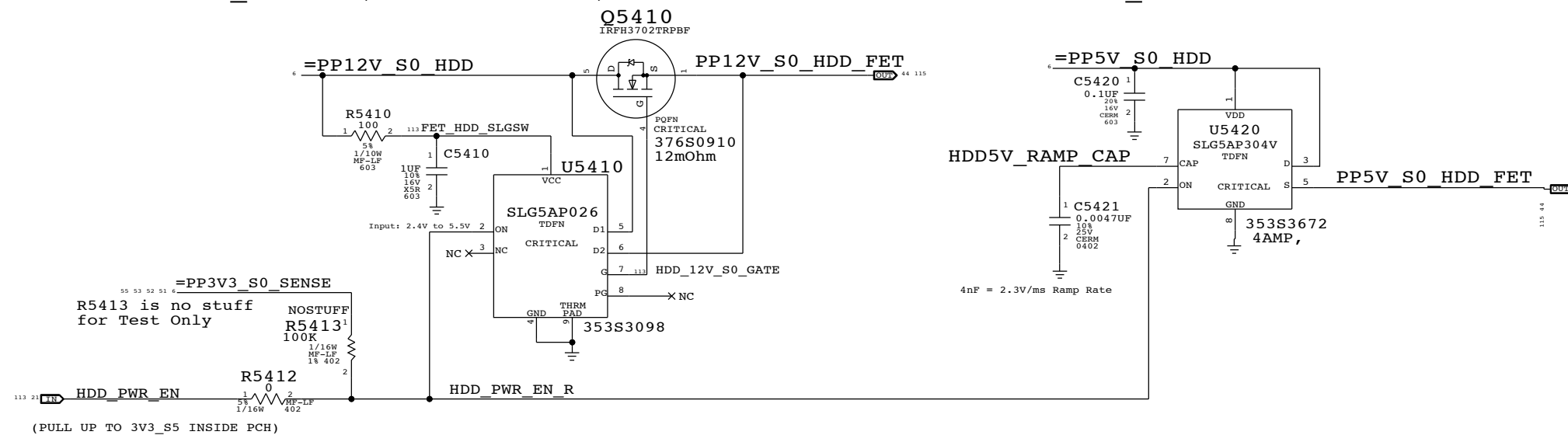
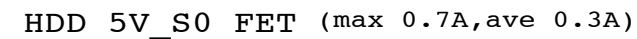
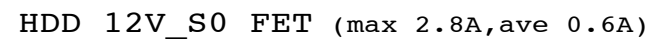
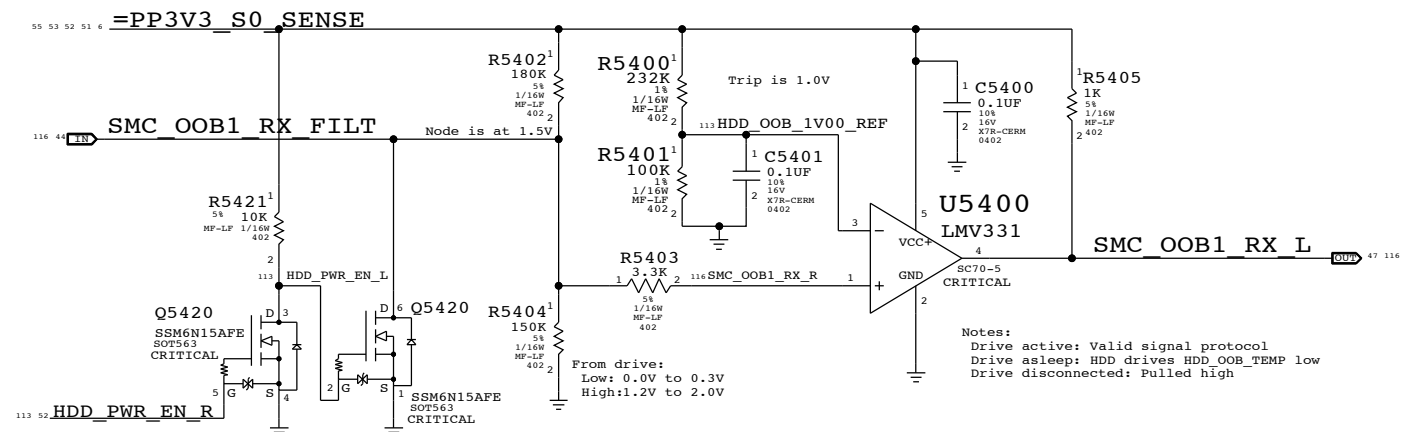
SYNC MASTER=D8_MLB		SYNC DATE=N/A	
PAGE TITLE			
SPI and Debug Connector			
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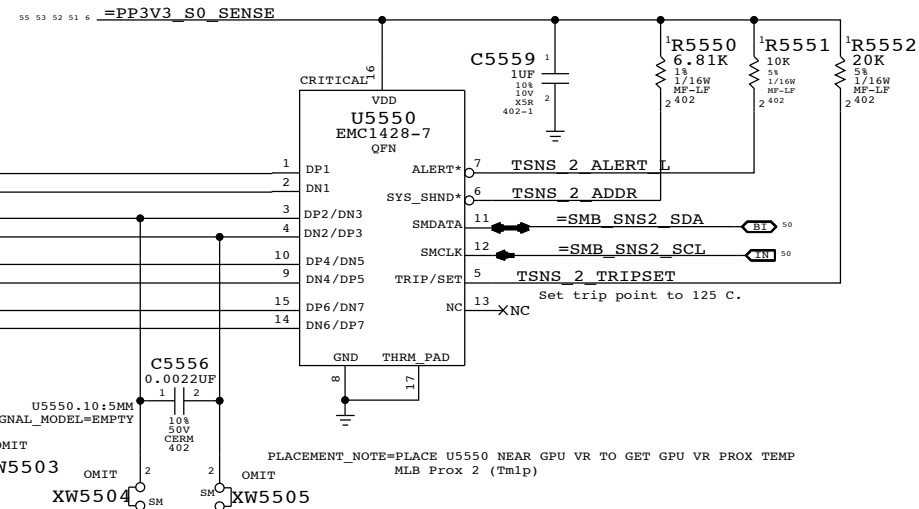
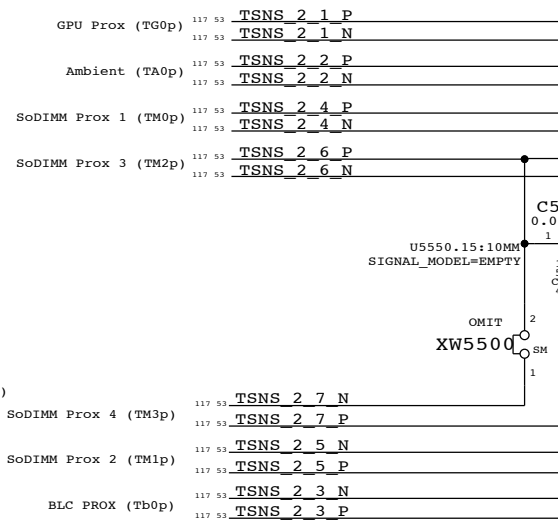
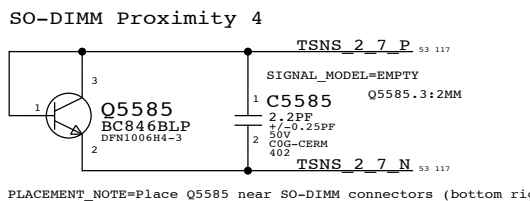
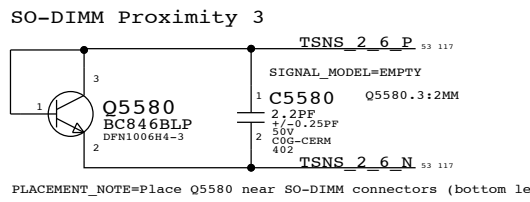
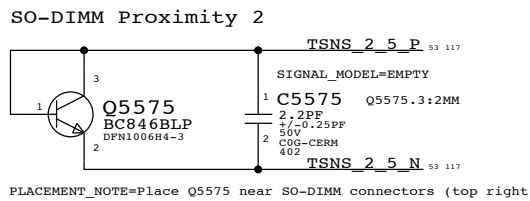
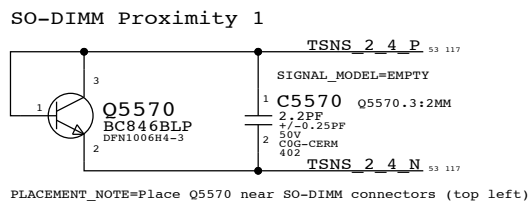
```
Temperature read from SATA power connector pin 11
```



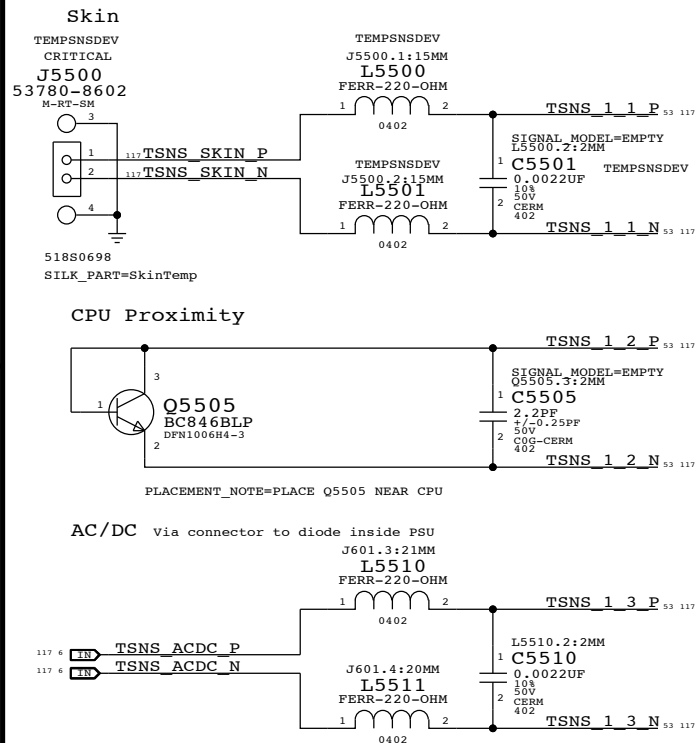
REMOVE R5413 AND SHORT R5412 AFTER HDD_PWR_EN WORKS

- - - - - SYNC-MASTER=DS - JERRY- - - - -		- - - - - SYNC-DATE=02/25/2012 - - - - -	
PAGE TITLE			
HDD/SSD Temp Sense			
 Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	7.0.0
		D	
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PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode



EMC1428-7: 6.8K PULL UP: I2C ADDRESS: WRITE: 0x92, READ: 0x93

[illegible]

5 53 52 51 0

=PP3V3 _S0 =SENSE

TEMPSENSDEV

1 C5590

0.01UF

10% 16V

X7R-CERM 0402

TEMPSENSDEV

1 R5590

10K

5% 1/16W

MF-LF 2402

U590

TMP006AIYZER

WCSP

ADR0

ADR1

DRDY*

TEMPSENSDEV

DGND AGND

A1 A2

50

50

=SMB_SNS3_SDA

C3

SDA

=SMB_SNS3_SCL

B3

SCL

C1

B1

C2

TSNS_3_DRDY_L

I2C Address (TMP006):

0xBA (Write)

0xBA (Read)

LCD Skim (TL2p)

NOTE - Follow TI layout guide(SBOU108.pdf) for

SYNC MASTER=D8 DOUG		SYNC DATE=06/07/2012	
PAGE TITLE			
Temperature Sensors			
 Apple Inc.		DRAWING NUMBER 051-9504	SIZE D
		REVISION 7.0.0	
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		SHEET 53 OF 117	

CRITICAL
L5600
220-OHM-1.4A

PP12V_S0_FAN

C5600
4.7UF
20%
16V
2 CERM
1206-1

C5601
0.01UF
20%
16V
2 X7R-CERM
0402

VOLTAGE=12V
MIN_LINE_WIDTH=0.5MM
MIN_NECK_WIDTH=0.25MM

518S0730

CRITICAL
Q5610
SSM3K15AMFVAPE
VESM

R5610
10K
5%
1/16W
MF-LF
402

SMC_FAN_0_CTL

FAN_0_PWM_FET

C5610
100PF
5%
50V
2 CERM
0402

CRITICAL
L5610
FERR-220-OHM

FAN_0_PWM_FILT

FAN_0_TACH_FILT

CRITICAL
J5600
53780-8604
M-RT-SM

Tach
GND
12V DC

PP3V3_S0_FAN

R5626
47K
5%
1/16W
MF-LF
402

SMC_FAN_0_TACH

FAN_0_TACH_FET

L5620
FERR-220-OHM

C5620
100PF
5%
50V
2 CERM
0402

U4900.L13:5MM
C5690
1000PF
10%
16V
2 X7R-CERM
0201

```

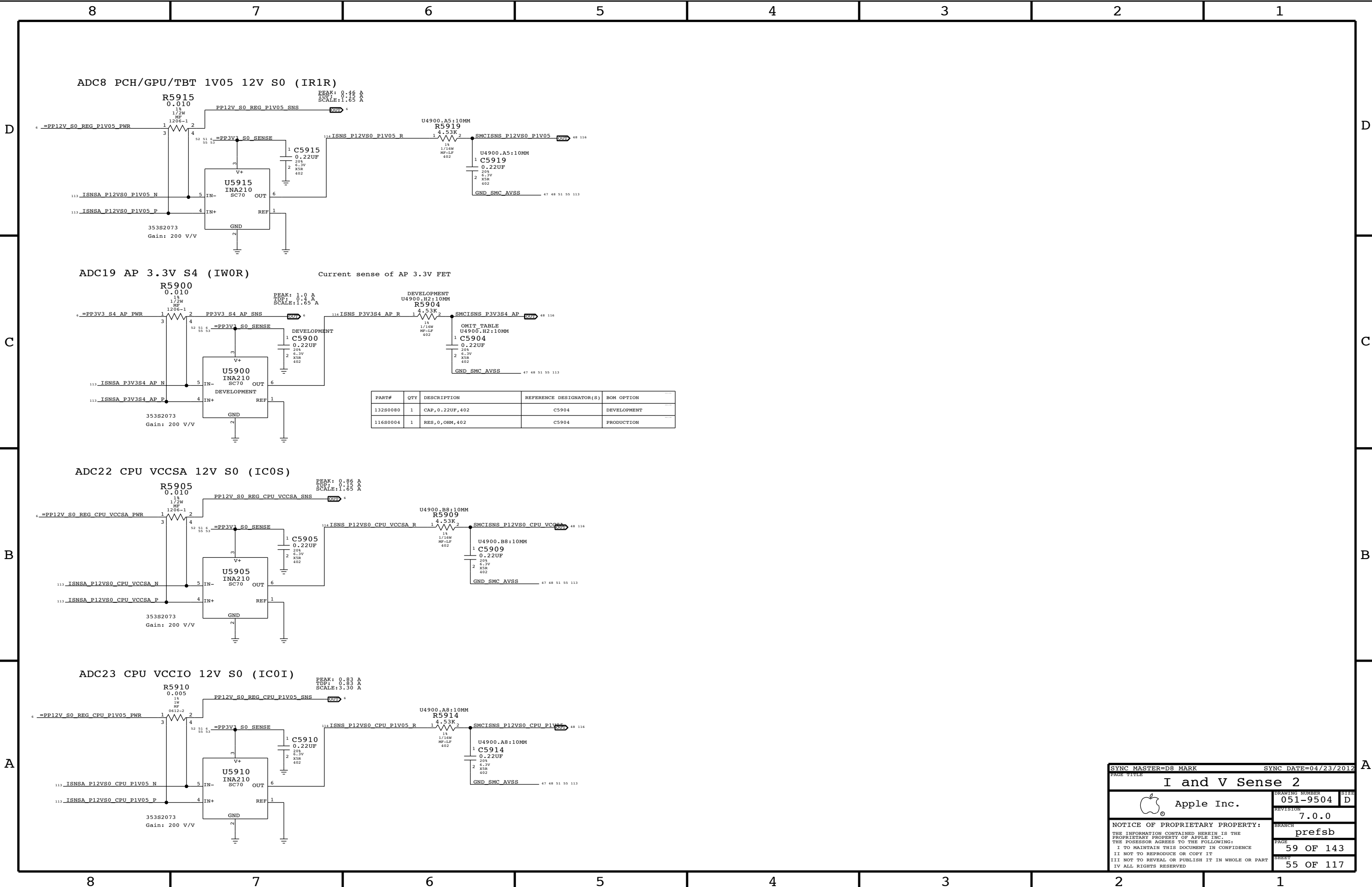
47 IN SMC_FAN_1_CTL == NC_SMC_FAN_1_CTL
== MAKE_BASE=TRUE NO_TEST=TRUE

47 OUT SMC_FAN_1_TACH == NC_SMC_FAN_1_TACH
== MAKE_BASE=TRUE NO_TEST=TRUE

```

The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q5610 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q5610 is at common and the SMC sinks current when Q5610 is on.

Otherwise, this is simply a pass-FET.
See RADAR: 10565825- D7: Need scematic and PCB file of fan(All Vendors).



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0080	1	CAP,0.22UF,402	C5904	DEVELOPMENT
116S0004	1	RES,0,0HM,402	C5904	PRODUCTION

SYNC MASTER=D8 MARK

SYNC DATE=04/23/2012

I and V Sense 2

 Apple Inc.

051-9504

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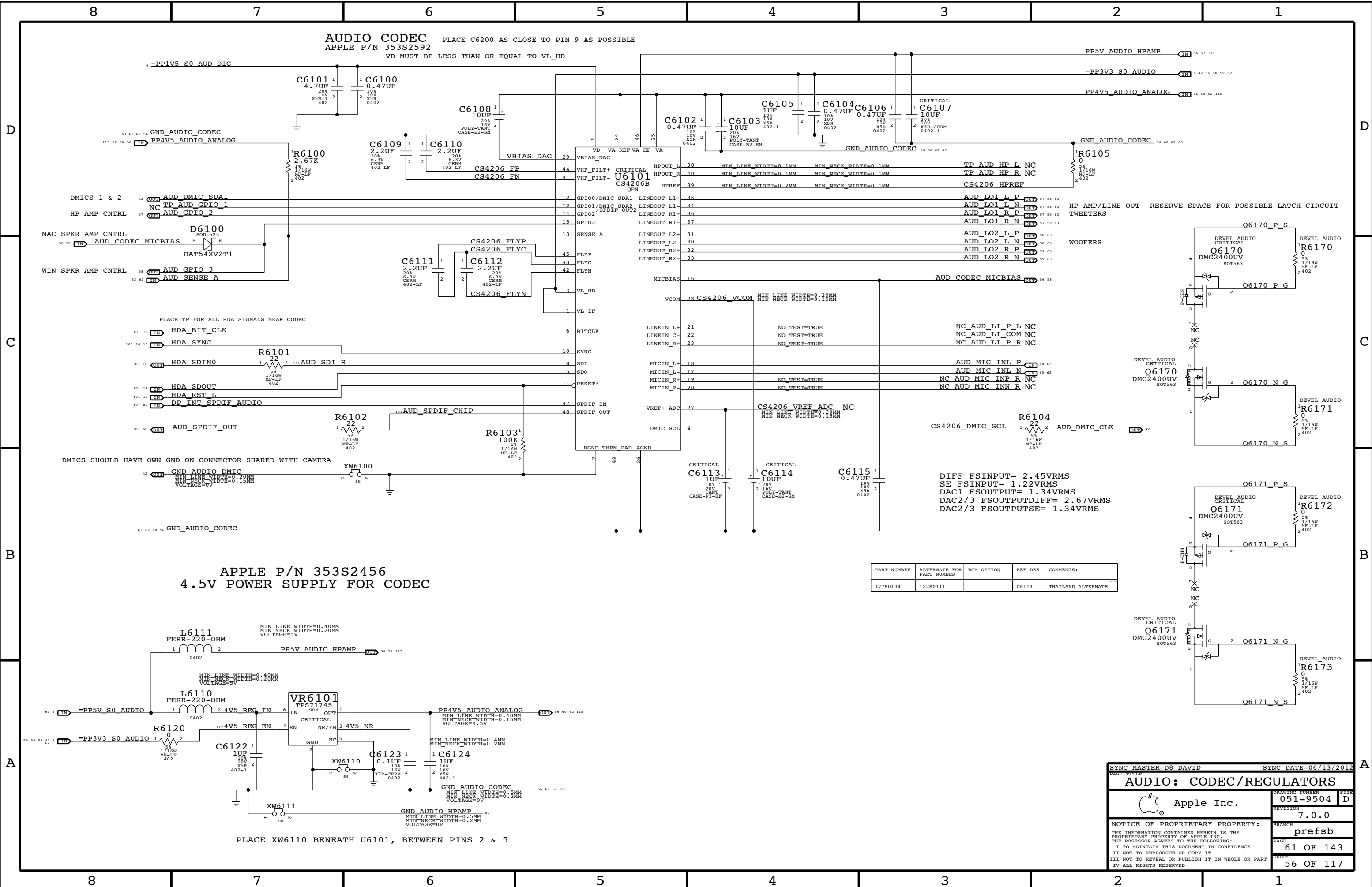
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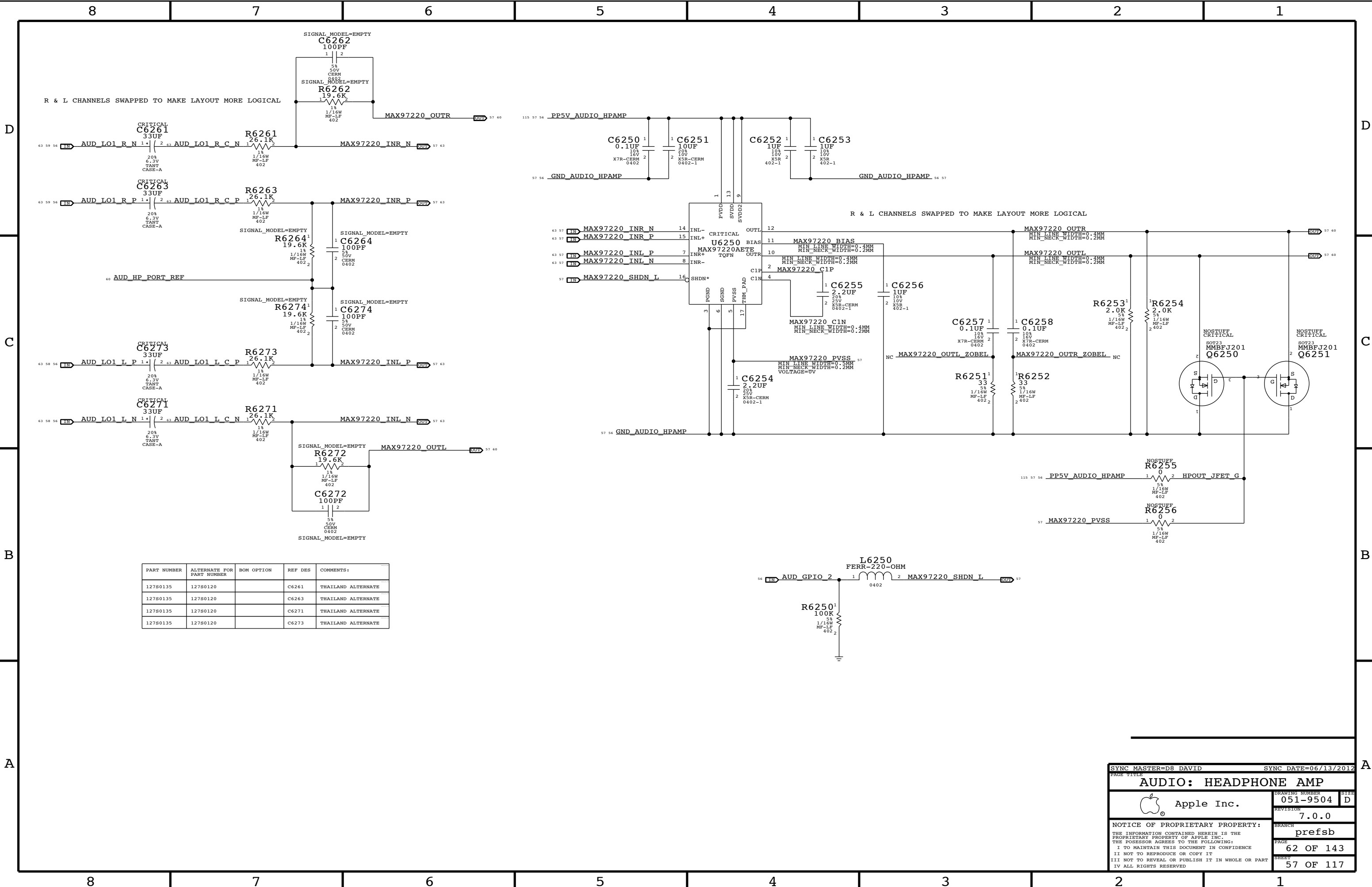
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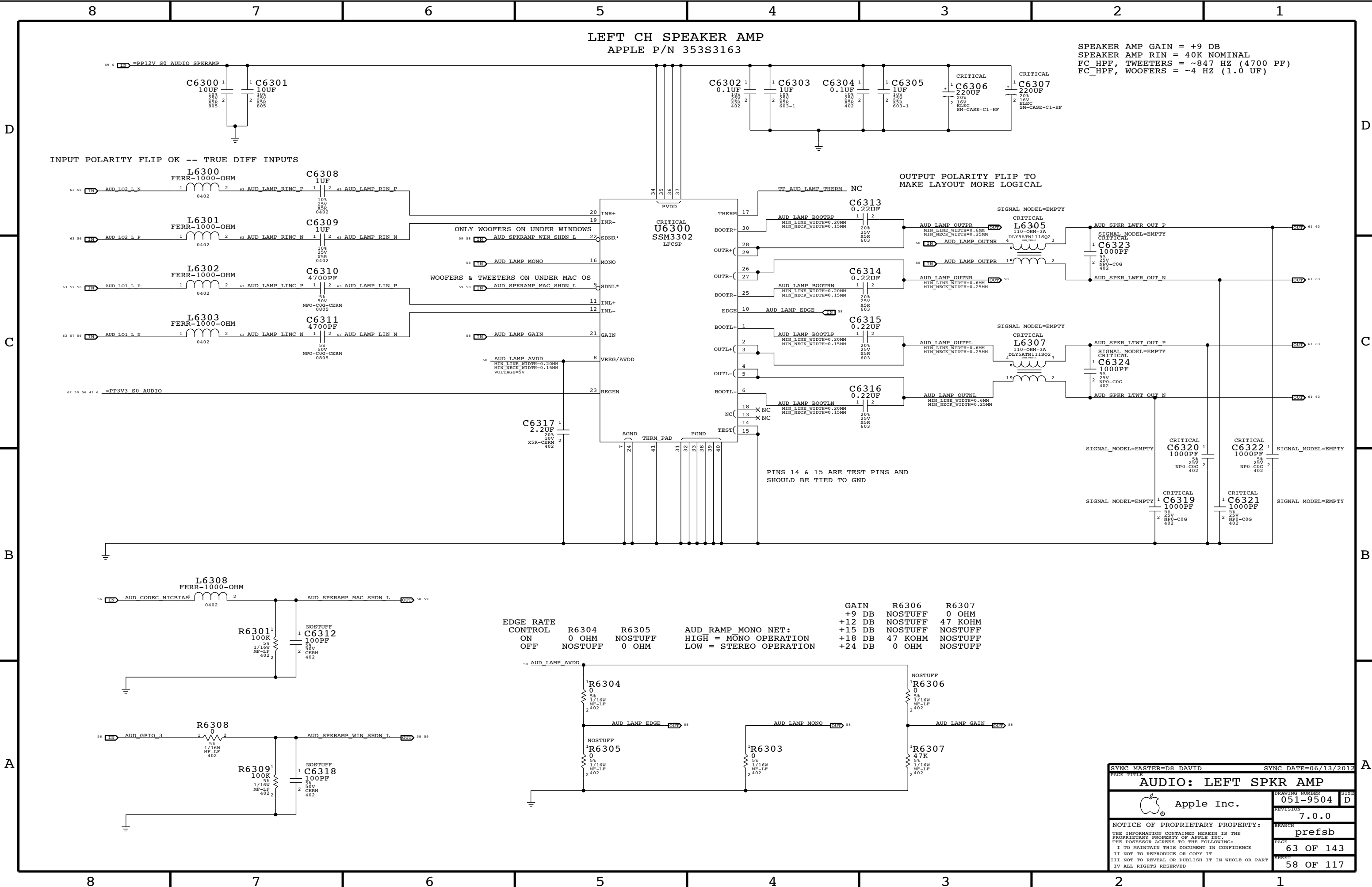


PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
127S0134	127S0111		C6113	THAILAND ALTERNATE

PAGE TITLE		SYNC DATE=06/13/2012	
AUDIO: CODEC/REGULATORS		DRAWING NUMBER	
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PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
12780135	12780120		C6261	THAILAND ALTERNATE
12780135	12780120		C6263	THAILAND ALTERNATE
12780135	12780120		C6271	THAILAND ALTERNATE
12780135	12780120		C6273	THAILAND ALTERNATE

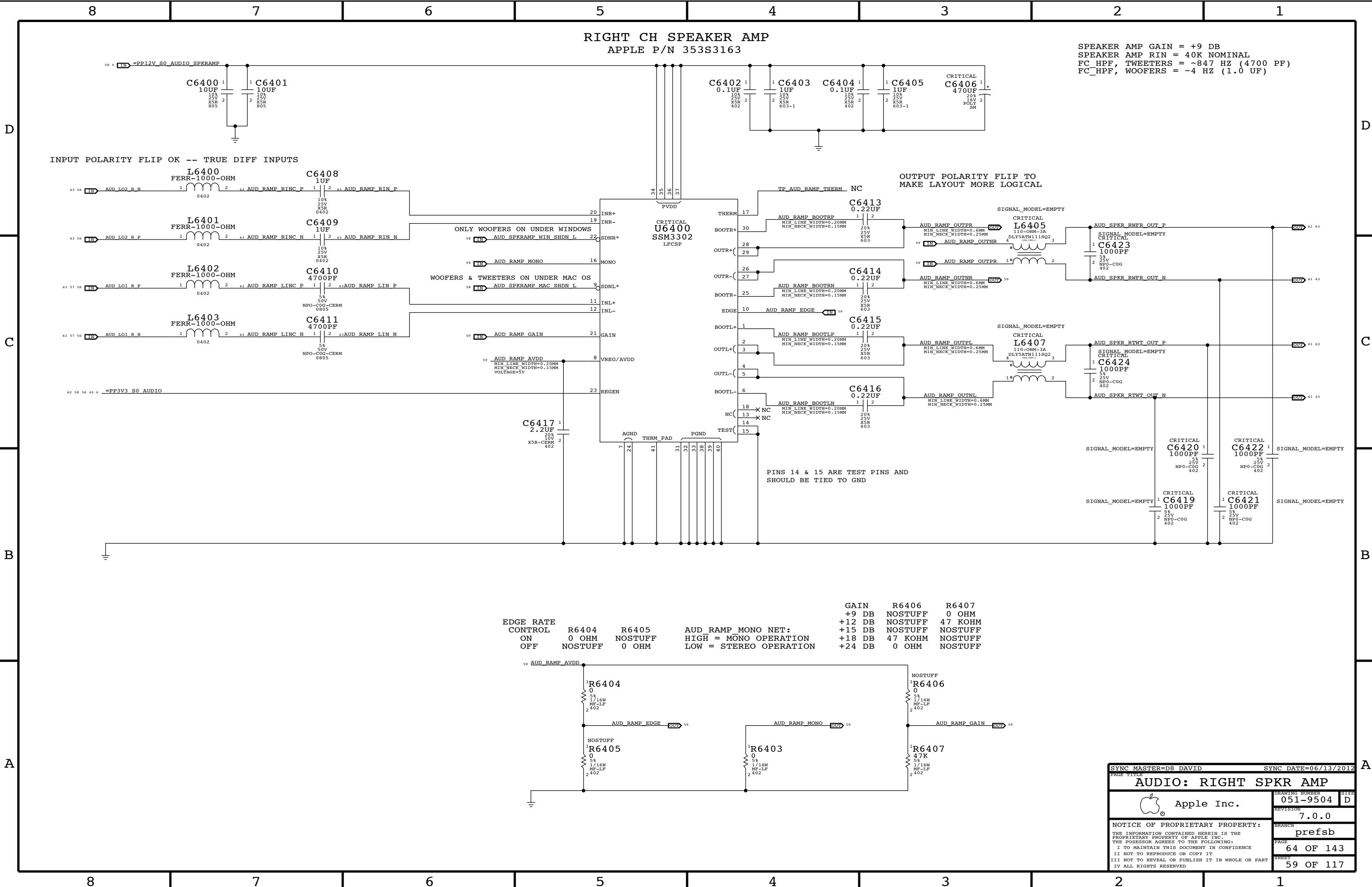


SPEAKER AMP GAIN = +9 DB
SPEAKER AMP RIN = 40K NOMINAL
FC HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HP, WOOFERS = ~4 HZ (1.0 UF)

PINS 14 & 15 ARE TEST PINS AND SHOULD BE TIED TO GND

EDGE RATE CONTROL ON OFF	R6304 0 OHM NOSTUFF	R6305 0 OHM NOSTUFF	AUD_RAMP_MONO NET: HIGH = MONO OPERATION LOW = STEREO OPERATION	GAIN +9 DB +12 DB +15 DB +18 DB +24 DB	R6306 NOSTUFF NOSTUFF NOSTUFF NOSTUFF NOSTUFF	R6307 0 OHM 47 KOHM NOSTUFF NOSTUFF NOSTUFF

SYNC MASTER=D8 DAVID		SYNC DATE=06/13/2012	
PAGE TITLE			
AUDIO: LEFT SPKR AMP			
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RIGHT CH SPEAKER AMP
APPLE P/N 353S3163

SPEAKER AMP GAIN = +9 DB
SPEAKER AMP RIN = 40K NOMINAL
FC HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HP, WOOFERS = ~4 HZ (1.0 UF)

INPUT POLARITY FLIP OK -- TRUE DIFF INPUTS

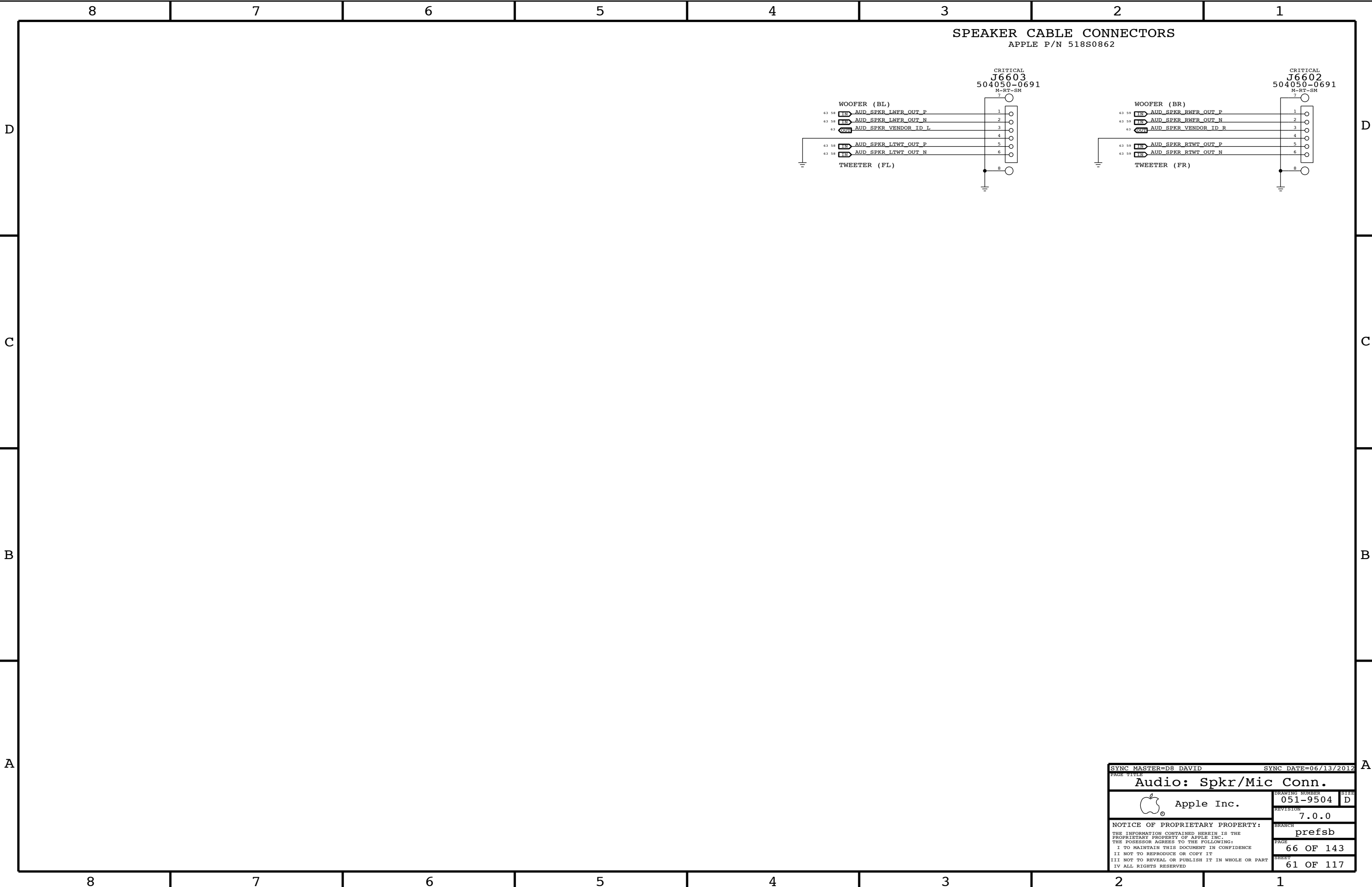
OUTPUT POLARITY FLIP TO
MAKE LAYOUT MORE LOGICAL

PINS 14 & 15 ARE TEST PINS AND
SHOULD BE TIED TO GND

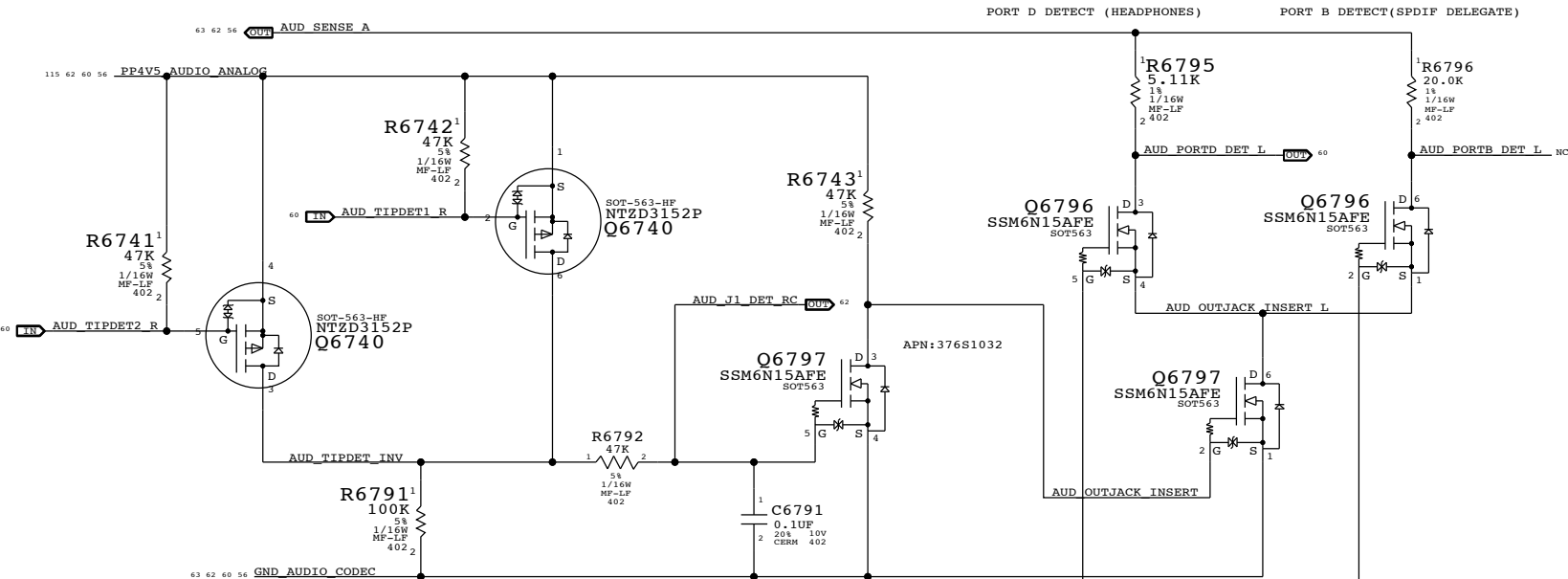
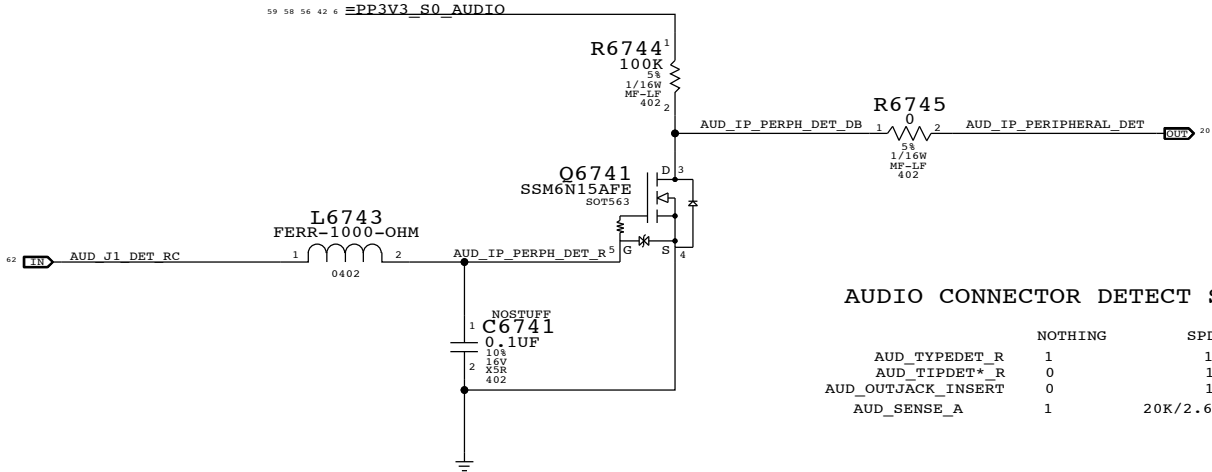
EDGE RATE	GAIN	R6406	R6407
CONTROL	+9 DB	NOSTUFF	0 OHM
ON	+12 DB	NOSTUFF	47 KOHM
OFF	+15 DB	NOSTUFF	NOSTUFF
	+18 DB	47 KOHM	NOSTUFF
	+24 DB	0 OHM	NOSTUFF

R6404	R6405	AUD_RAMP_MONO NET:
0 OHM	NOSTUFF	HIGH = MONO OPERATION
NOSTUFF	0 OHM	LOW = STEREO OPERATION

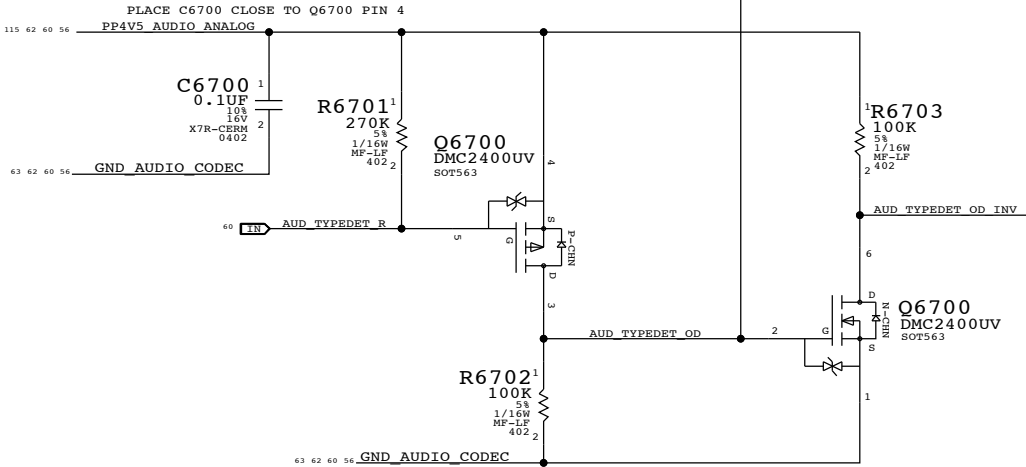
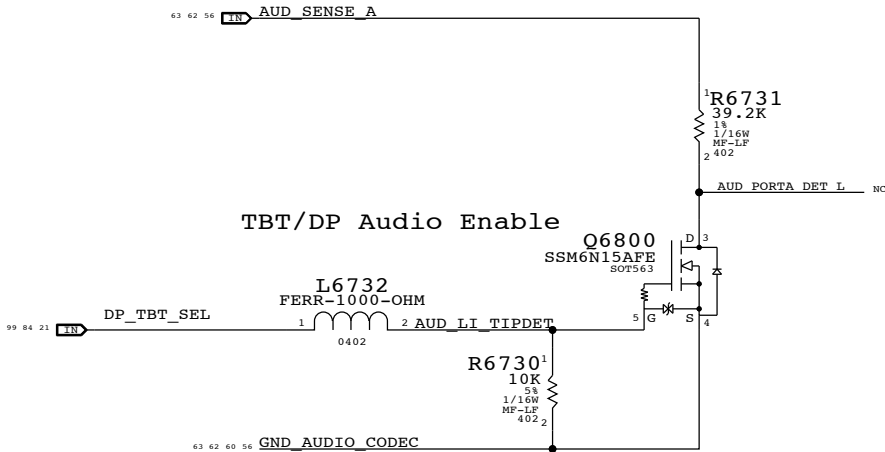
SYNC MASTER=D8 DAVID		SYNC DATE=06/13/2012	
PAGE TITLE			
AUDIO: RIGHT SPKR AMP			
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		REVISION	7.0.0
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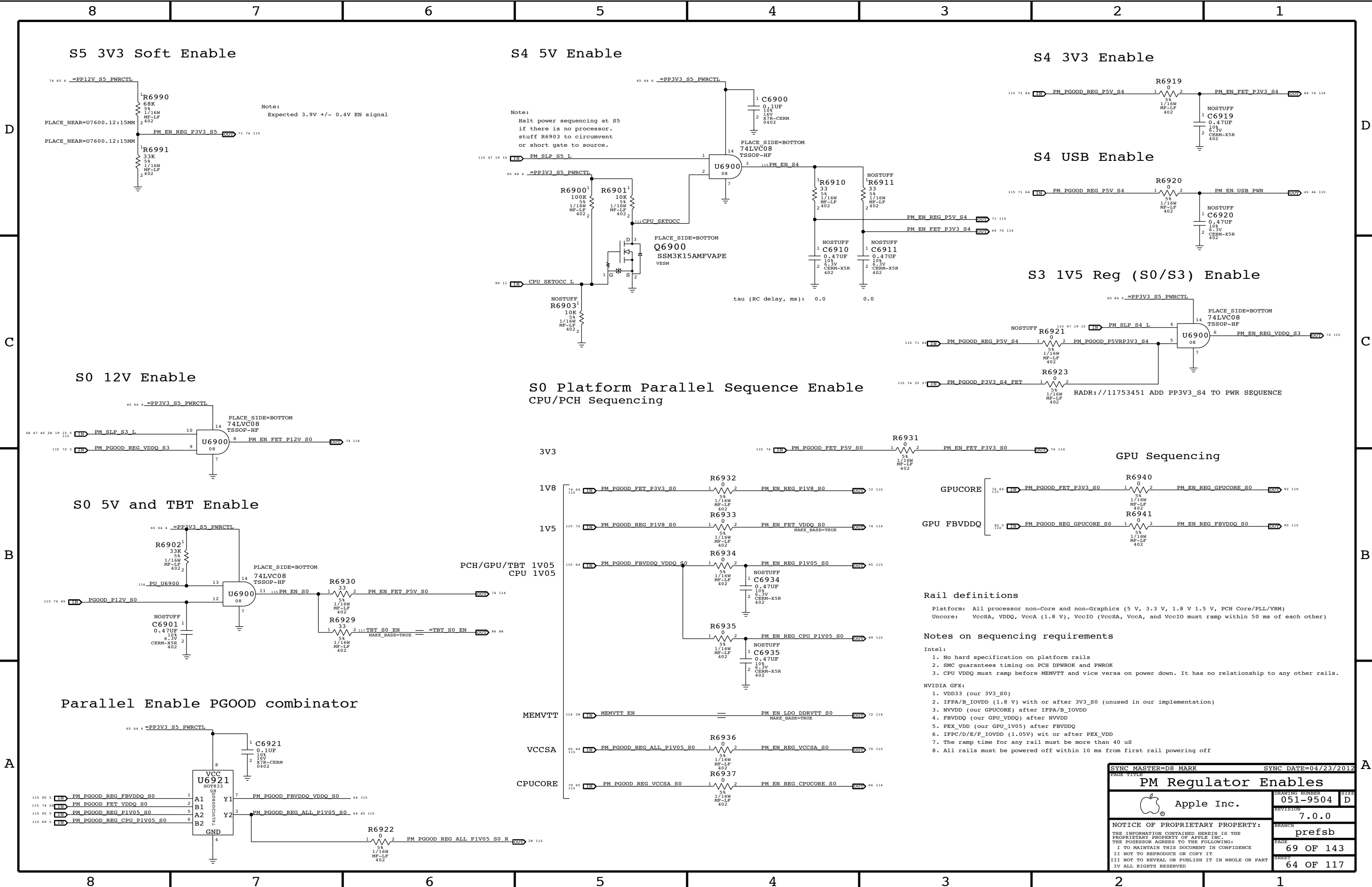
IPHS HS Detect Debounce CKT



LI Insert Detect (DETECT A)



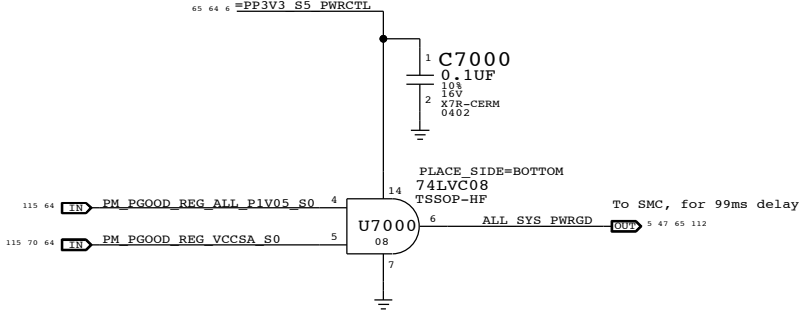
SYNC MASTER=D8 DAVID		SYNC DATE=06/13/2012	
PAGE TITLE			
AUDIO: Detects/Grounding			
 Apple Inc.	DRAWING NUMBER	051-9504	SIZE D
	REVISION	7.0.0	
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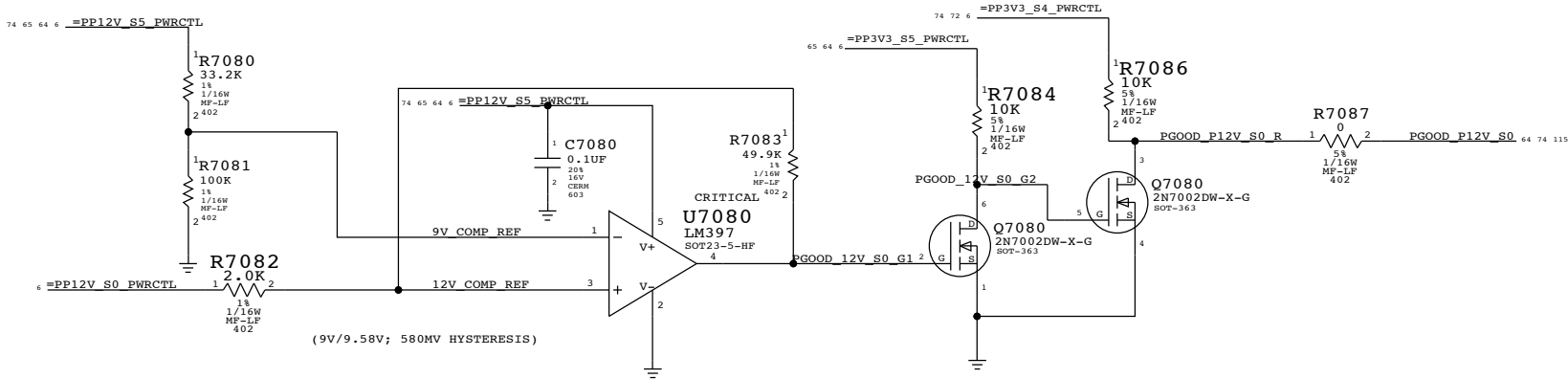
SYNC MASTER=D8 MARK		SYNC DATE=04/23/2012	
PAGE TITLE		PM Regulator Enables	
DRAWING NUMBER		051-9504	SIZE D
REVISION		7.0.0	BRANCH
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Platform Power Good Derive SMC ALL_SYS_PWRGD

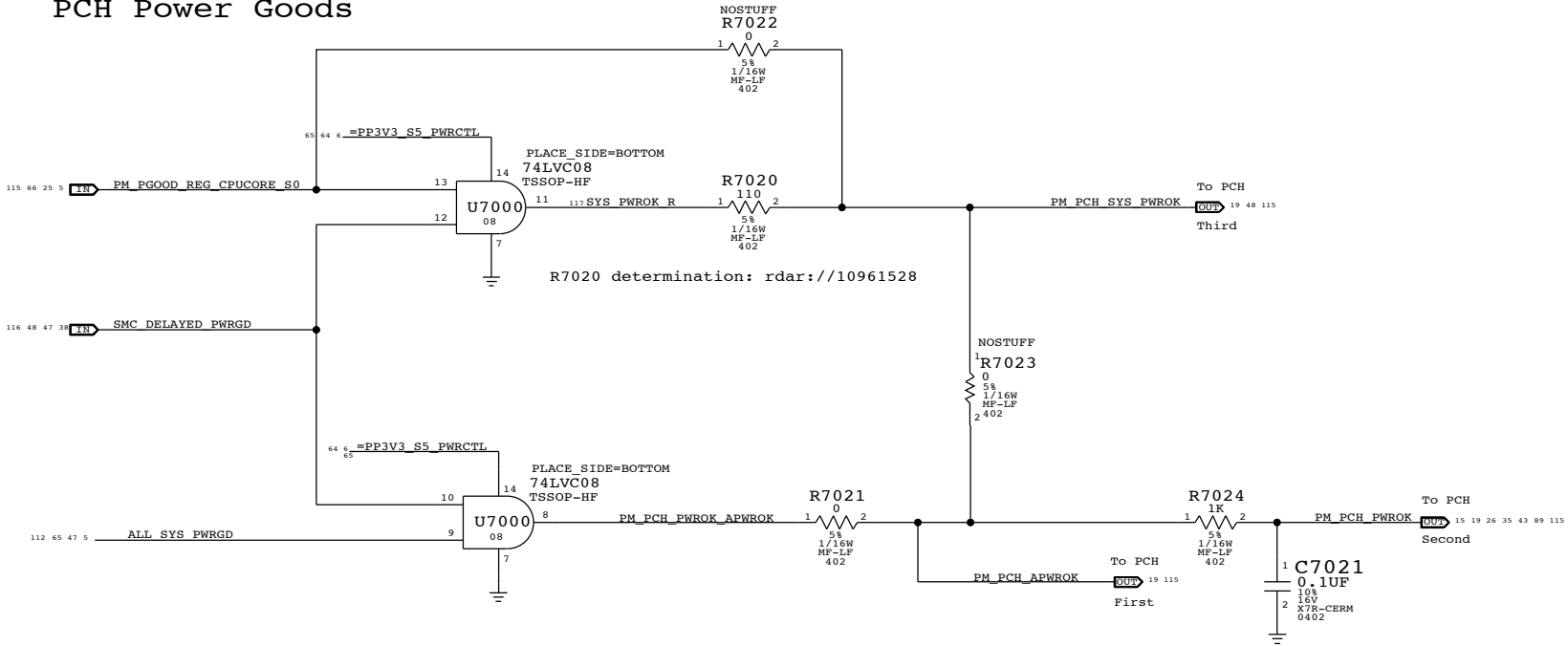
The end of the power sequence for S0 rails except CPU CORE.



PGOOD COMPARATORS FOR PP12V_S0



PCH Power Goods



radar://11043352 Need AND Gate to deassert PM_PCH_PWROK to PCH when unexpected power loss happens

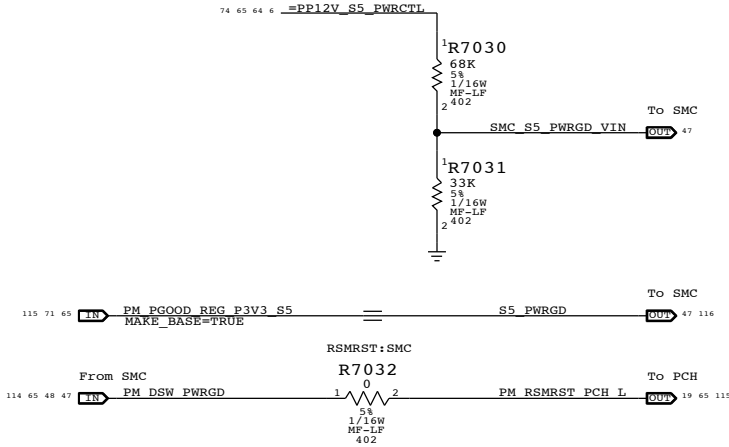
Resume Reset

Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

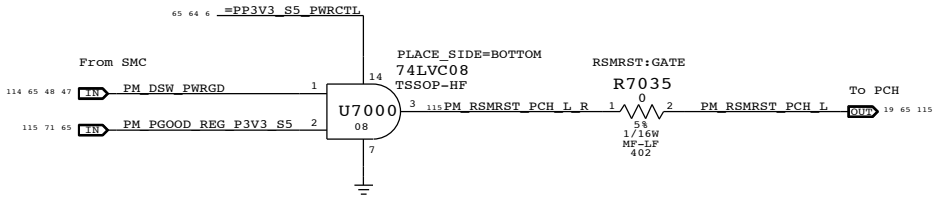
Note:
The iMac K70K72 designs does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

Requirements:
Power on:
Asserted at least 10 ms after all suspend well power is valid
Power off or loss of AC:
Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V
to allow PCH to switch suspend well to battery without excessive loading

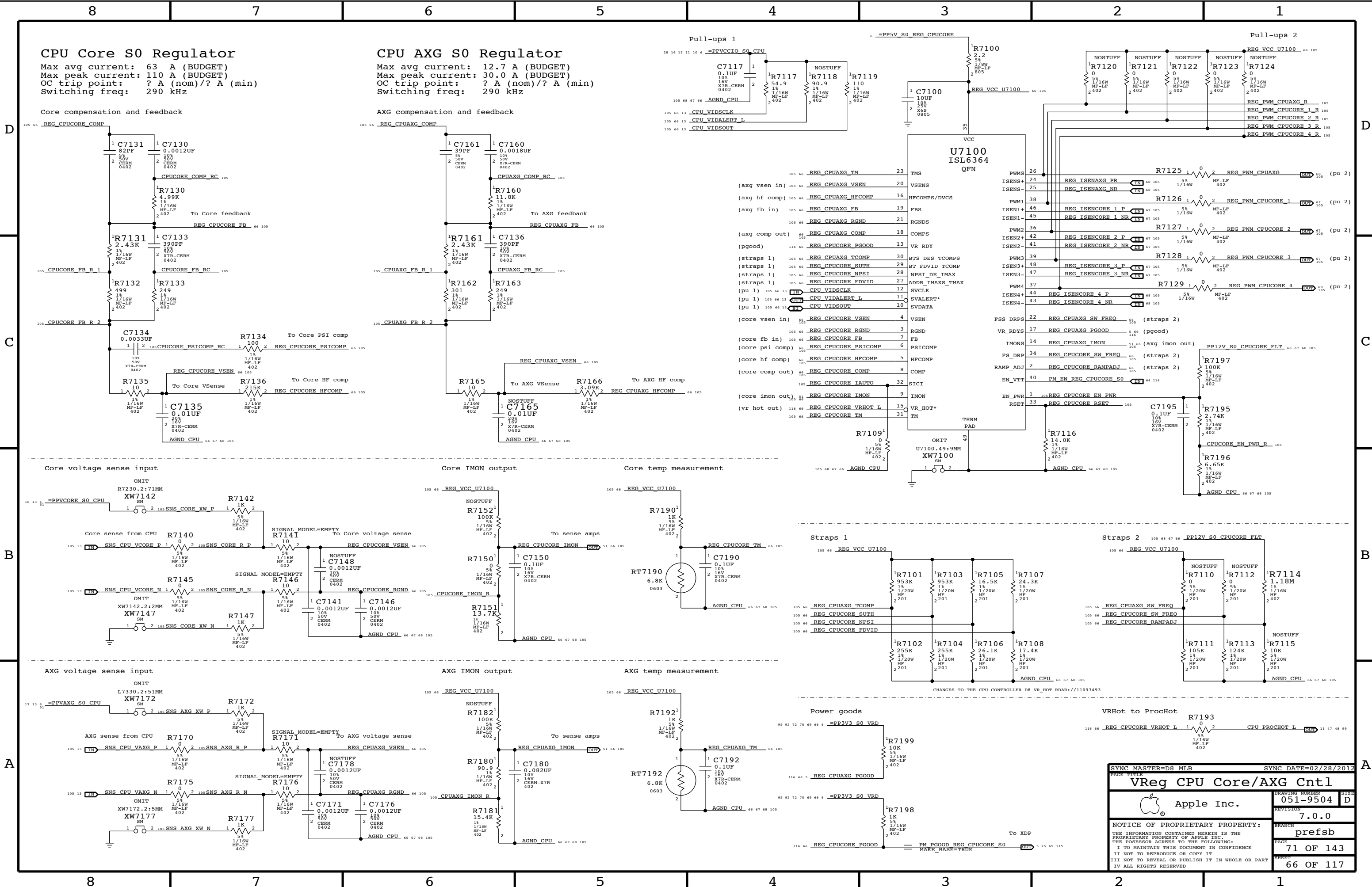
Primary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation.
SMC de-asserts RSMRST# (PM_DSW_PWRGD) when S5_PWRGD input is asserted and SMC_S5_PWRGD_VIN input is above comparator input level of 1.5 V.
SMC asserts RSMRST# (PM_DSW_PWRGD) when SMC_S5_PWRGD_VIN input drops from 1.8 V to 1.5 V (as implemented) when 12 V S5 rail drops to 10 V.



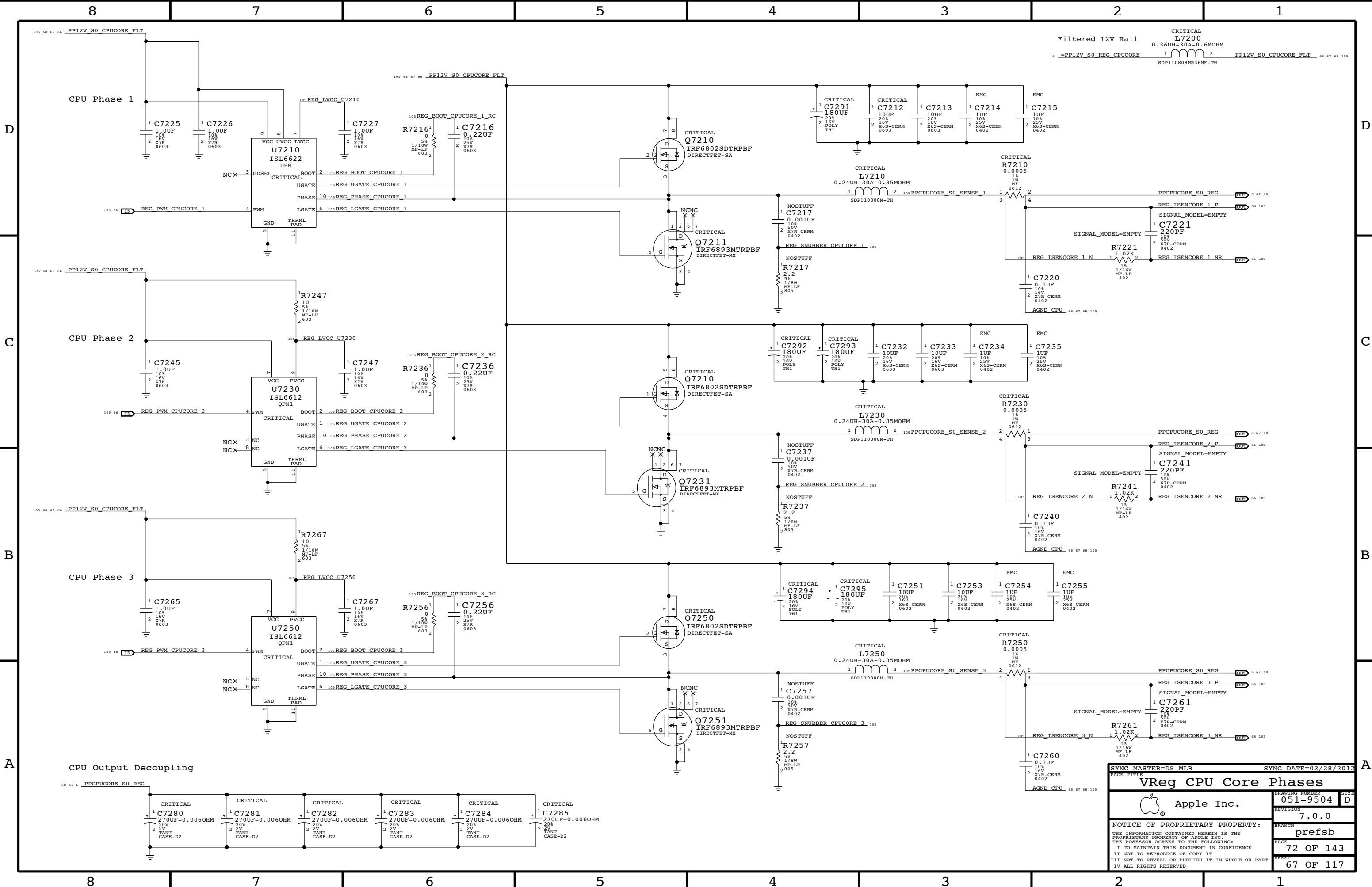
Secondary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.
RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



SYNC MASTER=D8 MARK		SYNC DATE=04/23/2012	
PAGE TITLE		PM Power Good	
Apple Inc.		DRAWING NUMBER	051-9504
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PAGE TITLE		VReg CPU Core/AXG Cntl	
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SYNC MASTER=D8_MLB

SYNC DATE=02/28/2012

VReg CPU Core Phases

Apple Inc.

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051-9504

SIZE

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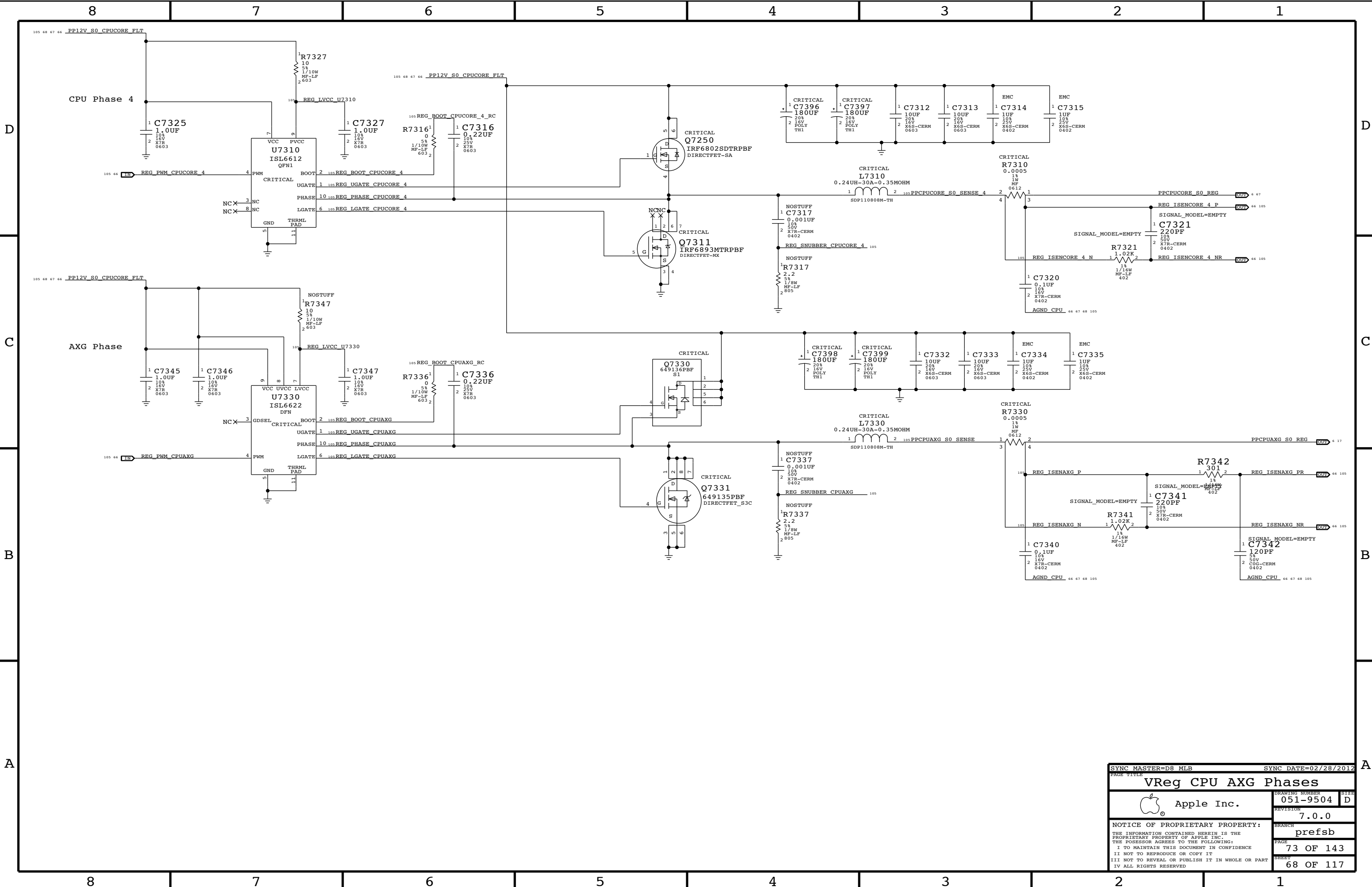
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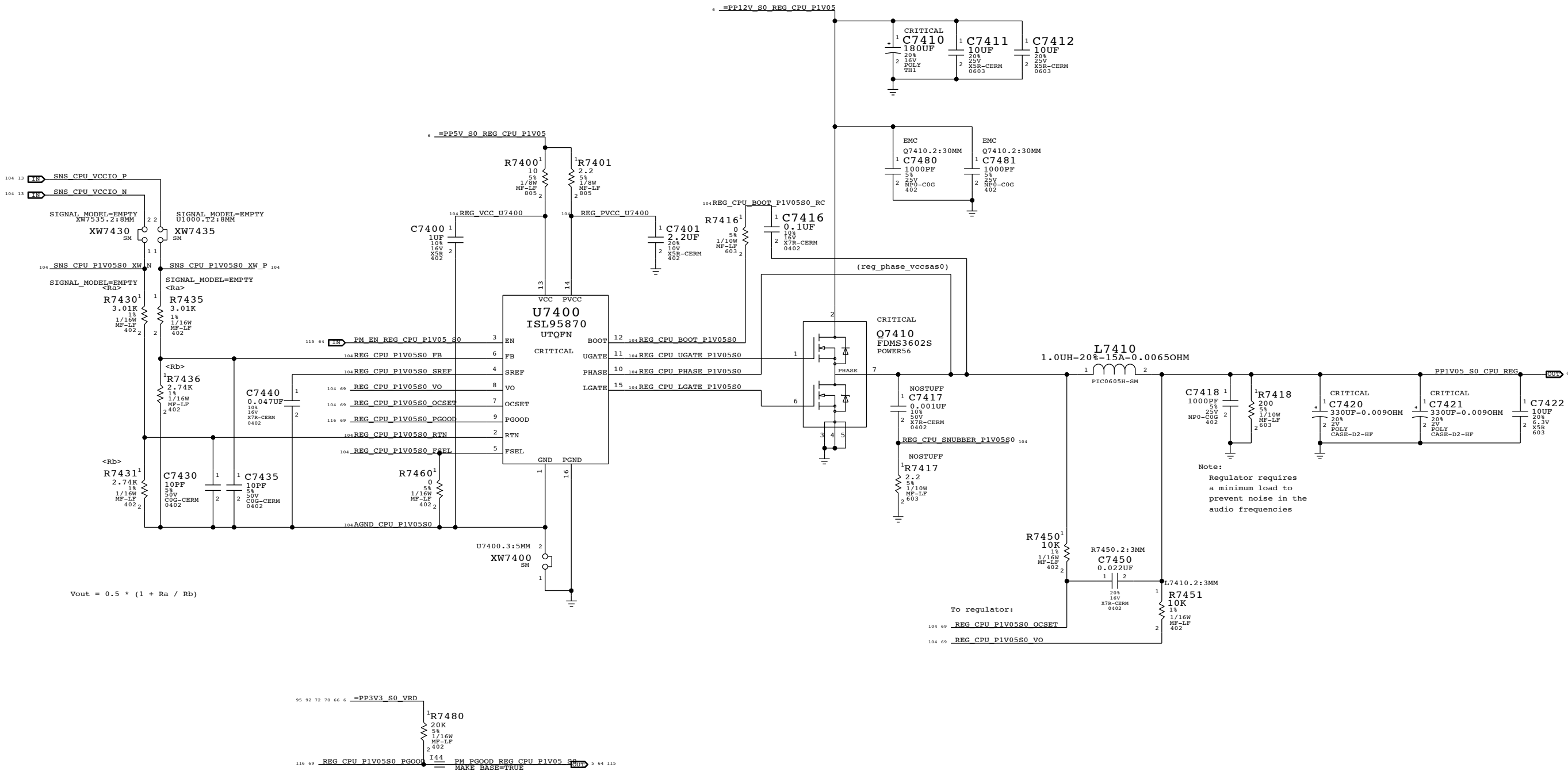
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 Apple Inc.		DRAWING NUMBER	SIZE
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		REVISION	7.0.0
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CPU VccIO (1.05V) S0 Regulator

Max avg current: 8.10 A (BUDGET)
Max peak current: 8.50 A (BUDGET)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 kHz

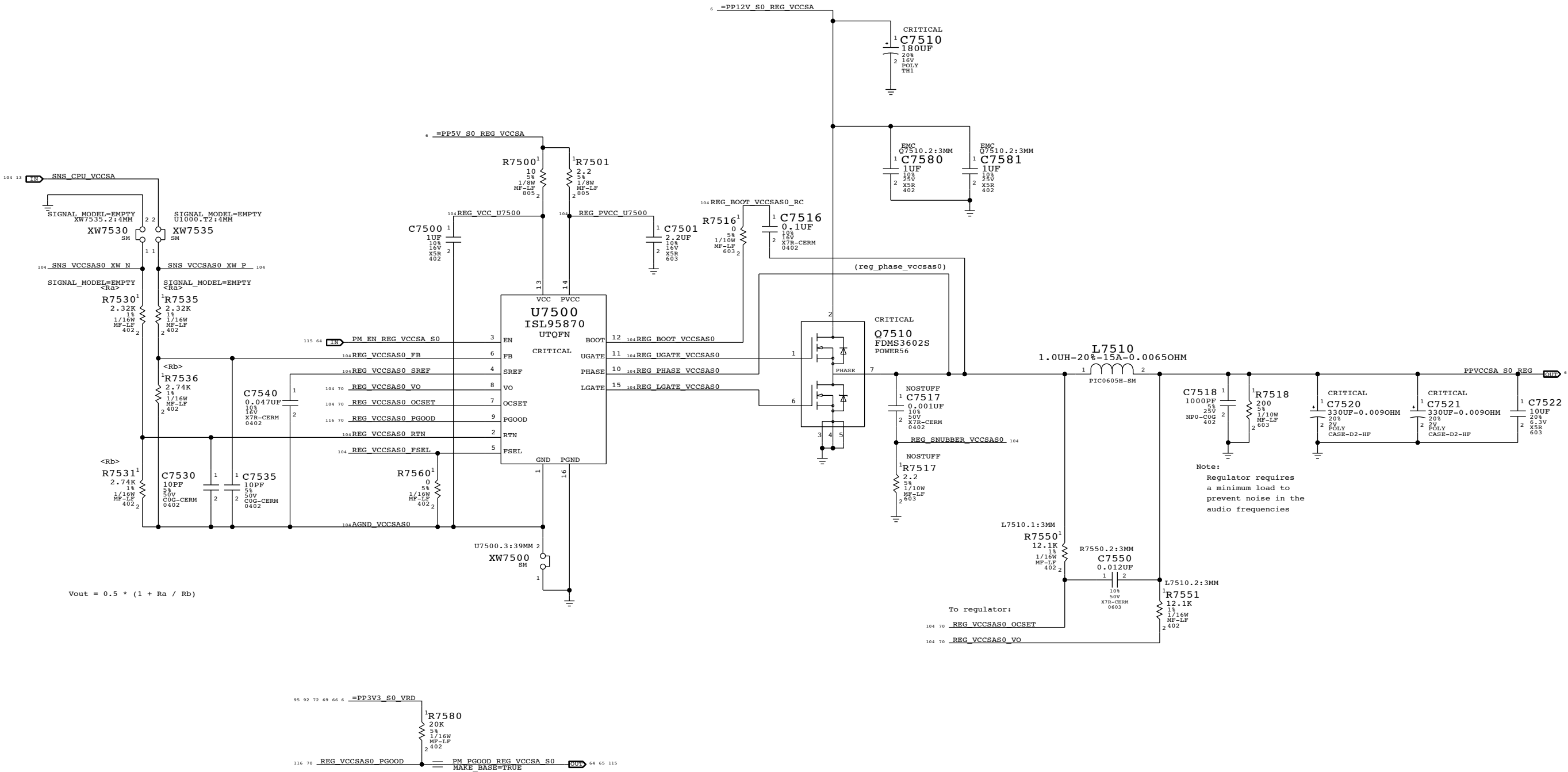


$$V_{out} = 0.5 * (1 + R_a / R_b)$$

SYNC MASTER=D8 KOSECOFF		SYNC DATE=02/25/2012	
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VReg CPU 1.05V S0			
 Apple Inc.	DRAWING NUMBER		051-9504
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CPU VccSA (0.925V) S0 Regulator

Max avg current: 12.07 A (BUDGET)
Max peak current: 30 A (BUDGET)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 kHz



SYNC MASTER=D8 KOSECOFF		SYNC DATE=02/25/2012	
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VReg VccSA S0			
 Apple Inc.	DRAWING NUMBER		SIZE
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A

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PAGE TITLE		VReg 3.3V S5/5V S4	
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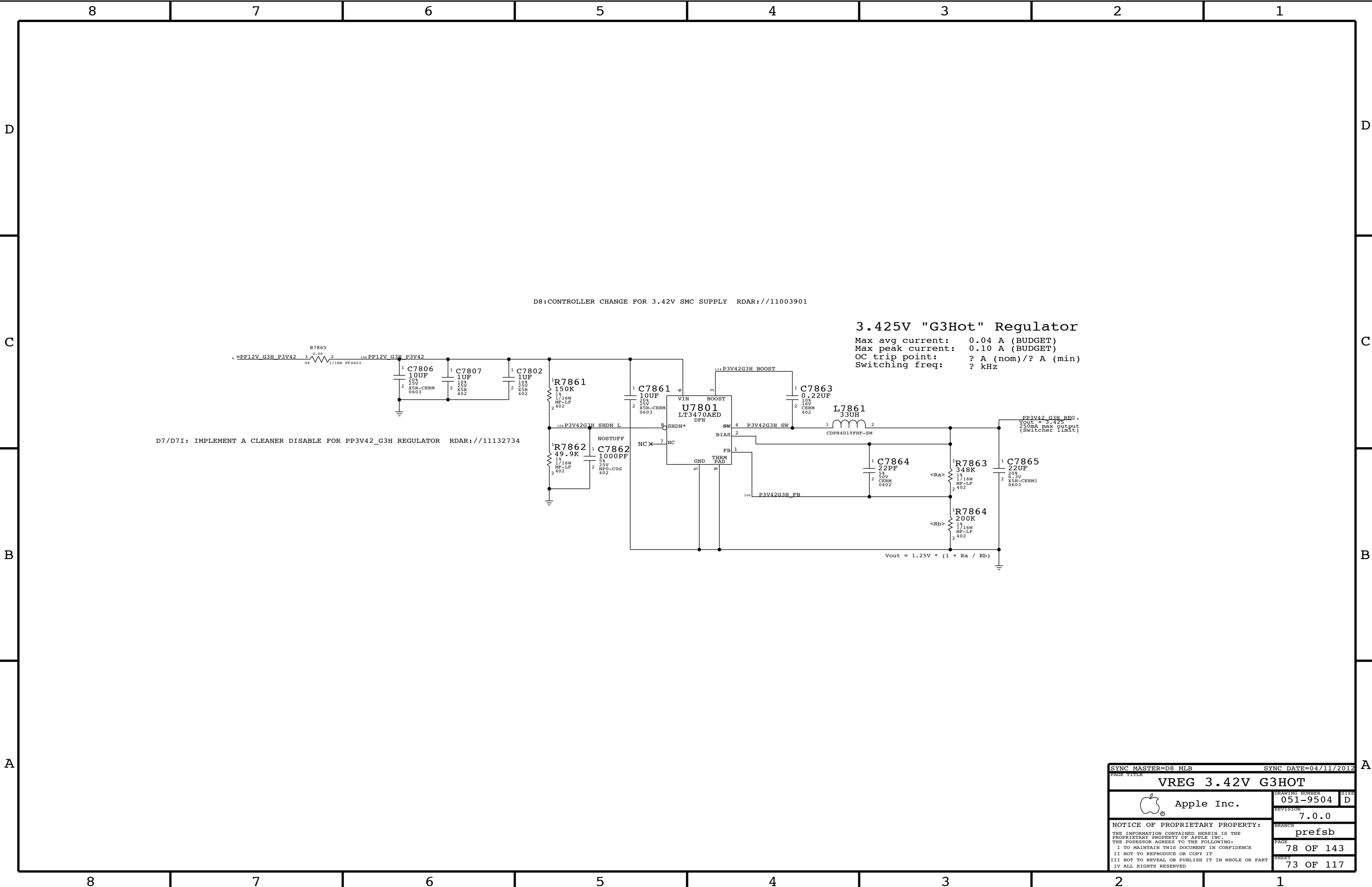
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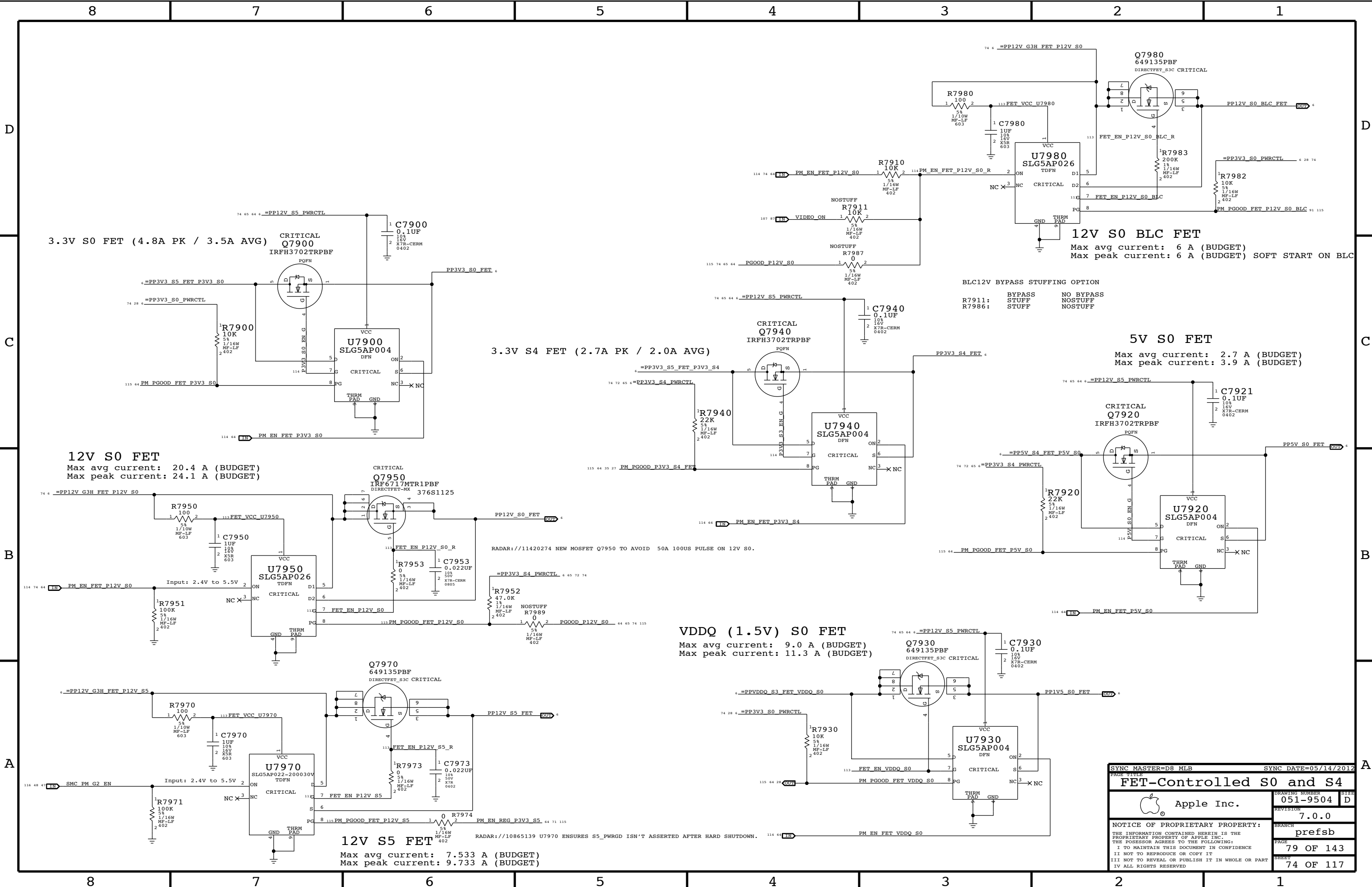


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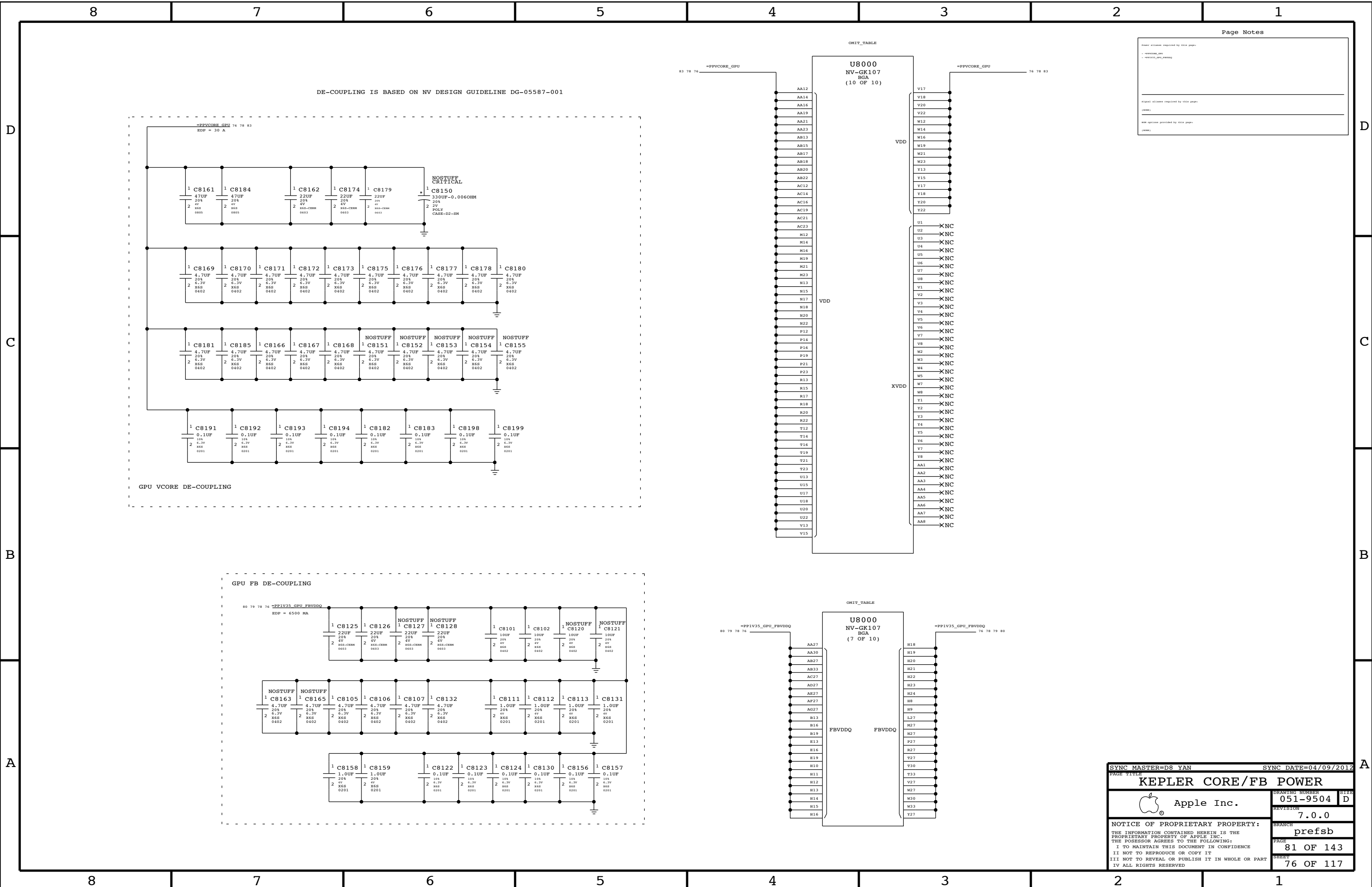
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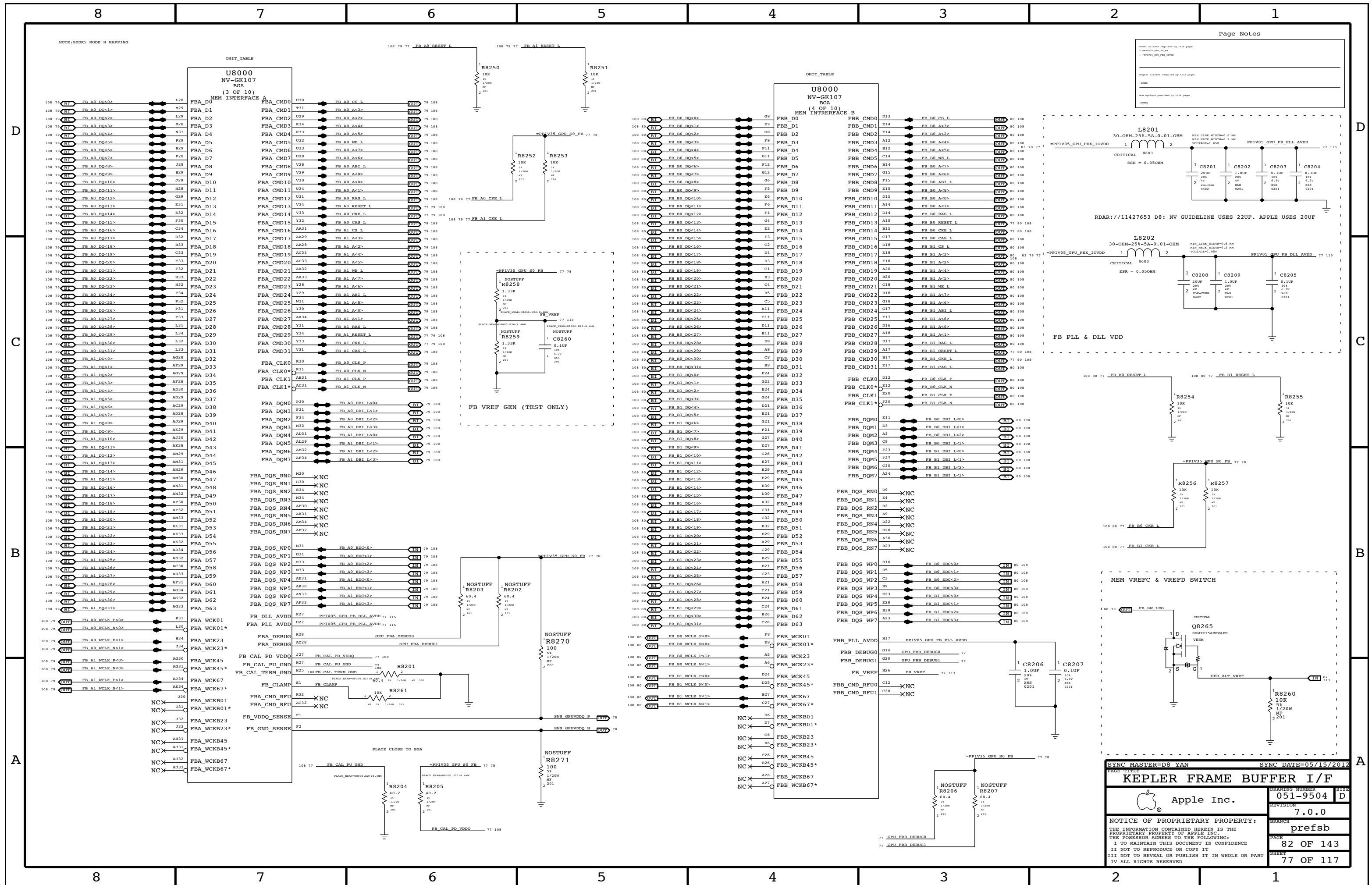
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SYNC MASTER=D8 MLB		SYNC DATE=05/14/2012	
PAGE TITLE		FET-Controlled S0 and S4	
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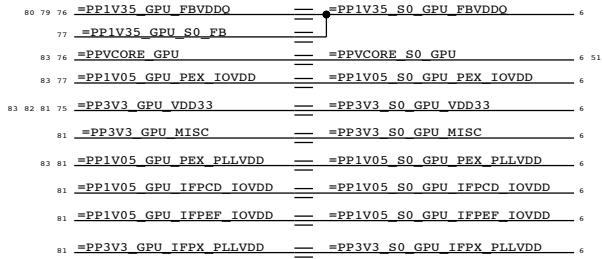
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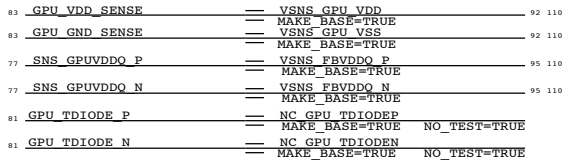
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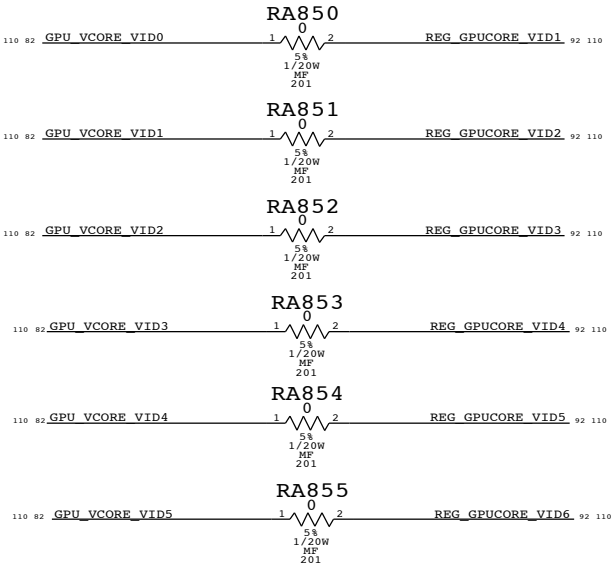
GPU POWER ALIAS



GPU SIGNAL & SENSE ALIAS

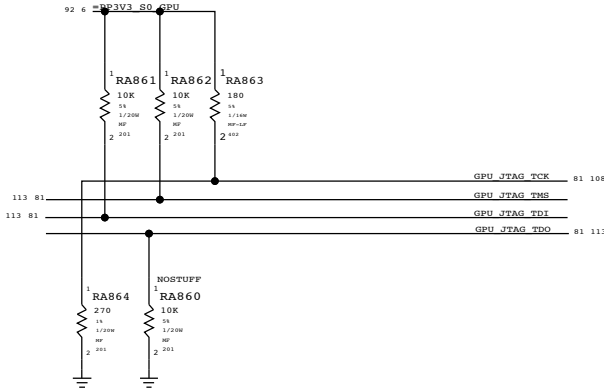


GPU VIDS ALIAS (VR VID0 IS TIED LOW)



THESE POWER ALIASES ARE CRAETED TO MATCH D8 GK104

PU/PD IS BASED ON RECOMMENDATION FROM NV FOR NVDIA GPU JTAG DEBUGGER



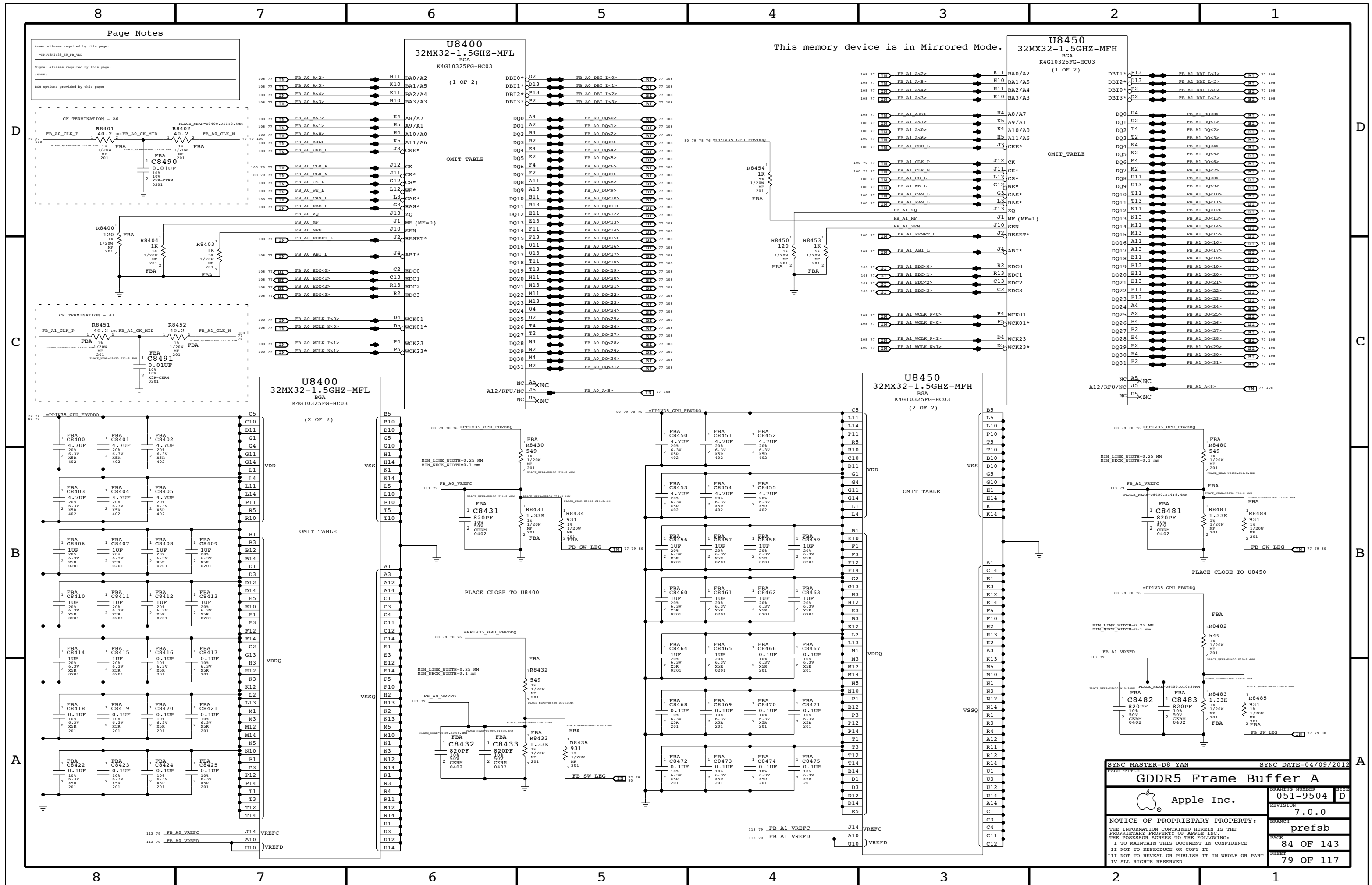
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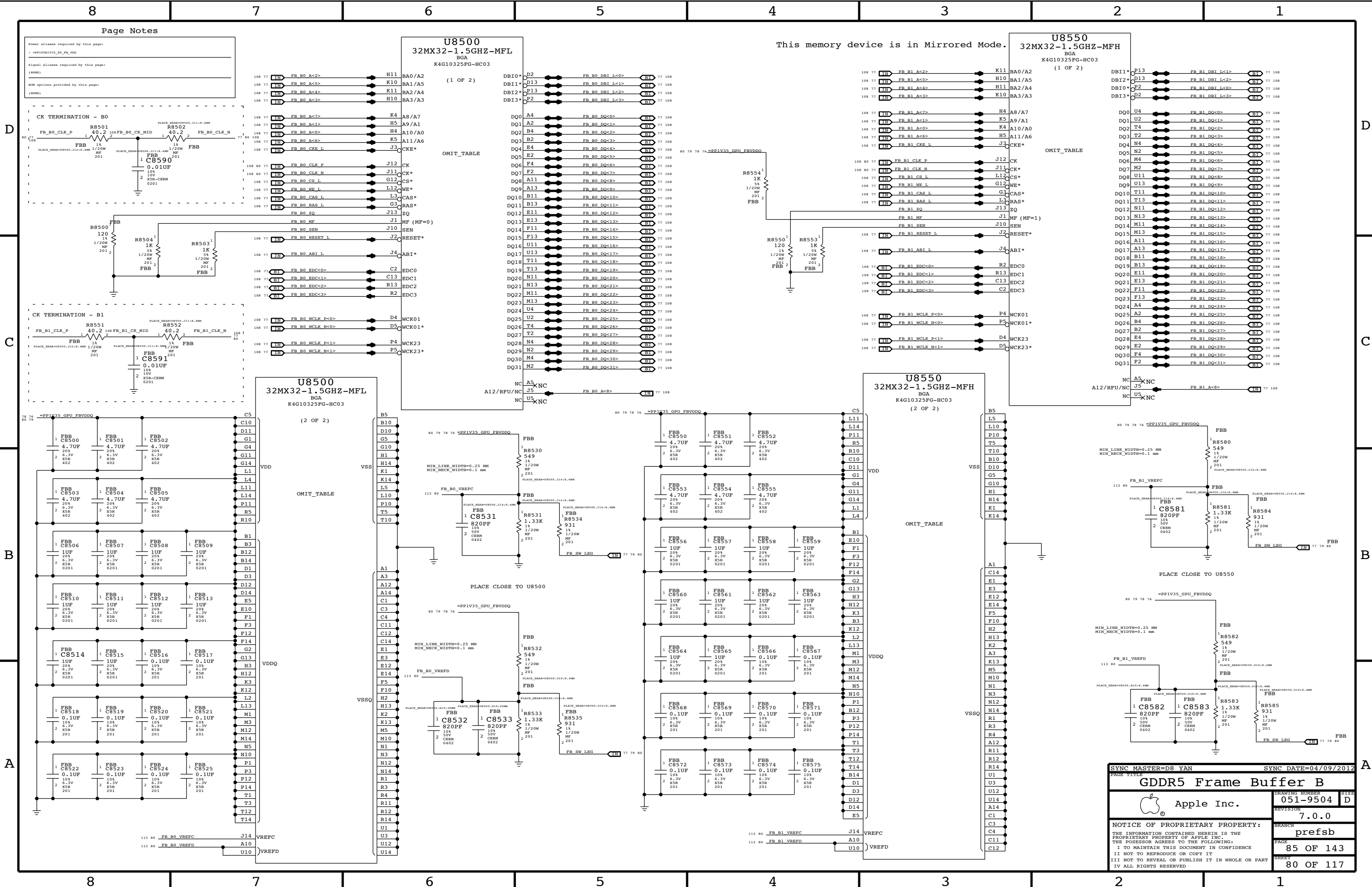
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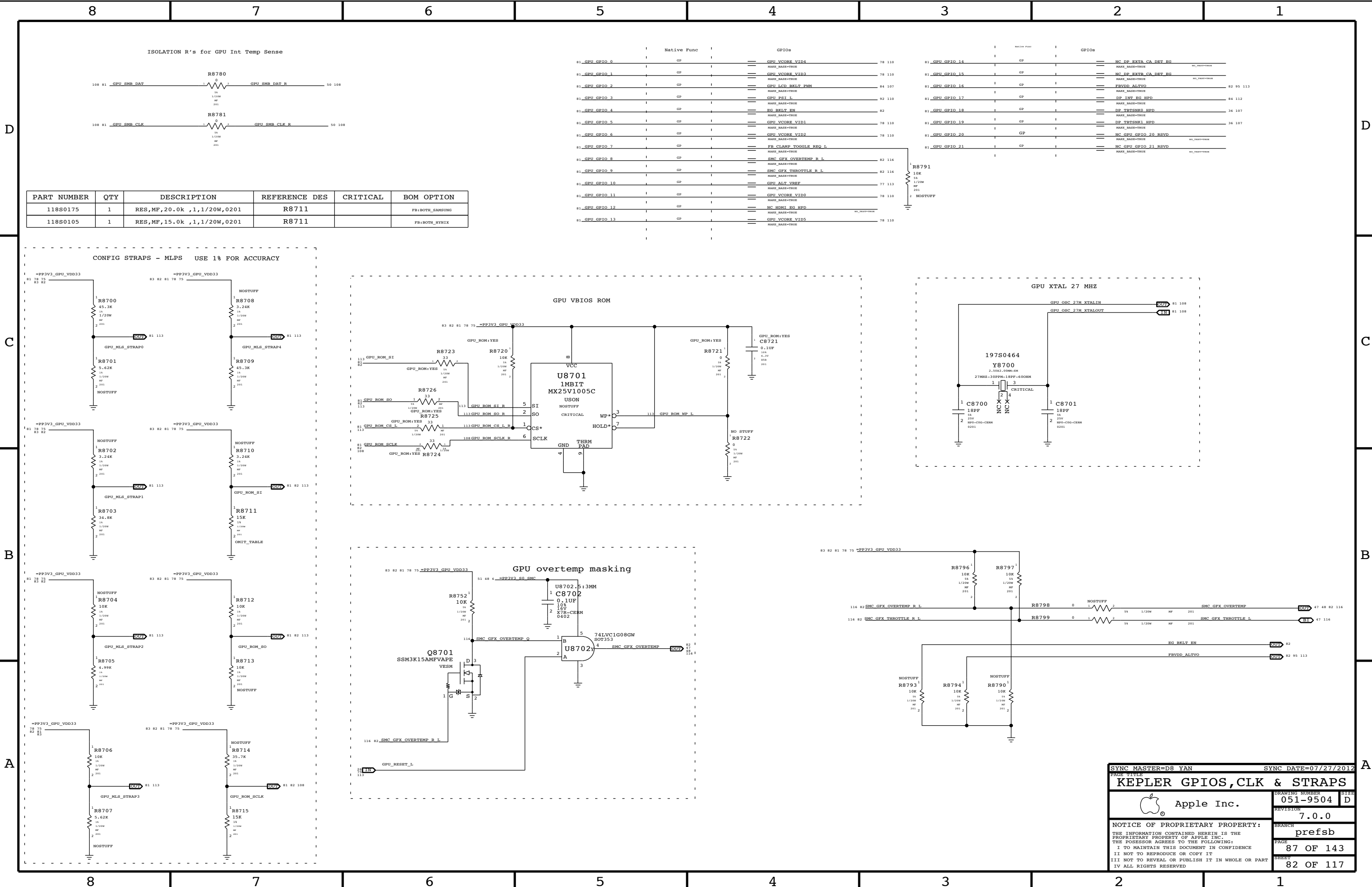
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GPU SIGNAL & POWER ALIASES			
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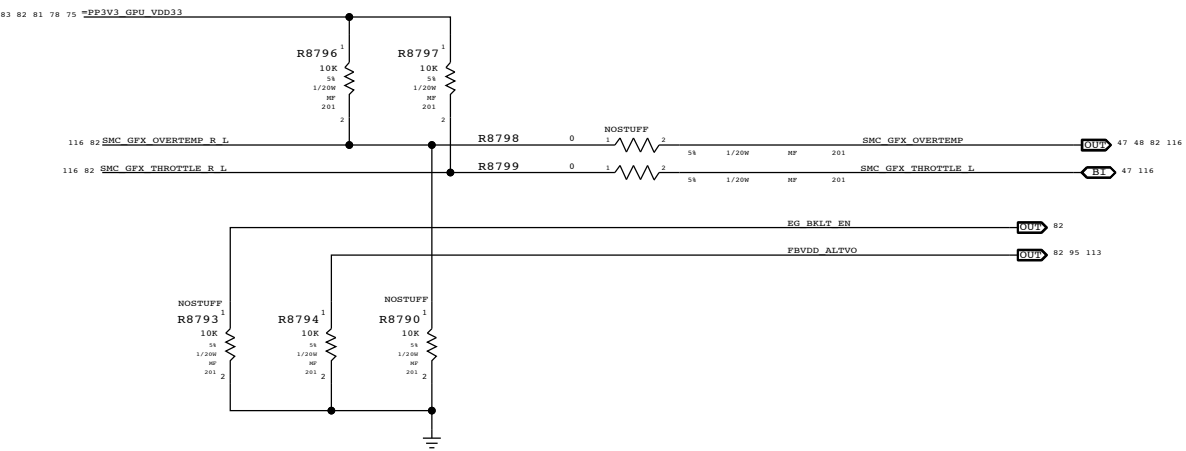
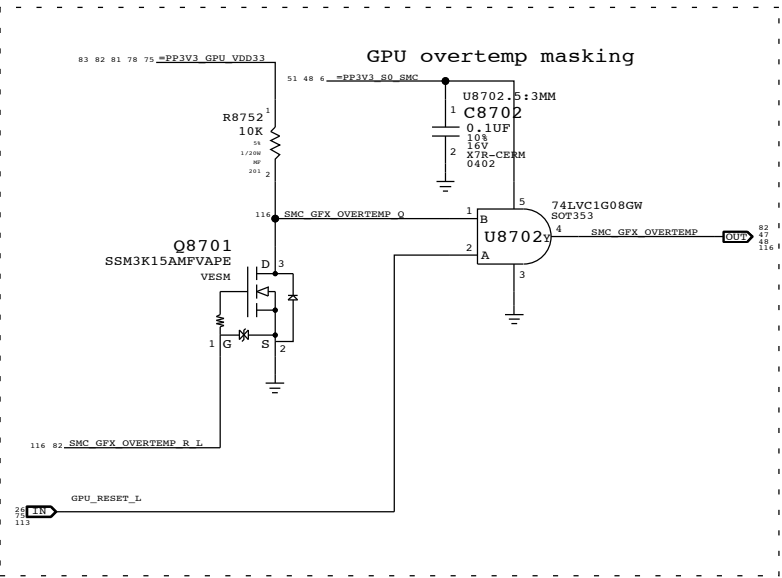
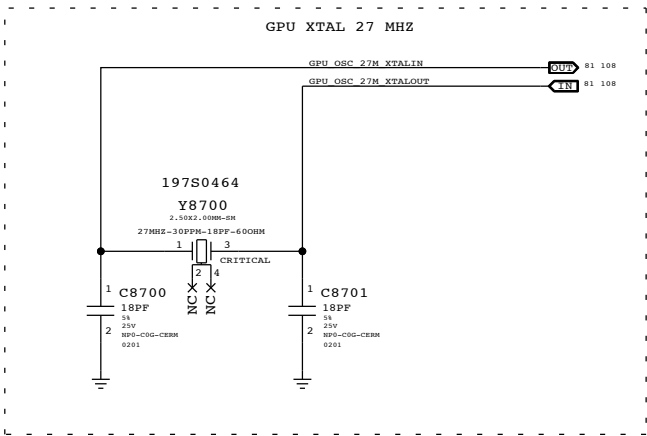
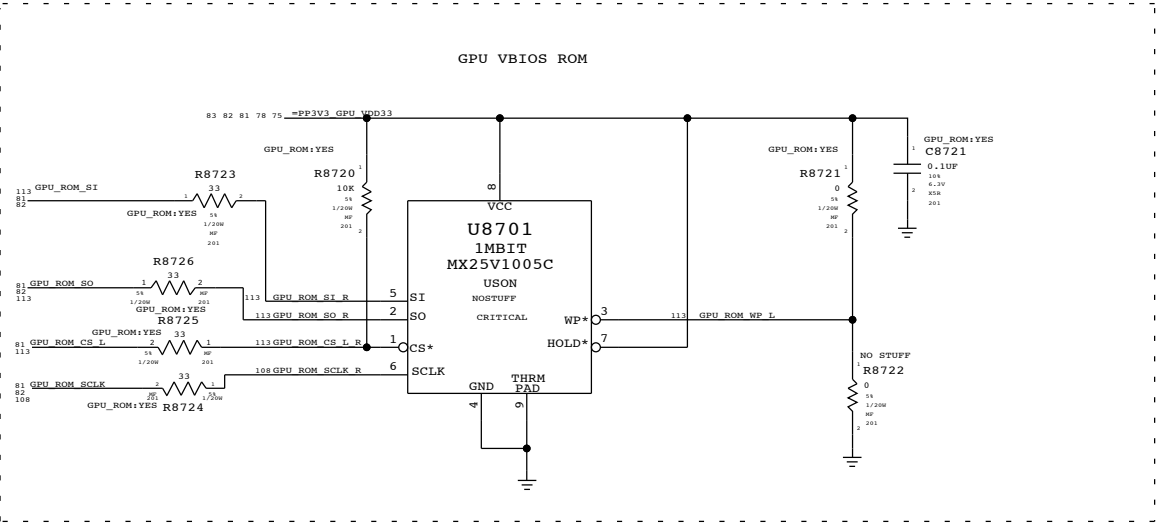
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0175	1	RES,MF,20.0k ,1,1/20W,0201	R8711		FB:BOTH_SAMSUNG
118S0105	1	RES,MF,15.0k ,1,1/20W,0201	R8711		FB:BOTH_HYNIX

	Native Func	GPIOs	
81_GPU_GPIO_0	GP	GPU_VCORE_VID4	78 110
81_GPU_GPIO_1	GP	MAKE_BASE-TRUE	
81_GPU_GPIO_2	GP	GPU_VCORE_VID3	78 110
81_GPU_GPIO_3	GP	MAKE_BASE-TRUE	
81_GPU_GPIO_4	GP	GPU_LCD_RKLT_EN	84 107
81_GPU_GPIO_5	GP	MAKE_BASE-TRUE	
81_GPU_GPIO_6	GP	GPU_FBI_L	92 110
81_GPU_GPIO_7	GP	MAKE_BASE-TRUE	
81_GPU_GPIO_8	GP	EG_RKLT_EN	82
81_GPU_GPIO_9	GP	GPU_VCORE_VID1	78 110
81_GPU_GPIO_10	GP	MAKE_BASE-TRUE	
81_GPU_GPIO_11	GP	GPU_VCORE_VID2	78 110
81_GPU_GPIO_12	GP	FB_CLAMP_TOGGLE_REQ_L	82 116
81_GPU_GPIO_13	GP	MAKE_BASE-TRUE	
81_GPU_GPIO_14	GP	SMC GFX OVERTEMP_R_L	82 116
81_GPU_GPIO_15	GP	SMC GFX THROTTLE_R_L	82 116
81_GPU_GPIO_16	GP	GPU_ALT_VREF	77 113
81_GPU_GPIO_17	GP	MAKE_BASE-TRUE	
81_GPU_GPIO_18	GP	GPU_VCORE_VID0	78 110
81_GPU_GPIO_19	GP	NC_HDMI_EG_HPD	84 112
81_GPU_GPIO_20	GP	MAKE_BASE-TRUE	
81_GPU_GPIO_21	GP	GPU_VCORE_VID5	78 110
81_GPU_GPIO_22	GP	MAKE_BASE-TRUE	

	GPIOs	
81_GPU_GPIO_14	GP	NC_DP_EXTA_CA_DET_EG
81_GPU_GPIO_15	GP	MAKE_BASE-TRUE
81_GPU_GPIO_16	GP	NC_DP_EXTRA_CA_DET_EG
81_GPU_GPIO_17	GP	MAKE_BASE-TRUE
81_GPU_GPIO_18	GP	FBVDD_ALTV0
81_GPU_GPIO_19	GP	MAKE_BASE-TRUE
81_GPU_GPIO_20	GP	DP_INF_EG_HPD
81_GPU_GPIO_21	GP	MAKE_BASE-TRUE
81_GPU_GPIO_22	GP	DP_TBTSNK0_HPD
81_GPU_GPIO_23	GP	MAKE_BASE-TRUE
81_GPU_GPIO_24	GP	DP_TBTSNK1_HPD
81_GPU_GPIO_25	GP	MAKE_BASE-TRUE
81_GPU_GPIO_26	GP	NC_GPU_GPIO_20_RSVD
81_GPU_GPIO_27	GP	MAKE_BASE-TRUE
81_GPU_GPIO_28	GP	NC_GPU_GPIO_21_RSVD
81_GPU_GPIO_29	GP	MAKE_BASE-TRUE



SYNC MASTER=D8_YAN

SYNC DATE=07/27/2012

KEPLER GPIOs,CLK & STRAPS

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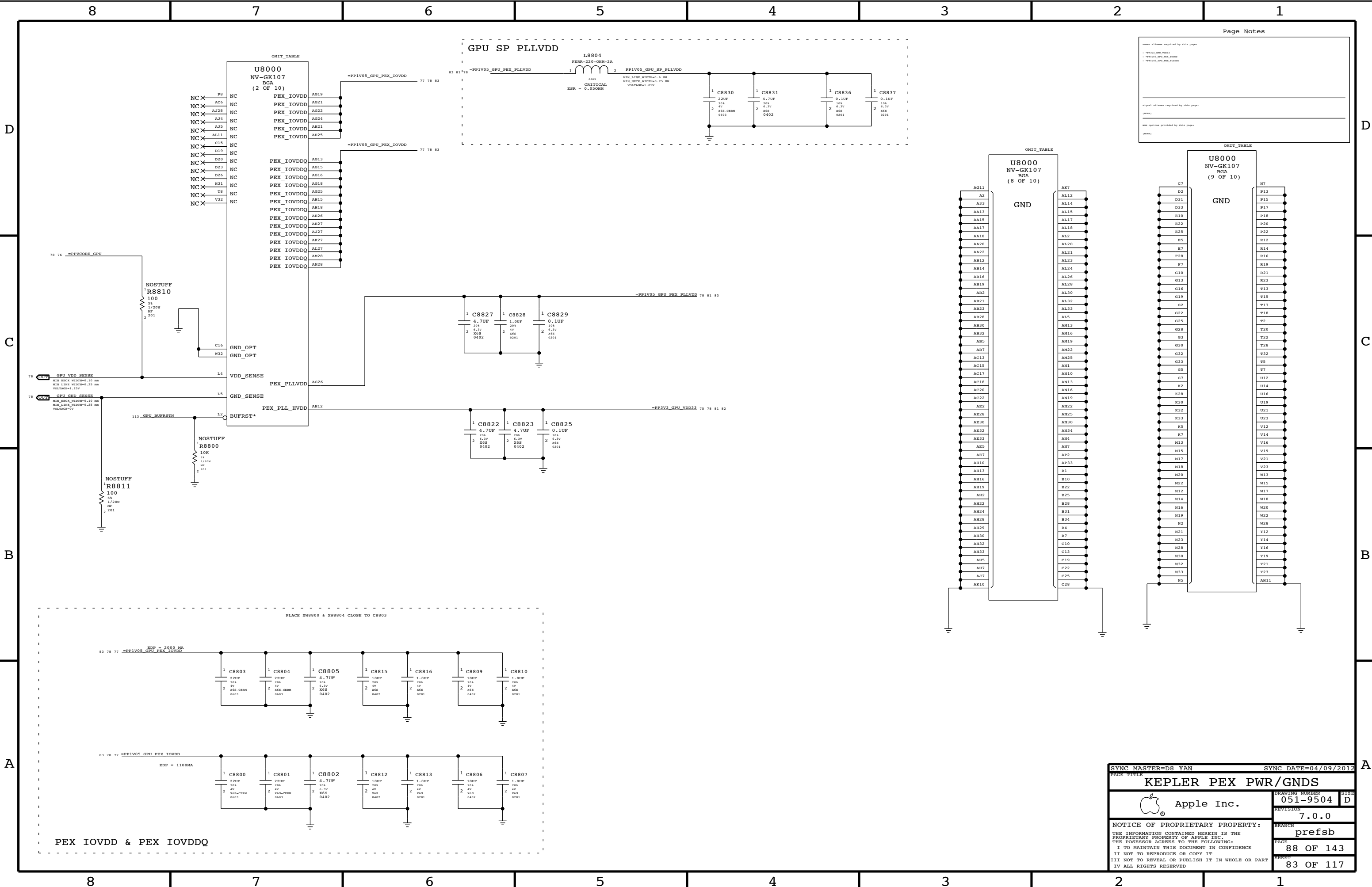
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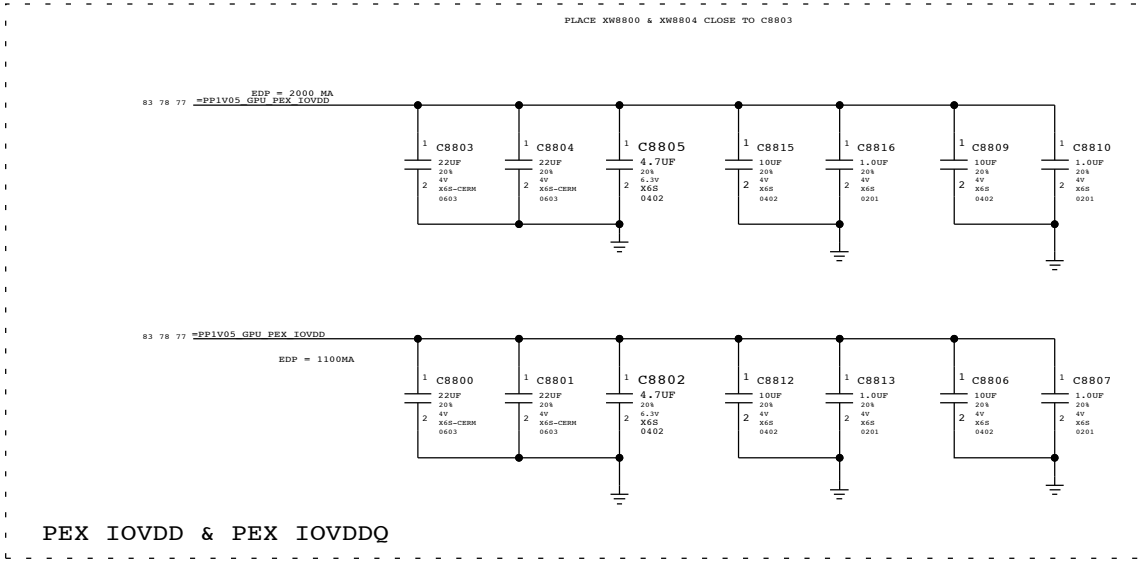
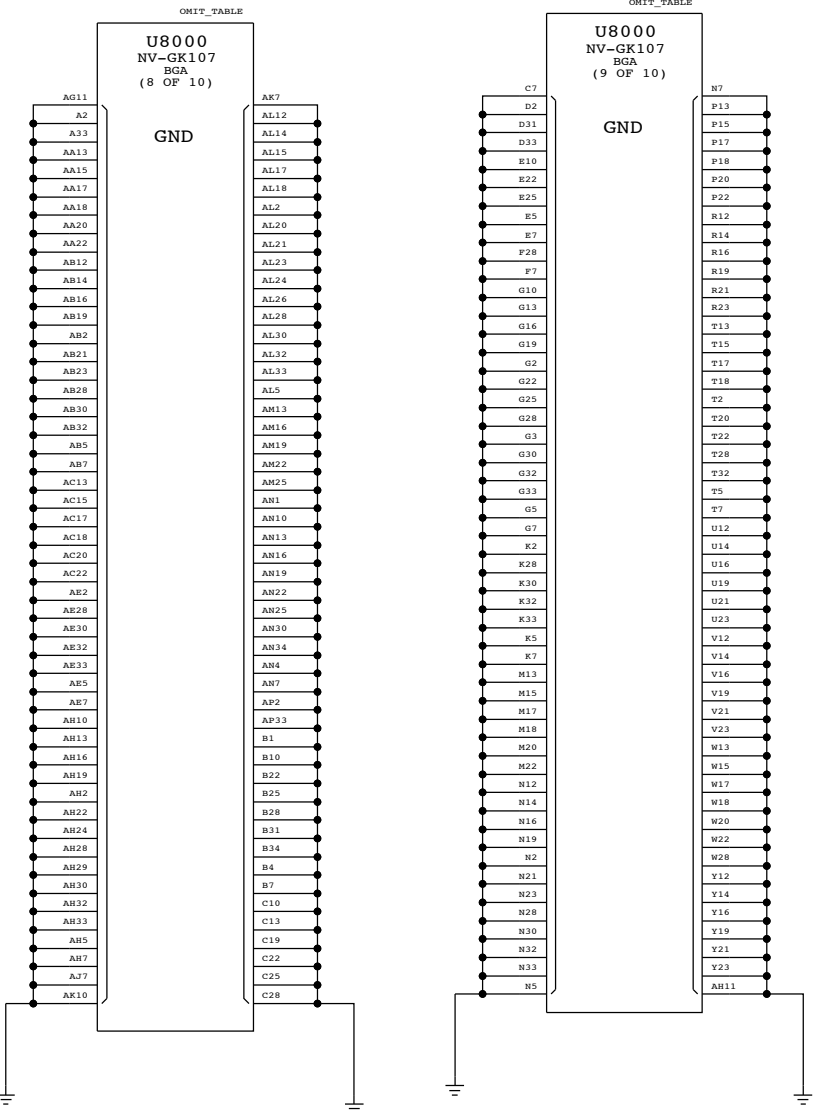
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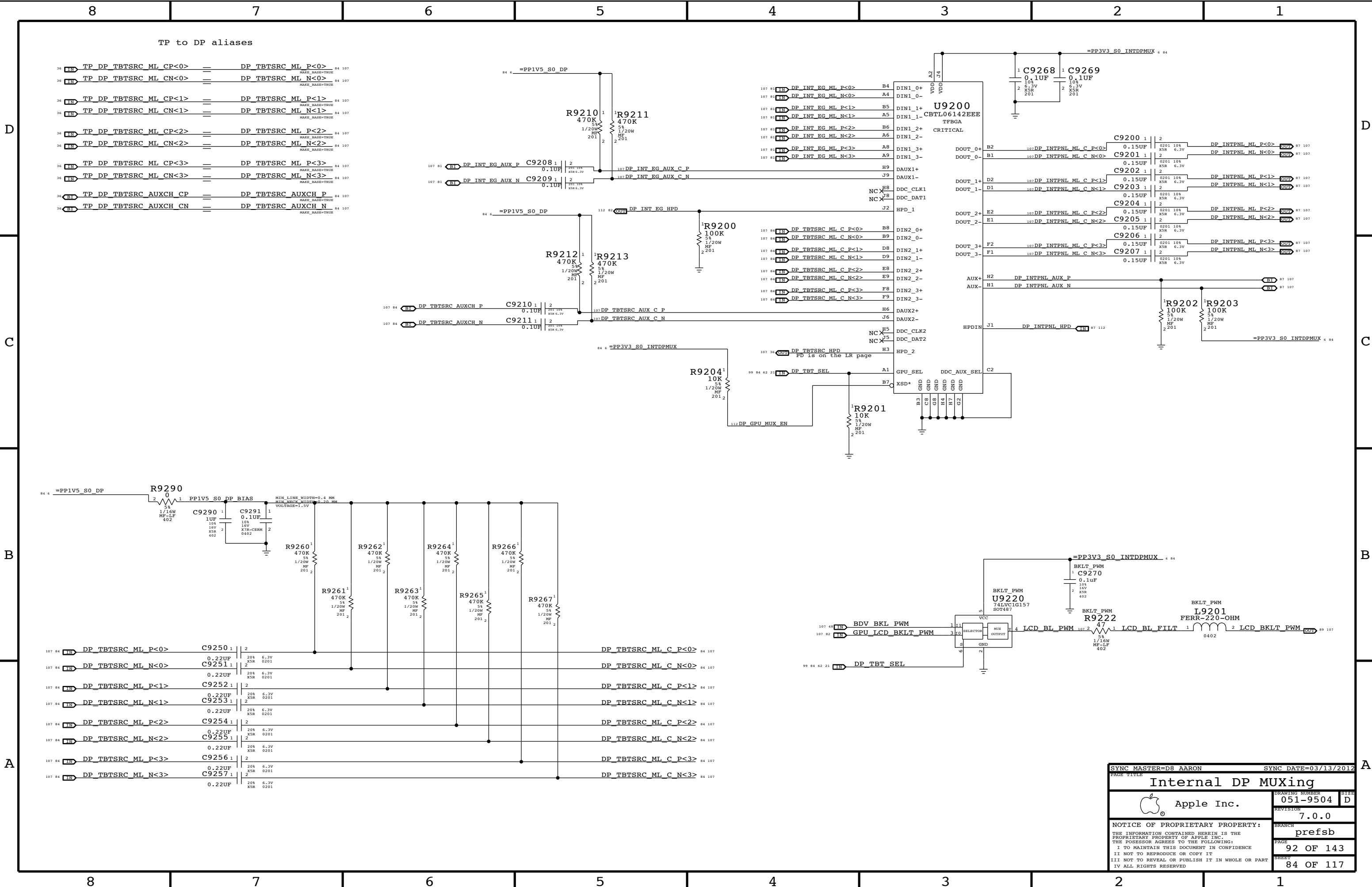
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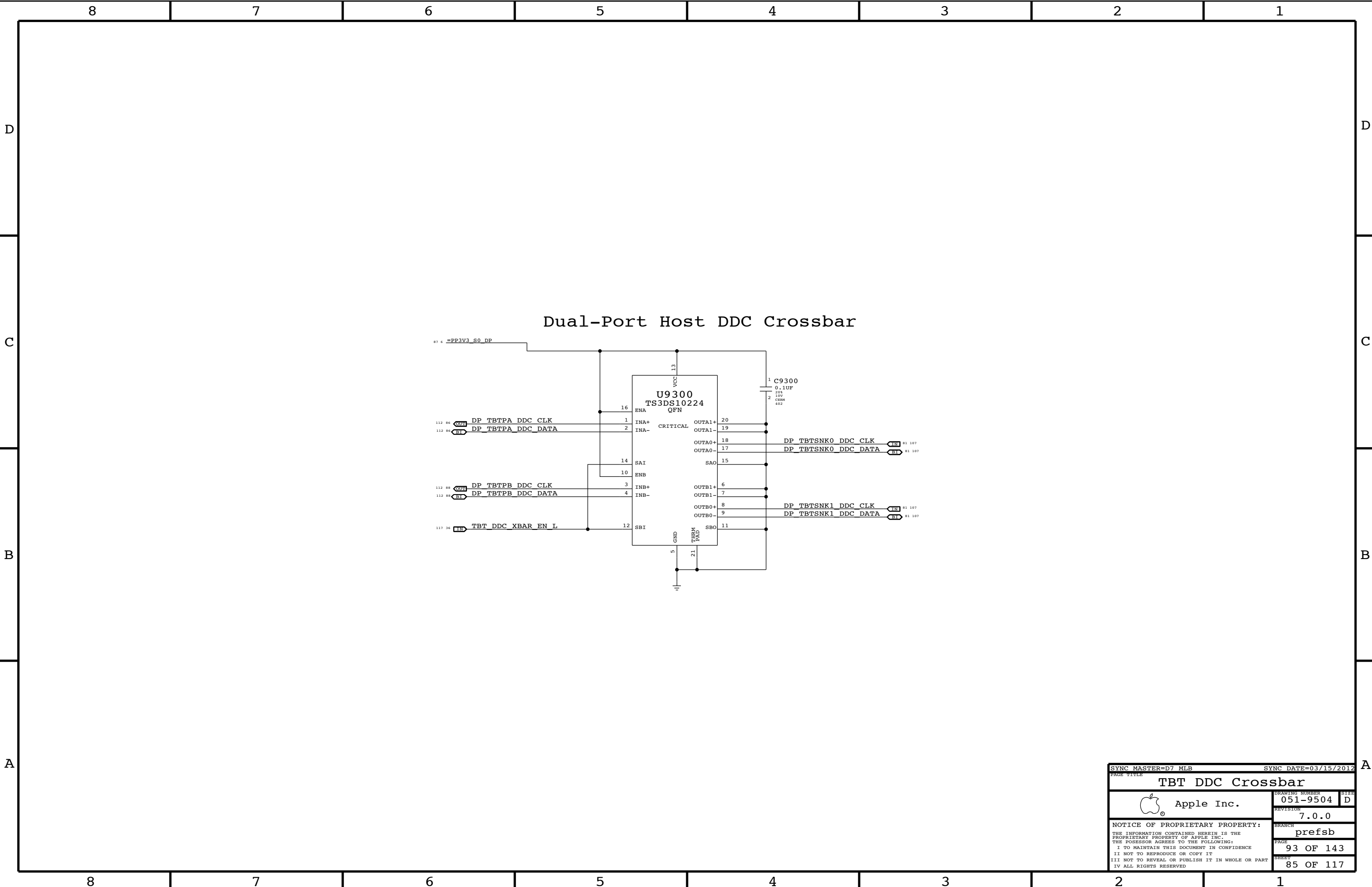


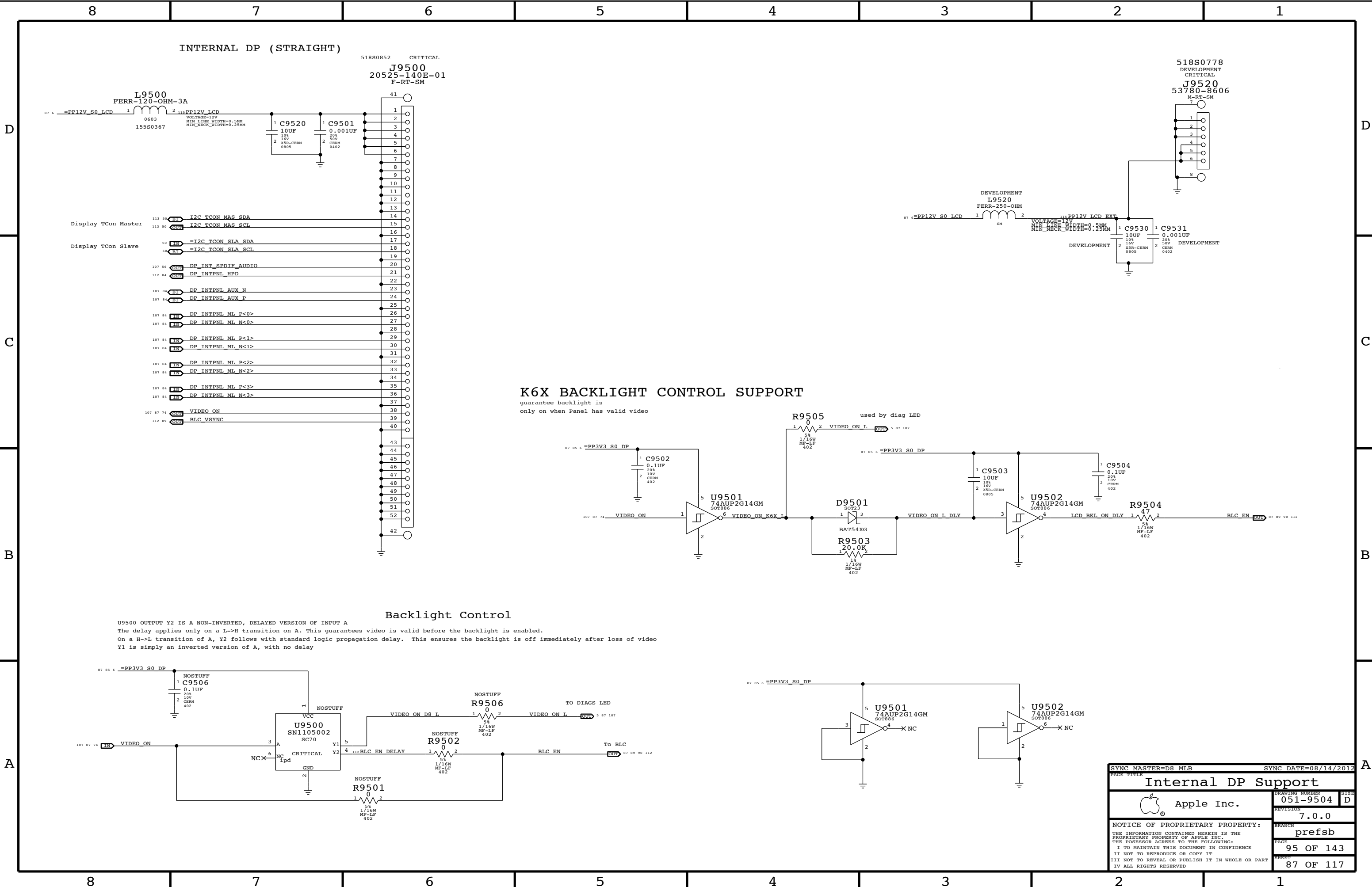
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- PEXIOVDDQ_VDDQ1	
- PEXIOVDDQ_VDDQ2	
- PEXIOVDDQ_VDDQ3	
Signal aliases required by this page:	
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Non options provided by this page:	
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KEPLER PEX PWR/GNDS			
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INTERNAL DP (STRAIGHT)

518S0852 CRITICAL
J9500
20525-140E-01
F-RT-SM

L9500
FERR-120-OHM-3A

0603
155S0367
VOLTAGE=12V
MIN_LINE_WIDTH=0.5MM
MIN_NECK_WIDTH=0.25MM

C9520
10UF
10%
10V
X5R-CERM
0805

C9501
0.001UF
20%
50V
CERM
0402

Display TCon Master

Display TCon Slave

- I2C TCON MAS_SDA
- I2C TCON MAS_SCL
- I2C TCON SLA_SDA
- I2C TCON SLA_SCL
- DP_INT_SPDIF_AUDIO
- DP_INTPNL_HPD
- DP_INTPNL_AUX_N
- DP_INTPNL_AUX_P
- DP_INTPNL_ML_P<0>
- DP_INTPNL_ML_N<0>
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- DP_INTPNL_ML_N<2>
- DP_INTPNL_ML_P<3>
- DP_INTPNL_ML_N<3>
- VIDEO_ON
- BLC_VSYNC

K6X BACKLIGHT CONTROL SUPPORT

guarantee backlight is
only on when Panel has valid video

Backlight Control

U9500 OUTPUT Y2 IS A NON-INVERTED, DELAYED VERSION OF INPUT A
The delay applies only on a L->H transition on A. This guarantees video is valid before the backlight is enabled.
On a H->L transition of A, Y2 follows with standard logic propagation delay. This ensures the backlight is off immediately after loss of video
Y1 is simply an inverted version of A, with no delay

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Internal DP Support			
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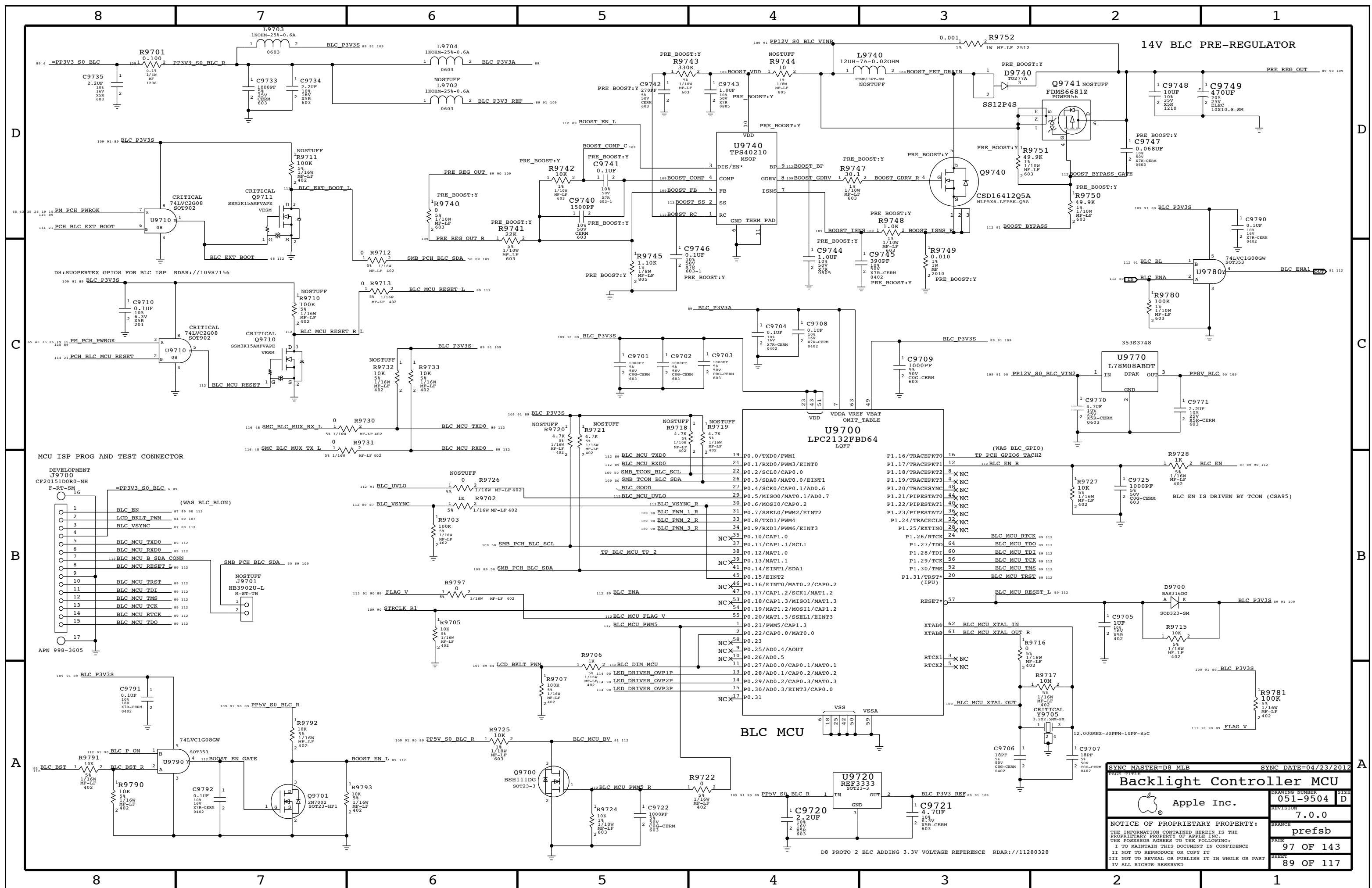


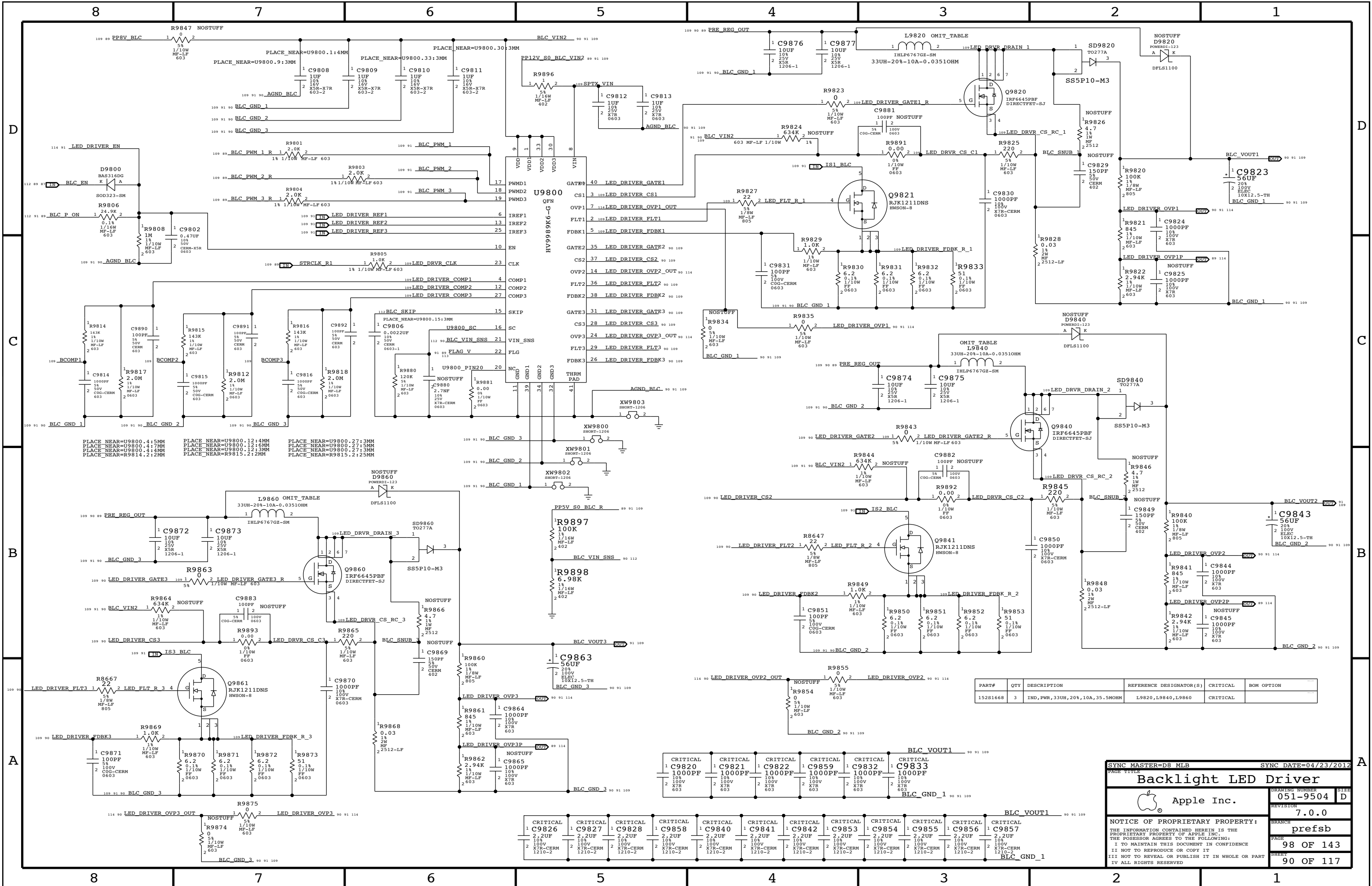
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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
15281668	3	IND,PWR,33UH,20%,10A,35.5MOHM	L9820,L9840,L9860	CRITICAL	

SYNC MASTER=D8 MLB

SYNC DATE=04/23/2012

Backlight LED Driver

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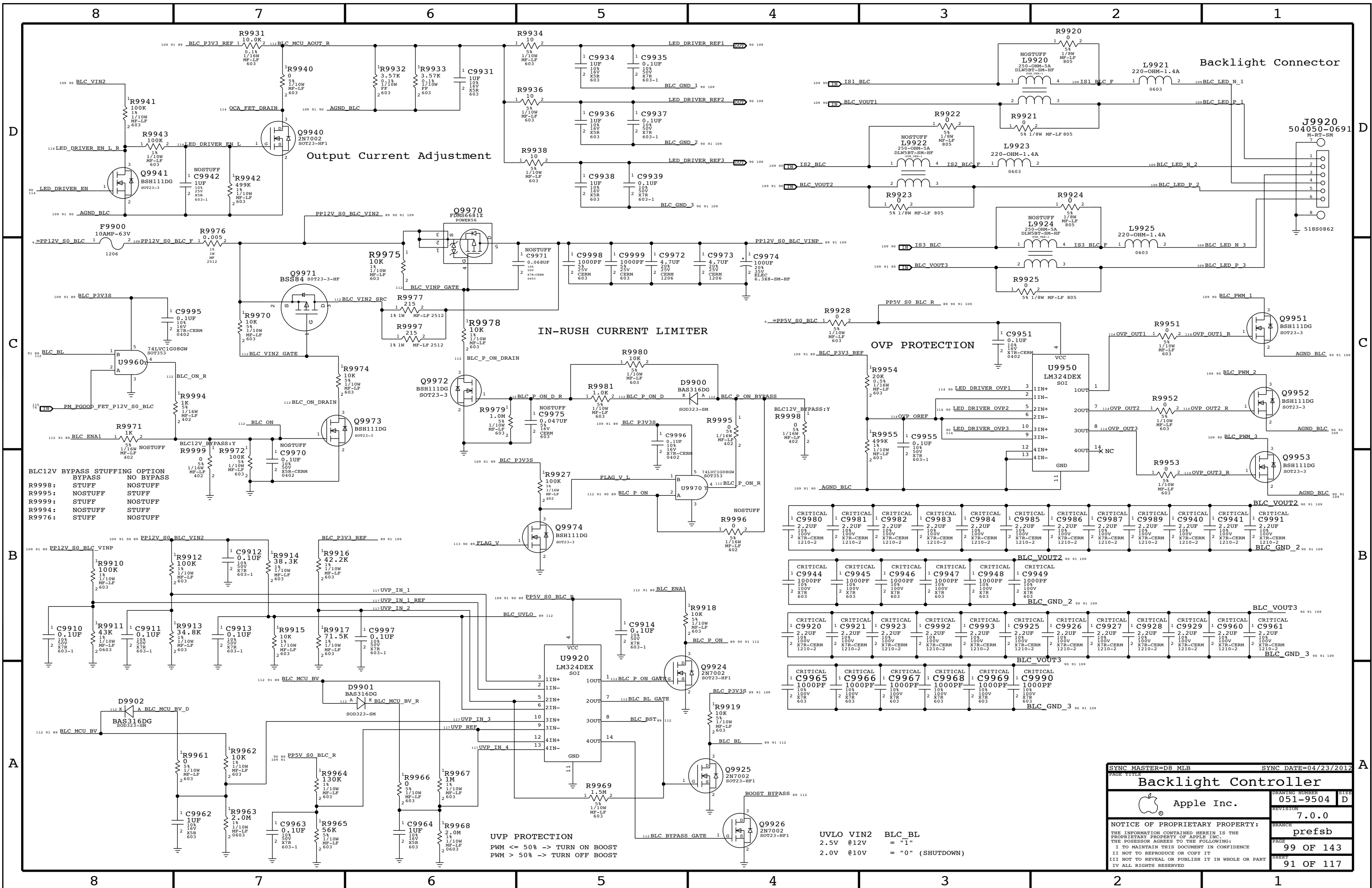
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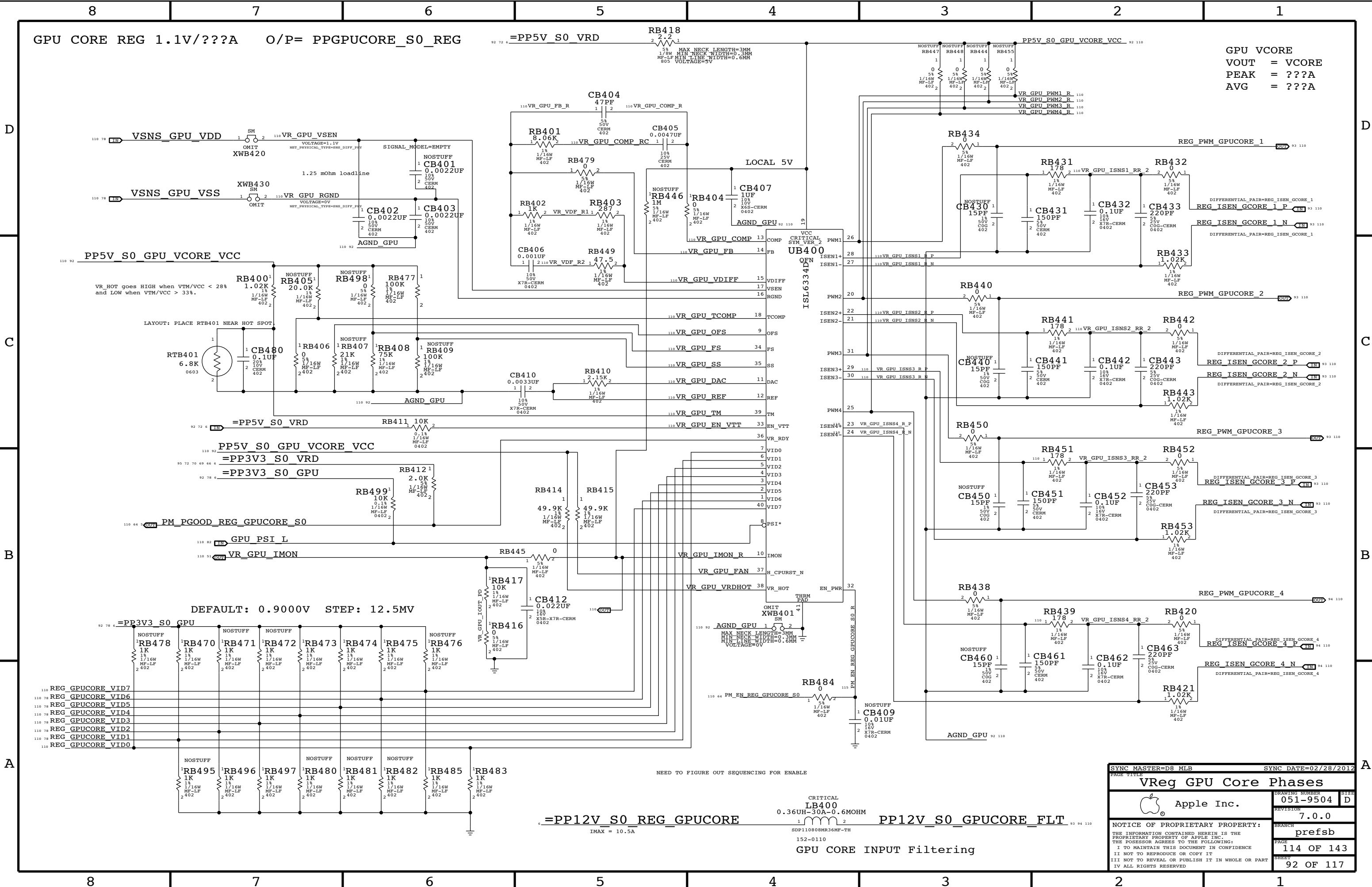
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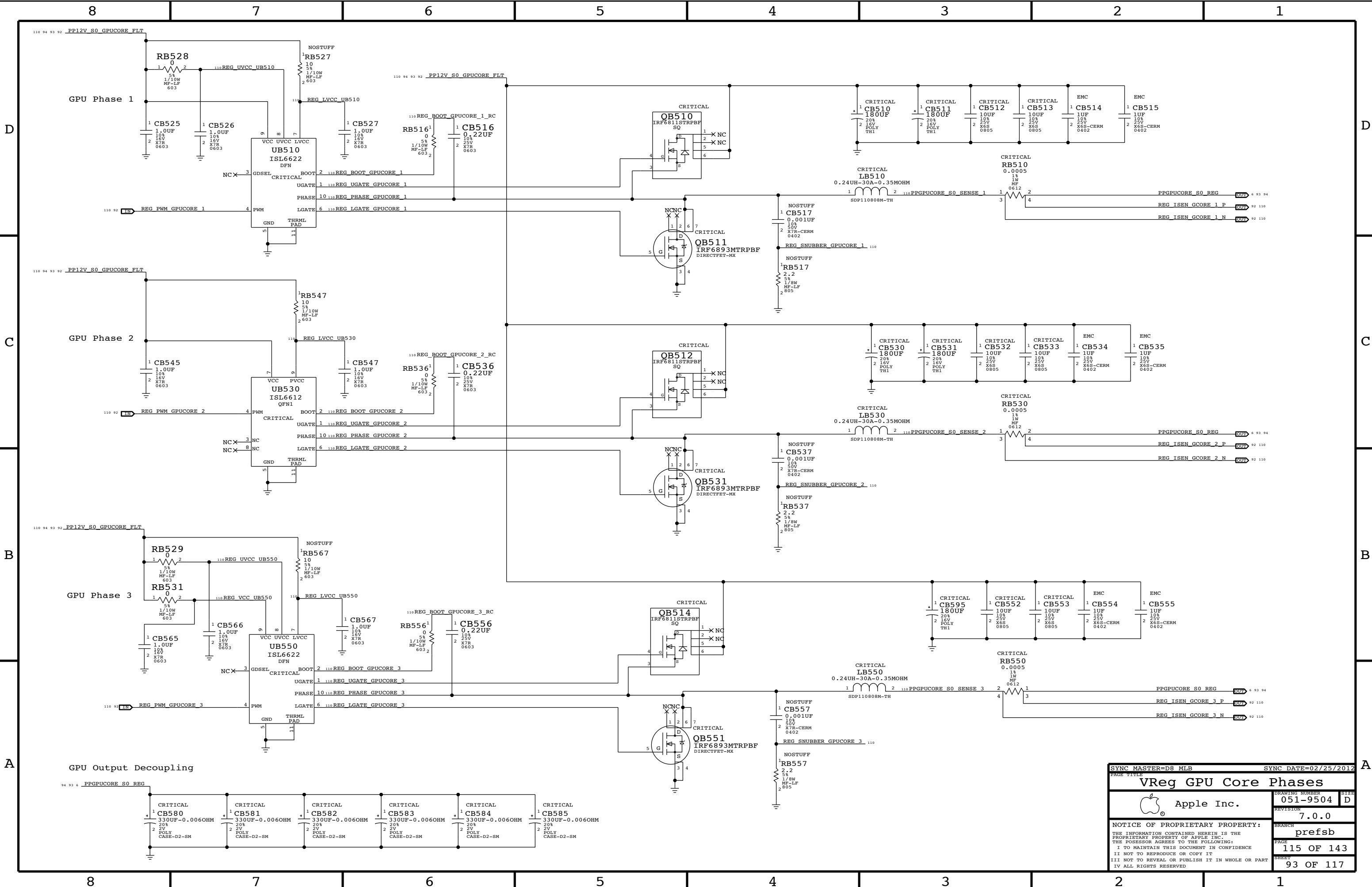


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Backlight Controller			
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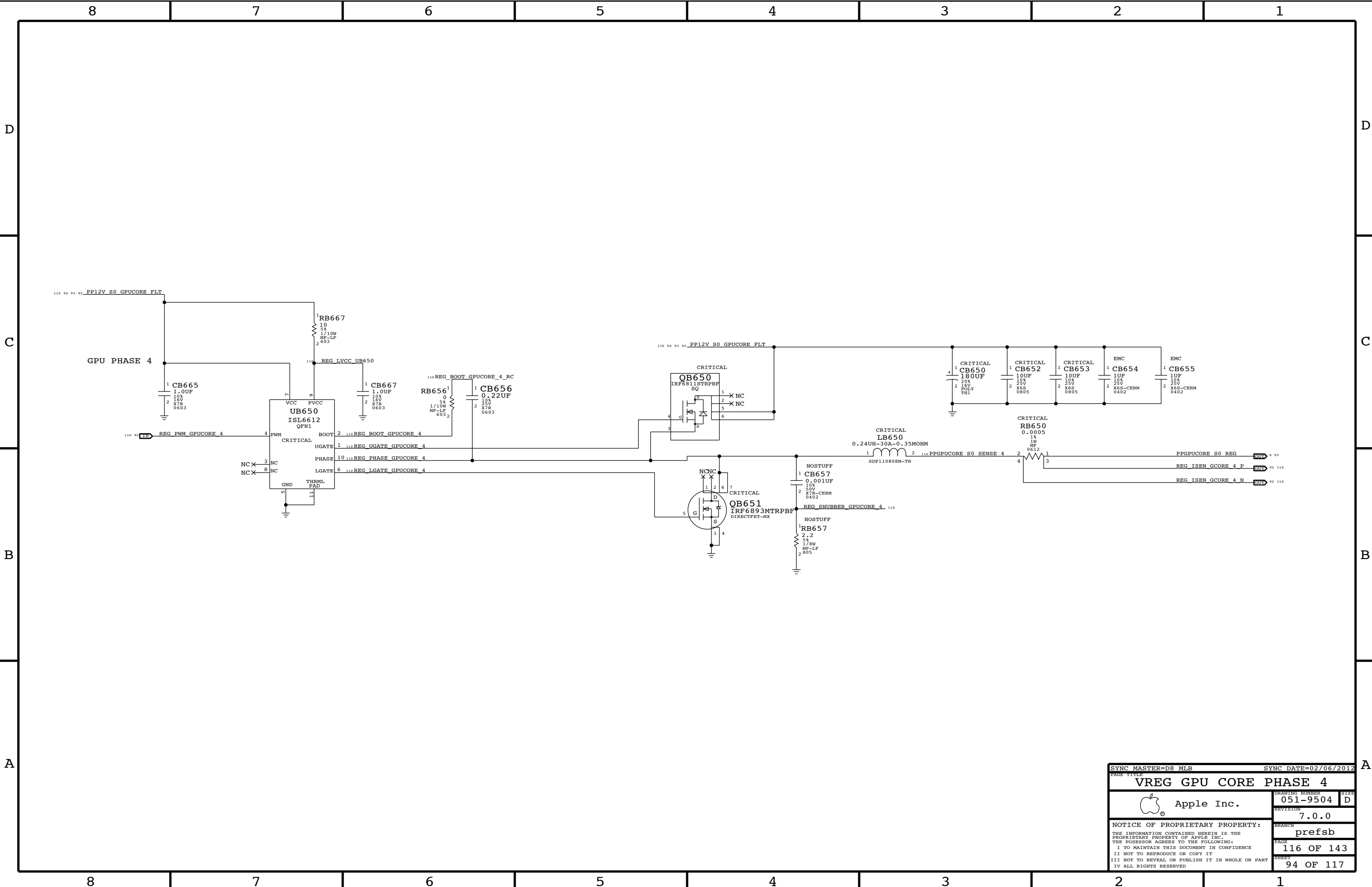
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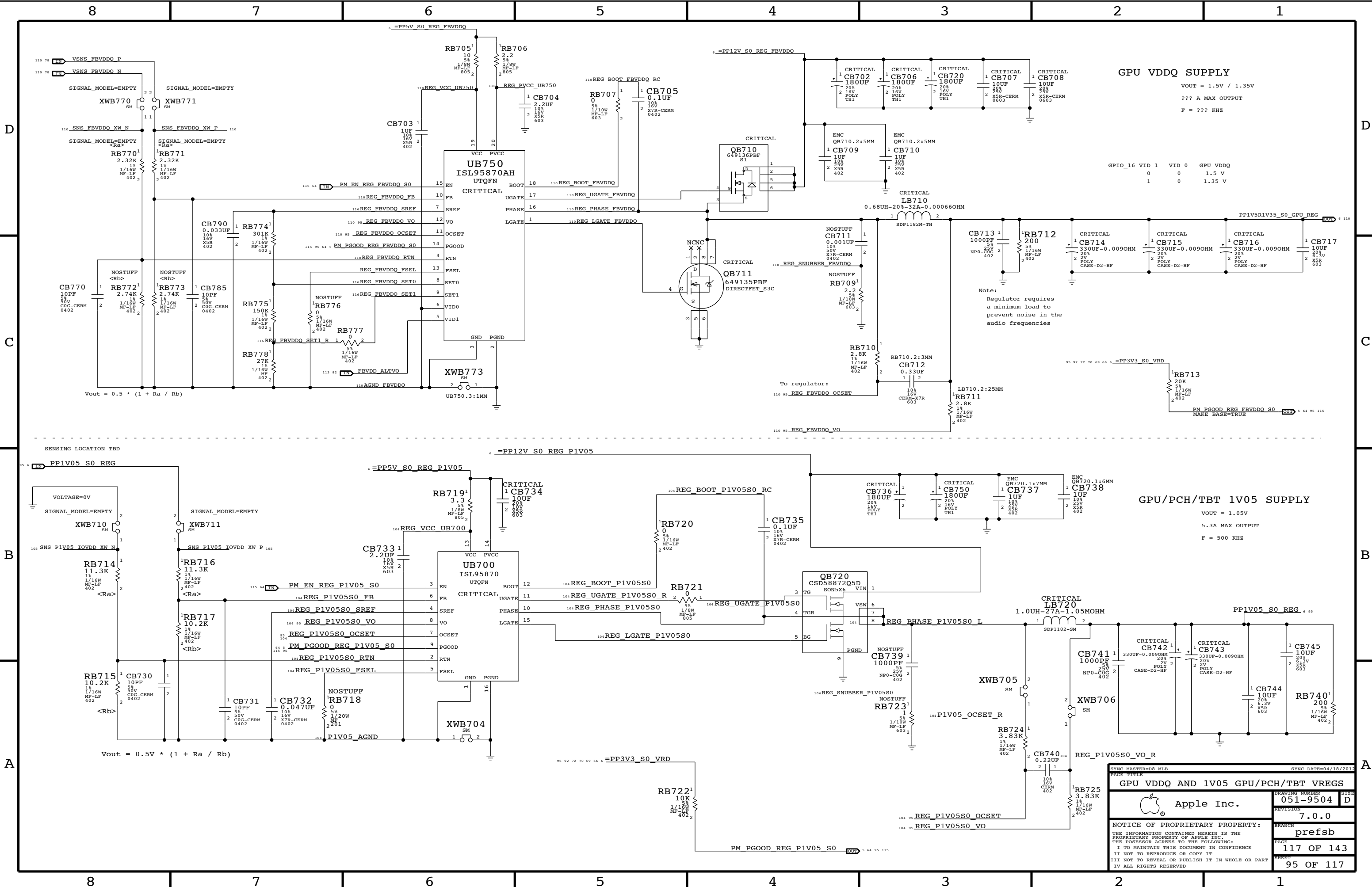


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GPU VDDQ SUPPLY

VOUT = 1.5V / 1.35V
??? A MAX OUTPUT
F = ??? KHZ

GPIO_16 VID 1 VID 0 GPU VDDQ
0 0 1.5 V
1 0 1.35 V

GPU/PCH/TBT 1V05 SUPPLY

VOUT = 1.05V
5.3A MAX OUTPUT
F = 500 KHZ

GPU VDDQ AND 1V05 GPU/PCH/TBT VREGS		DRAWING NUMBER	051-9504	SIZE	D
Apple Inc.		REVISION	7.0.0	BRANCH	prefsb
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PCI EXPRESS

PCIe-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE_PHY	*	PCIE_80D
CLK_PCIE_PHY	*	PCIE_90D
COMP_PCIE_PHY	*	COMP_SE

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_ISO	TOP,BOTTOM	=5X_DIELECTRIC	?
PCIE_ISO	*	=4X_DIELECTRIC	?
CLK_PCIE_ISO	*	=5:1_SPACING	?
COMP_PCIE_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE	*	*	PCIE_ISO
CLK_PCIE	*	*	CLK_PCIE_ISO
COMP_PCIE	*	*	COMP_PCIE_ISO

PCIe (PCH)

Electrical Constraint Set	Physical	Spacing
xl AirPort		
E59 PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
E60 PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
E61	PCIE_PHY	PCIE
E62	PCIE_PHY	PCIE
E63 PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
E64 PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
E65		
E66 PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
E67 PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
xl Caesar IV		
E68 PCIE_GEN2_R2D	PCIE_PHY	PCIE
E69 PCIE_GEN2_R2D	PCIE_PHY	PCIE
E70	PCIE_PHY	PCIE
E71	PCIE_PHY	PCIE
E72 PCIE_GEN2_D2R	PCIE_PHY	PCIE
E73 PCIE_GEN2_D2R	PCIE_PHY	PCIE
E74	PCIE_PHY	PCIE
E75	PCIE_PHY	PCIE
E76		
E77 PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
E78 PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE

PCIE (PCH - TBT)

Electrical Constraint Set		Physical	Spacing	
x4 Thunderbolt				
ES17	PCIE_GEN2_R2D_P1NV	PCIE_PHY	PCIE	PCIE TBT R2D P<2..0> no testpattern 36
ES18	PCIE_GEN2_R2D_P1NV	PCIE_PHY	PCIE	PCIE TBT R2D N<2..0> no testpattern 36
ES20	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE TBT R2D P<3> no testpattern 36
ES19	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE TBT R2D N<3> no testpattern 36
ES22		PCIE_PHY	PCIE	PCIE TBT R2D C P<3..0> no testpattern 18 36
ES21		PCIE_PHY	PCIE	PCIE TBT R2D C N<3..0> no testpattern 18 36
ES30		PCIE_PHY	PCIE	PCIE TBT D2R P<3..0> no testpattern 18 36
ES31		PCIE_PHY	PCIE	PCIE TBT D2R N<3..0> no testpattern 18 36
ES33	PCIE_GEN2_D2R_P1NV	PCIE_PHY	PCIE	PCIE TBT D2R C P<1..0> no testpattern 36
ES32	PCIE_GEN2_D2R_P1NV	PCIE_PHY	PCIE	PCIE TBT D2R C N<1..0> no testpattern 36
ES34	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE TBT D2R C P<2> no testpattern 36
ES35	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE TBT D2R C N<2> no testpattern 36
ES36	PCIE_GEN2_D2R_P1NV	PCIE_PHY	PCIE	PCIE TBT D2R C P<3> no testpattern 36
ES37	PCIE_GEN2_D2R_P1NV	PCIE_PHY	PCIE	PCIE TBT D2R C N<3> no testpattern 36
ES38	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE CLK100M TBT P
ES39	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE CLK100M TBT N

PCIe (CPU)

[illegible]

PCIe (CPU)

[illegible]

SYNCH MASTER=D6 AARON		SYNCH DATE=03/13/2012	
PAGE TITLE			
CPU PCIe Constraints			
 Apple Inc.		DRAWING NUMBER	051-9504
		SIZE	D
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D

C

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B

A

Table	Trace	Design	Iso	Design	Comments
6-4	10	11.81	-	15.75	Using PCIe guidelines

Table	Trace	Design	Iso	Design	Comments
6-4	10	11.81	-	15.75	Using PCIe guidelines

Section	Imp	Design	Iso	Design	Comments
1.5	45-65	50	-	15.75	Isolation is for JTAG clocks. All signals default are 50 Ohm SE.

Section	Imp	Design	Iso	Design	Comments
1.5	45-65	50	-	15.75	Isolation is for JTAG clocks. All signals default are 50 Ohm SE.

FDI

XDP

X170

I190

T23

I23	
I23	

PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCH_ISO	*	=6.5X_DIELECTRIC	?	CLK_PCH	*	*	CLK_PCH_ISO
COMP_PCH_ISO	*	=2:1_SPACING	?	COMP_PCH	*	*	COMP_PCH_ISO
PCH_ISO	*	=3:1_SPACING	?	PCH	*	*	PCH_ISO

PCI

PCI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_PCI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

PCI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCI_ISO	*	=3.6X_DIELECTRIC	?	CLK_PCI	*	*	CLK_PCI_ISO

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LPC_ISO	*	=1.5:1_SPACING	?	LPC	*	*	LPC_ISO
CLK_LPC_ISO	*	=3.6X_DIELECTRIC	?	CLK_LPC	*	*	CLK_LPC_ISO

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
HDA_ISO	*	=2X_DIELECTRIC	?	HDA	*	*	HDA_ISO

Crystal

Crystal-specific Physical Rules

[illegible]

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XTAL_ISO	*	=4X_DIELECTRIC	?	XTAL	*	*	XTAL_ISO

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SPI_ISO	*	=2X_DIELECTRIC	?	SPI	*	*	SPI_ISO

PCI

Electrical Constraint Set	Physical	Spacing	
PCI Clock			
	CLK_PCT_50S	CLK_PCT	PCH_CLK33M_PCTIIN 18 26
	CLK_PCT_50S	CLK_PCT	PCH_CLK33M_PCTIOUT 20 26

LPC

Electrical Constraint Set	Physical	Spacing	
LPC			
RE30	LPC_50S	LPC	LPC_AD<3..0> 18 47 49
RE30	LPC_50S	LPC	LPC_R_AD<3..0> 18
RE30	LPC_50S	LPC	LPC_FRAME_L 18 47 49
RE30	LPC_50S	LPC	LFRAME_L 18
LPC Clocks			
RE30	CLK_LPC_50S	CLK_LPC	LPC_CLK33M LPCPLUS 26 49
RE40	CLK_LPC_50S	CLK_LPC	LPC_CLK33M LPCPLUS_R 20 26
RE40	CLK_LPC_50S	CLK_LPC	LPC_CLK33M SMC 26 47
RE40	CLK_LPC_50S	CLK_LPC	LPC_CLK33M SMC_R 20 26

PCH Clocks

Electrical Constraint Set	Physical	Spacing	
PCH Reference Clock			
RE15	CLK_PCH_50S	CLF_PCH	SYSCLK_CLK25M_SB 26
RE16	CLK_PCH_50S	CLF_PCH	PCH_CLK25M_XTALIN 18 26
PCH Ref Clock Comp			
RE17	PCH_50S	COMP_PCH	PCH_XCLK_RCOMP 18
PCH RTC 32K			
RE18	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1 18 26
RE19	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2 18 26
RE20	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R 26
SMC 32K			
RE21	CLK_PCH_50S	CLF_PCH	PM_CLK32K_SUSCLK_R 19 48
RE22	CLK_PCH_50S	CLF_PCH	SMC_CLK32K 47 48

25 MHz Reference Clocks

Electrical Constraint Set	Physical	Spacing	
25M Reference Crystal			
RE30	CLK_XTAL	XTAL	SYSCLK_CLK25M_X1 26
RE30	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2 26
RE30	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2_R 26
25M Reference Clocks			
RE30	CLK_ECH_50S	CLK_ECH	SYSCLK_CLK25M_ENET 26 39
RE30	CLK_ECH_50S	CLK_ECH	SYSCLK_CLK25M_ENET_R 26 39
RE30	CLK_ECH_50S	CLK_ECH	SYSCLK_CLK25M_TBT 26 36
RE30	CLK_ECH_50S	CLK_ECH	SYSCLK_CLK25M_TBT_R 26 36

HDA

Electrical Constraint Set	Physical	Spacing	
HDA			
H350	HDA_50S	HDA	HDA_BIT_CLK 18 56
H351	HDA_50S	HDA	HDA_BIT_CLK_R 18
H352	HDA_50S	HDA	HDA_RST_L 18 56
H353	HDA_50S	HDA	HDA_RST_R_L 18
H354	HDA_50S	HDA	HDA_SDOUT 18 56
H355	HDA_50S	HDA	HDA_SDOUT_R 15 18
H356	HDA_50S	HDA	HDA_SYNC 15 18 56
H357	HDA_50S	HDA	HDA_SYNC_R 18
H358	HDA_50S	HDA	HDA_SDIO 18 56
H359	HDA_50S	HDA	AUD_SDI_R 56
H316	HDA_50S	HDA	SPI_DESCRIPTOR_OVERRIDE_R 15
SPDIF			
H330		HDA	AUD_SPDIF_CHIP 56
H331		HDA	AUD_SPDIF_OUT 56 60

SPI Bootrom

Electrical Constraint Set	Physical	Spacing	
SPI ROM			
H393	SPI_50S	SPI	SPI_CLK_R 18 49
H392	SPI_50S	SPI	SPI_CLK 49
H392	SPI_50S	SPI	SPI_ALT_CLK 49
H400	SPI_50S	SPI	SPI_SMC_CLK 47 48
H409	SPI_50S	SPI	SPI_MLB_CLK 48 49
H392	SPI_50S	SPI	SPI_CS0_R_L 18 49
H395	SPI_50S	SPI	SPI_CS0_L 49
H396	SPI_50S	SPI	SPI_ALT_CS_L 49
H400	SPI_50S	SPI	SPI_SMC_CS_L 47 48
H397	SPI_50S	SPI	SPI_MLB_CS_L 48 49
H399	SPI_50S	SPI	SPI_MOSI_R 18 49
H400	SPI_50S	SPI	SPI_MOSI 49
H403	SPI_50S	SPI	SPI_ALT_MOSI 49
H398	SPI_50S	SPI	SPI_SMC_MOSI 47 48
H388	SPI_50S	SPI	SPI_MLB_MOSI 48 49
H400	SPI_50S	SPI	SPI_MISO 18 49
H403	SPI_50S	SPI	SPI_ALT_MISO 49
H399	SPI_50S	SPI	SPI_SMC_MISO 47 48
H388	SPI_50S	SPI	SPI_MLB_MISO 48 49
H400	SPI_50S	SPI	SPIROM_USE_MLB 21 49

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PAGE TITLE			
PCH and BR Constraints			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-9504	D
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USB

USB-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
USB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
USB2_PHY	*	USB_90D
USB3_PHY	*	USB_85D
USB_HUB_PHY	*	USB_50S

SMSC Hub Application Note 15.17

Single-ended impedance range from 45-80 ohm is acceptable

USB 2.0 Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB2_ISO	*	=4.4X_DIELECTRIC	?
USB2_ISO	TOP,BOTTOM	=4.4X_DIELECTRIC	?
USB2_CLK_ISO	*	=18X_DIELECTRIC	?
USB2_CLK_ISO	TOP,BOTTOM	=18X_DIELECTRIC	?
USB_HUB_ISO	*	=21X_SPACING	?

USB 2.0 Spacing Rules

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB2	*	*	USB2_ISO
USB2	*CLK*	*	USB2_CLK_ISO
USB2	DISPLAYPORT	*	USB2_CLK_ISO
USB2	*TBT*	*	USB2_CLK_ISO
USB2	*ENET*	*	USB2_CLK_ISO
USB2	*SD*	*	USB2_CLK_ISO
USB3	PCIE	*	USB3_CLK_ISO
USB3	SATA	*	USB3_CLK_ISO
USB_HUB	*	*	USB_HUB_ISO

CAMERA CONTROLLER

Camera Controller's SMIA Interface & MISC. Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMIA_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
CAM_SE	*	Y	0.2 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMIA_DIFF_PHY	*	SMIA_100D
CAM_PHY	*	CAM_SE

Camera Controller's SMIA Interface & MISC. Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMIA_DIFF_ISO	*	=6:1_SPACING	?
SMIA_DIFF2DIFF	*	=3:1_SPACING	?
CAM_ISO	*	=2:1_SPACING	?

USB 3.0 Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_ISO	*	=7.3X_DIELECTRIC	?
USB3_ISO	TOP,BOTTOM	=7.3X_DIELECTRIC	?
USB3_CLK_ISO	*	=18X_DIELECTRIC	?
USB3_CLK_ISO	TOP,BOTTOM	=18X_DIELECTRIC	?
USB3_USB3_ISO	*	=7.3X_DIELECTRIC	?
USB3_USB3_ISO	TOP,BOTTOM	=7.3X_DIELECTRIC	?
USB3_USB3_ISO	ISL10	=5X_DIELECTRIC	?

USB 3.0 Spacing Rules

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3	*	*	USB3_ISO
USB3	*CLK*	*	USB3_CLK_ISO
USB3	DISPLAYPORT	*	USB3_CLK_ISO
USB3	*TBT*	*	USB3_CLK_ISO
USB3	*ENET*	*	USB3_CLK_ISO
USB3	*SD*	*	USB3_CLK_ISO
USB3	PCIE	*	USB3_CLK_ISO
USB3	SATA	*	USB3_CLK_ISO
USB3	USB3	*	USB3_USB3_ISO

USB 3.0 and USB 2.0 Trixies Muxing

Electrical Constraint Set		Physical	Spacing	
External Port A (J4600)				
E490	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_P 45
E491	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_N 45
E492	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_F_P 20 45
E493	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_F_N 20 45
E494	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_P 20 45
E495	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_N 20 45
E496	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_F_P 45
E497	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_F_N 45
E498	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_C_P 45
E499	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_C_N 45
E490	USB2_MIXED_MOTO_CONN	USB2_PHY	USB2	USB_PCH_0_P 20 45
E491	USB2_MIXED_MOTO_CONN	USB2_PHY	USB2	USB_PCH_0_N 20 45
E492	USB2_MIXED_MOTO_CONN	USB2_PHY	USB2	USB2_EXTB_MUXED_P 45
E493	USB2_MIXED_MOTO_CONN	USB2_PHY	USB2	USB2_EXTB_MUXED_N 45
E494	USB2_MIXED_MOTO_CONN	USB2_PHY	USB2	USB2_EXTB_MUXED_F_P 45
E495	USB2_MIXED_MOTO_CONN	USB2_PHY	USB2	USB2_EXTB_MUXED_F_N 45
External Port B (J4610)				
E496	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_P 45
E497	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_N 45
E498	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_F_P 20 45
E499	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_F_N 20 45
E490	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_P 20 45
E491	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_N 20 45
E492	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_F_P 45
E493	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_F_N 45
E494	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_C_P 45
E495	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_C_N 45
E496	USB2_MIXED_CONN	USB2_PHY	USB2	USB_PCH_1_P 20 45
E497	USB2_MIXED_CONN	USB2_PHY	USB2	USB_PCH_1_N 20 45
E498	USB2_MIXED_CONN	USB2_PHY	USB2	USB_PCH_9_P 20 45
E499	USB2_MIXED_CONN	USB2_PHY	USB2	USB_PCH_9_N 20 45
E490	USB2_MIXED_CONN	USB2_PHY	USB2	USB2_EXTB_MUXED_P 45
E491	USB2_MIXED_CONN	USB2_PHY	USB2	USB2_EXTB_MUXED_N 45
E492	USB2_MIXED_CONN	USB2_PHY	USB2	USB2_EXTB_MUXED_F_P 45
E493	USB2_MIXED_CONN	USB2_PHY	USB2	USB2_EXTB_MUXED_F_N 45
External Port C (J4700)				
E494	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_P 46
E495	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_N 46
E496	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_F_P 20 46
E497	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_F_N 20 46
E498	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_P 20 46
E499	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_N 20 46
E490	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_F_P 46
E491	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_F_N 46
E492	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_C_P 46
E493	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_C_N 46
E494	USB2_CONN	USB2_PHY	USB2	USB_PCH_2_P 20 46
E495	USB2_CONN	USB2_PHY	USB2	USB_PCH_2_N 20 46
E496	USB2_CONN	USB2_PHY	USB2	USB2_EXTB_F_P 46
E497	USB2_CONN	USB2_PHY	USB2	USB2_EXTB_F_N 46
External Port D (J4710)				
E498	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_P 46
E499	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_N 46
E490	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_F_P 20 46
E491	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_F_N 20 46
E492	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_P 20 46
E493	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_N 20 46
E494	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_F_P 46
E495	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_F_N 46
E496	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_C_P 46
E497	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_C_N 46
E498	USB2_MIXED_CONN	USB2_PHY	USB2	USB_PCH_3_P 20 46
E499	USB2_MIXED_CONN	USB2_PHY	USB2	USB_PCH_3_N 20 46
E490	USB2_MIXED_CONN	USB2_PHY	USB2	USB_PCH_10_P 20 46
E491	USB2_MIXED_CONN	USB2_PHY	USB2	USB_PCH_10_N 20 46
E492	USB2_MIXED_CONN	USB2_PHY	USB2	USB2_EXTD_MUXED_P 46
E493	USB2_MIXED_CONN	USB2_PHY	USB2	USB2_EXTD_MUXED_N 46
E494	USB2_MIXED_CONN	USB2_PHY	USB2	USB2_EXTD_MUXED_F_P 46
E495	USB2_MIXED_CONN	USB2_PHY	USB2	USB2_EXTD_MUXED_F_N 46
Camera (U4200)				
E496	USB2_CONN_TNT	USB2_PHY	USB2	USB_CAMERA_P 42
E497	USB2_CONN_TNT	USB2_PHY	USB2	USB_CAMERA_N 42
PCH USB Compensation				
E498		PCH_S0S	COMP_PCH	PCH_USB_RBIA5 20

Electrical Constraint Set	Physical	Spacing	Voltage	
Camera Controller Local Ground				
157A	GND_PHY	GND	0V	CAM_AGND 42
157B	GND__PHY	GND	0V	CAM_PLLGND 42

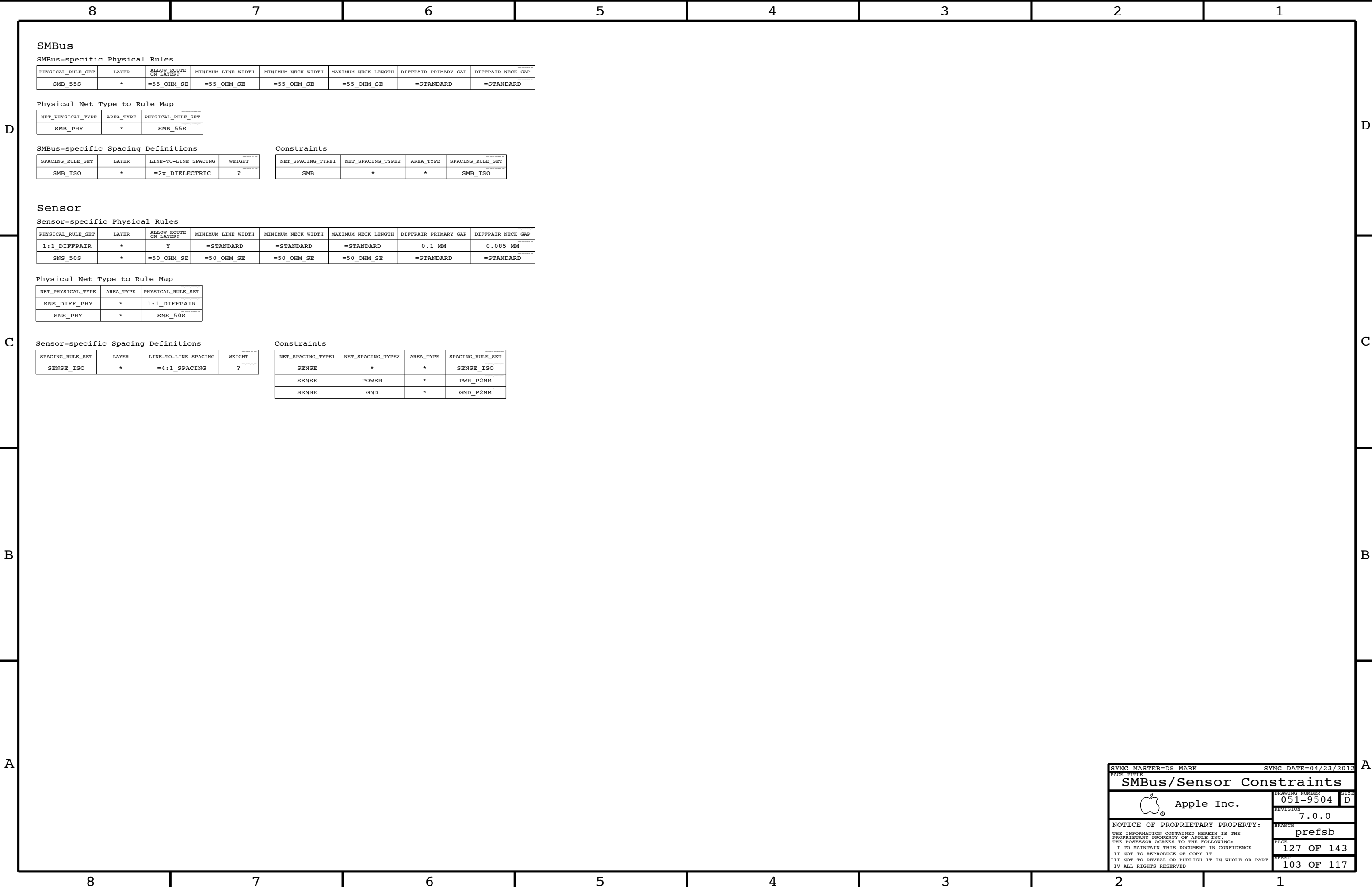
USB Hub

Electrical Constraint Set	Physical	Spacing	
USB 2.0 Hub			
REQ1 USB2_HUB_FCH	USB2_PHY	USB2	USB_FCH_7_P 26 27
REQ2 USB2_HUB_FCH	USB2_PHY	USB2	USB_FCH_7_N 26 27
REQ3 USB2_HUB_BT	USB2_PHY	USB2	USB_BT_P 27 35
REQ4 USB2_HUB_BT	USB2_PHY	USB2	USB_BT_N 27 35
REQ5 USB2_HUB_BT	USB2_PHY	USB2	USB_BT_MUX_P 35
REQ6 USB2_HUB_BT	USB2_PHY	USB2	USB_BT_MUX_N 35
REQ7	USB_HUB_PHY	USB_HUB	USB_HUB_2P 27
REQ8	USB_HUB_PHY	USB_HUB	USB_HUB_2N 27
USB 2.0 Hub Misc.			
REQ9	USB_HUB_PHY	USB_HUB	USB_HUB_RESET_L 27
REQ10	USB_HUB_PHY	USB_HUB	USB_HUB_VBUS_DET 27
REQ11	USB_HUB_PHY	USB_HUB	USB_HUB_NON_REMO 27
REQ12	USB_HUB_PHY	USB_HUB	USB_HUB_NON_REM1 27
REQ13	USB_HUB_PHY	USB_HUB	USB_HUB_HS_IND 27
USB 2.0 Hub Compensation			
REQ14	FCH_S0S	COMP_FCH	USB_HUB_RBIAS 27
USB 2.0 Hub Crystal			
REQ15	CLK_XTAL	XTAL	USB_HUB_XTAL1 27
REQ16	CLK_XTAL	XTAL	USB_HUB_XTAL2 27
REQ17	CLK_XTAL	XTAL	USB_HUB_XTAL2_R 27

Camera Controller

Electrical Constraint Set	Physical	Spacing	
SMIA			
HS00 SMIA_DATA	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_DATA P 42
HS00 SMIA_DATA	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_DATA N 42
HS00 SMIA_CLK	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_CLK P 42
HS00 SMIA_CLK	SMIA_DIFF_PHY	SMIA_DIFF	SMIA_CLK N 42
MISC			
HS00	SPT_50S	SPT	CAM_SF_CLK 42
HS00	SPT_50S	SPT	CAM_SF_CLK_R 42
HS00	SPT_50S	SPT	CAM_SF_DIN 42
HS00	SPT_50S	SPT	CAM_SF_DIN_R 42
HS00	SPT_50S	SPT	CAM_SF_CS_L 42
HS00	SPT_50S	SPT	CAM_SF_WP_L 42
HS00	SPT_50S	SPT	CAM_SF_DOUT 42
HS00	SPT_50S	SPT	CAM_SF_DOUT_R 42
HS00	SPT_50S	SPT	CAM_SF_HOLD_L 42
HS00	CAM_PHY	CAM	CAM_USB_VRES 42
HS00	CAM_PHY	CAM	MIPI_RESISTOR 42
HS00		EM	CAM_EXT_BOOT_L 43
HS00		EM	PCH_CAM_EXT_BOOT_R_L 21 43
HS00		EM	CAM_P1V2_RST_HOLDOFF 43
HS00		EM	CAM_P1V2_RST_HOLDOFF_L 43
I2C			
HS00 SMB_PHY	SMB		I2C_CAMSENSOR_SDA 42
HS00 SMB_PHY	SMB		I2C_CAMSENSOR_SCL 42
HS00 SMB_PHY	SMB		SMB_ALS_F_SDA 42
HS00 SMB_PHY	SMB		SMB_ALS_F_SCL 42
Camera Controller Crystal			
HS00 CLK_XTAL	XTAL		CAM_XTAL_IN 42
HS00 CLK_XTAL	XTAL		CAM_XTAL_OUT 42
HS00 CLK_XTAL	XTAL		CAM_XTAL_OUT_R 42

SYNCH MASTER=D8 KOSECOFF		SYNCH DATE=06/22/2012	
PAGE TITLE			
USB/Camera Constraints			
 Apple Inc.		DRAWING NUMBER 051-9504	
		SIZE D	
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D

D

C

C

B

B

A

A

DC-DC

Power-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
GND_F3MM	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
GND_P5MM	*	Y	0.500 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
POWER_P3MM	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P6MM	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P5MM	*	Y	0.500 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VR_CTL_PHY	*	POWER_P5MM
VR_CTL_PHY	BGA	STANDARD
VR_VID_PHY	*	POWER_50S
POWER_PHY	*	POWER_P6MM
GND_PHY	*	GND_P5MM

Constraints
Power and Common

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	POWER_ISO
GND	*	*	GND_ISO

Power-specific Spacing Definitions
Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
POWER_ISO	*	=STANDARD	?

DC-DC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SWNODE_ISO	*	=8:1_SPACING	1000
SWNODE_SW2SW	*	=1:1_SPACING	?
SWNODE_SW2PWR	*	=2:1_SPACING	?
SWNODE_SW2GND	*	=2:1_SPACING	?
SWNODE_LG2SW	*	=3:1_SPACING	?

DC-DC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
VR_CTL_ISO	*	=3:1_SPACING	?
VR_VID_ISO	*	=4X_DIELECTRIC	?

CPU VccSA

Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST
Input Bus					
E24	POWER_PHY	POWER	5V		REG_VCC_U7500
E25	POWER_PHY	POWER	5V		REG_PVCC_U7500
Local Ground					
E16	GND_PHY	GND	0V		AGND_VCCSA0
VCCIO					
E27	POWER_PHY	VR_SWITCH	12V	TRUE	REG_PHASE_VCCSA0
E28	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_BOOT_VCCSA0
E29	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_BOOT_VCCSA0_RC
E31	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_UGATE_VCCSA0
E30	VR_CTL_PHY	VR_LGATE	12V	TRUE	REG_LGATE_VCCSA0
E33	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_SNUBBER_VCCSA0
E32	VR_CTL_PHY	VR_CTL			REG_VCCSA0_OCSET
E35	VR_CTL_PHY	VR_CTL			REG_VCCSA0_VO
E34		SENSE			SNS_CPU_VCCSA
E37	VSNS_CPU_VCCSA	SNS_DIFF_PHY	SENSE		SNS_VCCSA0_XW_P
E36	VSNS_CPU_VCCSA	SNS_DIFF_PHY	SENSE		SNS_VCCSA0_XW_N
E38		SENSE			REG_VCCSA0_FB
E39		SENSE			REG_VCCSA0_RTN
E40	VR_CTL_PHY	VR_CTL			REG_VCCSA0_SREF
E41	VR_CTL_PHY	VR_CTL			REG_VCCSA0_FSEL
Output Bus					
E42	POWER_PHY	POWER	0.925V		PPVCCSA_S0

CPU VccIO PCH 1.05V S0

Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST
Input Bus					
E11	POWER_PHY	POWER	5V		REG_VCC_U7400
E12	POWER_PHY	POWER	5V		REG_PVCC_U7400
Local Ground					
E13	GND_PHY	GND	0V		AGND_CPU_P1V05S0
1.05V S0					
E15	POWER_PHY	VR_SWITCH	12V	TRUE	REG_CPU_PHASE_P1V05S0
E17	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_CPU_BOOT_P1V05S0
E16	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_CPU_BOOT_P1V05S0_RC
E18	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_CPU_UGATE_P1V05S0
E19	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_CPU_UGATE_P1V05S0_R
E10	VR_CTL_PHY	VR_LGATE	12V	TRUE	REG_CPU_LGATE_P1V05S0
E20	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_CPU_SNUBBER_P1V05S0
E21	VR_CTL_PHY	VR_CTL			REG_CPU_P1V05S0_OCSET
E22	VR_CTL_PHY	VR_CTL			REG_CPU_P1V05S0_VO
VSNS_CPU_VCCIO	SNS_DIFF_PHY	SENSE			SNS_CPU_VCCIO_P
E14	VSNS_CPU_VCCIO	SNS_DIFF_PHY	SENSE		SNS_CPU_VCCIO_N
E18	SNS_DIFF_PHY	SENSE			SNS_CPU_P1V05S0_XW_P
E19	SNS_DIFF_PHY	SENSE			SNS_CPU_P1V05S0_XW_N
E23		SENSE			REG_CPU_P1V05S0_FB
E24		SENSE			REG_CPU_P1V05S0_RTN
E21	VR_CTL_PHY	VR_CTL			REG_CPU_P1V05S0_SREF
E20	VR_CTL_PHY	VR_CTL			REG_CPU_P1V05S0_FSEL
Output Bus					
E22	POWER_PHY	POWER	1.05V		PP1V05_S0_CPU

PCH/GPU/TBT 1.05V S0

Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST
Input Bus					
E44	POWER_PHY	POWER	5V		REG_VCC_U8700
Local Ground					
E46	GND_PHY	GND	0V		P1V05_AGND
1.05V S0					
E47	POWER_PHY	VR_SWITCH	12V	TRUE	REG_PHASE_P1V05S0
E48	POWER_PHY	VR_SWITCH	12V	TRUE	REG_PHASE_P1V05S0_L
E49	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_BOOT_P1V05S0
E50	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_BOOT_P1V05S0_RC
E51	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_UGATE_P1V05S0
E52	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_UGATE_P1V05S0_R
E53	VR_CTL_PHY	VR_LGATE	12V	TRUE	REG_LGATE_P1V05S0
E54	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_SNUBBER_P1V05S0
E55	VR_CTL_PHY	VR_CTL			REG_P1V05S0_OCSET
E56	VR_CTL_PHY	VR_CTL			REG_P1V05S0_VO
E57	POWER_PHY	VR_SWITCH	12V		REG_P1V05S0_VO_R
E58	POWER_PHY	VR_SWITCH	12V		P1V05_OCSET_R
E61		SENSE			REG_P1V05S0_FB
E62		SENSE			REG_P1V05S0_RTN
E63	VR_CTL_PHY	VR_CTL			REG_P1V05S0_SREF
E64	VR_CTL_PHY	VR_CTL			REG_P1V05S0_FSEL
Output Bus					
E65	POWER_PHY	POWER	1.05V		PP1V05_S0_PCH
E66	POWER_PHY	POWER	1.05V		PP1V05_S0
PFT Switched					
E67	POWER_PHY	POWER	1.05V		PP1V05_TBTLIC
E68	POWER_PHY	POWER	1.05V		PP1V05_TBTCIO

SYNC MASTER=D8 MARK		SYNC DATE=02/10/2012	
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VReg Constraints			
 Apple Inc.		DRAWING NUMBER	051-9504
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CPU Core Phases				CPU AXG Phase and Core Controller																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
<table><tr><th>Electrical Constraint Set</th><th>Physical</th><th>Spacing</th><th>Voltage</th><th>DIDT</th><th>NO_TEST</th><th></th></tr><tr><td colspan="7">Input Bus</td></tr><tr><td>PP12V</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>PP12V_S0_CPUCORE FLT</td></tr><tr><td>REG_VCC_U7100</td><td>POWER_PHY</td><td>POWER</td><td>5V</td><td></td><td></td><td>REG_VCC_U7100</td></tr><tr><td colspan="7">Local Ground</td></tr><tr><td>AGND_CPU</td><td>GND_PHY</td><td>GND</td><td>0V</td><td></td><td></td><td>AGND_CPU</td></tr><tr><td colspan="7">Phase 1</td></tr><tr><td>REG_LVCC_U7210</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7210</td></tr><tr><td>REG_PWM_CPUCORE_1</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_1</td></tr><tr><td>REG_PWM_CPUCORE_1_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_1_R</td></tr><tr><td>REG_PHASE_CPUCORE_1</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUCORE_1</td></tr><tr><td>REG_BOOT_CPUCORE_1</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUCORE_1</td></tr><tr><td>REG_BOOT_CPUCORE_1_RC</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT_CPUCORE_1_RC</td></tr><tr><td>REG_UGATE_CPUCORE_1</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUCORE_1</td></tr><tr><td>REG_LGATE_CPUCORE_1</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUCORE_1</td></tr><tr><td>REG_SNUBBER_CPUCORE_1</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER_CPUCORE_1</td></tr><tr><td>PPCPUCORE_S0_SENSE_1</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUCORE_S0_SENSE_1</td></tr><tr><td>REG_ISENCORE_1_P</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_ISENCORE_1_P</td></tr><tr><td>REG_ISENCORE_1_N</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_ISENCORE_1_N</td></tr><tr><td>REG_ISENCORE_1_NR</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_ISENCORE_1_NR</td></tr><tr><td colspan="7">Phase 2</td></tr><tr><td>REG_LVCC_U7230</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7230</td></tr><tr><td>REG_PWM_CPUCORE_2</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_2</td></tr><tr><td>REG_PWM_CPUCORE_2_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_2_R</td></tr><tr><td>REG_PHASE_CPUCORE_2</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUCORE_2</td></tr><tr><td>REG_BOOT_CPUCORE_2</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUCORE_2</td></tr><tr><td>REG_BOOT_CPUCORE_2_RC</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT_CPUCORE_2_RC</td></tr><tr><td>REG_UGATE_CPUCORE_2</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUCORE_2</td></tr><tr><td>REG_LGATE_CPUCORE_2</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUCORE_2</td></tr><tr><td>REG_SNUBBER_CPUCORE_2</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER_CPUCORE_2</td></tr><tr><td>PPCPUCORE_S0_SENSE_2</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUCORE_S0_SENSE_2</td></tr><tr><td>REG_ISENCORE_2_P</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_ISENCORE_2_P</td></tr><tr><td>REG_ISENCORE_2_N</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_ISENCORE_2_N</td></tr><tr><td>REG_ISENCORE_2_NR</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_ISENCORE_2_NR</td></tr><tr><td colspan="7">Phase 3</td></tr><tr><td>REG_LVCC_U7250</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7250</td></tr><tr><td>REG_PWM_CPUCORE_3</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_3</td></tr><tr><td>REG_PWM_CPUCORE_3_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_3_R</td></tr><tr><td>REG_PHASE_CPUCORE_3</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUCORE_3</td></tr><tr><td>REG_BOOT_CPUCORE_3</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUCORE_3</td></tr><tr><td>REG_BOOT_CPUCORE_3_RC</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT_CPUCORE_3_RC</td></tr><tr><td>REG_UGATE_CPUCORE_3</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUCORE_3</td></tr><tr><td>REG_LGATE_CPUCORE_3</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUCORE_3</td></tr><tr><td>REG_SNUBBER_CPUCORE_3</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER_CPUCORE_3</td></tr><tr><td>PPCPUCORE_S0_SENSE_3</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUCORE_S0_SENSE_3</td></tr><tr><td>REG_ISENCORE_3_P</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_ISENCORE_3_P</td></tr><tr><td>REG_ISENCORE_3_N</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_ISENCORE_3_N</td></tr><tr><td>REG_ISENCORE_3_NR</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_ISENCORE_3_NR</td></tr><tr><td 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4</td></tr><tr><td>REG_LVCC_U7310</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7310</td></tr><tr><td>REG_PWM_CPUCORE_4</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_4</td></tr><tr><td>REG_PWM_CPUCORE_4_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_4_R</td></tr><tr><td>REG_PHASE_CPUCORE_4</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUCORE_4</td></tr><tr><td>REG_BOOT_CPUCORE_4</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUCORE_4</td></tr><tr><td>REG_BOOT_CPUCORE_4_RC</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT_CPUCORE_4_RC</td></tr><tr><td>REG_UGATE_CPUCORE_4</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUCORE_4</td></tr><tr><td>REG_LGATE_CPUCORE_4</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUCORE_4</td></tr><tr><td>REG_SNUBBER_CPUCORE_4</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER_CPUCORE_4</td></tr><tr><td>PPCPUCORE_S0_SENSE_4</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUCORE_S0_SENSE_4</td></tr><tr><td>REG_ISENCORE_4_P</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_ISENCORE_4_P</td></tr><tr><td>REG_ISENCORE_4_N</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_ISENCORE_4_N</td></tr><tr><td>REG_ISENCORE_4_NR</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_ISENCORE_4_NR</td></tr><tr><td>SNS_CORE_XW_P</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>SNS_CORE_XW_P</td></tr><tr><td>SNS_CORE_XW_N</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>SNS_CORE_XW_N</td></tr><tr><td>SNS_CORE_R_P</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>SNS_CORE_R_P</td></tr><tr><td>SNS_CORE_R_N</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>SNS_CORE_R_N</td></tr><tr><td>SNS_CPU_VAXG_P</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>SNS_CPU_VAXG_P</td></tr><tr><td>SNS_CPU_VAXG_N</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>SNS_CPU_VAXG_N</td></tr><tr><td>SNS_CPU_VCORE_P</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>SNS_CPU_VCORE_P</td></tr><tr><td>SNS_CPU_VCORE_N</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>SNS_CPU_VCORE_N</td></tr><tr><td>SNS_P1V05_IOVDD_XW_P</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>SNS_P1V05_IOVDD_XW_P</td></tr><tr><td>SNS_P1V05_IOVDD_XW_N</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>SNS_P1V05_IOVDD_XW_N</td></tr></table>				Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST		Input Bus							PP12V	POWER_PHY	POWER	12V			PP12V_S0_CPUCORE FLT	REG_VCC_U7100	POWER_PHY	POWER	5V			REG_VCC_U7100	Local Ground							AGND_CPU	GND_PHY	GND	0V			AGND_CPU	Phase 1							REG_LVCC_U7210	POWER_PHY	POWER	12V			REG_LVCC_U7210	REG_PWM_CPUCORE_1	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1	REG_PWM_CPUCORE_1_R	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1_R	REG_PHASE_CPUCORE_1	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_1	REG_BOOT_CPUCORE_1	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_1	REG_BOOT_CPUCORE_1_RC	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_1_RC	REG_UGATE_CPUCORE_1	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_1	REG_LGATE_CPUCORE_1	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_1	REG_SNUBBER_CPUCORE_1	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_1	PPCPUCORE_S0_SENSE_1	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_1	REG_ISENCORE_1_P	ISNS_CPU_CORE	SNS_DIFF_PHY				REG_ISENCORE_1_P	REG_ISENCORE_1_N	ISNS_CPU_CORE	SNS_DIFF_PHY				REG_ISENCORE_1_N	REG_ISENCORE_1_NR	ISNS_CPU_CORE	SNS_DIFF_PHY				REG_ISENCORE_1_NR	Phase 2							REG_LVCC_U7230	POWER_PHY	POWER	12V			REG_LVCC_U7230	REG_PWM_CPUCORE_2	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2	REG_PWM_CPUCORE_2_R	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2_R	REG_PHASE_CPUCORE_2	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_2	REG_BOOT_CPUCORE_2	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_2	REG_BOOT_CPUCORE_2_RC	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_2_RC	REG_UGATE_CPUCORE_2	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_2	REG_LGATE_CPUCORE_2	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_2	REG_SNUBBER_CPUCORE_2	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_2	PPCPUCORE_S0_SENSE_2	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_2	REG_ISENCORE_2_P	ISNS_CPU_CORE	SNS_DIFF_PHY				REG_ISENCORE_2_P	REG_ISENCORE_2_N	ISNS_CPU_CORE	SNS_DIFF_PHY				REG_ISENCORE_2_N	REG_ISENCORE_2_NR	ISNS_CPU_CORE	SNS_DIFF_PHY				REG_ISENCORE_2_NR	Phase 3							REG_LVCC_U7250	POWER_PHY	POWER	12V			REG_LVCC_U7250	REG_PWM_CPUCORE_3	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3	REG_PWM_CPUCORE_3_R	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3_R	REG_PHASE_CPUCORE_3	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_3	REG_BOOT_CPUCORE_3	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_3	REG_BOOT_CPUCORE_3_RC	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_3_RC	REG_UGATE_CPUCORE_3	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_3	REG_LGATE_CPUCORE_3	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_3	REG_SNUBBER_CPUCORE_3	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_3	PPCPUCORE_S0_SENSE_3	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_3	REG_ISENCORE_3_P	ISNS_CPU_CORE	SNS_DIFF_PHY				REG_ISENCORE_3_P	REG_ISENCORE_3_N	ISNS_CPU_CORE	SNS_DIFF_PHY				REG_ISENCORE_3_N	REG_ISENCORE_3_NR	ISNS_CPU_CORE	SNS_DIFF_PHY				REG_ISENCORE_3_NR	Phase 4							REG_LVCC_U7310	POWER_PHY	POWER	12V			REG_LVCC_U7310	REG_PWM_CPUCORE_4	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_4	REG_PWM_CPUCORE_4_R	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_4_R	REG_PHASE_CPUCORE_4	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_4	REG_BOOT_CPUCORE_4	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_4	REG_BOOT_CPUCORE_4_RC	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_4_RC	REG_UGATE_CPUCORE_4	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_4	REG_LGATE_CPUCORE_4	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_4	REG_SNUBBER_CPUCORE_4	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_4	PPCPUCORE_S0_SENSE_4	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_4	REG_ISENCORE_4_P	ISNS_CPU_CORE	SNS_DIFF_PHY				REG_ISENCORE_4_P	REG_ISENCORE_4_N	ISNS_CPU_CORE	SNS_DIFF_PHY				REG_ISENCORE_4_N	REG_ISENCORE_4_NR	ISNS_CPU_CORE	SNS_DIFF_PHY				REG_ISENCORE_4_NR	SNS_CORE_XW_P	ISNS_CPU_CORE	SNS_DIFF_PHY				SNS_CORE_XW_P	SNS_CORE_XW_N	ISNS_CPU_CORE	SNS_DIFF_PHY				SNS_CORE_XW_N	SNS_CORE_R_P	ISNS_CPU_CORE	SNS_DIFF_PHY				SNS_CORE_R_P	SNS_CORE_R_N	ISNS_CPU_CORE	SNS_DIFF_PHY				SNS_CORE_R_N	SNS_CPU_VAXG_P	ISNS_CPU_CORE	SNS_DIFF_PHY				SNS_CPU_VAXG_P	SNS_CPU_VAXG_N	ISNS_CPU_CORE	SNS_DIFF_PHY				SNS_CPU_VAXG_N	SNS_CPU_VCORE_P	ISNS_CPU_CORE	SNS_DIFF_PHY				SNS_CPU_VCORE_P	SNS_CPU_VCORE_N	ISNS_CPU_CORE	SNS_DIFF_PHY				SNS_CPU_VCORE_N	SNS_P1V05_IOVDD_XW_P	ISNS_CPU_CORE	SNS_DIFF_PHY				SNS_P1V05_IOVDD_XW_P	SNS_P1V05_IOVDD_XW_N	ISNS_CPU_CORE	SNS_DIFF_PHY				SNS_P1V05_IOVDD_XW_N	<table><tr><th>Electrical Constraint Set</th><th>Physical</th><th>Spacing</th><th>Voltage</th><th>DIDT</th><th>NO_TEST</th><th></th></tr><tr><td colspan="7">AXG</td></tr><tr><td>REG_LVCC_U7330</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7330</td></tr><tr><td>REG_PWM_CPUAXG</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUAXG</td></tr><tr><td>REG_PWM_CPUAXG_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUAXG_R</td></tr><tr><td>REG_PHASE_CPUAXG</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUAXG</td></tr><tr><td>REG_BOOT_CPUAXG</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUAXG</td></tr><tr><td>REG_BOOT_CPUAXG_RC</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT_CPUAXG_RC</td></tr><tr><td>REG_UGATE_CPUAXG</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUAXG</td></tr><tr><td>REG_LGATE_CPUAXG</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUAXG</td></tr><tr><td>REG_SNUBBER_CPUAXG</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER_CPUAXG</td></tr><tr><td>PPCPUAXG_S0_SENSE</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUAXG_S0_SENSE</td></tr><tr><td>REG_ISENAXG_P</td><td>ISNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_ISENAXG_P</td></tr><tr><td>REG_ISENAXG_N</td><td>ISNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>REG_ISENAXG_N</td></tr><tr><td>REG_ISENAXG_PR</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_ISENAXG_PR</td></tr><tr><td>REG_ISENAXG_NR</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_ISENAXG_NR</td></tr><tr><td>SNS_AXG_R_P</td><td>ISNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>SNS_AXG_R_P</td></tr><tr><td>SNS_AXG_R_N</td><td>ISNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>SNS_AXG_R_N</td></tr><tr><td>SNS_AXG_XW_P</td><td>ISNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>SNS_AXG_XW_P</td></tr><tr><td>SNS_AXG_XW_N</td><td>ISNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td></td><td></td><td></td><td>SNS_AXG_XW_N</td></tr><tr><td colspan="7">ISL6364</td></tr><tr><td>REG_CPUCORE_COMP</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_COMP</td></tr><tr><td>CPUCORE_COMP_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_COMP_RC</td></tr><tr><td>REG_CPUCORE_FB</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_FB</td></tr><tr><td>CPUCORE_FB_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_FB_RC</td></tr><tr><td>CPUCORE_FB_R_1</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_FB_R_1</td></tr><tr><td>CPUCORE_FB_R_2</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_FB_R_2</td></tr><tr><td>CPUCORE_PSICOMP_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_PSICOMP_RC</td></tr><tr><td>REG_CPUCORE_PSICOMP</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_PSICOMP</td></tr><tr><td>REG_CPUCORE_HFCOMP</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_HFCOMP</td></tr><tr><td>REG_CPUCORE_VSEN</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_CPUCORE_VSEN</td></tr><tr><td>REG_CPUCORE_RGND</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_CPUCORE_RGND</td></tr><tr><td>REG_CPUCORE_IMON</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_IMON</td></tr><tr><td>CPUCORE_IMON_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_IMON_R</td></tr><tr><td>REG_CPUCORE_TM</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_TM</td></tr><tr><td>REG_CPUCORE_SUTH</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_SUTH</td></tr><tr><td>REG_CPUCORE_NPSI</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_NPSI</td></tr><tr><td>REG_CPUCORE_FDIVID</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_FDIVID</td></tr><tr><td>REG_CPUCORE_IAUTO</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_IAUTO</td></tr><tr><td>REG_CPUCORE_SW_FREQ</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_SW_FREQ</td></tr><tr><td>REG_CPUCORE_RAMPADJ</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_RAMPADJ</td></tr><tr><td>REG_CPUCORE_EN_PWR</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_EN_PWR</td></tr><tr><td>CPUCORE_EN_PWR_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_EN_PWR_R</td></tr><tr><td>REG_CPUCORE_RSET</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_RSET</td></tr><tr><td>REG_CPUAXG_COMP</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_COMP</td></tr><tr><td>CPUAXG_COMP_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_COMP_RC</td></tr><tr><td>REG_CPUAXG_FB</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_FB</td></tr><tr><td>CPUAXG_FB_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_FB_RC</td></tr><tr><td>CPUAXG_FB_R_1</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_FB_R_1</td></tr><tr><td>CPUAXG_FB_R_2</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_FB_R_2</td></tr><tr><td>REG_CPUAXG_HFCOMP</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_HFCOMP</td></tr><tr><td>REG_CPUAXG_VSEN</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_CPUAXG_VSEN</td></tr><tr><td>REG_CPUAXG_RGND</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_CPUAXG_RGND</td></tr><tr><td>REG_CPUAXG_IMON</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_IMON</td></tr><tr><td>CPUAXG_IMON_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_IMON_R</td></tr><tr><td>REG_CPUAXG_TM</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_TM</td></tr><tr><td>REG_CPUAXG_TCOMP</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_TCOMP</td></tr><tr><td>REG_CPUAXG_SW_FREQ</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_SW_FREQ</td></tr><tr><td>CPU_VIDSCLK</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td><td>CPU_VIDSCLK</td></tr><tr><td>CPU_VIDSCLK_R</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td><td>CPU_VIDSCLK_R</td></tr><tr><td>CPU_VIDALERT_L</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td><td>CPU_VIDALERT_L</td></tr><tr><td>CPU_VIDALERT_R_L</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td><td>CPU_VIDALERT_R_L</td></tr><tr><td>CPU_VIDSOUT</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td><td>CPU_VIDSOUT</td></tr><tr><td>CPU_VIDSOUT_R</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td><td>CPU_VIDSOUT_R</td></tr><tr><td colspan="7">Output Bus</td></tr><tr><td>PPVCORE_S0_CPU</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPVCORE_S0_CPU</td></tr><tr><td>PPVAXG_S0</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPVAXG_S0</td></tr></table>				Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST		AXG							REG_LVCC_U7330	POWER_PHY	POWER	12V			REG_LVCC_U7330	REG_PWM_CPUAXG	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG	REG_PWM_CPUAXG_R	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG_R	REG_PHASE_CPUAXG	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUAXG	REG_BOOT_CPUAXG	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUAXG	REG_BOOT_CPUAXG_RC	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUAXG_RC	REG_UGATE_CPUAXG	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUAXG	REG_LGATE_CPUAXG	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUAXG	REG_SNUBBER_CPUAXG	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUAXG	PPCPUAXG_S0_SENSE	POWER_PHY	POWER	1.1V			PPCPUAXG_S0_SENSE	REG_ISENAXG_P	ISNS_CPU_AXG	SNS_DIFF_PHY				REG_ISENAXG_P	REG_ISENAXG_N	ISNS_CPU_AXG	SNS_DIFF_PHY				REG_ISENAXG_N	REG_ISENAXG_PR		SENSE				REG_ISENAXG_PR	REG_ISENAXG_NR		SENSE				REG_ISENAXG_NR	SNS_AXG_R_P	ISNS_CPU_AXG	SNS_DIFF_PHY				SNS_AXG_R_P	SNS_AXG_R_N	ISNS_CPU_AXG	SNS_DIFF_PHY				SNS_AXG_R_N	SNS_AXG_XW_P	ISNS_CPU_AXG	SNS_DIFF_PHY				SNS_AXG_XW_P	SNS_AXG_XW_N	ISNS_CPU_AXG	SNS_DIFF_PHY				SNS_AXG_XW_N	ISL6364							REG_CPUCORE_COMP	VR_CTL_PHY	VR_CTL				REG_CPUCORE_COMP	CPUCORE_COMP_RC	VR_CTL_PHY	VR_CTL				CPUCORE_COMP_RC	REG_CPUCORE_FB	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FB	CPUCORE_FB_RC	VR_CTL_PHY	VR_CTL				CPUCORE_FB_RC	CPUCORE_FB_R_1	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_1	CPUCORE_FB_R_2	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_2	CPUCORE_PSICOMP_RC	VR_CTL_PHY	VR_CTL				CPUCORE_PSICOMP_RC	REG_CPUCORE_PSICOMP	VR_CTL_PHY	VR_CTL				REG_CPUCORE_PSICOMP	REG_CPUCORE_HFCOMP	VR_CTL_PHY	VR_CTL				REG_CPUCORE_HFCOMP	REG_CPUCORE_VSEN		SENSE				REG_CPUCORE_VSEN	REG_CPUCORE_RGND		SENSE				REG_CPUCORE_RGND	REG_CPUCORE_IMON	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IMON	CPUCORE_IMON_R	VR_CTL_PHY	VR_CTL				CPUCORE_IMON_R	REG_CPUCORE_TM	VR_CTL_PHY	VR_CTL				REG_CPUCORE_TM	REG_CPUCORE_SUTH	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SUTH	REG_CPUCORE_NPSI	VR_CTL_PHY	VR_CTL				REG_CPUCORE_NPSI	REG_CPUCORE_FDIVID	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FDIVID	REG_CPUCORE_IAUTO	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IAUTO	REG_CPUCORE_SW_FREQ	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SW_FREQ	REG_CPUCORE_RAMPADJ	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RAMPADJ	REG_CPUCORE_EN_PWR	VR_CTL_PHY	VR_CTL				REG_CPUCORE_EN_PWR	CPUCORE_EN_PWR_R	VR_CTL_PHY	VR_CTL				CPUCORE_EN_PWR_R	REG_CPUCORE_RSET	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RSET	REG_CPUAXG_COMP	VR_CTL_PHY	VR_CTL				REG_CPUAXG_COMP	CPUAXG_COMP_RC	VR_CTL_PHY	VR_CTL				CPUAXG_COMP_RC	REG_CPUAXG_FB	VR_CTL_PHY	VR_CTL				REG_CPUAXG_FB	CPUAXG_FB_RC	VR_CTL_PHY	VR_CTL				CPUAXG_FB_RC	CPUAXG_FB_R_1	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_1	CPUAXG_FB_R_2	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_2	REG_CPUAXG_HFCOMP	VR_CTL_PHY	VR_CTL				REG_CPUAXG_HFCOMP	REG_CPUAXG_VSEN		SENSE				REG_CPUAXG_VSEN	REG_CPUAXG_RGND		SENSE				REG_CPUAXG_RGND	REG_CPUAXG_IMON	VR_CTL_PHY	VR_CTL				REG_CPUAXG_IMON	CPUAXG_IMON_R	VR_CTL_PHY	VR_CTL				CPUAXG_IMON_R	REG_CPUAXG_TM	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TM	REG_CPUAXG_TCOMP	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TCOMP	REG_CPUAXG_SW_FREQ	VR_CTL_PHY	VR_CTL				REG_CPUAXG_SW_FREQ	CPU_VIDSCLK	VR_VID_PHY	VR_VID				CPU_VIDSCLK	CPU_VIDSCLK_R	VR_VID_PHY	VR_VID				CPU_VIDSCLK_R	CPU_VIDALERT_L	VR_VID_PHY	VR_VID				CPU_VIDALERT_L	CPU_VIDALERT_R_L	VR_VID_PHY	VR_VID				CPU_VIDALERT_R_L	CPU_VIDSOUT	VR_VID_PHY	VR_VID				CPU_VIDSOUT	CPU_VIDSOUT_R	VR_VID_PHY	VR_VID				CPU_VIDSOUT_R	Output Bus							PPVCORE_S0_CPU	POWER_PHY	POWER	1.1V			PPVCORE_S0_CPU	PPVAXG_S0	POWER_PHY	POWER	1.1V			PPVAXG_S0
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REG_CPUCORE_EN_PWR	VR_CTL_PHY	VR_CTL				REG_CPUCORE_EN_PWR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
CPUCORE_EN_PWR_R	VR_CTL_PHY	VR_CTL				CPUCORE_EN_PWR_R																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
REG_CPUCORE_RSET	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RSET																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
REG_CPUAXG_COMP	VR_CTL_PHY	VR_CTL				REG_CPUAXG_COMP																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
CPUAXG_COMP_RC	VR_CTL_PHY	VR_CTL				CPUAXG_COMP_RC																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
REG_CPUAXG_FB	VR_CTL_PHY	VR_CTL				REG_CPUAXG_FB																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
CPUAXG_FB_RC	VR_CTL_PHY	VR_CTL				CPUAXG_FB_RC																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
CPUAXG_FB_R_1	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
CPUAXG_FB_R_2	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
REG_CPUAXG_HFCOMP	VR_CTL_PHY	VR_CTL				REG_CPUAXG_HFCOMP																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
REG_CPUAXG_VSEN		SENSE				REG_CPUAXG_VSEN																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
REG_CPUAXG_RGND		SENSE				REG_CPUAXG_RGND																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
REG_CPUAXG_IMON	VR_CTL_PHY	VR_CTL				REG_CPUAXG_IMON																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
CPUAXG_IMON_R	VR_CTL_PHY	VR_CTL				CPUAXG_IMON_R																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
REG_CPUAXG_TM	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TM																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
REG_CPUAXG_TCOMP	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TCOMP																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
REG_CPUAXG_SW_FREQ	VR_CTL_PHY	VR_CTL				REG_CPUAXG_SW_FREQ																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
CPU_VIDSCLK	VR_VID_PHY	VR_VID				CPU_VIDSCLK																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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CPU_VIDALERT_L	VR_VID_PHY	VR_VID				CPU_VIDALERT_L																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
CPU_VIDALERT_R_L	VR_VID_PHY	VR_VID				CPU_VIDALERT_R_L																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
CPU_VIDSOUT	VR_VID_PHY	VR_VID				CPU_VIDSOUT																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
CPU_VIDSOUT_R	VR_VID_PHY	VR_VID				CPU_VIDSOUT_R																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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PPVCORE_S0_CPU	POWER_PHY	POWER	1.1V			PPVCORE_S0_CPU																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
PPVAXG_S0	POWER_PHY	POWER	1.1V			PPVAXG_S0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														

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Constraints

BLC High Voltage Output

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_HV	BLC_HV	*	BLC_CTL_ISO
BLC_HV	*	*	BLC_HV_ISO

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC__PHASE	*	*	PHASE_ISO
BLC_PHASE	BLC_PHASE	*	PHASE_SW2SW
BLC_PHASE	POWER	*	PHASE_SW2PWR
BLC_PHASE	GND	*	PHASE_SW2GND

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_CTL	*	*	BLC_CTL_ISO

BLC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE SET
BLC_CTL	*	*	BLC_CTL_ISO

	Physical	Spacing	Voltage	DIDT	NO_TEST	
Input	Bus					
R450	POWER_PHY	POWER	12V		PP12V_S0_BLC_VIN2	89 90 91
R477	POWER_PHY	POWER	12V		PP12V_S0_BLC_VINP	89 91
R480	POWER_PHY	POWER	14V		PRE_REG_OUT	89 90 91
R481	POWER_PHY	POWER	3.3V		BLC_P3V3S	89 91
R484	POWER_PHY	POWER	3.3V		BLC_P3V3_REF	89 91
R485	POWER_PHY	POWER	3.3V		BLC_P3V3	
R486	POWER_PHY	POWER	3.3V		PP3V3_S0_BLC_R	89
R489	POWER_PHY	POWER	8V		SPTX_VIN	
R490	POWER_PHY	POWER	8V		PP8V_BLC	89 90
R491	POWER_PHY	POWER	8V		BLC_VIN2	89 91
R492	POWER_PHY	POWER	14V		BOOST_FET_DRAIN	89 90 91
R493	POWER_PHY	POWER	12V		BOOST_VDD	89
R495	POWER_PHY	POWER	5V		PP5V_S0_BLC_R	89 90 91
R496	POWER_PHY	POWER	12V		PP12V_S0_BLC_F	91
Local Ground						
R499	BLC_CTL_PHY	BLC_PHASE	0V		BLC_GND_1	90 91
R498	BLC_CTL_PHY	BLC_PHASE	0V		BLC_GND_2	90 91
R498	BLC_CTL_PHY	BLC_PHASE	0V		BLC_GND_3	90 91
R491	GND_PHY	GND	0V		AGND_BLC	90 91
Backlight						
R455	BLC_CTL_PHY	BLC_PHASE			LED_DRIVER_GATE1	90
R462	BLC_CTL_PHY	BLC_PHASE			LED_DRIVER_GATE1_R	90
R459	BLC_CTL_PHY	BLC_PHASE			LED_DRIVER_GATE2	90
R459	BLC_CTL_PHY	BLC_PHASE			LED_DRIVER_GATE2_R	90
R460	BLC_CTL_PHY	BLC_PHASE			LED_DRIVER_GATE3	90
R458	BLC_CTL_PHY	BLC_PHASE			LED_DRIVER_GATE3_R	90
R455	BLC_CTL_PHY	BLC_CTL			LED_DRVR_CS_RC_1	90
R458	BLC_CTL_PHY	BLC_CTL			LED_DRVR_CS_RC_2	90
R459	BLC_CTL_PHY	BLC_CTL			LED_DRVR_CS_RC_3	90
R461	BLC_CTL_PHY	BLC_CTL			LED_DRIVER_CS1	90
R460	BLC_CTL_PHY	BLC_CTL			LED_DRIVER_CS2	90
R460	BLC_CTL_PHY	BLC_CTL			LED_DRIVER_CS3	90
R462	BLC_CTL_PHY	BLC_CTL			LED_DRVR_CS_C1	90
R462	BLC_CTL_PHY	BLC_CTL			LED_DRVR_CS_C2	90
R462	BLC_CTL_PHY	BLC_CTL			LED_DRVR_CS_C3	90
R463	BLC_CTL_PHY	BLC_HV	80V		LED_DRIVER_FDBK_R_1	90
R463	BLC_CTL_PHY	BLC_HV	80V		LED_DRIVER_FDBK_R_2	90
R463	BLC_CTL_PHY	BLC_HV	80V		LED_DRIVER_FDBK_R_3	90
R460	BLC_CTL_PHY	BLC_CTL	80V		LED_DRIVER_FDBK1	90
R463	BLC_CTL_PHY	BLC_CTL	80V		LED_DRIVER_FDBK2	90
R462	BLC_CTL_PHY	BLC_CTL	80V		LED_DRIVER_FDBK3	90
R465	BLC_CTL_PHY	BLC_CTL			BLC_PWM_1_R	89 90
R466	BLC_CTL_PHY	BLC_CTL			BLC_PWM_2_R	89 90
R466	BLC_CTL_PHY	BLC_CTL			BLC_PWM_3_R	89 90
R466	BLC_CTL_PHY	BLC_CTL			BLC_PWM_1	90 91
R466	BLC_CTL_PHY	BLC_CTL			BLC_PWM_2	90 91
R466	BLC_CTL_PHY	BLC_CTL			BLC_PWM_3	90 91
R466	BLC_CTL_PHY	BLC_CTL			LED_DRIVER_REF1	90 91
R466	BLC_CTL_PHY	BLC_CTL			LED_DRIVER_REF2	90 91
R466	BLC_CTL_PHY	BLC_CTL			LED_DRIVER_REF3	90 91
R465	BLC_CTL_PHY	BLC_CTL			LED_DRIVER_COMP1	90
R465	BLC_CTL_PHY	BLC_CTL			LED_DRIVER_COMP2	90
R466	BLC_CTL_PHY	BLC_CTL			LED_DRIVER_COMP3	90
R466	BLC_CTL_PHY	BLC_CTL			BCOMP1	90
R466	BLC_CTL_PHY	BLC_CTL			BCOMP2	90
R466	BLC_CTL_PHY	BLC_CTL			BCOMP3	90
R466	BLC_CTL_PHY	BLC_CTL			LED_DRIVER_FLT1	90
R466	BLC_CTL_PHY	BLC_CTL			LED_DRIVER_FLT2	90
R466	BLC_CTL_PHY	BLC_CTL			LED_DRIVER_FLT3	90
R466	BLC_CTL_PHY	BLC_CTL			LED_FLT_R_1	90
R466	BLC_CTL_PHY	BLC_CTL			LED_FLT_R_2	90
R466	BLC_CTL_PHY	BLC_CTL			LED_FLT_R_3	90
R466	BLC_CTL_PHY	BLC_CTL			PRE_REG_OUT_R	89
R466	BLC_CTL_PHY	BLC_CTL			BOOST_FB	89
R466	BLC_CTL_PHY	BLC_CTL			BOOST_COMP	89
R466	BLC_CTL_PHY	BLC_CTL			BOOST_COMP_C	89
R466	BLC_CTL_PHY	BLC_CTL		TRUE	BOOST_GDRV	89
R466	BLC_CTL_PHY	BLC_CTL		TRUE	BOOST_GDRV_R	89
R466	BLC_CTL_PHY	BLC_CTL			BOOST_ISNS	89
R466	BLC_CTL_PHY	BLC_CTL			BOOST_ISNS_R	89
R459	BLC_CTL_PHY	BLC_HV	80V		LED_DRVR_DRAIN_1	90
R460	BLC_CTL_PHY	BLC_HV	80V		LED_DRVR_DRAIN_2	
R460	BLC_CTL_PHY	BLC_HV	80V		LED_DRVR_DRAIN_3	90
OUTPUT BUS						
R480	POWER_BLC_RET	BLC_HV	80V		IS1_BLC_F	91
R480	POWER_BLC_RET	BLC_HV	80V		IS2_BLC_F	91
R480	POWER_BLC_RET	BLC_HV	80V		IS3_BLC_F	91
R480	POWER_BLC_RET	BLC_HV	80V		IS1_BLC	90 91
R480	POWER_BLC_RET	BLC_HV	80V		IS2_BLC	90 91
R480	POWER_BLC_RET	BLC_HV	80V		IS3_BLC	90 91
R480	POWER_BLC_RET	BLC_HV	80V		BLC_LED_P_1	91
R480	POWER_BLC_RET	BLC_HV	80V		BLC_LED_N_1	91
R480	POWER_BLC_RET	BLC_HV	80V		BLC_LED_P_2	91
R480	POWER_BLC_RET	BLC_HV	80V		BLC_LED_N_2	91
R480	POWER_BLC_RET	BLC_HV	80V		BLC_LED_P_3	91
R480	POWER_BLC_RET	BLC_HV	80V		BLC_LED_N_3	91
R480	POWER_BLC	BLC_HV	80V		BLC_VOUT1	90 91
R480	POWER_BLC	BLC_HV	80V		BLC_VOUT2	90 91
R480	POWER_BLC	BLC_HV	80V		BLC_VOUT3	90 91

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GPU VREG CONSTRAINTS

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GPU VREG CONSTRAINTS

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Caesar IV (Ethernet/SD)

CIV-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
ENET_COMP_PHY	*	ENET_50S
ENET_DIFF_PHY	*	ENET_100D
SD_PHY	*	SD_50S

CIV-specific Spacing Definitions
Ethernet

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF_ISO	*	=6:1_SPACING	?
ENET_DIFF2DIFF	*	=3:1_SPACING	?
ENET_TRANS_ISO	*	1.27 MM	?
COMP_ENET_ISO	*	=4:1_SPACING	?

Constraints
Ethernet

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_DIFF	*	*	ENET_DIFF_ISO
ENET_DIFF	ENET_DIFF	*	ENET_DIFF2DIFF
ENET_TRANS	*	*	ENET_TRANS_ISO
COMP_ENET	*	*	COMP_ENET_ISO
ENET_TRANS	ENET_TRANS	*	ENET_DIFF2DIFF

2 kV isolation

SD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_ISO	*	=3:1_SPACING	?

SD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SD	*	*	SD_ISO

VENI, VIDI, VICI!

Electrical Constraint Set	Physical	Spacing	
Ethernet			
R496	ENET_MDI_NORM	ENET_DIFF_PHY	ENET_DIFF
R497	ENET_MDI_NORM	ENET_DIFF_PHY	ENET_DIFF
R498		ENET_DIFF_PHY	ENET_TRANS
R499		ENET_DIFF_PHY	ENET_TRANS
R500			ENET_TRANS
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AUTO-CONSTRAINTS PG 2

ENETCONN_*

Spacing	Netname
GENERIC ISO	ENETCONN_TCT

ENET_*

Physical	Spacing	Netname
	GENERIC ISO	ENET_ACT
	GENERIC ISO	ENET_ASF_GPIO
	GENERIC ISO	ENET_CLKREQ_L
	PM	ENET_CLKREQ_L_O
	PM	ENET_CR_1V8_EN
	PM	ENET_CR_1V8_EN_R
	PM	ENET_CR_3V3_EN_L
	PM	ENET_CR_3V3_EN_L_R
	PM	ENET_CR_PWREN
XDP_PHY	XDP	ENET_LOW_PWR
	PM	ENET_PWR_EN_L
	PM	ENET_PWR_EN_L_R
	PM	ENET_RESET_L
	PM	ENET_SD_RESET_L
	GENERIC ISO	ENET_SR_DISABLE
	GENERIC ISO	ENET_SR_LX
	PM	ENET_WAKE_L

FAN_*

Spacing	Netname
GENERIC ISO	FAN_0_PWM_FET
GENERIC ISO	FAN_0_PWM_FILT
GENERIC ISO	FAN_0_TACH_FET
GENERIC ISO	FAN_0_TACH_FILT

FBVDD_*

Spacing	Netname
GENERIC ISO	FBVDD_ALTV0

FB_*

Spacing	Netname
GENERIC ISO	FB_A0_VREFC
GENERIC ISO	FB_A0_VREFD
GENERIC ISO	FB_A1_VREFC
GENERIC ISO	FB_A1_VREFD
GENERIC ISO	FB_B0_VREFC
GENERIC ISO	FB_B0_VREFD
GENERIC ISO	FB_B1_VREFC
GENERIC ISO	FB_B1_VREFD
GENERIC ISO	FB_VREF

FET_*

Physical	Spacing	Voltage	Netname
	GENERIC ISO		FET_EN_P12V_S0
	GENERIC ISO		FET_EN_P12V_S0_BLC
	GENERIC ISO		FET_EN_P12V_S0_BLC_R
	GENERIC ISO		FET_EN_P12V_S0_R
	GENERIC ISO		FET_EN_P12V_S5
	GENERIC ISO		FET_EN_P12V_S5_R
	GENERIC ISO		FET_EN_VDDQ_S0
	GENERIC ISO		FET_HDD_SLGSW
POWER_PHY	POWER	12V	FET_VCC_U7950
POWER_PHY	POWER	12V	FET_VCC_U7970
POWER_PHY	POWER	12V	FET_VCC_U7980

FLAG_*

Spacing	Netname
GENERIC ISO	FLAG_V

G3_*

Spacing	Netname
GENERIC ISO	G3_POWERON_L

GND_*

Spacing	Netname
SENSE	GND_SMC_AVSS

GPU_*

Physical	Spacing	Netname
	GENERIC ISO	GPU_ALT_VREF
	GENERIC ISO	GPU_BUFRSTN
	GENERIC ISO	GPU_IFPAB_PLLVDD
	GENERIC ISO	GPU_IFPA_IOVDD
	GENERIC ISO	GPU_IFPB_IOVDD
	GENERIC ISO	GPU_IFPC_IOVDD
XDP_PHY	XDP	GPU_JTAG_TDI
XDP_PHY	XDP	GPU_JTAG_TDO
XDP_PHY	XDP	GPU_JTAG_TMS
XDP_PHY	XDP	GPU_JTAG_TRST_L
	GENERIC ISO	GPU_MLS_STRAP0
	GENERIC ISO	GPU_MLS_STRAP1
	GENERIC ISO	GPU_MLS_STRAP2
	GENERIC ISO	GPU_MLS_STRAP3
	GENERIC ISO	GPU_MLS_STRAP4
	GENERIC_ISO	GPU_RESET_L
SPI_50S	SPI	GPU_ROM_CS_L
SPI_50S	SPI	GPU_ROM_CS_L_R
SPI_50S	SPI	GPU_ROM_SI
SPI_50S	SPI	GPU_ROM_SI_R
SPI_50S	SPI	GPU_ROM_SO
SPI_50S	SPI	GPU_ROM_SO_R
SPI_50S	SPI	GPU_ROM_WP_L

HDD_*

Spacing	Netname
GENERIC ISO	HDD_12V_S0_GATE
GENERIC ISO	HDD_OOB_1V00_REF
PM	HDD_PWR_EN
GENERIC ISO	HDD_PWR_EN_L
GENERIC ISO	HDD_PWR_EN_R

I2C_*

Physical	Spacing	Netname
SMB_PHY	SMB	I2C_TCON_MAS_SCL
SMB_PHY	SMB	I2C_TCON_MAS_SDA

IFPD_*

Spacing	Netname
GENERIC ISO	IFPD_RSET

IFPEF_*

Spacing	Netname
GENERIC ISO	IFPEF_RSET

ISNSA_*

Electrical	Physical	Spacing	Netname
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VG3H_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VG3H_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_P1V05_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_P1V05_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_VCCSA_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_VCCSA_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_FBVDDQ_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_FBVDDQ_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_HDD_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_HDD_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_P1V05_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_P1V05_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V05S0_PCH_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V05S0_PCH_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V5S0_CPU_MEM_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V5S0_CPU_MEM_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S0_SSD_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S0_SSD_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S4_AP_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S4_AP_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P5VS0_HDD_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P5VS0_HDD_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_PVDDQS3_DDR_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_PVDDQS3_DDR_P

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E44

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E49E51I52X31

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	Physical	Spacing	Voltage	Netname
09	POWER_PHY	POWER	12V	PP3V3RHV_SW_TBTPWR 86
10	POWER_PHY	POWER	12V	PP3V3RHV_SW_TBTPWR 88

PP3V3_*				
	Physical	Spacing	Voltage	Netname
PP3V3_DMIC_CONN	POWER PHY	POWER	3.3V	42
PP3V3_G3H_AVREF_SMC	POWER PHY	POWER	3.3V	47 48
PP3V3_G3H_BT_FET	POWER PHY	POWER	3.3V	35
PP3V3_G3H_BT_FLT	POWER PHY	POWER	3.3V	35
PP3V3_G3H_SMC_USBMUX_R	POWER PHY	POWER	3.3V	45
PP3V3_G3H_SMC_VDDA	POWER PHY	POWER	3.3V	47
PP3V3_G3_RTC	POWER PHY	POWER	3.3V	6 26
PP3V3_GPU_IFPX_PL1VDD	POWER PHY	POWER	3.3V	81
PP3V3_PVDQOS3_ISNS	POWER PHY	POWER	3.3V	51
PP3V3_S0_PCH_VCCA_DAC_F	POWER PHY	POWER	3.3V	17 22
PP3V3_S0_SSD_FLT	POWER PHY	POWER	3.3V	44
PP3V3_S0_SW_SD_PWR	POWER PHY	POWER	3.3V	41
PP3V3_S4_ALS_F	POWER PHY	POWER	3.3V	42
PP3V3_S4_AP_FET	POWER PHY	POWER	3.3V	35
PP3V3_S4_AP_FLT	POWER PHY	POWER	3.3V	35
PP3V3_S4_ENET_FET_AVDDH	POWER PHY	POWER	3.3V	39
PP3V3_S4_ENET_FET_BIASVDDH	POWER PHY	POWER	3.3V	39
PP3V3_S4_ENET_FET_SRDVD	POWER PHY	POWER	3.3V	39
PP3V3_S4_ENET_FET_XTALVDDH	POWER PHY	POWER	3.3V	39
PP3V3_S4_USB_HUB_VDD	POWER PHY	POWER	3.3V	27
PP3V3_S5_XDP_R	POWER PHY	POWER	3.3V	25

PP4V5_*			
Physical	Spacing	Voltage	Netname
POWER PHY	POWER	4.5V	PP4V5_AUDIO_ANALOG

PP5V_*				
	Physical	Spacing	Voltage	Netname
PP5V	POWER_PHY	POWER	5V	PP5V_AUDIO_HPAMP
PP5V	POWER_PHY	POWER	5V	PP5V_S0_HDD_FET
PP5V	POWER_PHY	POWER	5V	PP5V_S0_PCH_V5REF
PP5V	POWER_PHY	POWER	5V	PP5V_S4_EXT_A
PP5V	POWER_PHY	POWER	5V	PP5V_S4_EXT_A_F
PP5V	POWER_PHY	POWER	5V	PP5V_S4_EXT_B
PP5V	POWER_PHY	POWER	5V	PP5V_S4_EXT_B_F
PP5V	POWER_PHY	POWER	5V	PP5V_S4_EXT_C
PP5V	POWER_PHY	POWER	5V	PP5V_S4_EXT_C_F
PP5V	POWER_PHY	POWER	5V	PP5V_S4_EXT_D
PP5V	POWER_PHY	POWER	5V	PP5V_S4_EXT_D_F
PP5V	POWER_PHY	POWER	5V	PP5V_S5_PCH_V5REFSUS

	Physical	Spacing	Voltage	Netname	
POWER	POWER_PHY	POWER	5V	PP5V_AUDIO_HPAMP	56 57
POWER	POWER_PHY	POWER	5V	PP5V_S0_HDD_FET	44 52
POWER	POWER_PHY	POWER	5V	PP5V_S0_PCH_V5REF	24
POWER	POWER_PHY	POWER	5V	PP5V_S4_EXT_A	45
POWER	POWER_PHY	POWER	5V	PP5V_S4_EXTB_F	45
POWER	POWER_PHY	POWER	5V	PP5V_S4_EXTB	45
POWER	POWER_PHY	POWER	5V	PP5V_S4_EXTB_F	45
POWER	POWER_PHY	POWER	5V	PP5V_S4_EXTC	46
POWER	POWER_PHY	POWER	5V	PP5V_S4_EXTC_F	46
POWER	POWER_PHY	POWER	5V	PP5V_S4_EXTD	46
POWER	POWER_PHY	POWER	5V	PP5V_S4_EXTD_F	46
POWER	POWER_PHY	POWER	5V	PP5V_S5_PCH_V5REFSUS	24

U12	POWER_PHY	POWER	5V	PP5V_S4_EXTD_F	46
U120	POWER_PHY	POWER	5V	PP5V_S5_PCH_V5REFSUS	24

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PPHV_*

	Physical	Spacing	Voltage	Netname	
18	POWER_PHY	POWER	12V	PPHV_SW_TBTAPWR	86
19	POWER_PHY	POWER	12V	PPHV_SW_TBTBPWR	88

S5_*

	Spacing	Netname	
183	PM	S5_PWRGD	47 65

SMCISNS_*

	Physical	Spacing	Netname	
148	SNS_PHY	SENSE	SMCISNS_CPUAXG	48 51
149	SNS_PHY	SENSE	SMCISNS_CPUCORE	48 51
150	SNS_PHY	SENSE	SMCISNS_GPUCORE	48 51
151	SNS_PHY	SENSE	SMCISNS_P12VG3H	48 51
152	SNS_PHY	SENSE	SMCISNS_P12VS0_CPU_P1V05	48 55
153	SNS_PHY	SENSE	SMCISNS_P12VS0_CPU_VCCSA	48 55
154	SNS_PHY	SENSE	SMCISNS_P12VS0_FBVDDQ	48 51
155	SNS_PHY	SENSE	SMCISNS_P12VS0_HDD	48 51
156	SNS_PHY	SENSE	SMCISNS_P12VS0_P1V05	48 55
157	SNS_PHY	SENSE	SMCISNS_P1V05S0_PCH	48 51
158	SNS_PHY	SENSE	SMCISNS_P1V5S0_CPU_MEM	48 51
159	SNS_PHY	SENSE	SMCISNS_P3V3S0_SSD	48 51
160	SNS_PHY	SENSE	SMCISNS_P3V3S4_AP	48 55
161	SNS_PHY	SENSE	SMCISNS_P5VS0_HDD	48 51
162	SNS_PHY	SENSE	SMCISNS_PVDDQS3_DDR	48 51

PPVBATT_*

	Physical	Spacing	Voltage	Netname	
18	POWER_PHY	POWER	3.3V	PPVBATT_G3_RTC	26
19	POWER_PHY	POWER	3.3V	PPVBATT_G3_RTC_R	26

SATALED_*

	Spacing	Netname	
183	GENERIC_ISO	SATALED_L	44

SDCONN_*

	Spacing	Netname	
183	GENERIC_ISO	SDCONN_DETECT	41
184	GENERIC_ISO	SDCONN_ILIM	41
185	GENERIC_ISO	SDCONN_OC_L	41

SMCVSNS_*

	Physical	Spacing	Netname	
162	SNS_PHY	SENSE	SMCVSNS_CPUAXG	48 51
163	SNS_PHY	SENSE	SMCVSNS_CPUCORE	48 51
164	SNS_PHY	SENSE	SMCVSNS_GPUCORE	48 51
165	SNS_PHY	SENSE	SMCVSNS_P12VG3H	48 51
166	SNS_PHY	SENSE	SMCVSNS_P1V05S0_PCH	48 51
167	SNS_PHY	SENSE	SMCVSNS_P1V5S0_CPU_MEM	48 51
168	SNS_PHY	SENSE	SMCVSNS_P3V3S0	48 51
169	SNS_PHY	SENSE	SMCVSNS_P5VS0_HDD	48 51
170	SNS_PHY	SENSE	SMCVSNS_PVDDQS3_DDR	48 51

PU_*

	Spacing	Netname	
18	GENERIC_ISO	PU_U6900	64

SD_*

	Spacing	Netname	
183	GENERIC_ISO	SD_DETECT_LVL	41

PWR_*

	Spacing	Netname	
18	PM	PWR_BTN	6 48
19	PM	PWR_BTN_R	6

SLG_*

	Spacing	Netname	
183	PM	SLG_ENET_RESET_L	41
184	PM	SLG_ENET_RESET_R_L	41

SMC_*

	Physical	Spacing	Netname	
187		GENERIC_ISO	SMC_ACDC_ID	6 48
188		GENERIC_ISO	SMC_ACDC_ID_R	6
189		PM	SMC_ASSERT_RTCRST	48
190		GENERIC_ISO	SMC_BLC_MUX_RX_L	48 89
191		GENERIC_ISO	SMC_BLC_MUX_TX_L	48 89
192		GENERIC_ISO	SMC_CPU_CATERR_L	47 48
193	CPU_PHY	CPU	SMC_CPU_PECI	48
194		PM	SMC_DELAYED_PWRGD	38 47 48 65
195		GENERIC_ISO	SMC_DP_HPD_L	47 48
196	CLK_XTAL	XTAL	SMC_EXTAL	47 48
197		GENERIC_ISO	SMC_FAN_0_CTL	47 54
198		GENERIC_ISO	SMC_FAN_0_TACH	47 54
199		GENERIC_ISO	SMC_GFX_OVERTEMP	47 48 82
200		GENERIC_ISO	SMC_GFX_OVERTEMP_Q	82
201		GENERIC_ISO	SMC_GFX_OVERTEMP_R_L	82
202		PM	SMC_GFX_THROTTLE_L	47 82
203		PM	SMC_GFX_THROTTLE_R_L	82
204		PM	SMC_LRESET_L	26 47
205		PM	SMC_MANUAL_RST_L	48
206		PM	SMC_ONOFF_L	47 48
207		SENSE	SMC_OOB1_RX_CN	44
208		SENSE	SMC_OOB1_RX_FILT	44 52
209		SENSE	SMC_OOB1_RX_L	47 52
210		SENSE	SMC_OOB1_RX_R	52
211		SENSE	SMC_OOB1_TX_L	47 48
212		SENSE	SMC_OOB2_RX_L	44 48
213		SENSE	SMC_OOB2_TX_L	44 48
214	CPU_PHY	CPU	SMC_PECI_L	47 48
215	CPU_PHY	CPU	SMC_PECI_L_R	48
216		PM	SMC_PME_S4_WAKE_L	35 47 48
217		GENERIC_ISO	SMC_PM_G2_EN	47 48 74
218		PM	SMC_PM_PCH_SYS_PWROK	47 48
219		CPU	SMC_PROCHOT	47 48
220		PM	SMC_RESET_L	47 48 49

REG_*

	Physical	Spacing	Netname	
18		PM	REG_CPUAXG_PGOOD	5 66
19		PM	REG_CPUCORE_PGOOD	66
190	CPU_PHY	CPU	REG_CPUCORE_VRHOT_L	66
191		PM	REG_CPU_P1V05S0_PGOOD	69
192	VR_CTL_PHY	VR_CTL	REG_FBVDDQ_SET0	95
193	VR_CTL_PHY	VR_CTL	REG_FBVDDQ_SET1	95
194	VR_CTL_PHY	VR_CTL	REG_FBVDDQ_SET1_R	95
195	VR_CTL_PHY	VR_CTL	REG_FBVDDQ_SREF	95
196		PM	REG_P1V8S0_PGOOD	72
197		PM	REG_P3V3S5_PGOOD	71
198		PM	REG_P5VS4_PGOOD	71
199		PM	REG_VCCSA0_PGOOD	70
200		PM	REG_VDDQS3_PGOOD	72

SMBUS_*

	Physical	Spacing	Netname	
183	SMB_PHY	SMB	SMBUS_PCH_CLK	18 50
184	SMB_PHY	SMB	SMBUS_PCH_CLK_R	50
185	SMB_PHY	SMB	SMBUS_PCH_DATA	18 50
186	SMB_PHY	SMB	SMBUS_PCH_DATA_R	50
187	SMB_PHY	SMB	SMBUS_SMC_0_S0_SCL	47 50
188	SMB_PHY	SMB	SMBUS_SMC_0_S0_SDA	47 50
189	SMB_PHY	SMB	SMBUS_SMC_1_S0_SCL	47 50
190	SMB_PHY	SMB	SMBUS_SMC_1_S0_SDA	47 50
191	SMB_PHY	SMB	SMBUS_SMC_2_S4_SCL	47 50
192	SMB_PHY	SMB	SMBUS_SMC_2_S4_SDA	47 50
193	SMB_PHY	SMB	SMBUS_SMC_3_SCL	47 50
194	SMB_PHY	SMB	SMBUS_SMC_3_SDA	47 50
195	SMB_PHY	SMB	SMBUS_SMC_4_ASF_SCL	47 48 50
196	SMB_PHY	SMB	SMBUS_SMC_4_ASF_SDA	47 48 50
197	SMB_PHY	SMB	SMBUS_SMC_5_G3H_SCL	47 48 50
198	SMB_PHY	SMB	SMBUS_SMC_5_G3H_SDA	47 48 50

RTC_*

	Spacing	Netname	
183	PM	RTC_RESET_L	18 26 48
184	PM	RTC_RESET_L_R	48

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F17	SMB_PHY	SMB	SML_PCH_0_CLK	18 50
F18	SMB_PHY	SMB	SML_PCH_0_DATA	18 50
F19	SMB_PHY	SMB	SML_PCH_1_CLK	18 50
F20	SMB_PHY	SMB	SML_PCH_1_DATA	18 50

C

121	GENERIC ISO	SPI_DESCRIPTOR_OVERRIDE L	15 47
SSD_*			
Spacing		Netname	
122	GENERIC ISO	SSD_CLKREQ L	15 18

B

	Spacing	Netname
122	GENERIC ISO	SSD CLKREQ L 15 18

A

Spacing	Netname
PM	SYS PWROK R

123 65

A

	Physical	Spacing	Voltage	Netname	
#24		GENERIC ISO	3.3V	TBT_A_BIAS	86
#25	TBT_GEN_55S	TBT_GEN		TBT_A_CONFIG1_RC	86
#26		PM		TBT_A_HV_EN	36 86
#27		GENERIC ISO	3.3V	TBT_B_BIAS	88
#28	TBT_GEN_55S	TBT_GEN		TBT_B_CONFIG1_RC	88
#29		PM		TBT_B_HV_EN	36 88
#30		GENERIC ISO		TBT_CLKREQ_ISOL_L	
#31		GENERIC ISO		TBT_CLKREQ_L	15 38
#32		GENERIC ISO		TBT_DDC_XBAR_EN_L	38
#33		GENERIC ISO		TBT_EN_CIO_PWR	38
#34		PM		TBT_EN_CIO_PWR_L	36 38
#35		GENERIC ISO		TBT_EN_LC_ISOL	38
#36		GENERIC ISO		TBT_EN_LC_PWR	36 38
#37		GENERIC ISO		TBT_PCH_CLKREQ_L	15 21
#38		PM		TBT_PWR_EN	15 26
#39		PM		TBT_PWR_EN_PCH	18 26
#40		PM		TBT_PWR_ON_POC_RST_L	36 38
#41		PM		TBT_PWR_REQ_L	15 20
#42		PM		TBT_S0_EN	64
#43		PM		TBT_SW_RESET_L	21 38

	Electrical	Physical	Spacing	Netname	
R14	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_1_N	5.3
R15	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_1_P	5.3
R16	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_2_N	5.3
R17	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_2_P	5.3
R18	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_3_N	5.3
R19	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_1_3_P	5.3
R20	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_1_N	5.3
R21	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_1_P	5.3
R22	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_2_N	5.3
R23	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_2_P	5.3
R24	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_3_N	5.3
R25	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_3_P	5.3
R26	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_4_N	5.3
R27	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_4_P	5.3
R28	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_5_N	5.3
R29	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_5_P	5.3
R30	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_6_N	5.3
R31	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_6_P	5.3
R32	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_7_N	5.3
R33	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_2_7_P	5.3
R34	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_ACDC_N	6 5.3
R35	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_ACDC_P	6 5.3
R36	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_SKIN_N	5.3
R37	SNS_TEMP	SNS_DIFF_PHY	SENSE	TSNS_SKIN_P	5.3

	Spacing	Netname
U00	PM	UVP_IN_1
U05	PM	UVP_IN_1_REF
U10	PM	UVP_IN_2
U15	PM	UVP_IN_3
U20	PM	UVP_IN_4
U25	PM	UVP_REF

	Spacing	Netname
U00	PM	UVP_IN_1
U05	PM	UVP_IN_1_REF
U10	PM	UVP_IN_2
U15	PM	UVP_IN_3
U20	PM	UVP_IN_4
U25	PM	UVP_REF

	Spacing	Netname
E74	GENERIC_ISO	VREFMRGN_CA_SODIMMA_EN
E75	GENERIC_ISO	VREFMRGN_CA_SODIMMB_EN

	Spacing	Netname
E76	GENERIC ISO	VTTCLAMP EN
R77	GENERIC ISO	VTTCLAMP L

	Spacing	Netname
E78	GENERIC ISO	WOL EN